ओएनजीसी ONGC	**ENGINEERING SECTION – ENGINEERING & PROJECTS EXCELLENCE CENTRE**	**ADDENDUM**	**DATE**	23.04.2026
			PROJECT NAME:	BRRP
			PAGE	**1 OF 1**

Reference No: Bridge Refurbishment & Replacement Project of MH Asset (BRRP), Tender No.: ZW1PC26002

Sl.No.	Vol.No./ Spec. No.	Modification
1	Technical SCC	Revised Technical SCC is attached.
2	A-3 Proforma for Price Schedule	Revised A3 attached.
3	Suggested Vendor list	Latest SVL attached (Rev 12.3)
4	Electrical, Piping, Structure & Instrumentation drawings and Specification	Documents pertaining to Electrical, piping, Structure and Instrumentation section are added.

ANNEXURE - 1: TENTATIVE DETAILED ENGINEERING DELIVERABLE

ANALYSIS REQUIREMENT OF BRIDGE, DECK						Design Basis & Construction Methodology	Reports	AFC Drawings	Fabrication Drawing	As-Built Drawings	Tender Drawing (Strenthening only for reference)
Sr. No	**BRIDGE / PLATFORM**	**Surveys**	**New / Replace / Strenthening**	**Analysis Type**	**Remarks**						
1	ICW-ICG UD	1. Pre-Engineering Survey reports for all bridges, platforms. 2. NDT survey reports for all platforms. 3. Laser scanning Survey reports, if any, performed by contractor. 4. Pre-Construction Survey reports for all bridges, platforms. 5. Post-Construction Survey reports for all bridges, platforms.	Strenthening	Global Bridge	Bridge	1. Design basis to be prepared as per ONGC design criteria attacthed in the tender for approval 2. Construction / Installation Methodology for each bridge, landing strenthening shall be submitted seperately for approval	Approval	Approval	Records	Records	Refer Annexure -2 with TQ reply.
2	ICW-ICG LD		New	Global Bridge, Lift, transportation	Bridge		Approval	Approval	Records	Records	
3	ICG-ICP UD		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
4	ICG-ICP LD		New	Global Bridge, Lift, transportation	Bridge		Approval	Approval	Records	Records	
5	BHS-SLQ		Replacement	Global Bridge, Lift, transportation, Demolition & Removal	Bridge		Approval	Approval	Records	Records	
6	SLQ-WIS		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
7	NQO-NQD		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
8	WIN-NC		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
9	SCA-SC1		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
10	SCA-SCF		Strenthening	Global Bridge	Bridge		Approval	Approval	Records	Records	
11	ICP Platform		Strenthening	Global Deck Inplace	Landing and Global Deck		Approval	Approval	Records	Records	
12	ICG Platform		Strenthening	Global Deck Inplace	Landing and Global Deck		Approval	Approval	Records	Records	
13	ICW Platform		Strenthening	Global Deck Inplace	Landing and Global Deck		Approval	Approval	Records	Records	
14	BHS Platform		Strenthening	Global Deck Inplace	Landing and Global Deck		Approval	Approval	Records	Records	
15	SLQ Platform		Strenthening	Global Deck Inplace	Landing and Global Deck		Approval	Approval	Records	Records	
16	WIS Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
17	NOQ Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
18	NQD Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
19	WIN Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
20	NC Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
21	SC1 Platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
22	SCA platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	
23	SCF platform		Strenthening	Deck Local Inplace	Landing		Approval	Approval	Records	Records	

Note: The provided list is not complete. It is contractors responsibility to develop and provide all documentation, drawings, technical notes, sketches, etc required to complete the works.

Sr. No	DRAWING NUMBER	Rev No	DESCRIPTION	BRIDGE / PLATFORM	No. OF SHEETS
1	B774-000-83-43-0001	0	General Notes	Common	2
2	B774-001-83-43-0001	0	ICW-ICG Upper Deck drawing – Strengthening and modifications.	ICW-ICG Upper Deck	
3	B774-002-83-43-0001	A	ICP-ICG Upper Deck drawing – Strengthening and modifications.	ICP-ICG Upper Deck	
4	B774-002-83-43-0002	A	ICP-ICG Upper Deck drawing – Strengthening and modifications.	ICP-ICG Upper Deck	
5	B774-003-83-43-0001	0	NQO-NQD Upper Deck drawing – Strengthening and modifications.	WIS-SLQ Lower deck Bridge	
6	B774-003-83-43-0002	0	WIS Landing Area - Modification	WIS Platform	
7	B774-004-83-43-0001	0	WIS-SLQ Upper Deck drawing – Strengthening and modifications.	NQO-NQD Bridge	
8	B774-004-83-43-0002	A	NQO Landing Area - Modification	NQO Platform	
9	B774-004-83-43-0003	A	NQO Landing Area - Modification	NQO Platform	
10	B774-004-83-43-0004	A	NQD Landing Area - Modification	NQD Platform	
11	B774-004-83-43-0005	A	NQD Landing Area - Modification	NQD Platform	
12	B774-005-83-43-0001	0	WIN-NC Bridge drawing – Strengthening and modifications.	WIN-NC Bridge	
13	B774-005-83-43-0002	0	WIN Landing Area - Modification	WIN Platform	
14	B774-005-83-43-0003	0	WIN Landing Area - Modification	WIN Platform	
15	B774-005-83-43-0004	0	NC Landing Area -	NC Platform	

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			Modification		
16	B774-007-83-43-0001	0	SC1-SCA Bridge drawing – Strengthening and modifications.	SC1-SCA Bridge	
17	B774-007-83-43-0002	A	SC1 Landing Area - Modification	SC1 Platform	
18	B774-007-83-43-0003	A	SCA Landing Area - Modification	SCA Platform	
19	B774-008-83-43-0001	0	SCA-SCF Bridge drawing – Strengthening and modifications.	SC1-SCA Bridge	
20	B774-008-83-43-0003	0	SCF Flare boom, Tripod Modification	SCF Platform	
21	B774-101-83-43-0001	0	ICW-ICG Lower Deck bridge drawings	ICW-ICG Lower Deck	
22	B774-101-83-43-0002	A	ICW Landing Area - Modification	ICW Platform	
23	B774-101-83-43-0003	A	ICW Landing Area - New Extensions	ICW Platform	
24	B774-101-83-43-0004	A	ICG Landing Area - Modification	ICG Platform	
25	B774-102-83-43-0001	A	ICP-ICG Lower Deck bridge drawings	ICP-ICG Lower Deck	2
26	B774-102-83-43-0002	A	ICP Landing Area - Modification	ICP Platform	
27	B774-102-83-43-0003	A	ICP Landing Area - New Extensions	ICP Platform	
28	B774-102-83-43-0004	A	ICG Landing Area - Modification	ICG Platform	
29	B774-103-83-43-0001	0	BHS-SLQ Lower Deck bridge drawings	BHS-SLQ Lower Deck (New)	
30	B774-103-83-43-0002	A	BHS Landing Area - Modification	BHS Platform	
31	B774-103-83-43-0003	A	BHS Landing Area - New Extensions	BHS Platform	
32	B774-103-83-43-0004	0	SLQ Landing Area - Modification	SLQ Platform	
33	B774-103-83-43-0005	0	SLQ Landing Area -	SLQ Platform	

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			New Extensions		
34	B774-103-83-43-0006	0	BHS – SLQ Bridge Demolition Drawings	BHS-SLQ Lower Deck (Existing)	

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B774-000-83-43-0001

S.NO.	GENERAL
1.	THE NOTES CONTAINED IN THIS DRAWING PERTAIN TO ALL DRAWINGS LISTED UNDER DECK DRAWINGS AND MODULE DRAWINGS IN DRAWING INDEX. IF A CONFLICT OCCURS BETWEEN ANY OF THESE NOTES AND THE DRAWINGS , THE NOTES GIVEN IN INDIVIDUAL DRAWINGS SHALL GOVERN.
2.	ALL MAIN STRUCTURAL MEMBERS INDICATED ON THESE DRAWINGS HAVE BEEN DETERMINED FROM DESIGN WORK PERFORMED IN CONFORMANCE WITH THE FOLLOWING: a) STRUCTURAL DESIGN CRITERIA DESCRIBED IN BID PACKAGE. b) LATEST EDITION OF API RECOMMENDED PRACTICE 2A WSD FOR PLANNING, DESIGNING AND CONSTRUCTING FIXED OFFSHORE PLATFORMS. c) LATEST EDITION OF AISC SPECIFICATION FOR THE DESIGN, FABRICATION AND ERECTION OF STRL. STEEL FOR MODULES. d) CRITERIA FOR EARTHQUAKE DESIGN AS PER IS:1893/API RP 2A/API RP 2EQ e) API RECOMMENDED PRACTICE 2L FOR PLANNING, DESIGNING AND CONSTRUCTING HELIPORTS FOR FIXED OFFSHORE PLATFORMS AND ICAO GUIDELINES. f) OTHER SPECIFIC REQUIREMENTS OF THE BID PACKAGE.
3.	ALL DIMENSIONS SHOWN ON THESE DRAWINGS ARE IN MILLIMETRES AND ALL LEVELS SHOWN ARE IN METRES W.R.T CHART DATUM.
4.	FOR TUBULAR SECTIONS, UB SECTIONS, INDIAN ROLLED SECTIONS AND PLATES, SIZES ARE SHOWN IN MM AND FOR W SHAPE AND AMERICAN STANDARD CHANNELS, THE SIZES ARE SHOWN IN INCHES.
5.	THE FOLLOWING ABBREVIATIONS HAVE BEEN USED IN THESE DRAWINGS: C/C : CENTER TO CENTER COND : CONDUCTOR CHQD. : CHEQUERED ℄ / CL : CENTER LINE COL : COLUMN DET : DETAIL DIA : DIAMETER DIAPH : DIAPHRAGM DK : DECK DWG/DRG : DRAWING EA : EACH EL : ELEVATION EQ : EQUAL EXT : EXTERIOR EXTN : EXTENSIONS FL : FLANGE FS : FAR SIDE GALV : GALVANISED GRAT : GRATING GS : GENERAL SPECIFICATIONS HD : HEAD HR : HANDRAIL HXG : HEXAGONAL ICAO : INTERNATIONAL CIVIL AVIATION ORGANIZATION INT : INTERNAL ISMB : BEAM SECTION AS PER IS ISMC : CHANNEL SECTION AS PER IS JKT : JACKET KSI : KIPS PER SQ. INCH L : ANGLE MIN : MINIMUM MECH : MECHANICAL MM : MILLIMETRES Mpa : MEGA PASCAL NB : NOMINAL BORE NOS. : NUMBERS NS : NEAR SIDE NTS : NOT TO SCALE ø : OUTER DIAMETER O/O : OUT TO OUT OPP : OPPOSITE PC : PIECE ℗ : PLATE PL : PLATE RAD : RADIUS REF : REFER REQD : REQUIRED SCH : SCHEDULE SEC : SECTION SHT/SH : SHEET SHW : SHOWN SPEC : SPECIFICATION SPCS : SPACES STIFF : STIFFENER STRL : STRUCTURAL STD : STANDARD THK/TH : THICK T.O.G. : TOP OF GRATING T.O.P. : TOP OF PLATE T.O.S. : TOP OF STEEL TYP : TYPICAL UB : UNIVERSAL BEAMS UNO : UNLESS NOTED OTHERWISE W : WIDE FLANGE W.P. : WORK POINT WSD : WORKING STRESS DESIGN

S.NO.	MATERIAL
1.	ALL MATERIALS INCLUDING STEEL, CEMENT AND TIMBER SHALL BE IN ACCORDANCE WITH MATERIAL SPEC. NO. 6-60-0011. i.e.STANDARD SPECIFICATION FOR STRUCTURAL MATERIALS
2.	HELIDECK PLATING SHALL BE AS PER SPECIFICATION 6-60-0011. i.e.STANDARD SPECIFICATION FOR STRUCTURAL MATERIALS
3.	EXCEPT WHERE SPECIFICALLY MENTIONED OTHERWISE IN THE DRAWINGS, THE FOLLOWING MARKINGS HAVE BEEN USED TO IDENTIFY THE GRADE OF STEEL MATERIAL AS PER MATERIAL SPEC. NO. 6-60-0011 i.e.STANDARD SPECIFICATION FOR STRUCTURAL MATERIALS I. FOR TUBULAR AND PLATES: a) TUBULARS USED IN JOINT CANS (25MM TH. AND ABOVE) FOR DECK LEGS AND CRANE PEDESTAL SHALL CONFIRM TO API SPEC. 2H (MIN. Fy =345 Mpa) OR EQUIVALENT (INTERMEDIATE STRENGTH STEEL WITH THROUGH THICKNESS PROPERTY) b) TUBULAR UPTO 406 mm DIA AND HANDRAIL SHALL BE SEAMLESS AS PER API SPEC 5L Grade B (MIN. Fy =240 Mpa.) OR EQUIVALENT. c) TUBULARS 457 mm DIA AND ABOVE TO BE USED IN IN DECK LEGS/ BRACES INCLUDING CRANE PEDESTAL AND JOINT CANS LESS THAN 25MM THICK SHALL BE ROLLED FROM PLATES AS PER API SPEC 2H Grade 50 (MIN. Fy =345 Mpa.) OR EQUIVALENT. II. FOR BEAMS: a) WIDE FLANGE BEAMS OF SIZE W 12 AND ABOVE SHALL CONFORM TO ASTM A 572 Grade 50 OR EQUIVALENT. b) WIDE FLANGE BEAMS OF SIZE LESS THEN W 12 SHALL CONFIRM TO ASTM A36 OR EQUIVALENT. c) ALL INDIAN SECTIONS SHALL BE OF MATERIAL CONFORMING TO IS:2062 Grade B.
4.	MATERIAL GRADE FOR ALL BEAM EXTENSION PLATES, WEB STIFFENERS BOX PLATES SHALL BE OF THE SAME GRADE WITH MATCHING PROPERTIES AS THE PARENT MATERIAL TO WHICH THEY ARE CONNECTED.
5.	ALL HANDRAILS & SOCKETS SHALL BE AS PER SPEC. NO. 6-60-0011 AND STRUCTURAL DESIGN BASIS.
6.	ALL GRATING AND STAIR TREADS SHALL BE GALVANISED 30 MM SERRATED BAR GRATING AND DETAILS AS PER SPEC. NO. 6-60-0011 i.e.STANDARD SPECIFICATION FOR STRUCTURAL MATERIALS
7.	ALL BOLTS SHALL CONFORM TO ASTM A193 Grade B7 OR EQUIVALENT AND NUTS SHALL CONFORM TO ASTM A194 Grade 2H AND SHALL BE HOT DIPPED GALVANISED.

S.NO.	FABRICATION AND WELDING
1.	ALL FABRICATION SHALL CONFORM TO THE FABRICATION SPEC NO. 6-60-0013 i.e.STANDARD SPECIFICATION FOR FABRICATION AND ASSEMBLY OF OFFSHORE STRUCTURES/COMPONENTS.
2.	ONLY FIGURED DIMENSIONS SHALL BE FOLLOWED. DIMENSIONS NOT SHOWN SHALL BE COMPUTED BUT NOT SCALED FROM THE DRAWINGS.
3.	WELDING AND WELD INSPECTION SHALL BE DONE IN ACCORDANCE WITH SPECIFICATION AND PROCEDURE.
4.	ALL MEMBERS SHALL INTERSECT AT A COMMON CENTER LINE WORK POINT (W.P.) AT ANY GIVEN JOINT UNLESS SHOWN OTHERWISE.
5.	ALL DECK BEAMS SHALL BE FLUSH AT TOP OF STEEL (T.O.S.) ELEVATION UNLESS OTHERWISE NOTED.
6.	DECK BEAM SPLICES IF USED, SHALL BE STAGGERED SO THAT THE JOINTS IN ANY TWO CONSECUTIVE BEAMS DO NOT OCCUR IN THE SAME LINE.
7.	UNLESS NOTED OTHERWISE THE FOLLOWING CONNECTIONS SHALL BE OF FULL PENETRATION WELDS AND WELDED ALL AROUND. a) ALL BEAM AND PIPE SPLICES. b) ALL BEAM FLANGE EXTENSIONS. c) ALL TRUSS MEMBER CONNECTIONS. d) ALL FRAMING BEAM CONNECTIONS. e) ALL OFFSHORE JOINTS. f) ALL TUBULAR CONNECTIONS. ALL OTHER CONNECTIONS SHALL BE FULL STRENGTH WELD (U.N.O)
8.	MINIMUM SIZE OF FILLET WELDS SHALL BE 6 MM UNLESS NOTED OTHERWISE.
9.	DECK PLATING SHALL BE CONTINUOUSLY SEAL WELDED TO THE DECK BEAM FLANGES AND TUBULARS. SEAL WELDING SIZE SHALL BE 4MM.
10.	COAMING ANGLES SHALL BE CONTINUOUSLY SEAL WELDED TO THE DECK PLATING.
11.	COPE DECK PLATING AND SEAL WELD ALL AROUND AT INTERSECTION OF STRUCTURAL MEMBERS.
12.	WALL PLATING SHALL BE CONTINUOUSLY SEAL WELDED TO THE SUPPORTS.
13.	UNLESS NOTED OTHERWISE ALL GRATINGS SHALL BE STITCH WELDED TO SUPPORTING BEAMS.
14.	ALL STITCH WELD SHALL BE MINIMUM 75mm LENGTH.
15.	ALL STRUCTURAL MEMBERS INCLUDING STIFFENERS SHALL BE CUT TO SUIT AND COPE TO FIT UNLESS OTHERWISE SHOWN.
16.	ALL REMOVABLE OR OFFSHORE INSTALLED MEMBERS SHALL BE TEST FITTED AT YARD BEFORE LOAD OUT AND SHALL BE REMOVED AFTER TEST FITTING & MARKED AND DESPATCHED SEPARATELY.
17.	STRUCTURE SHALL BE PAINTED AS PER TENDER SPECIFICATION.
18.	PERIMETERS AROUND CUTOUTS AND PENETRATIONS THROUGH FLOOR CHEQUERED PLATES AND GRATING TO BE REINFORCED.
19.	HOLES PROVIDED IN HANDRAILS DURING GALVANISATION SHALL BE SEALED BY AN APPROVED METHOD AFTER INSTALLATION.
20.	FOR WELDING REFER TENDER SPECIFICATION.
21.	ALL HANDRAIL, SOCKET, GRATING, STAIR TREADS SHALL BE HOT DIP GALVANISED AS PER SPECIFICATION 7-60-801 AND 7-60-802.
22.	FOR TYPICAL WELDING DETAILS REFER STD. NO. 7-60-0807. i.e. DECK WELDING DETAILS.

S.NO.	TRANSPORTATION AND INSTALLATION
1.	ALL COMPONENTS OF THE STRUCTURES SHALL BE PLANNED ASSUMING THE AVAILABILITY OF THE FOLLOWING MINIMUM CONSTRUCTION EQUIPMENT. a) MARINE SPREAD INCLUDING DERRICK BARGE WITH ADEQUATE CRANE CAPACITY. b) CARGO BARGES OF ADEQUATE SIZE/ STABILITY/STRENGTH.
2.	LIFTING EYE SHOWN IN THE DRAWINGS ARE TENTATIVE AND ONLY INDICATE THE PREFERRED SYSTEM.
3.	VERIFICATION OF ALL LIFTING EYES, SPREDER FRAMES LAUNCHING LUG ETC.SHALL BE CARRIED OUT BASED ON APPROVED PROCEDURES FOR FABRICATION/ LOADOUT/ TRANSPORTATION/ INSTALLATION AND STRUCTURAL DESIGN BASIS.
4.	ALL TEMPORARY MEMBERS/COMPONENTS, IF REQUIRED, TO FACILITATE FABRICATION, LOAD-OUT TRANSPORTATION AND INSTALLATION OF THE STRUCTURES SHALL BE PROVIDED AS PER DESIGN.
5.	INSTALLATION AIDS, SUCH AS COUNTER WEIGHTS AND BUMPER GUIDES RIGGING PLATFORMS AND OTHER TEMPORARY MEMBERS NOT INCLUDED IN THE DRAWINGS, SHALL BE DESIGNED AND PROVIDED AS PER REQUIREMENTS.
6.	TEMPORARY TIE DOWN & LIFT BRACES/ COMPONENTS, IF REQUIRED TO FACILITATE FABRICATION, LOADOUT, TRANSPORTION AND INSTALLATION SHALL BE PROVIDED IN ACCORDANCE WITH DESIGN AND SPECIFICATIONS.
7.	PERMANENT MEMBERS SHALL BE CLEANED OF WELD SPLATTER AND TOUCH UP PAINTED AFTER REMOVAL OF ANY TEMPORARY BRACE/COMPONENTS.
8.	DAMAGE OF ANY KIND TO THE PERMANENT STRUCTURES BEFORE AND DURING INSTALLATION AND PRIOR TO HANDOVER SHALL BE COMPLETELY REPAIRED BY AN APPROVED METHOD AND TO THE FULL SATISFACTION OF THE COMPANY.
9.	ALL MEMBERS MARKED "FIELD INSTALLED" SHALL BE INSTALLED "OFFSHORE" AFTER MAKING NECESSARY ADJUSTMENT IN LEVELS, ORIENTATION ETC. TO SUIT SITE CONDITION.

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES:

REV.	DATE	REVISIONS	BY	CHKD	APPD.	EE/PC
0	29.01.26	ISSUED FOR TENDER	MUNISH	AA	DP	

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OIL AND NATURAL GAS CORPORATION LIMITED

CONSULTANCY SERVICES FOR ENGINEERING DOCUMENT GENERATION FOR WELL PLATFORMS

GENERAL NOTES FOR DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT	DWG. NO.	REV.
1:1	B774	000	83	43	0001	0

Sht 01 of 02

3-1641-0501 REV.2 A1-841 x 594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और इन्हें इस पूर्वोक्त किसी अन्य प्रयोजन के अलावा इनका कोई अन्य प्रयोग उधार दिए जाएँगे, न प्रतिलिपि किए जाएँगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप से दी गई सहमति से होगा ।
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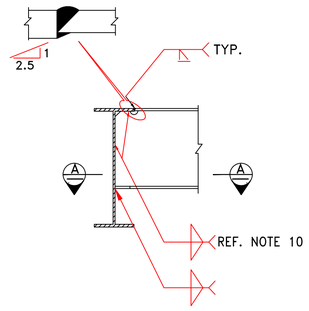
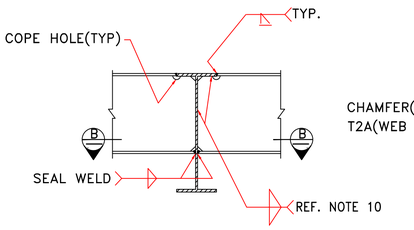
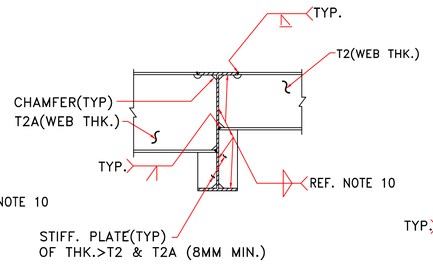
B818-002-83-11161



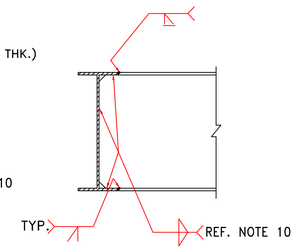
TYP. WELDING DETAIL AT END SUPPORTS OF SECONDARY BEAMS



TYP. WELDING DETAIL OF TWO EQUAL SECONDARY BEAMS

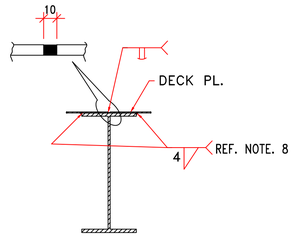


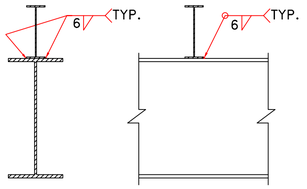
TYP. WELDING DETAIL OF UNEQUAL SECONDARY BEAMS



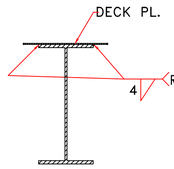
TYP. WELDING DETAIL OF BEAMS OF EQUAL SIZE
(WHEN THERE IS NO TUBULAR CONNECTED AT THE JOINT)

TYP. WELDING DETAILS OF SECONDARY BEAM

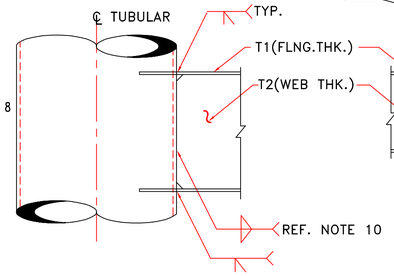




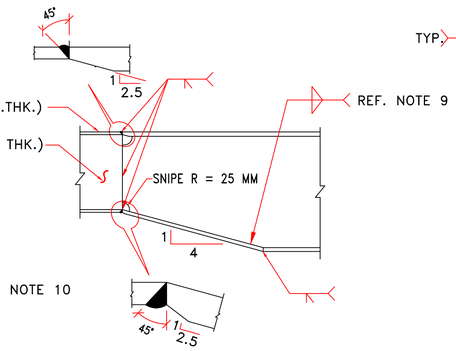
CONNECTION DETAIL OF SECONDARY BEAM TO MAIN BEAM



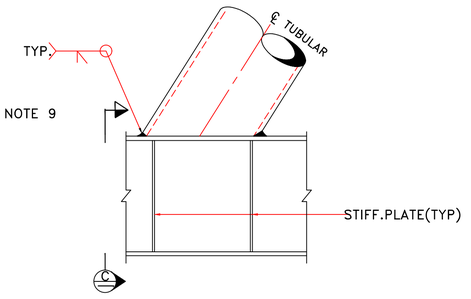
TYP. WELDING DETAIL OF DECK PLATE TO BEAM



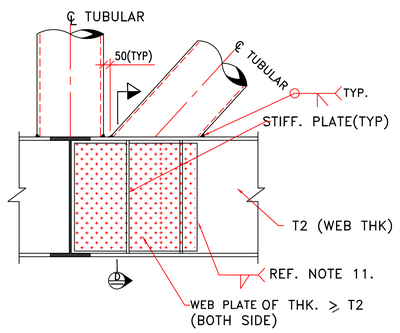
TYP. WELDING DETAIL OF BEAM TO TUBULAR



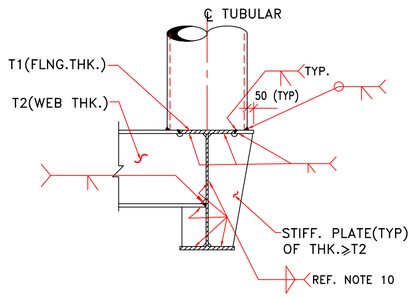
TYP. WELDING DETAIL OF SPLICE FOR BEAMS OF VARYING DEPTH



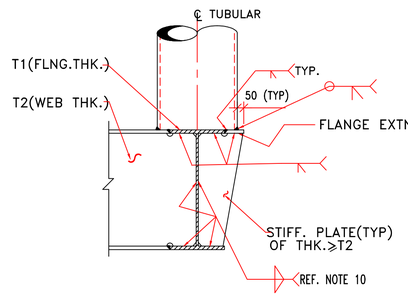
TYP. WELDING DETAIL OF TUBULAR ON BEAM



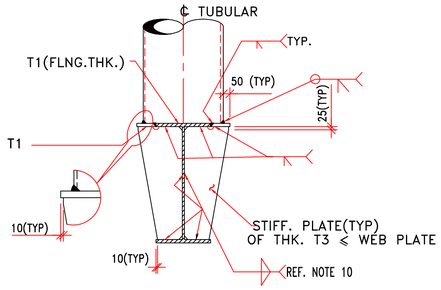
TYP. WELDING DETAIL OF WEB PLATE TO BEAM AND STIFFENER PLATE



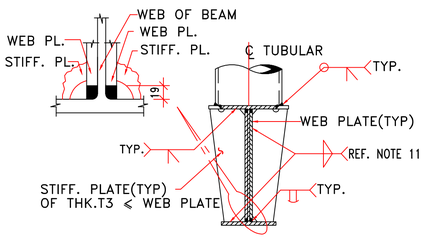
TYP. WELDING DETAIL OF BEAMS OF UNEQUAL SIZE
(WHEN THERE IS TUBULAR CONNECTED AT THE JOINT)



TYP. WELDING DETAIL OF BEAMS OF EQUAL SIZE
(WHEN THERE IS TUBULAR CONNECTED AT THE JOINT)



SECTION C



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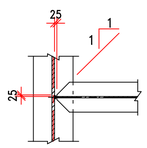

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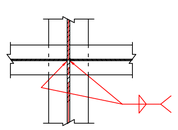
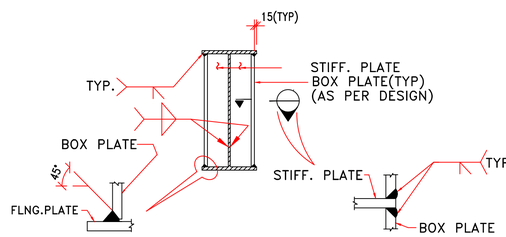
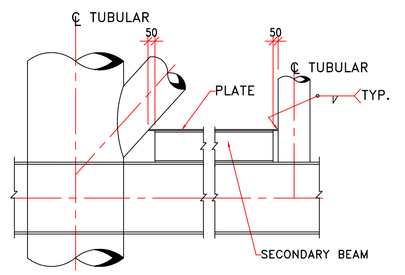


SECTION B



TYP. WELDING DETAIL OF BOX PLATE TO BEAM



TYPICAL SECONDARY BEAM TERMINATION DETAIL

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES:

1. FOR GENERAL NOTES REFER STD. NO. 7-60-0605.
2. ALL WELDING DETAILS TO COMPLY WITH AWS D.1.1 LATEST EDITION, API-RP-2A AND APPROVED WELDING PROCEDURE. ALL FPW SHALL BE BACK GOUGED AND BACK WELDED.
3. FOR TUBULAR TO TUBULAR CONNECTION REFER STD. NO. 7-60-0806.
4. UNLESS NOTED OTHERWISE, THE FOLLOWING CONNECTIONS SHALL BE OF FULL PENETRATION WELDS AND WELDED ALLROUND:
   a) ALL BEAMS ARE PIPE SPLICES
   b) ALL BEAM FLANGE EXTENSIONS
   c) ALL TUBULAR CONNECTIONS
   d) ALL TRUSS MEMBER CONNECTIONS (TUBULAR TO BEAM)
5. MINIMUM SIZE OF FILLET WELDS SHALL BE 6mm UNLESS OTHERWISE SHOWN ON THE DESIGN DRAWINGS (ONLY SEAL WELD SHALL BE OF MIN 4mm)
6. WELD DETAILS DESCRIBED IN THIS DRAWING ARE APPLICABLE UNLESS EXPLICITLY NOTED OTHERWISE ON THE DESIGN DRAWINGS.
7. COPE HOLES/CHAMFERS/SNIPES SHALL BE USED TO ALLOW CONTINUITY OF FULL PENETRATION BUTT WELDS DEPENDING ON FABRICATION REQUIREMENT.
8. CONTINUOUS SEAL WELDING (4mm) SHALL BE APPLIED BETWEEN DECK PLATING AND SECONDARY BEAMS.
9. THE CHOICE OF SINGLE/DOUBLE GROOVE AND WELD POSITION CAN VARY DEPENDING ON WELDING SEQUENCE IT SHALL CONFIRM TO THE WELDING PROCEDURE AND A.W.S. D.1.1.
10. THE DOUBLE FILLET WELD CONNECTING
    a) WEB OF BEAM TO WEB OF BEAM
    b) WEB OF BEAM TO TUBULAR
    c) STIFFENER TO WEB
    d) STIFFENER TO NON LOAD BEARING FLANGE SHALL BE OF SIZE 0.7xCONNECTING ELEMENT SIZE OR 6mm WHICHEVER IS HIGHER.
11. THE FILLET WELD CONNECTING THE WEB PLATE TO THE BEAM AND STIFFENER PLATE WITH WEB PLATE SHALL BE OF SIZE EQUAL TO 0.7 TIMES THE WEB PLATE THICKNESS OR STIFFENER PLATE THICKNESS RESPECTIVELY SUBJECT TO A MINIMUM OF 6mm.

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEMPC
0	29.01.202	ISSUED FOR TENDER	MUNISH	AA	DP	

इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ONGC ओएनजीसी — ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

CONSULTANCY SERVICES FOR ENGINEERING DOCUMENT GENERATION FOR WELL PLATFORMS

DECK WELDING DETAILS

SCALE	JOB NO.	UNIT	DIVN	DEPT	DWG. NO.	REV.
1:1	B774	000	83	43	0001	0

Sht 02 of 02

3-1641-0501 REV.2 A1-841 x 594

इस ड्राइंग अथवा इसमें दिये गए डिज़ाइन व संकेतिकरण इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। इन्हें इन्हें पूरा/पूर्णत: किसी प्रयोजन के लिए प्रयोग में नहीं लाया जाएगा, न प्रकाशित की जाएगी, न उधार दिए जाएंगे, न प्रतिलिपि किए जाएंगे और न ही तीसरे और किसी प्रयोग के अलावा इसका इस्तेमाल किया जाएगा। इसका इस्तेमाल केवल अन्य प्रयोग हेतु और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा।

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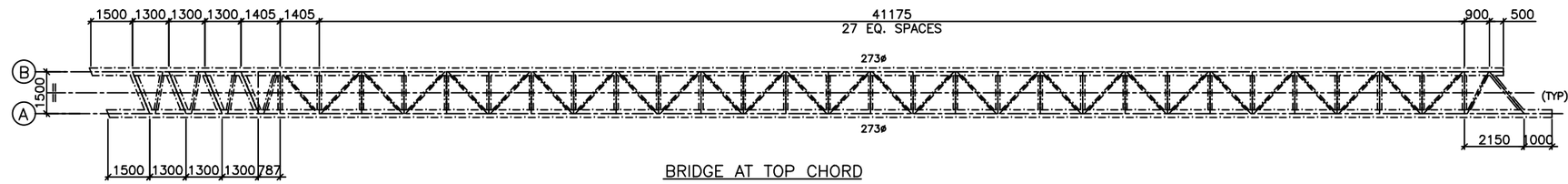
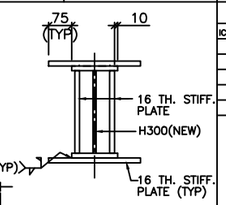
B774-001-83-43-0001



BRIDGE AT TOP CHORD



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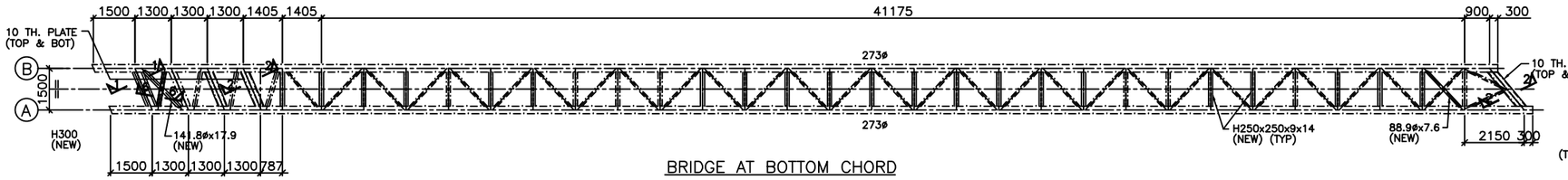
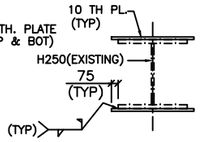
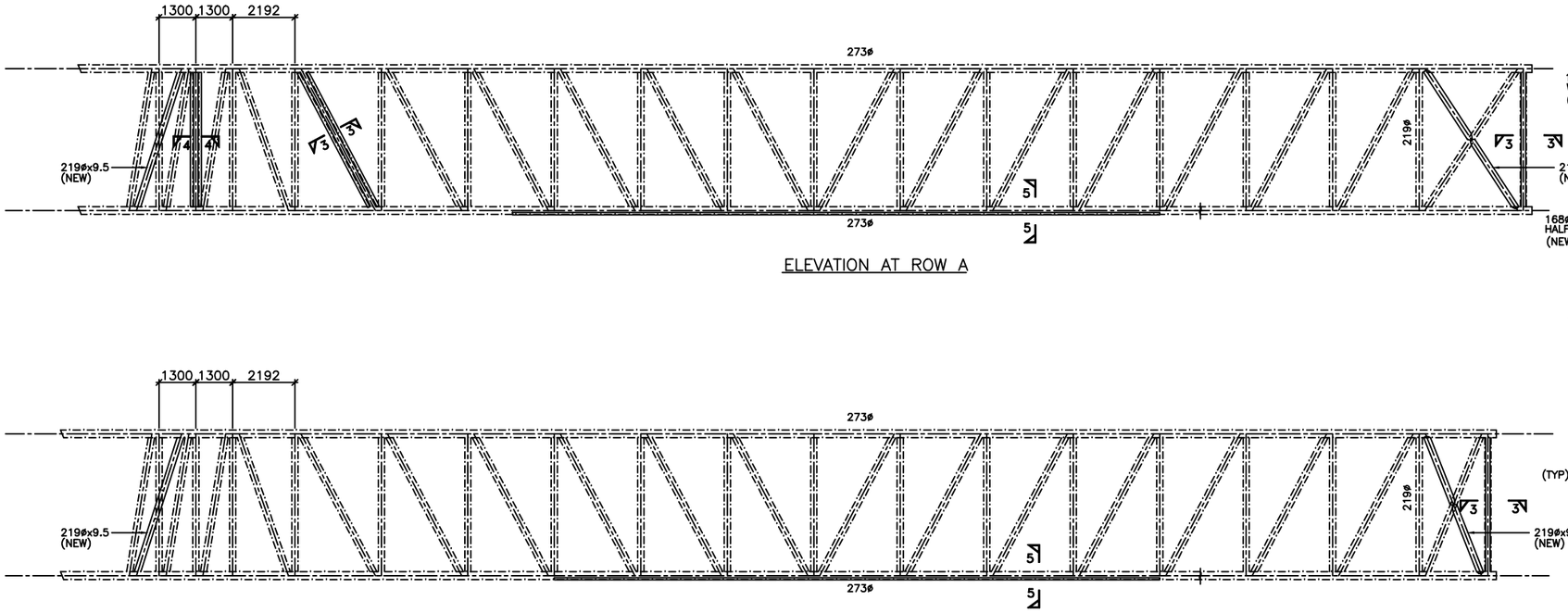


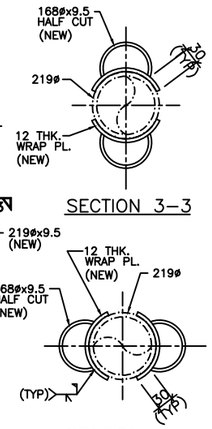
BRIDGE AT BOTTOM CHORD



SECTION 2–2



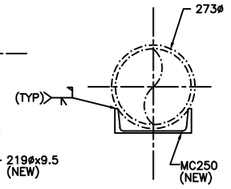
ELEVATION AT ROW A



SECTION 3–3

SECTION 4–4

ELEVATION AT ROW B



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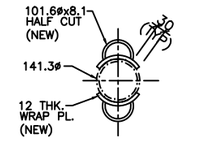
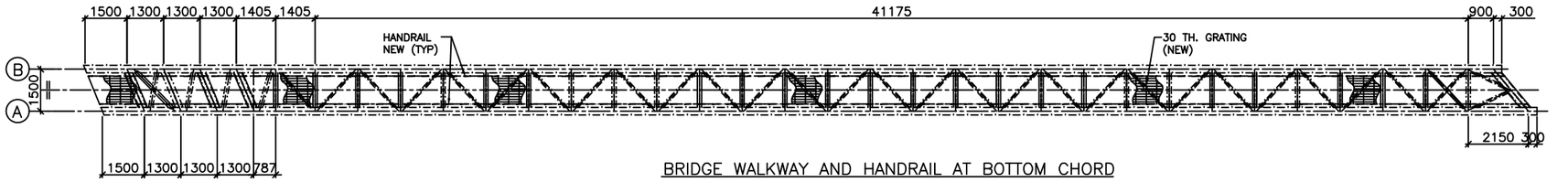


SECTION 6–6



BRIDGE WALKWAY AND HANDRAIL AT BOTTOM CHORD

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICG-DG-MB-301 TO 309	ICW-ICG BRIDGE DRAWINGS

NOTES :-

- ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE-ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

‡ H250x250x9x14
H300x300x10x15

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	08.10.25	ISSUED FOR TENDER	MUNISH	MM	DS	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF ICW-ICG BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	001	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए है और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सीमित और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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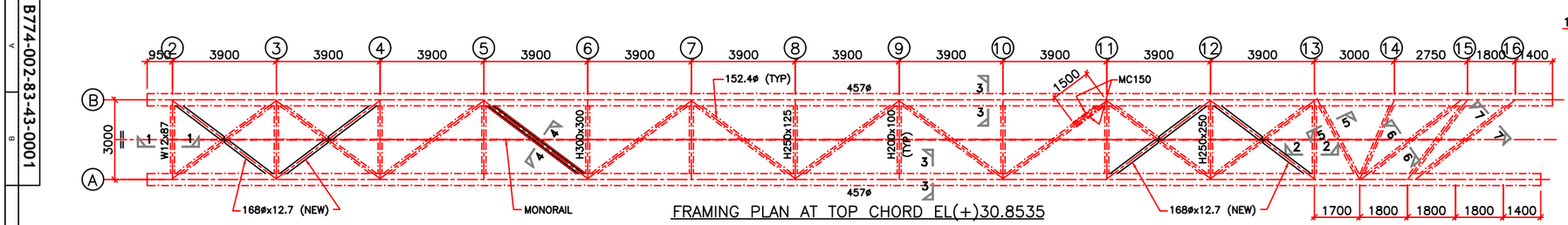
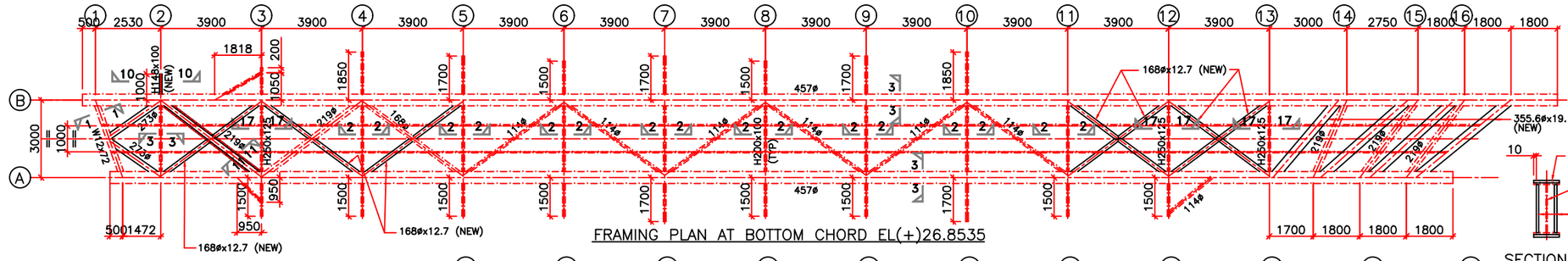
B774-002-83-43-0001



FRAMING PLAN AT TOP CHORD EL(+)30.8535



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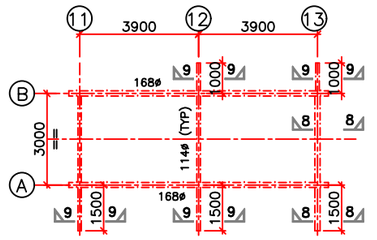

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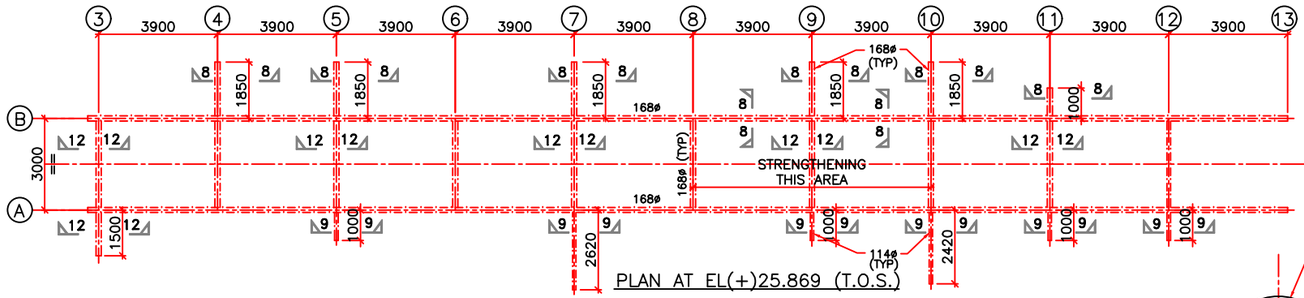

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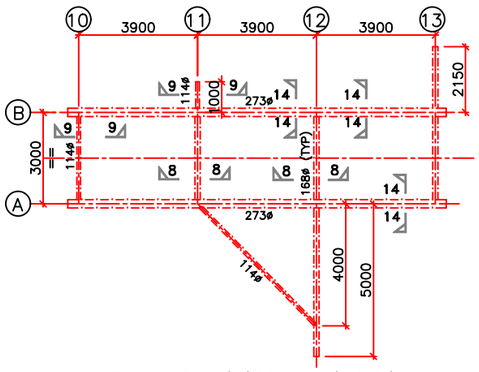

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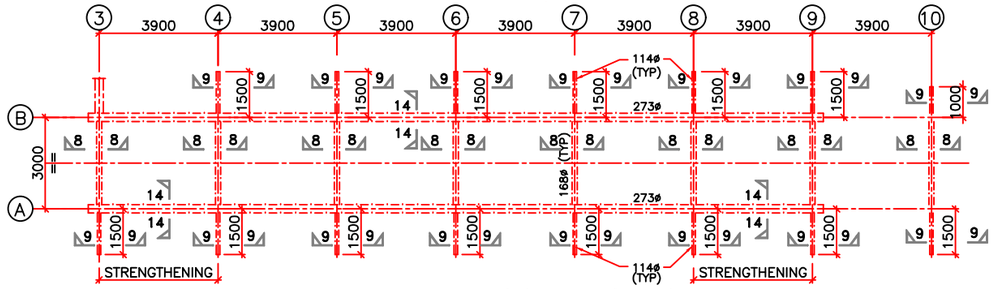

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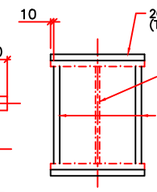

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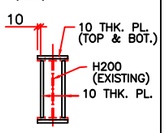

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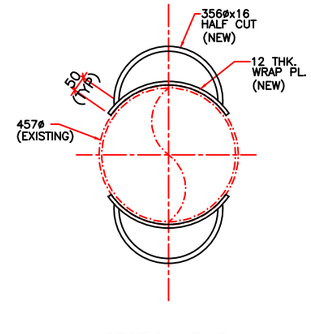

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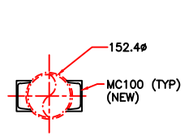

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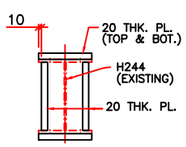
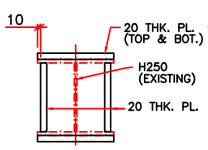


SECTION 5–5



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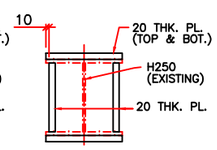

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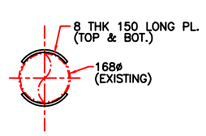

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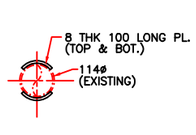

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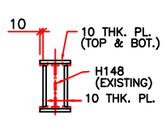

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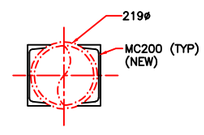

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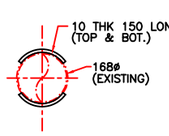

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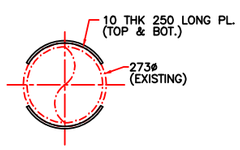

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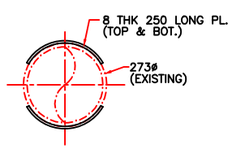

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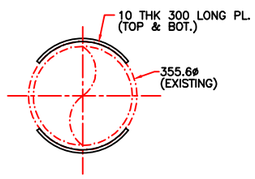
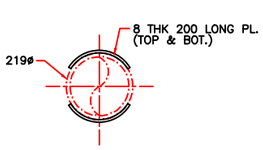
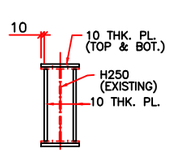


SECTION 15–15



SECTION 16–16



SECTION 17–17

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

* ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE–ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	10.10.25	ISSUED FOR TENDER	MUNISH	NS	DP	

ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

vWk; y ,.M uspqjy xSLk dkjiksjs'ku fyfeVsM
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF ICP-ICG BRIDGE TOP DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	002	83	43	0001	A

3-1641-0501 REV.2 A1-841x594

izLrqr vkjs[k ,oa blesa fufgr fMtkbu bathfu;Lkz bafMk;k fyfeVsM dh Lkaifÿr gSA ;s dsoy fufonk izfØ;k ,oa fu"iknu ds mn~ns'; ls gh tkjh fd;s x;s gSa vkSj m/kkjdrkZ us ;g L;oa'kZ;r :i esa nh xbZ gS fd u rks mUgsa iqu% eqfnzr fd;k tk;sxk u gh fdlh rhljs i{k dks bldk izdVhdj.k fd;k tk;sxk ;k izlkfjr fd;k tk;sxk] tc rd fd bl izdkj dk izdVhdj.k] izlkj ;k mi;ksx ds fy, dEiuh ls fyf[kr vuqefr izkIr u dj yh tk;sA
Lkkjr ljdkj ds lwpuk izkS|ksfxdh ea=ky; ds vUrxZr xfBr bathfu;fjax fMtkbu lsaVj (bathfu;lZ bafM;k fyfeVsM) ds dkWihjkbV ds varxZr ;g izLrko vkSj mlesa nh xbZ lHkh fMtkbuksa ,oa fooj.kksa ij bathfu;lZ bafM;k fyfeVsM dk LokfeRo gSA ;s lHkh fMtkbu ,oa fooj.k dsoy mlh izkf/kd`r ;kstuk ds fy, gSa ftlds fy, bUgsa rS;kj fd;k x;k gSA bUgsa fdlh vU; ;kstuk ds fy, ;k fdlh vU; mn~ns'; ls ;k fdlh vU; O;fDr }kjk bLrseky ugha fd;k tk ldrk gSA
Lkkjr ljdkj ds lwpuk izkS|ksfxdh ea=ky; ds vUrxZr xfBr bathfu;fjax fMtkbu lsaVj (bathfu;lZ bafM;k fyfeVsM) ds dkWihjkbV ds varxZr ;g izLrko vkSj mlesa nh xbZ lHkh fMtkbuksa ,oa fooj.kksa ij bathfu;lZ bafM;k fyfeVsM dk LokfeRo gSA ;s lHkh fMtkbu ,oa fooj.k dsoy mlh izkf/kd`r ;kstuk ds fy, gSa ftlds fy, bUgsa rS;kj fd;k x;k gSA bUgsa fdlh vU; ;kstuk ds fy, ;k fdlh vU; mn~ns'; ls ;k fdlh vU; O;fDr }kjk bLrseky ugha fd;k tk ldrk gSA

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izLrqr vkjs[k ,oa bl esa fufgr fMt+kbu bathfu;fjax lacaf/kr tkudkjh xksiuh; ,oa xksiuh; izd`fr dh gS tks fd vkbZ ,l bZ bafM;k bathfu;fjax fyfeVsM dh laifRr gSA ;g fdlh Hkh rjhds ls iquZ:Rikfnr ;k izdkf'kr ugha fd;k tk ldrk gS ;fn ,l bZ bafM;k bathfu;fjax fyfeVsM dh fyf[kr vuqefr ugha yh tkrh gSA
Lkeqfpr lhesa ds Hkhrj bl izk:i esa fn[kk;s x;s lHkh fMt+kbu ,oa fMVsYl ds fy, ,l bZ bafM;k bathfu;fjax fyfeVsM dh laifRr gSA ;s fMt+kbu ,oa fMVsYl fdlh Hkh rjhds ls iquZ:Rikfnr] izdkf'kr ;k izfrfyfir ugha fd;s tk ldrs gSa ;fn ,l bZ bafM;k bathfu;fjax fyfeVsM dh fyf[kr vuqefr ugha yh tkrh gSA
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izLrqr fMt+kbu ,oa fMVsYl fufgr fMt+kbu bathfu;fjax lacaf/kr tkudkjh xksiuh; ,oa xksiuh; izd`fr dh gS tks fd vkbZ ,l bZ bafM;k bathfu;fjax fyfeVsM dh laifRr gSA ;g fdlh Hkh rjhds ls iquZ:Rikfnr ;k izdkf'kr ugha fd;k tk ldrk gS ;fn ,l bZ bafM;k bathfu;fjax fyfeVsM dh fyf[kr vuqefr ugha yh tkrh gSA
Lkeqfpr lhesa ds Hkhrj bl izk:i esa fn[kk;s x;s lHkh fMt+kbu ,oa fMVsYl ds fy, ,l bZ bafM;k bathfu;fjax fyfeVsM dh laifRr gSA ;s fMt+kbu ,oa fMVsYl fdlh Hkh rjhds ls iquZ:Rikfnr] izdkf'kr ;k izfrfyfir ugha fd;s tk ldrs gSa ;fn ,l bZ bafM;k bathfu;fjax fyfeVsM dh fyf[kr vuqefr ugha yh tkrh gSA
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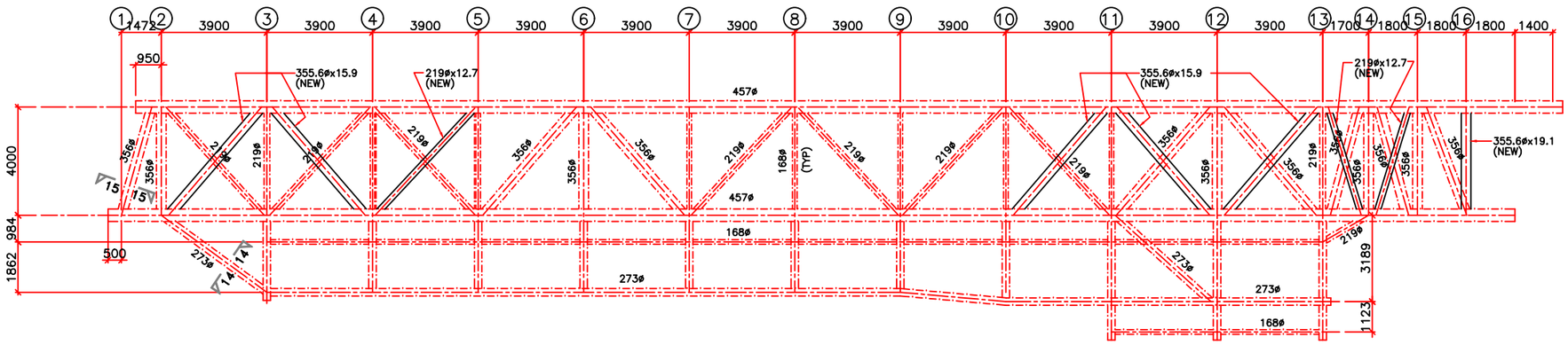
B774-002-83-43-0002



ELEVATION AT ROW A

ELEVATION AT ROW B

REF. DWG. NO. | REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. THE DRAWING SHALL BE READ IN CONJUNCTION WITH SHT 1 OF 2

- ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE-ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	10.10.25	ISSUED FOR TENDER	MUNISH	NS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

vWk;y ,.M uspqjy xSLk dkjiksjs'ku fyfeVsM
OIL AND NATURAL GAS CORPORATION LIMITED

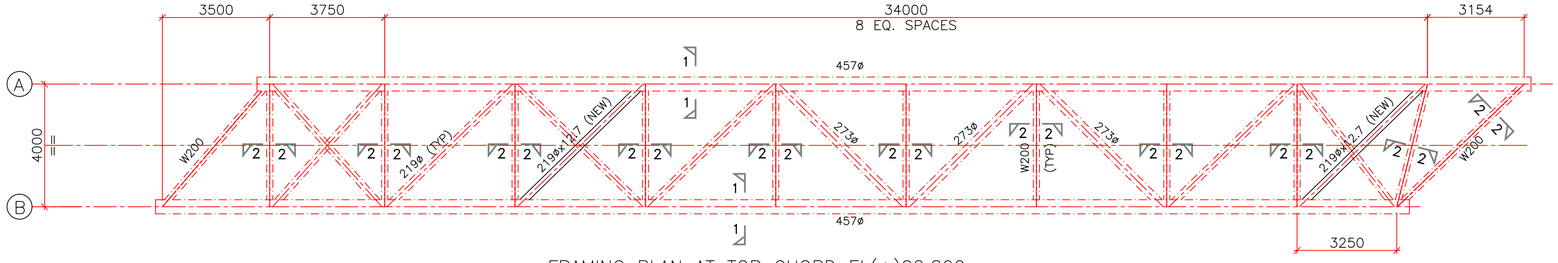
REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF ICP-ICG BRIDGE TOP DECK

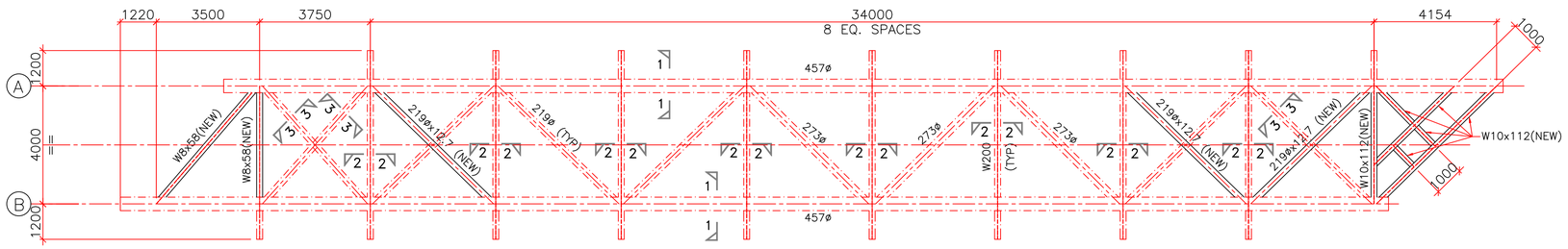
SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	002	83	43	0002	A

3-1641-0501 REV.2 A1-841x594

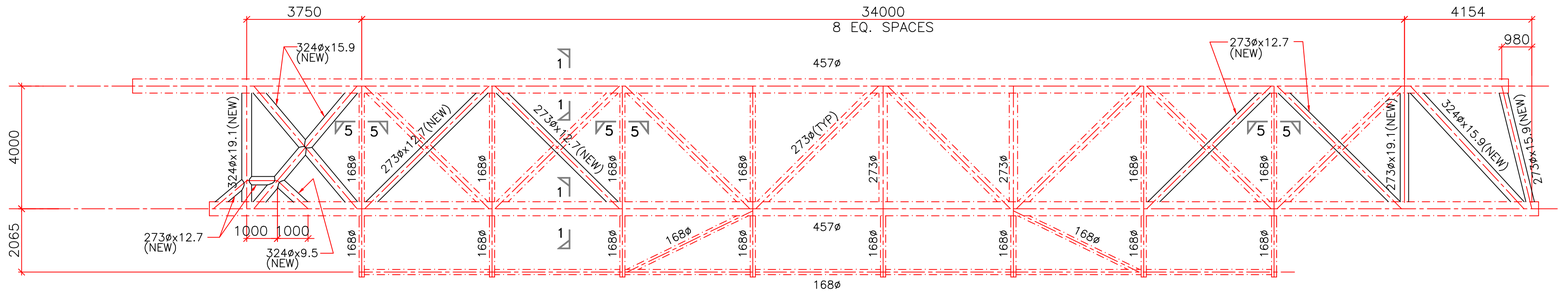
B774-003-83-43-0001



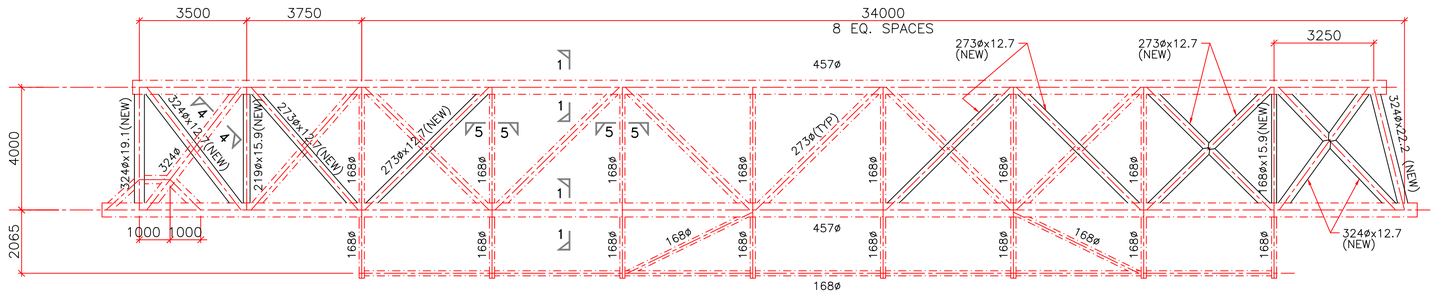
FRAMING PLAN AT TOP CHORD EL(+)22.800



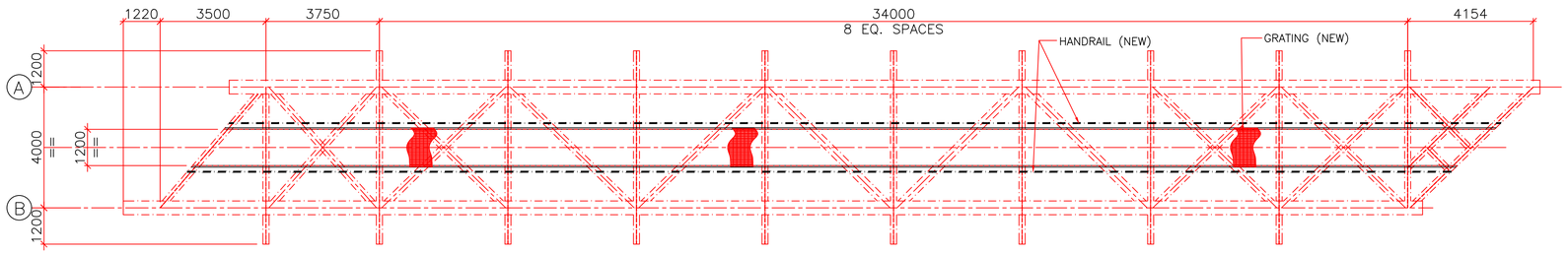
FRAMING PLAN AT BOTTOM CHORD EL(+)18.800



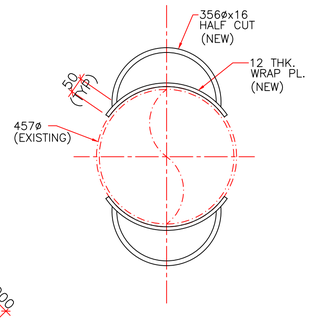
ELEVATION AT ROW A



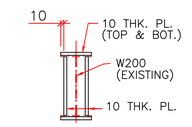
ELEVATION AT ROW B



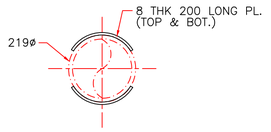
WALKWAY LEVEL



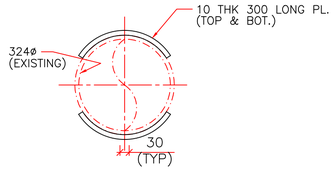
SECTION 1–1



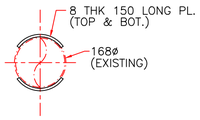
SECTION 2–2



SECTION 3–3



SECTION 4–4



SECTION 5–5

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR TENDER	MUNISH	RS	DP	

इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF NQO-NQD BRIDGE TOP DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	003	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उपयोगकर्ता ने यह स्पष्ट समझौता किया है कि न तो उन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रतिलिपि किए जाएंगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उपयोगकर्ता को लिखित रूप में दी गई सहमति से होगा।
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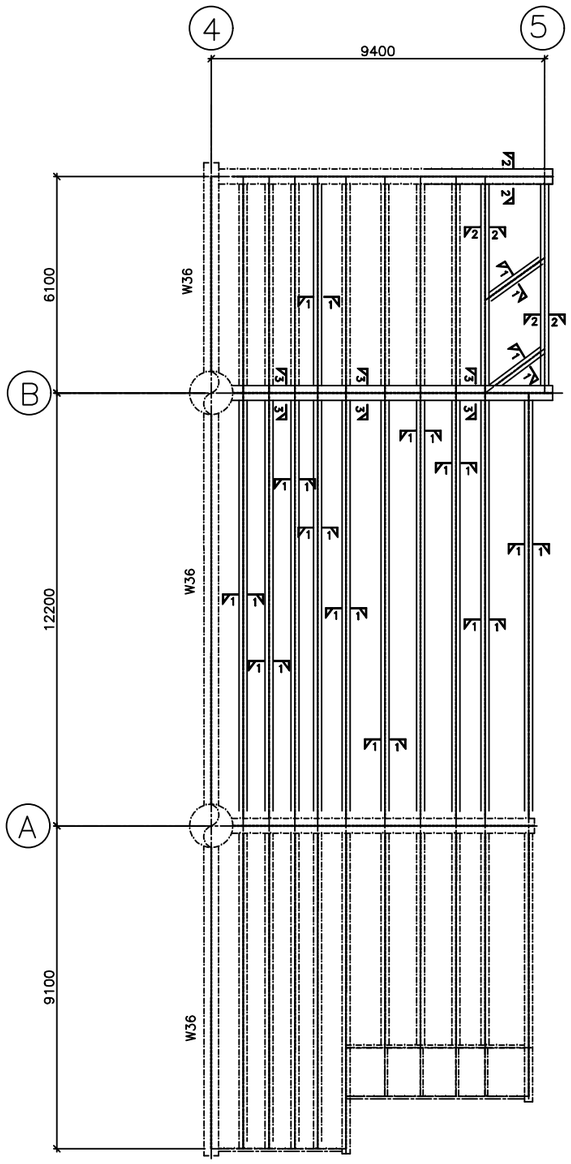
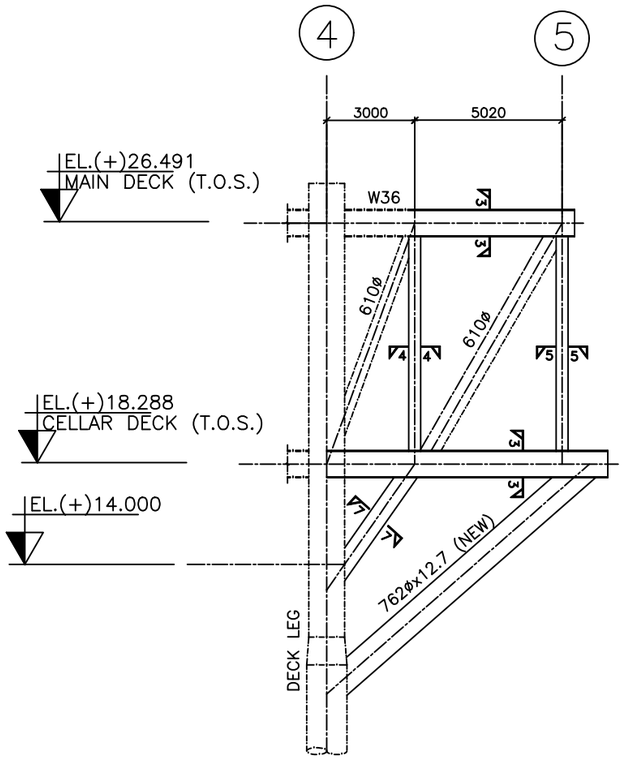
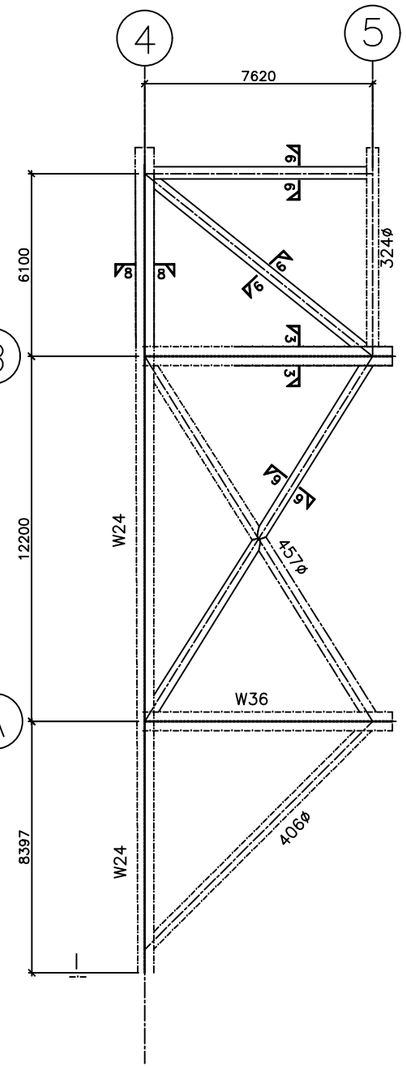
B774-003-83-43-0002



PART PLAN OF CELLAR DECK AT EL(+)18.288 (T.O.S.)



PART ELEVATION OF DECK ROW B



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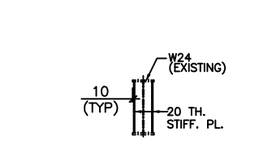
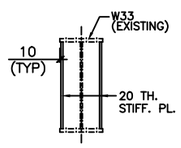
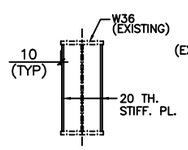


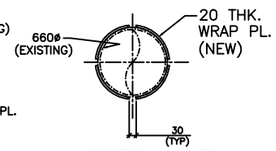
SECTION 1–1



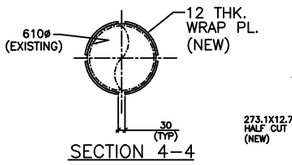
SECTION 2–2



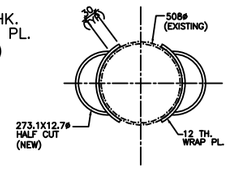
SECTION 3–3



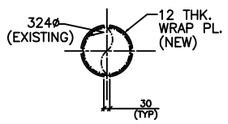
SECTION 7–7



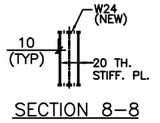
SECTION 4–4



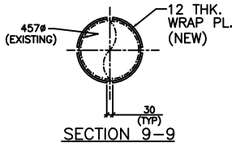
SECTION 5–5



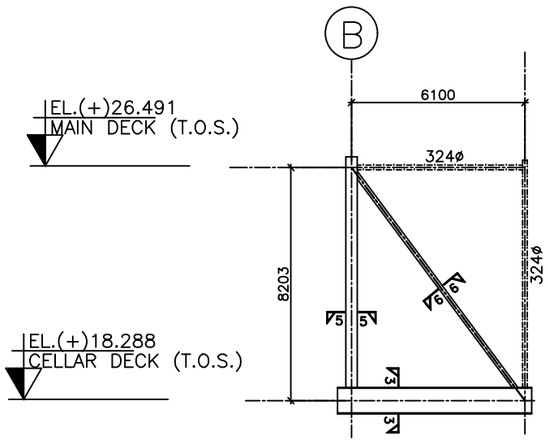
SECTION 6–6



SECTION 8–8



SECTION 9–9



PART ELEVATION OF DECK ROW 5

REF. DWG. NO.	REFERENCE DRAWING TITLE
1980-40-E-501	MAIN DECK FRAMING
1980-40-E-502	MAIN DECK PLATING,GRATING & H.R LAYOUT
1980-40-E-503	CELLAR DECK FRAMING
1980-40-E-504	CELLAR DECK PLATING,GRATING,H.R LAYOUT
1980-40-E-506	DECK TRUSS ROW A & B
1980-40-E-508	DECK TRUSS ROW 3 & 4

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
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3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7-60-0802.
9. FOR HANDRAIL REFER EIL STD. 7-60-0801.

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- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	01.04.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF WIS DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	003	83	43	0002	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न पुनर्वितरित किए जाएंगे और न ही संपत्ति और निजी प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा।
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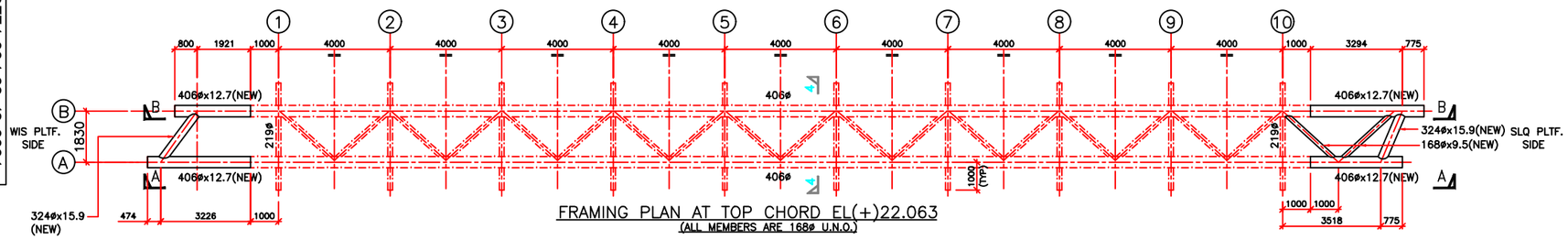
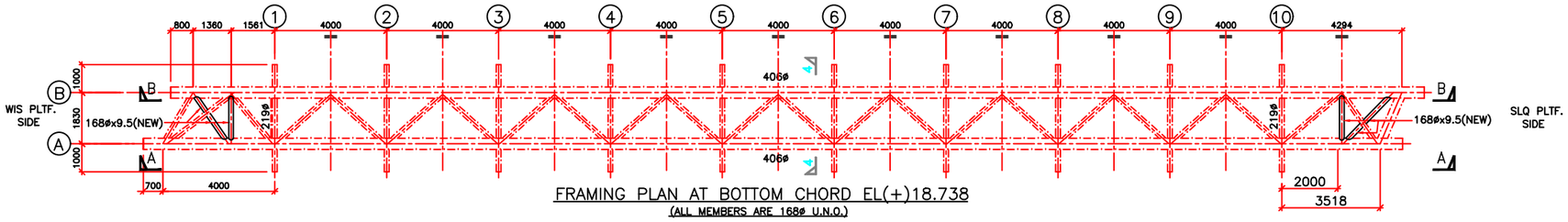
B774-004-83-43-0001



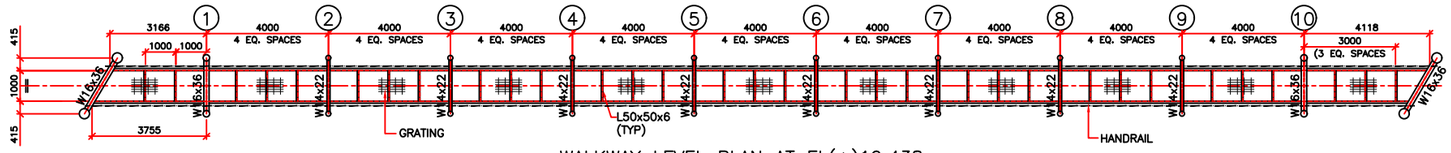
FRAMING PLAN AT TOP CHORD EL(+)22.063

(ALL MEMBERS ARE 168ø U.N.O.)



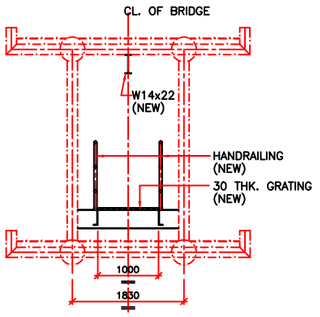
FRAMING PLAN AT BOTTOM CHORD EL(+)18.738

(ALL MEMBERS ARE 168ø U.N.O.)

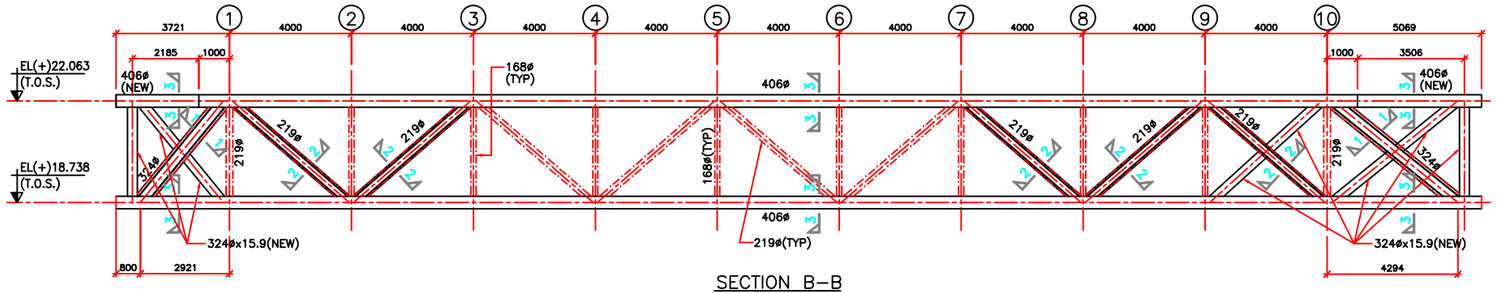


WALKWAY LEVEL PLAN AT EL(+)19.438

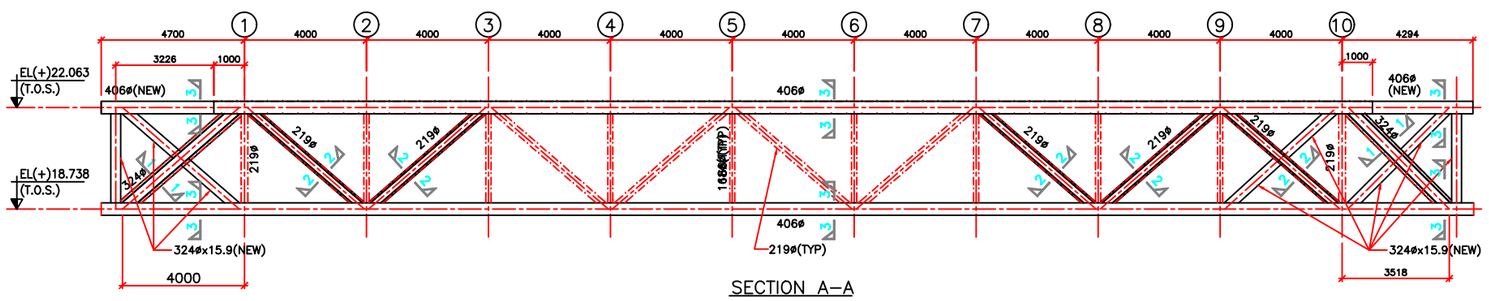
(ALL MEMBERS ARE C10x20 U.N.O.)



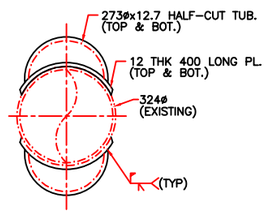
SECTION 4–4



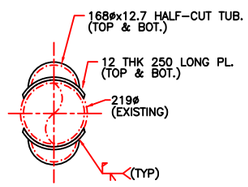
SECTION B–B



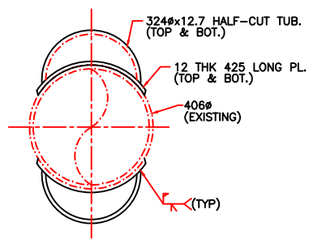
SECTION A–A



SECTION 1–1



SECTION 2–2



SECTION 3–3

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR TENDER	MUNISH	MM	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SLQ-WIS BRIDGE TOP DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	004	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

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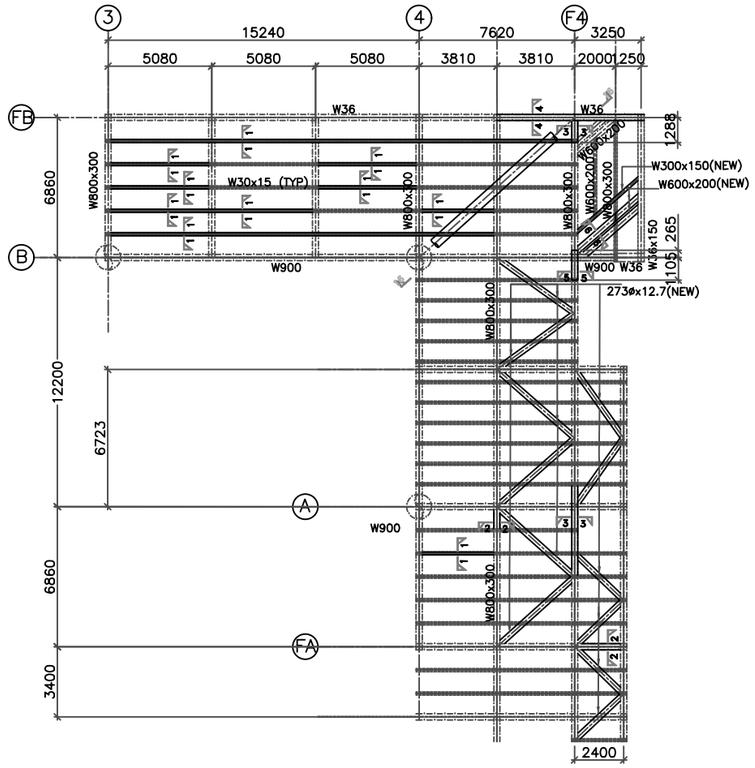
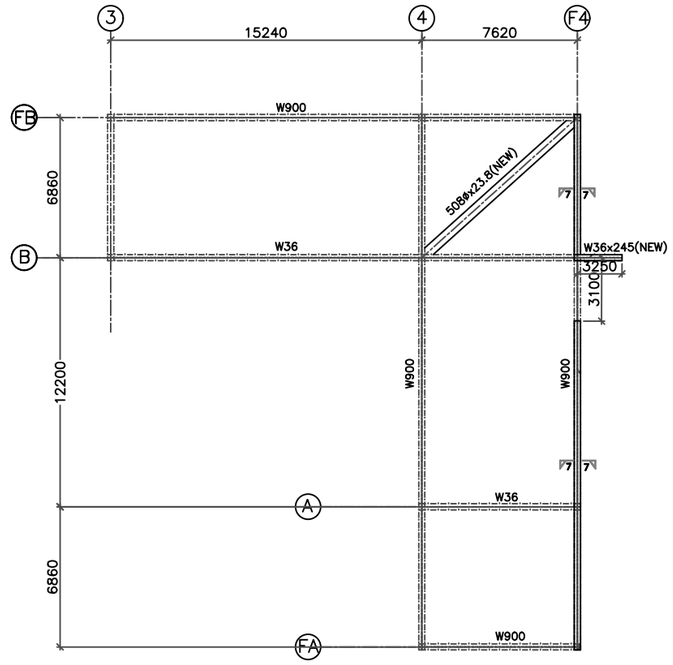
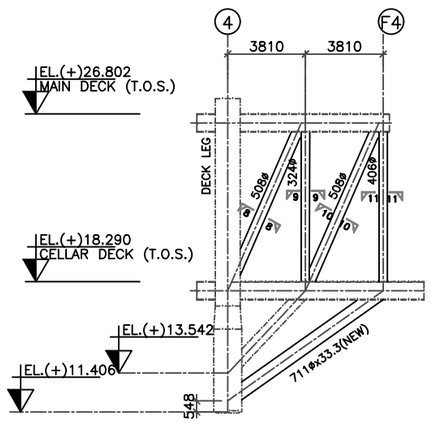
B774-004-83-43-0002



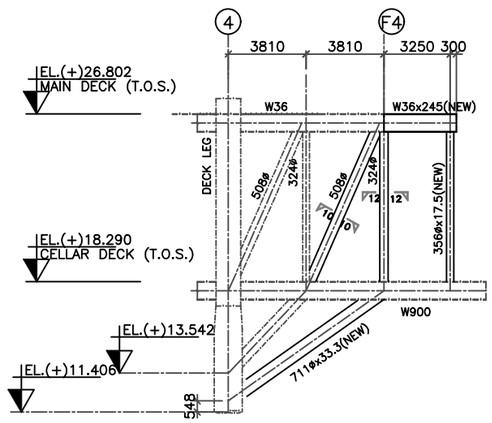
PART PLAN OF CELLAR DECK AT EL(+)18.290 (T.O.S.)



PART PLAN OF MAIN DECK AT EL(+)26.802 (T.O.S.)



PART ELEVATION OF DECK ROW A



PART ELEVATION OF DECK ROW B

REF. DWG. NO.	REFERENCE DRAWING TITLE
NQO-090-043 TO NQO-090-071	NQO DECK DRAWINGS

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	02.04.26	ISSUED FOR TENDER	MUNISH	AP	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF NQO DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	004	83	43	0002	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रतिलिपि किए जाएंगे और न ही संपत्ति और निजी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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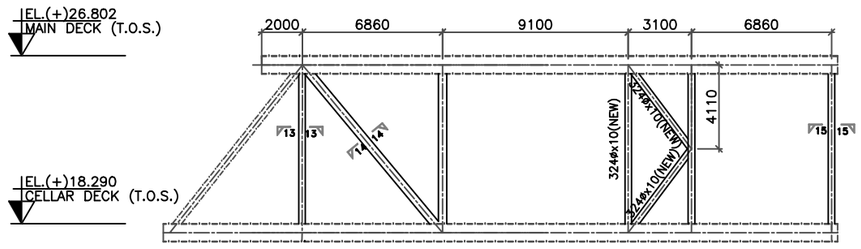
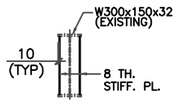
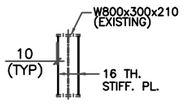
B774-004-83-43-0003



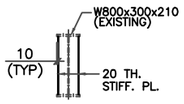
ELEVATION OF DECK ROW F4



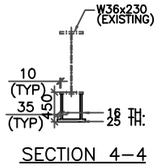
SECTION 1–1



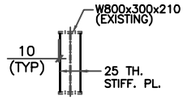
SECTION 2–2



SECTION 3–3



SECTION 4–4



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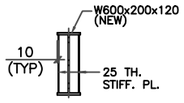
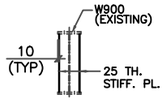


SECTION 6–6



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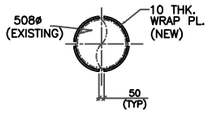
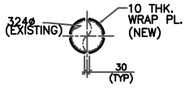
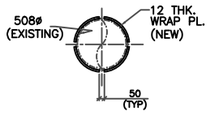


SECTION 8–8



SECTION 9–9



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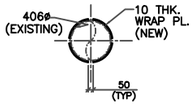
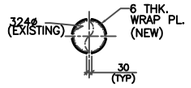


SECTION 11–11



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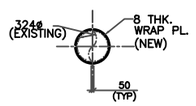
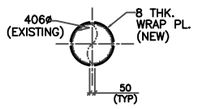
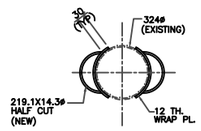


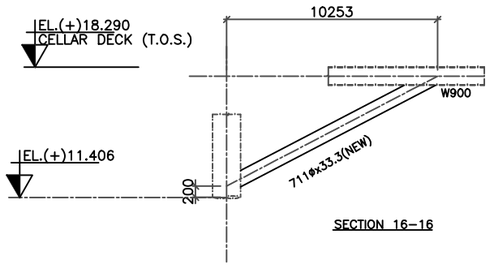
SECTION 13–13



SECTION 14–14



SECTION 15–15



SECTION 16–16

REF. DWG. NO.	REFERENCE DRAWING TITLE
NQO–090–043 TO NQO–090–071	NQO DECK DRAWINGS

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEMPC
A	02.04.26	ISSUED FOR TENDER	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF NQO DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	004	83	43	0003	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रतिलिप किए जाएंगे और न ही सांविदा और निजी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।

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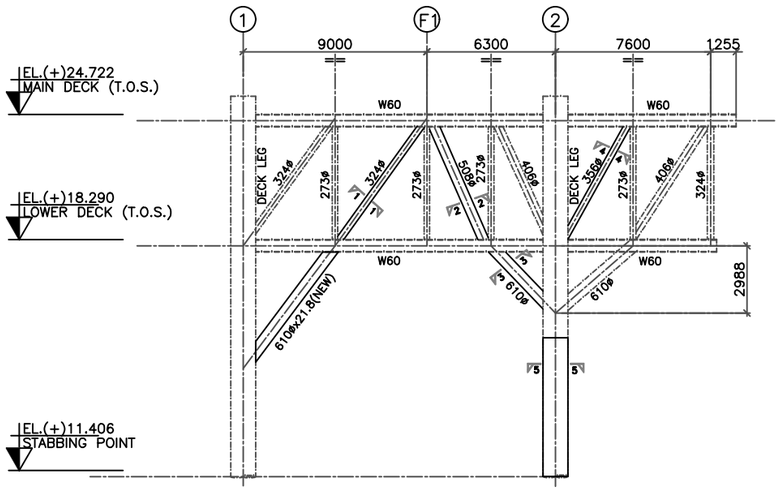
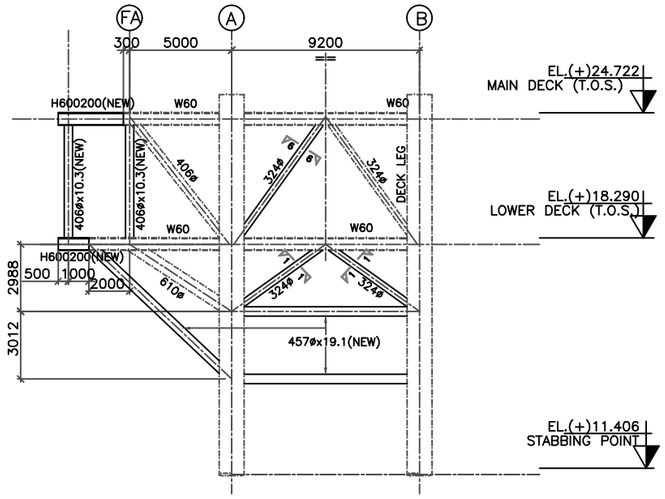
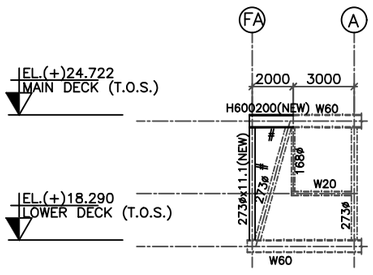
B774-004-83-43-0004



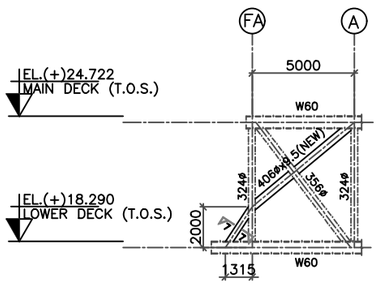
ELEVATION OF DECK ROW A



PART ELEVATION OF DECK ROW 2



PART ELEVATION OF DECK ROW F1



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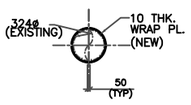
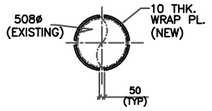


SECTION 1–1



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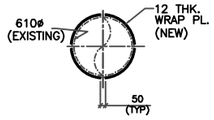

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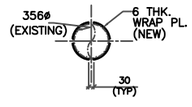

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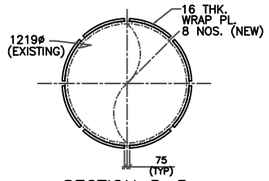

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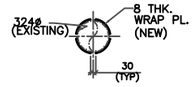

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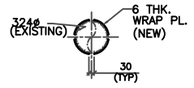

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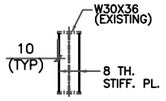

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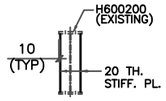

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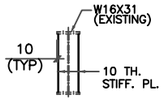

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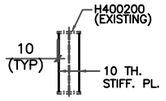

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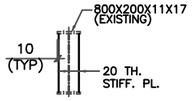

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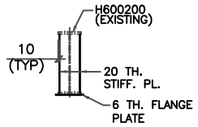

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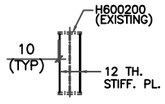

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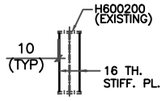

SECTION 15–15

REF. DWG. NO.	REFERENCE DRAWING TITLE
NQD–090–005 TO 022	NQD DECK DRWAINGS

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

- ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE–ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)
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FIELD NOTES

THE MEMBER MARKED SHALL BE PLACED BEFORE THE REMOVAL OF MEMBERS TO BE REPLACED AND THE SAME SHALL BE REMOVED AFTER THE REPLACEMENT OF MEMBERS AT BRIDGE LANDING AREA.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	02.04.26	ISSUED FOR TENDER	MUNISH	AP	DP	

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OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF NQD DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	004	83	43	0004	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये न उधार दिए जाएंगे, न प्रकाशित किए जाएंगे और न ही संपित और किसी प्रयोग के लिए किसी अन्य प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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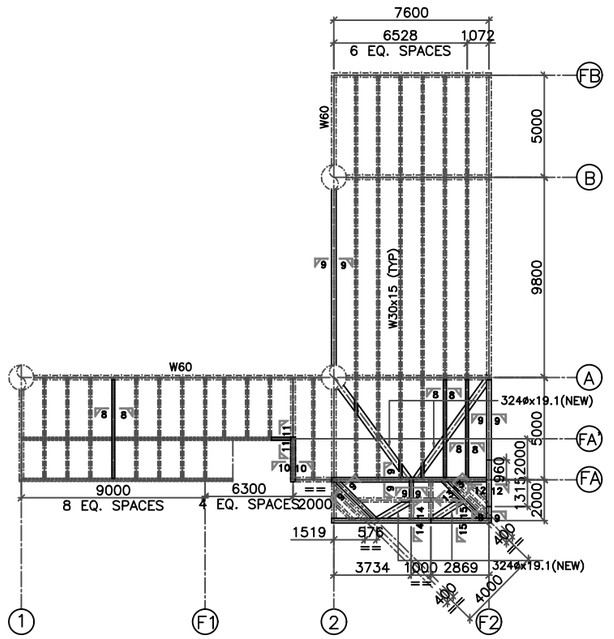
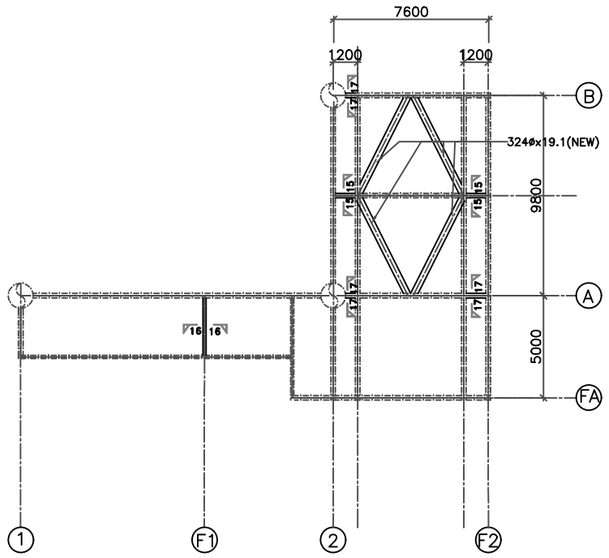


PART PLAN OF CELLAR DECK EL(+)18.290(T.O.S.)



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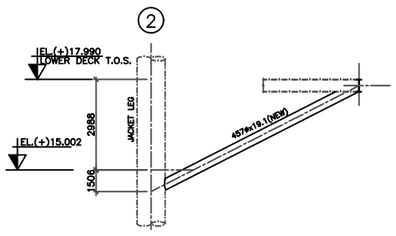
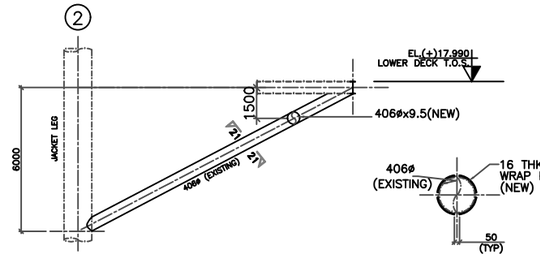


SECTION 19–19



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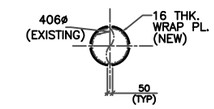

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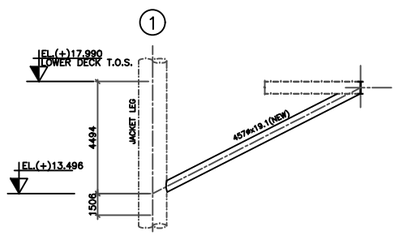

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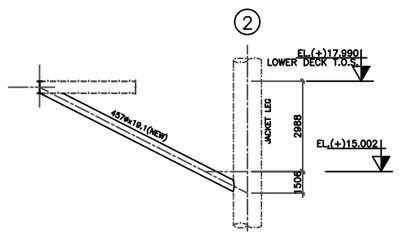
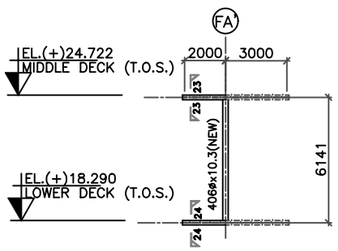
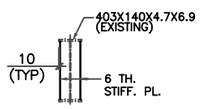


SECTION 22–22

PART PLAN OF CELLAR DECK EL(+)18.290(T.O.S.)



PART ELEVATION OF DECK ROW F1'



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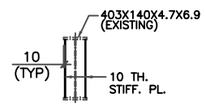

SECTION 24–24

REF. DWG. NO.	REFERENCE DRAWING TITLE
NQD–090–005 TO 022	NQD DECK DRWAINGS

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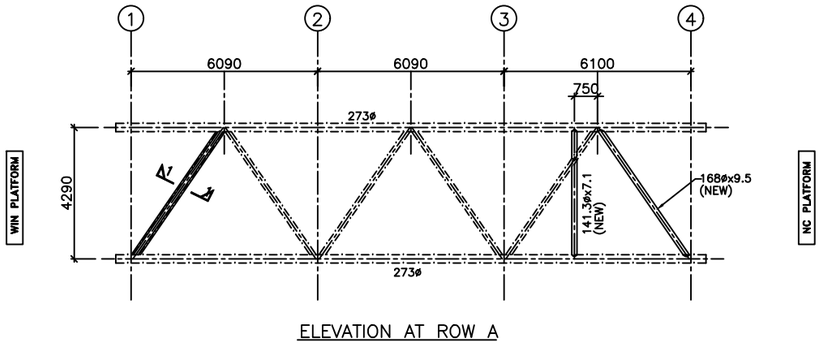
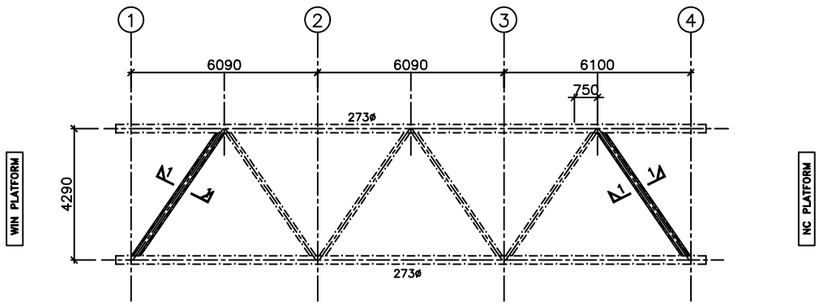
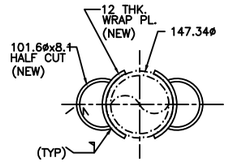


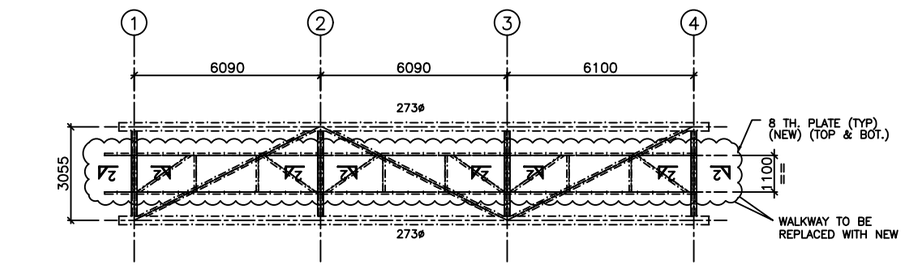
ELEVATION AT ROW A



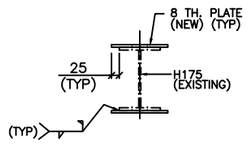
ELEVATION AT ROW B



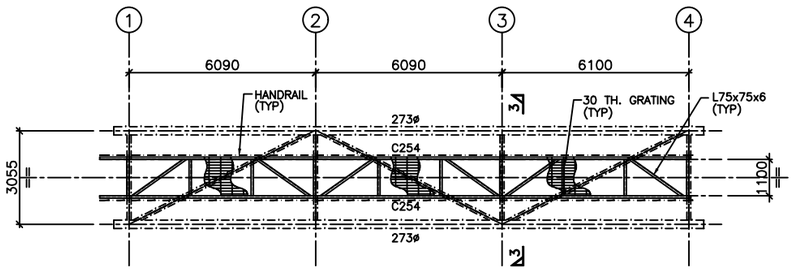
SECTION 1–1



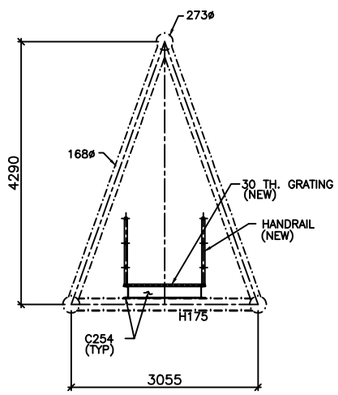
BRIDGE AT BOTTOM CHORD



SECTION 2–2



GRATING AND HANDRAIL LAYOUT (NEW)



SECTION 3–3

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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ISSUED FOR TENDER

C254x89x36

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	08.10.25	ISSUED FOR TENDER	MUNISH	MM	DS	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF WIN-NC BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	005	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की सम्पत्ति है। ये मात्र उधार दिए गए है और न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।

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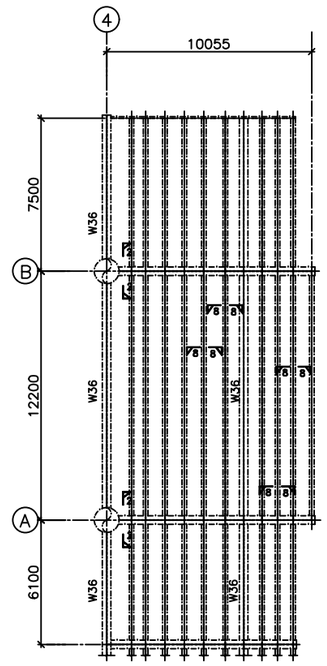
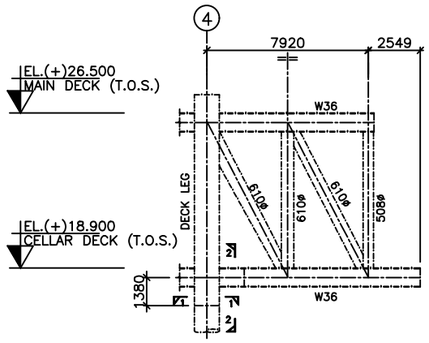
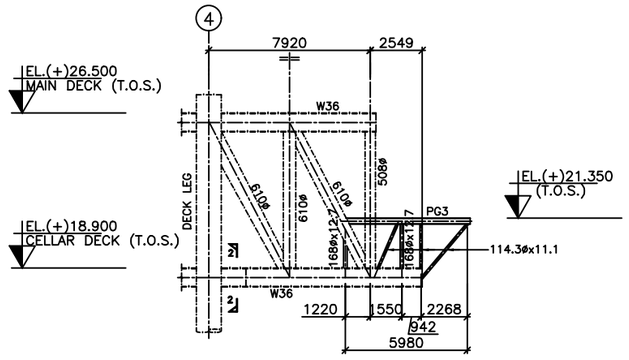
B774-005-83-43-0002



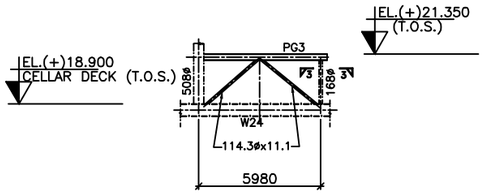
PART PLAN OF CELLAR DECK AT EL(+)18.900 (T.O.S.)



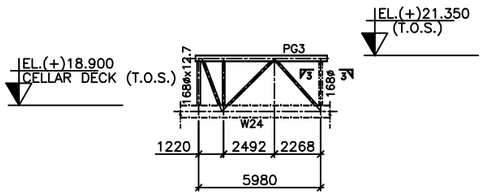
PART ELEVATION OF DECK ROW A



PART ELEVATION OF DECK ROW B



PART ELEVATION OF DECK ROW B2



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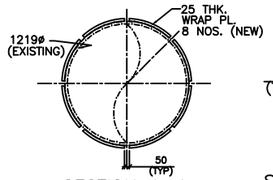
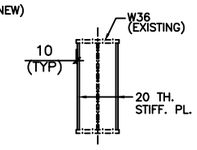
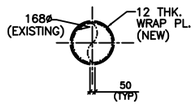


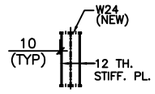
SECTION 1–1



SECTION 2–2



SECTION 3–3



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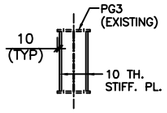
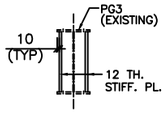
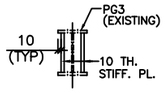


SECTION 4–4



SECTION 5–5



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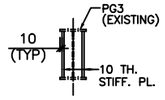

SECTION 7–7

REF. DWG. NO.	REFERENCE DRAWING TITLE
5502–SEPCD–01	EAST CELLAR DECK FRAMING PLAN
5502–SAPTR–02	TRUSS ELEVATION ROW A
5502–SAPTR–03	TRUSS ELEVATION ROW B
11026–APG–STR–DWG–0221	CELLAR DECK –STRENGTHENING PLAN

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
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   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	13.03.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF WIN DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	005	83	43	0002	0

प्रस्तुत आरेख एवं इसमे निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड को संरक्षित है। ये मात्र उधार दिए गए है और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हे पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सीमित और निजी प्रयोग उसार देने वाले द्वारा उधारकर्ता को लिखित रूप से दी गई सहमति से होगा।
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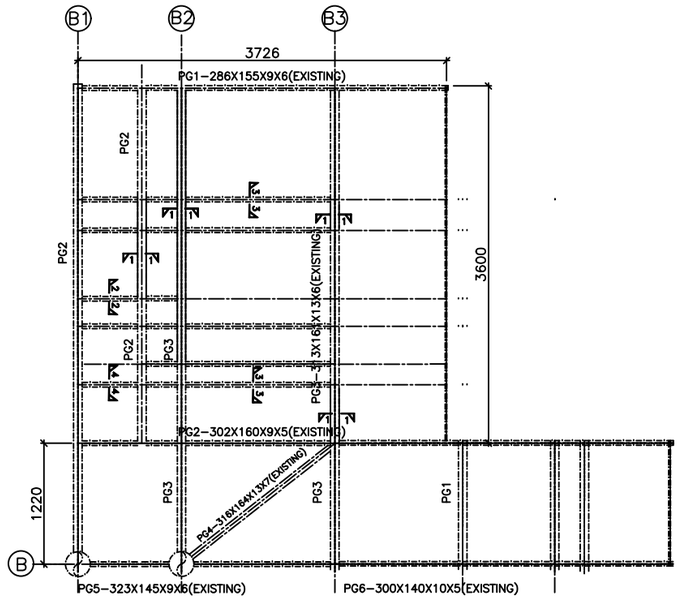
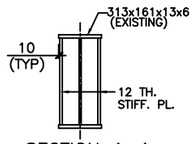
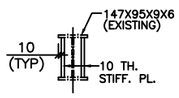
B774-005-83-43-0003



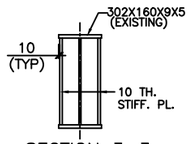
BRIDGE LANDING AREA FRAMING PLAN AT EL(+)21.350 (T.O.S)



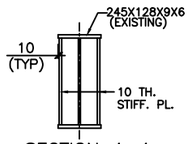
SECTION 1–1



SECTION 2–2



SECTION 3–3



SECTION 4–4

REF. DWG. NO.	REFERENCE DRAWING TITLE
5502–SEPCD–01	EAST CELLAR DECK FRAMING PLAN
5502–SAPTR–02	TRUSS ELEVATION ROW A
5502–SAPTR–03	TRUSS ELEVATION ROW B
11026–APC–STR–DWG–0221	CELLAR DECK –STRENGTHENING PLAN

NOTES :-

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	13.03.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF WIN DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	005	83	43	0003	0

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए है और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न पुनर्वितरित किए जाएंगे और न ही संपत्ति और किसी प्रयोग के अलावा इसका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।

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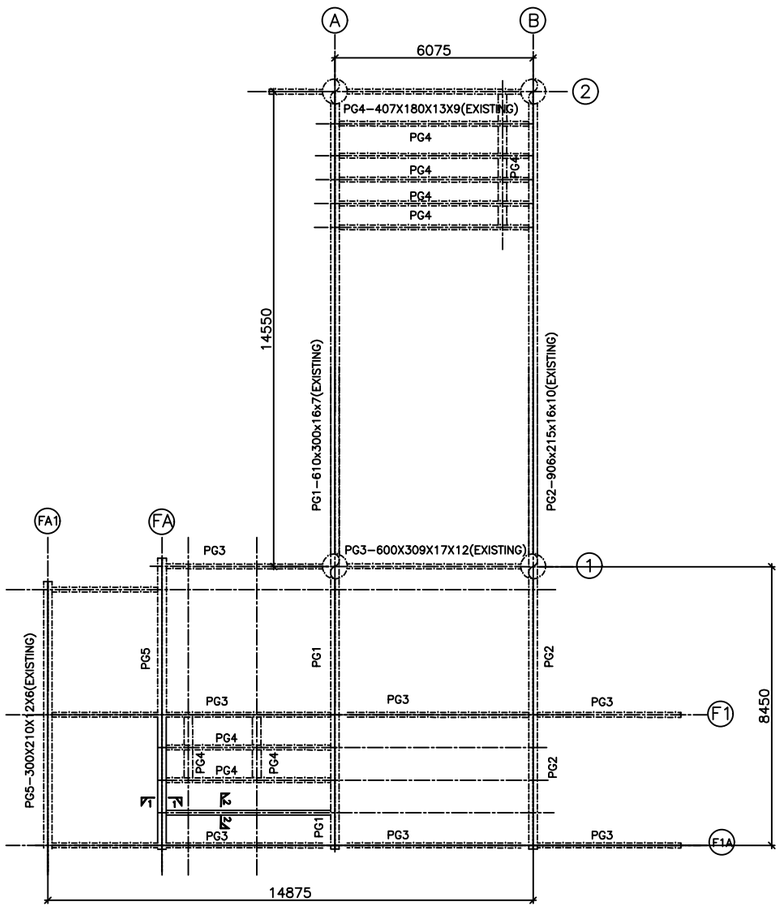
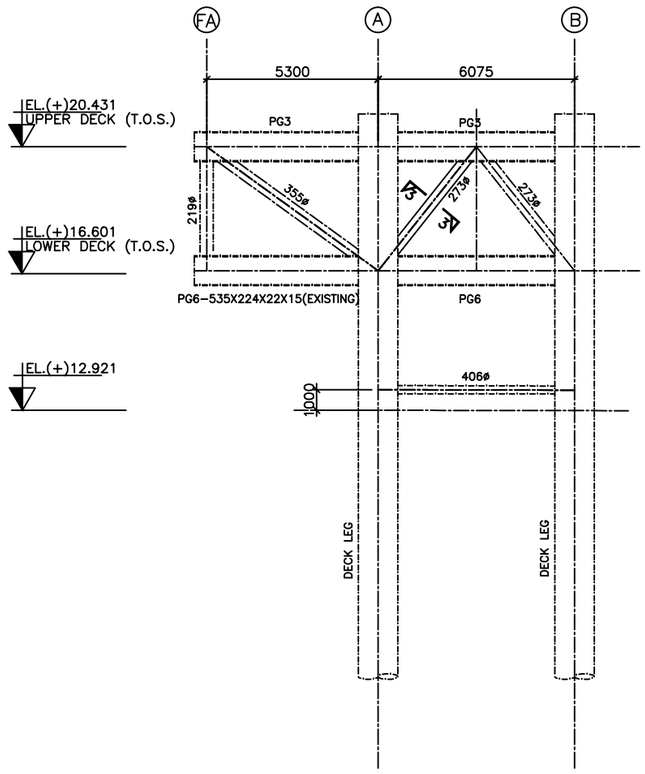
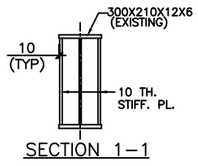


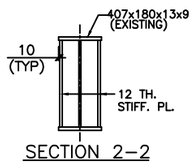
PART PLAN OF MAIN DECK EL(+)20.431(T.O.S.)



ELEVATION OF DECK ROW 1



SECTION 1–1



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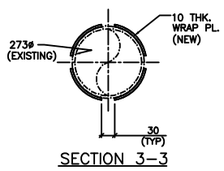

SECTION 3–3

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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ISSUED FOR TENDER

- EPCC CONTRACTOR SHALL PERFORM THE REQUIRED RESIDUAL ENGINEERING FOR REMOVAL OF CONTROL ROOM AT NC PLATFORM(MAIN DECK) TO FACILITATE THE MODIFICATION OF BRIDGE (WIN–NC) & LANDING MODIFICATION.
- MODIFICATION MARKED HERE WITH ARE BASED ON PRELIMINARY NDT RESULT & SITE OBSERVATION.IT IS EPCC CONTRACTOR RESPONSIBILITY TO CONDUCT NDT FOR COMPLETE PLATFORM AND MODIFY AS REQUIRED.

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	01.04.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF NC DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	005	83	43	0004	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की सम्पत्ति है। ये मात्र उधार दिए गए है और उपकर्मों ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही तृतीय पक्ष में दी गई सहमति से होगा ।
किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उपकर्मों को लिखित रूप में दी गई सहमति से होगा ।
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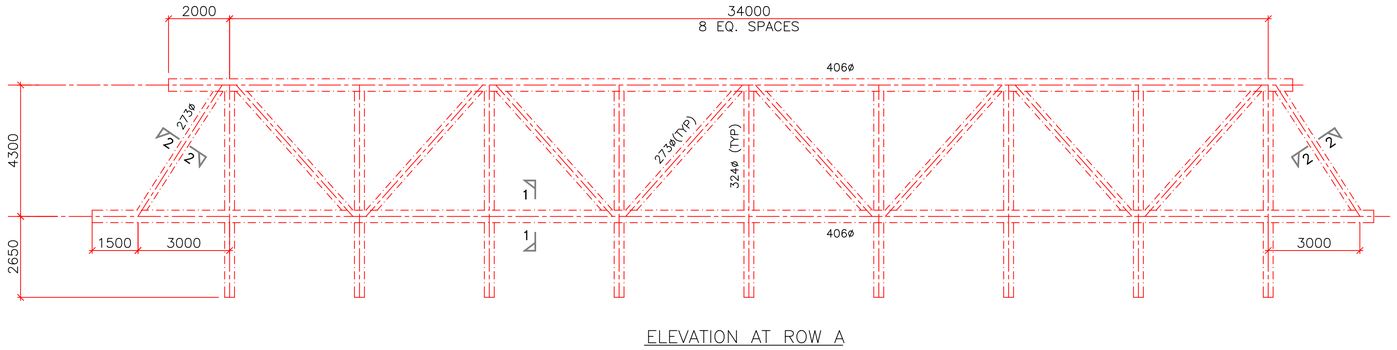
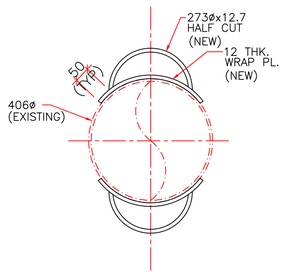
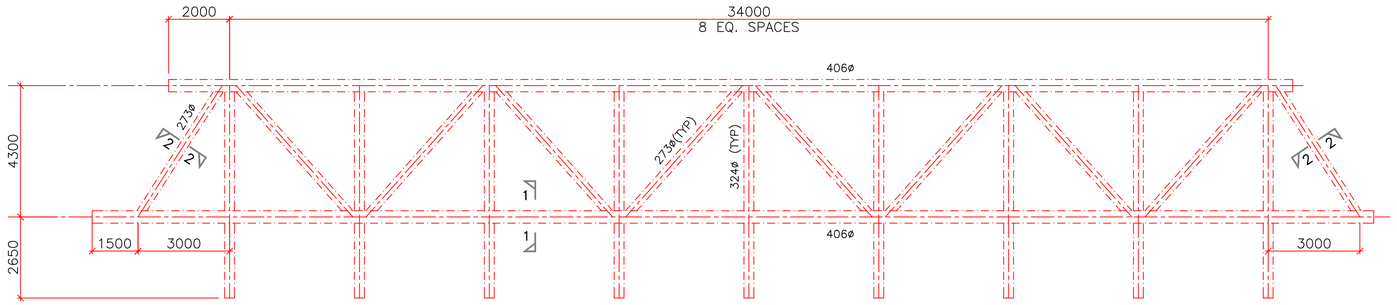
B774-007-83-43-0001



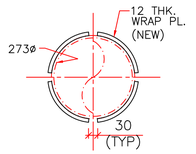
ELEVATION AT ROW A



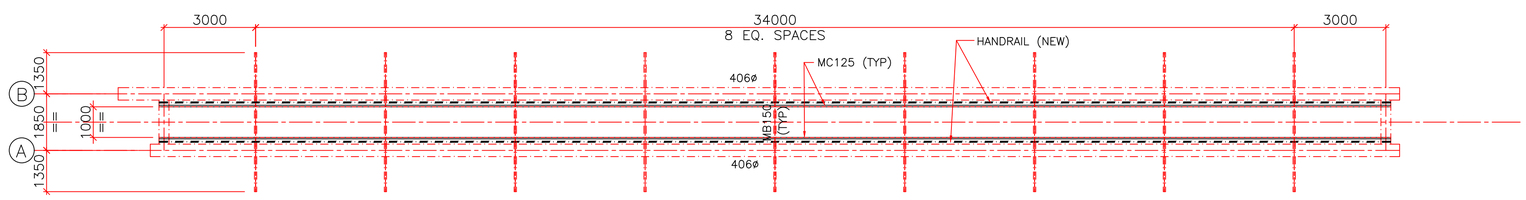
SECTION 1–1



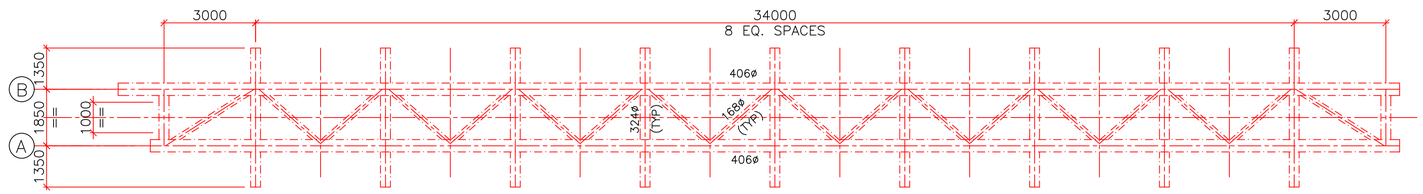
ELEVATION AT ROW B



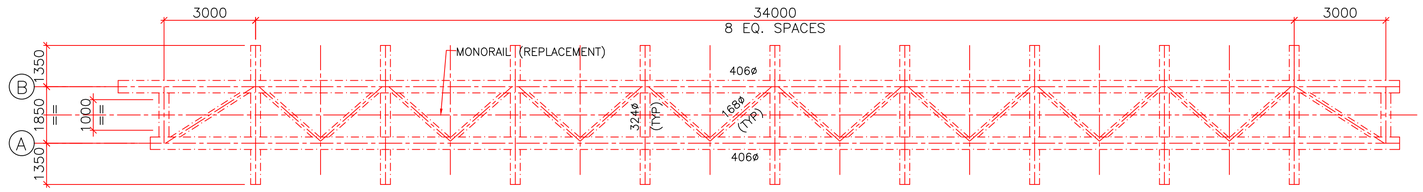
SECTION 2–2



WALKWAY LEVEL



FRAMING PLAN AT TOP CHORD EL(+)23.350



FRAMING PLAN AT TOP CHORD EL(+)19.050

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR TENDER	MUNISH	MM	DS	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SCI-SCA BRIDGE TOP DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	007	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए है और उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कार्य अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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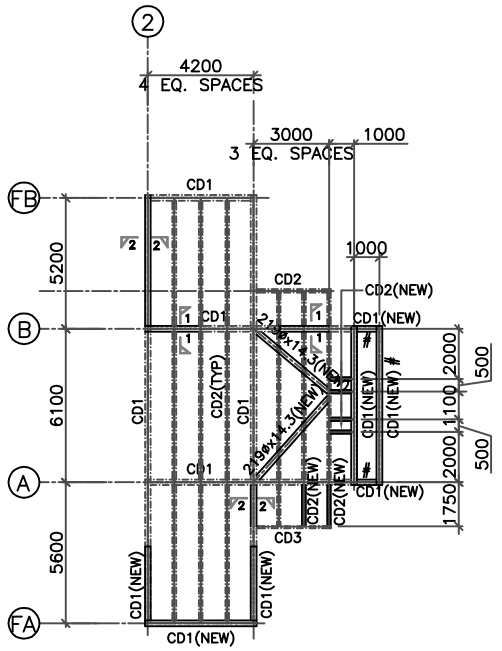
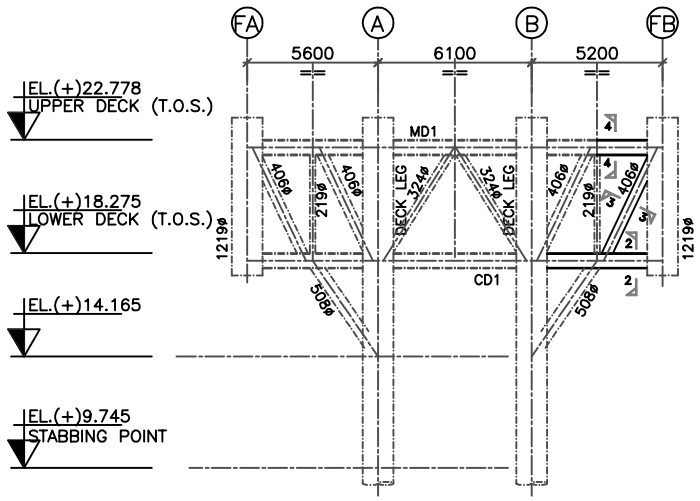
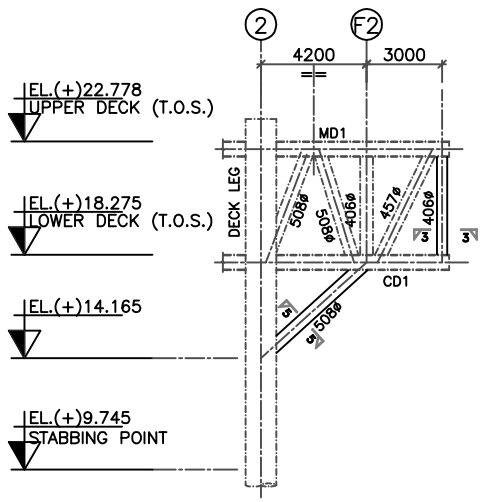
B774-007-83-43-0002



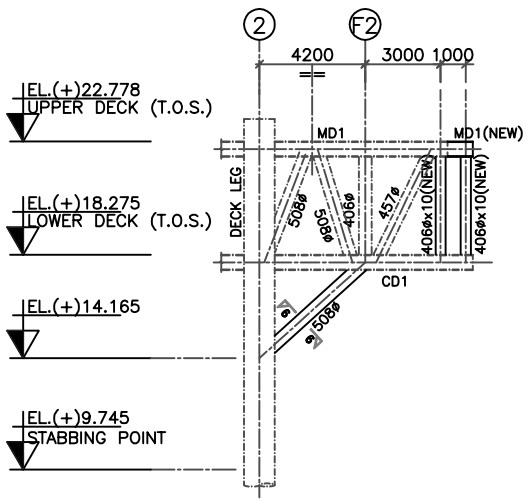
PART PLAN OF CELLAR DECK EL(+)18.275(T.O.S.)



ELEVATION OF DECK ROW 2



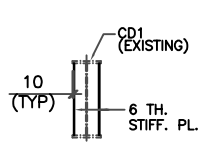
PART ELEVATION OF DECK ROW A



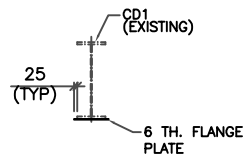
PART ELEVATION OF DECK ROW B

CD1	610X225X12X19
CD2	360X142X8X13
MD1	613X335X12X16

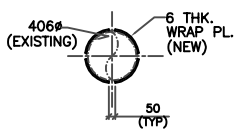
* THE MEMBER SIZES AND THICKNESS ARE MEASURED VALUES FROM NDT. EPC CONTRACTOR SHALL VERIFY ALL THE MEMBER SIZES DURING PRE-ENGINEERING SURVEY.



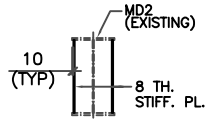
SECTION 1-1



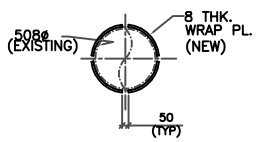
SECTION 2-2



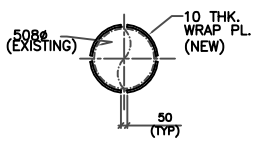
SECTION 3-3



SECTION 4-4



SECTION 5-5



SECTION 6-6

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7-60-0802.
9. FOR HANDRAIL REFER EIL STD. 7-60-0801.

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* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)
* THE BRIDGE SHALL BE LIFTED USING JACK-UP AND MEMBERS SHALL BE REPLACED AT THE BRIDGE LANDING AREA.

FIELD NOTES

THE MEMBER MARKED SHALL BE PLACED BEFORE THE REMOVAL OF MEMBERS TO BE REPLACED AND THE SAME SHALL BE REMOVED AFTER THE REPLACEMENT OF MEMBERS AT BRIDGE LANDING AREA.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	13.03.26	ISSUED FOR TENDER	MUNISH	AP	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

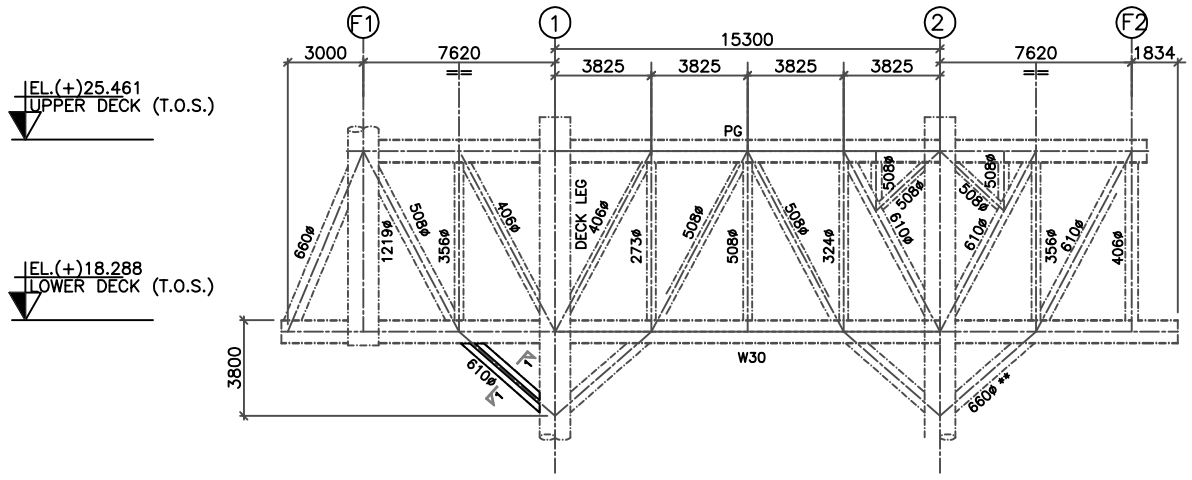
REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SC-1 DECK

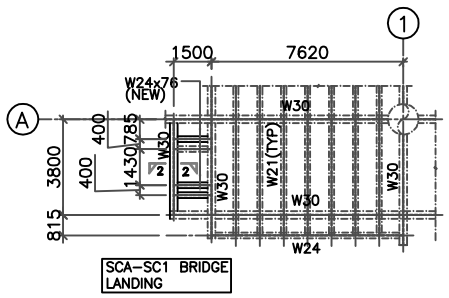
SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	007	83	43	0002	A

3-1641-0501 REV.2 A1-841x594

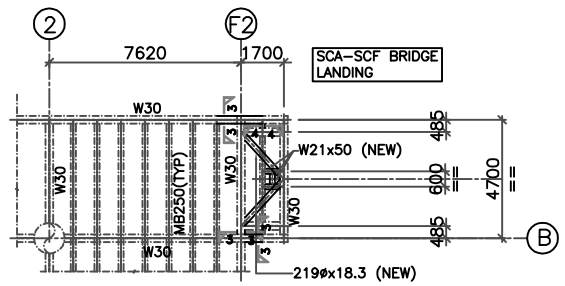
B774-007-83-43-0003



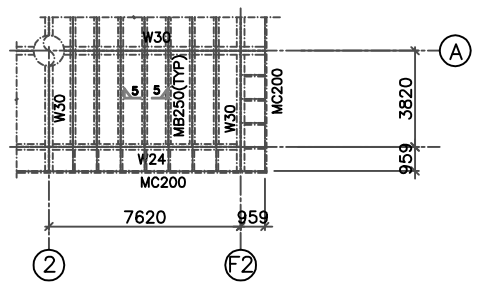
ELEVATION OF DECK ROW B



PART PLAN OF CELLAR DECK EL(+)18.050



PART PLAN OF CELLAR DECK EL(+)18.050



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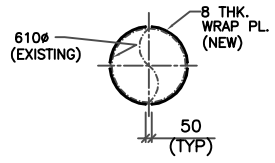
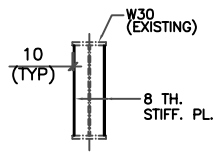
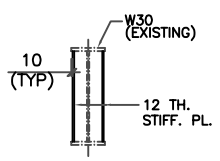


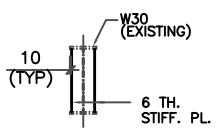
SECTION 1–1



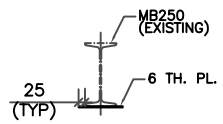
SECTION 2–2



SECTION 3–3



SECTION 4–4



SECTION 5–5

REF. DWG. NO.	REFERENCE DRAWING TITLE

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FIELD NOTES

\# THE MEMBER MARKED SHALL BE PLACED BEFORE THE REMOVAL OF MEMBERS TO BE REPLACED AND THE SAME SHALL BE REMOVED AFTER THE REPLACEMENT OF MEMBERS AT BRIDGE LANDING AREA.
** THE MEMBER SIZE AND THICKNESS SPECIALLY (DUE TO MISMATCH) SHALL BE VERIFIED DURING DETAILED ENGINEERING FOR THE MARKED MEMBER BEFORE FINALIZING THE MODIFICATIONS.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	13.03.26	ISSUED FOR TENDER	MUNISH	AP	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SCA DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	007	83	43	0003	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः तुलित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रकाशित किए जाएंगे और न ही सीमित और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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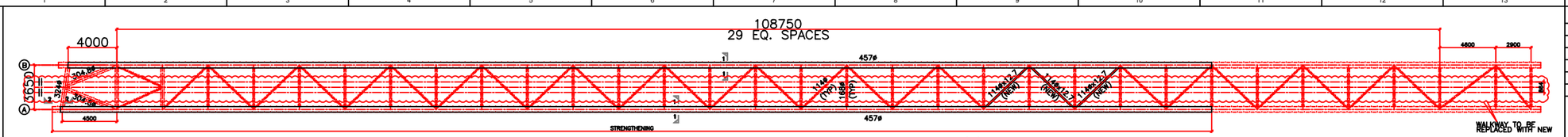
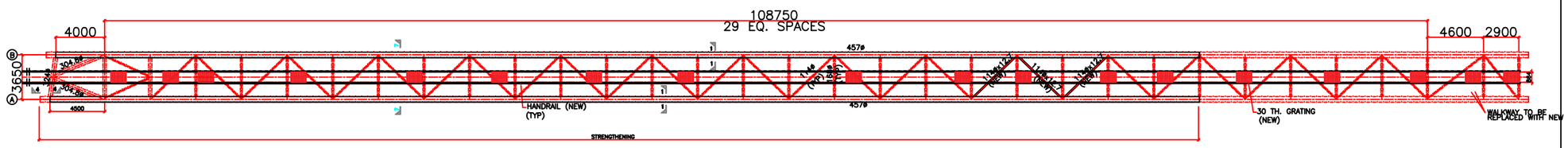
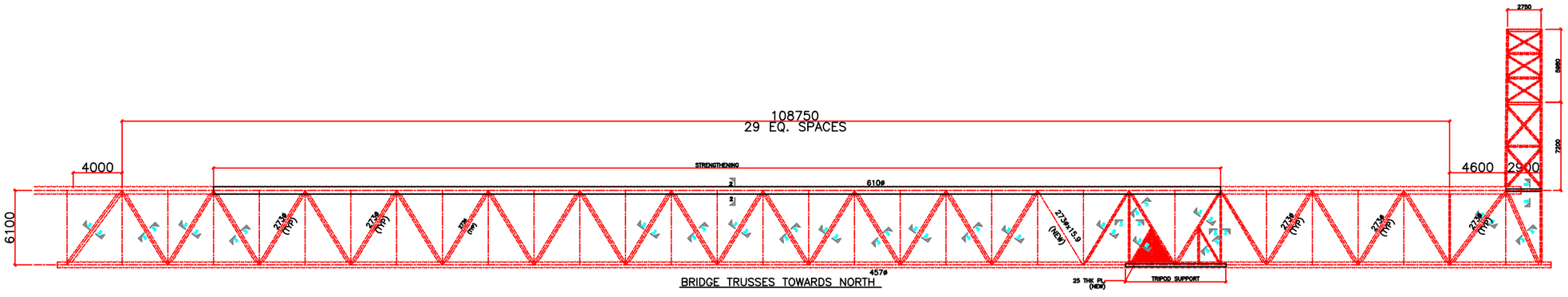
B774-008-83-43-001



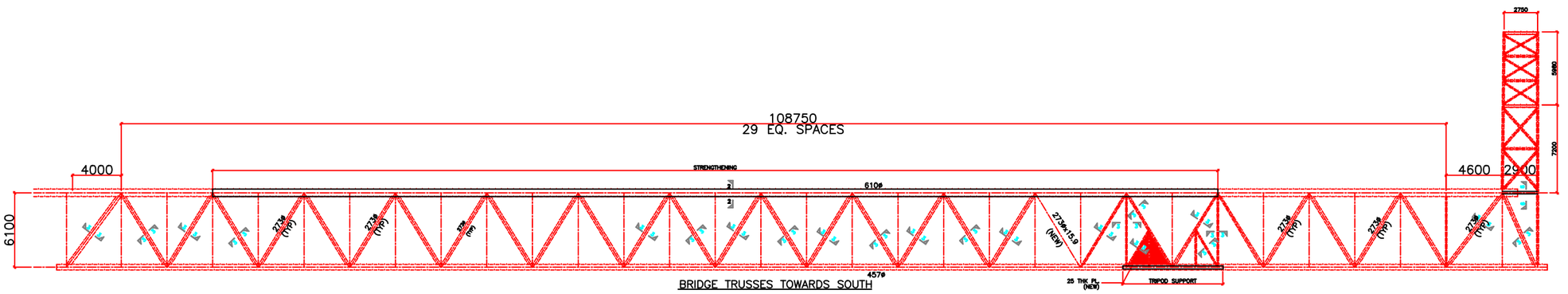
FRAMING PLAN AT BOTTOM CHORD EL(+)18.500



BRIDGE WALKWAY AND HANDRAIL AT BOTTOM CHORD



BRIDGE TRUSSES TOWARDS NORTH



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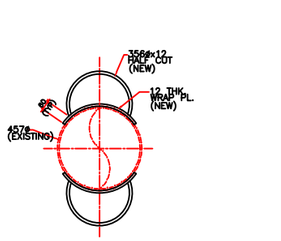

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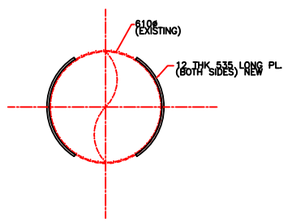

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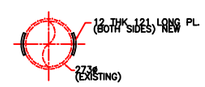
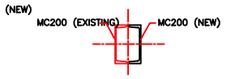


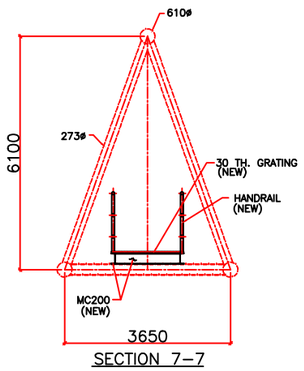
SECTION 3–3

SECTION 4–4

SECTION 5–5



SECTION 6–6



SECTION 7–7

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

- ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE–ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR TENDER	MUNISH	AMS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SCA-SCF BRIDGE TOP DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	008	83	43	0001	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रतिलिपि किए जाएंगे और न ही सहित और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।

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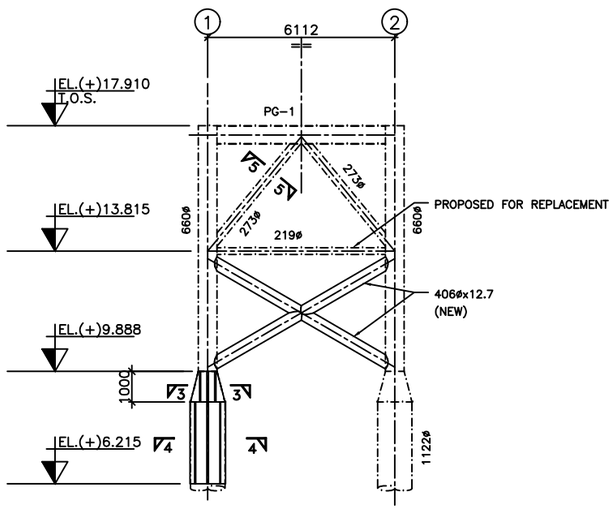
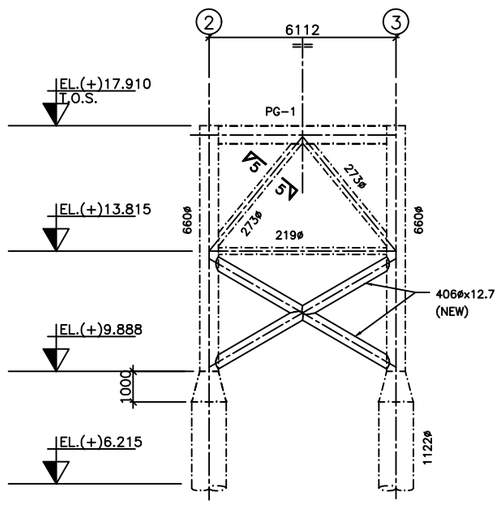
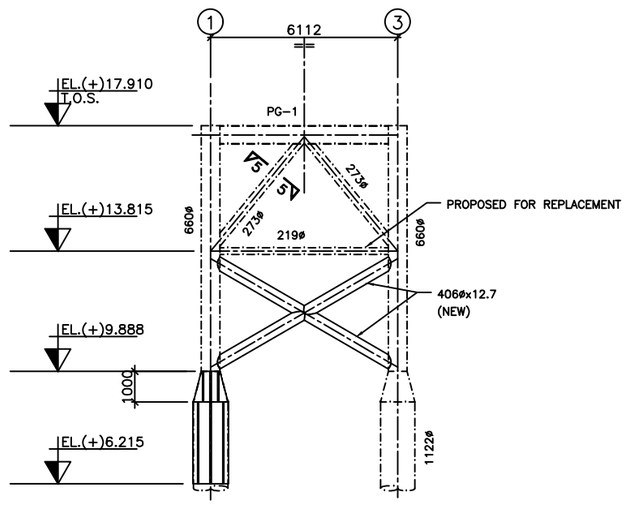
B774-008-83-43-0003



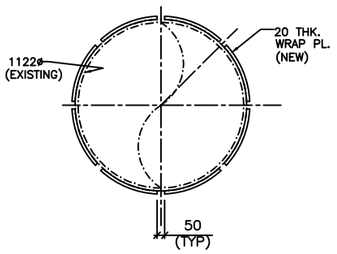
VIEW ON GRID 1 & 2



VIEW ON GRID 2 & 3



VIEW ON GRID 1 & 3



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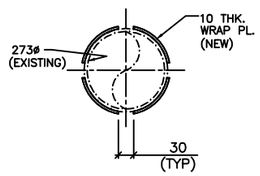

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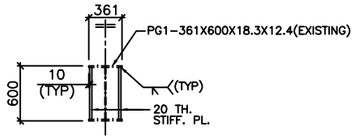

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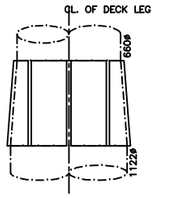
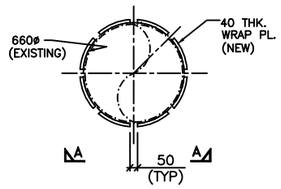


VIEW A–A
STRENGTHENING STABBING
POINT AT ROW B



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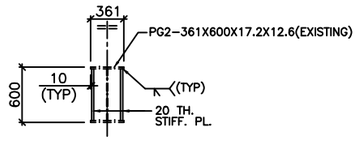
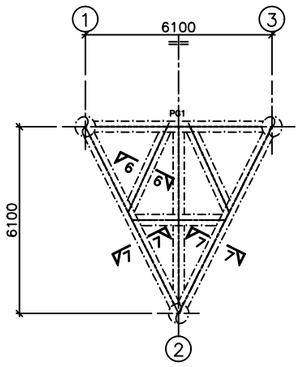


SECTION 7–7



TOP VIEW OF TRIPOD

REF. DWG. NO.	REFERENCE DRAWING TITLE
SAPL0038–STRL–L–332	SCF FLARE JACKET STEEL LAYOUT ELEVATION & PLAN VIEW

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.
10. SOME MEMBERS OF TRIPOD FALL IN PURVIEW OF WAVE SLAMMING DETAIL ENGINEERING CONTRACTOR SHALL CONSIDER THE APPROPRIATE LOADING AND FINALIZE DESIGN DURING EXECUTION STAGE.

- ALL DETAILS, DIMENSIONS, LEVELS & MEMBER SIZES IN THE DRAWING ARE INDICATIVE AND ARE PROVIDED FOR BIDDERS UNDERSTANDING OF SCOPE OF WORK ONLY.ALL DETAILS SHOWN IN THE DRAWING SHALL BE VERIFIED BY CONTRACTOR DURING PRE–ENGINEERING SURVEY & DETAIL ENGINEERING SHALL BE CARRIED OUT BY CONTRACTOR FOR EXECUTION OF WORKS.
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REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	13.03.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

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OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SCF FLARE JACKET

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	008	83	43	003	0

3-1641-0501 REV.2 A1-841x594

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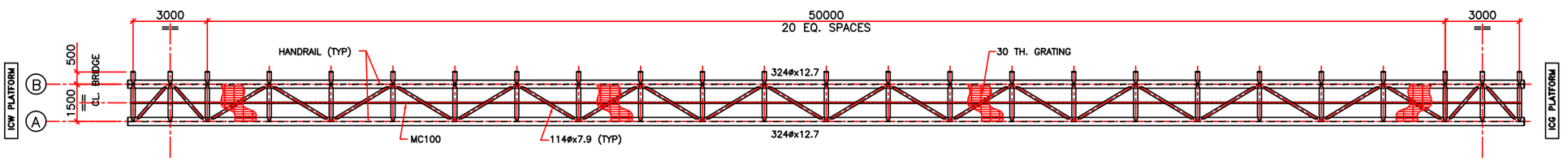
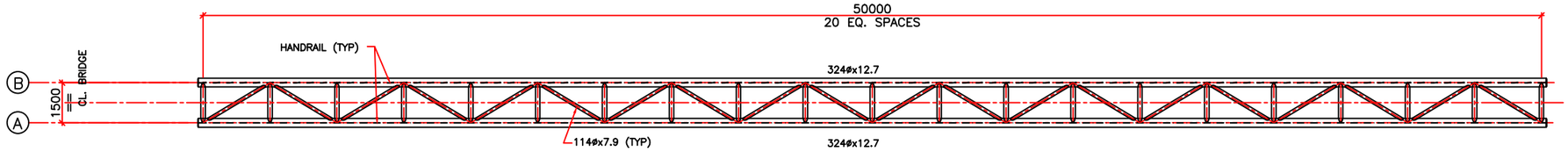


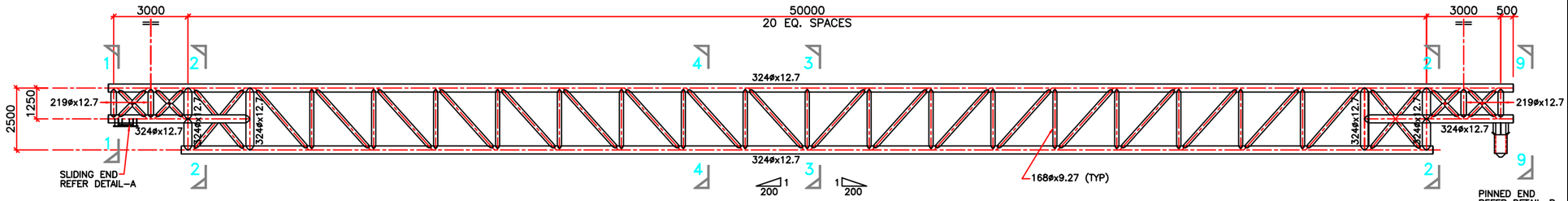
BRIDGE WALKWAY AND HANDRAIL LAYOUT AT TOP CHORD

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



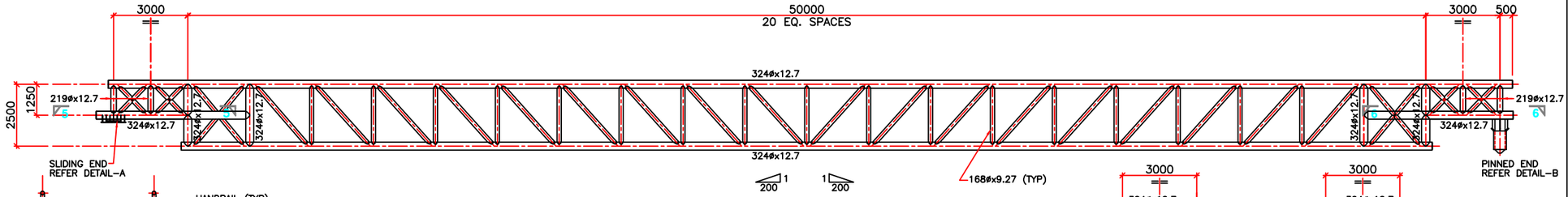
BRIDGE AT BOTTOM CHORD

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



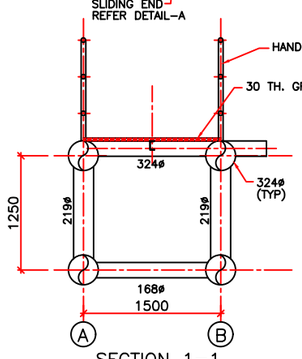
ELEVATION AT ROW A

ALL MEMBERS ARE 168ø x9.5 (U.N.O.)



ELEVATION AT ROW B

ALL MEMBERS ARE 168ø x9.5 (U.N.O.)



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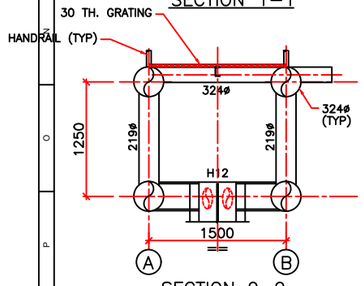

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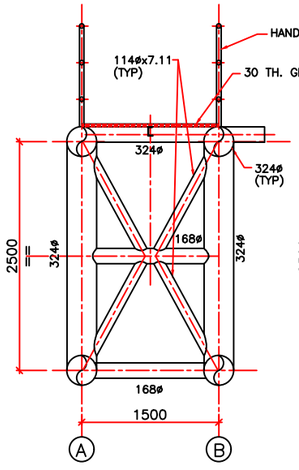

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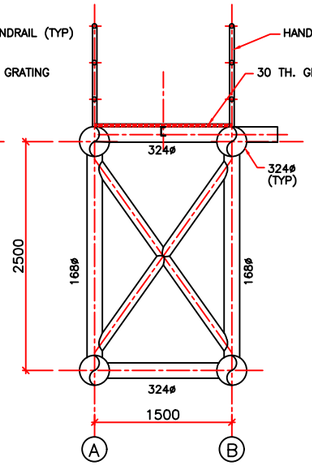

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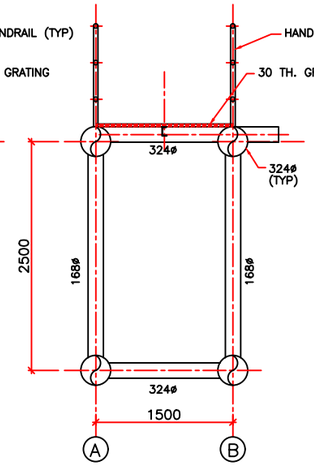

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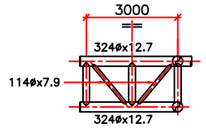
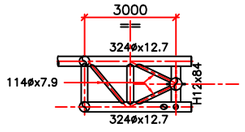


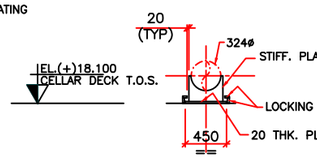
SECTION 5–5

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



SECTION 6–6

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



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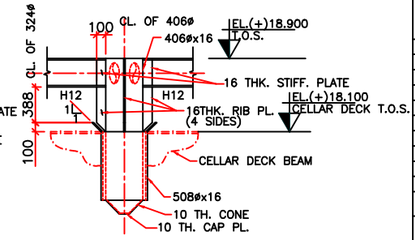

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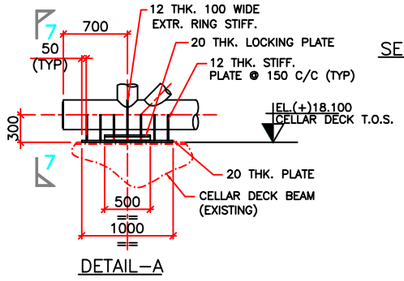
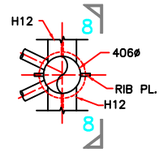


DETAIL–A



DETAIL–B

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICG–DG–MD–104	LOWER DECK FRAMING PLAN
ICG–DG–MD–105	ELEVATION ROW A & B
ICG–DG–MD–106	ELEVATION ROW 1 & 2
ICG–DG–MD–107	ELEVATION ROW 3 & 4
ICW–70–052	LOWER DECK FRAMING PLAN
ICW–70–056	ICW PLATFORM DECK ELEVATION AT ROW A & B
ICW–70–057	ICW PLATFORM DECK ELEVATION AT ROW 1,2,3 & 4

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   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC NO 6001–F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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ISSUED FOR TENDER

FOR TENDER PURPOSE ONLY

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR BIDS	MUNISH	AA	DP	

ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

एम एन संपत्तियों की मौजूदा पुल का नवीकरण/प्रदलाव	REFURBISHMENT/REPLACEMENT OF EXISTING BRIDGE AT MH ASSET
आई सी डबलू से आई सी जी के निचली छत पुल का सामान्य विन्यास एवं ब्यौरा	G.A. AND DETAILS OF ICW TO ICG LOWER DECK BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	101	83	43	0001	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रतिलिपि किए जाएंगे और न ही सीमित और निर्धा प्रयोग के अलावा इनका किसी अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा।
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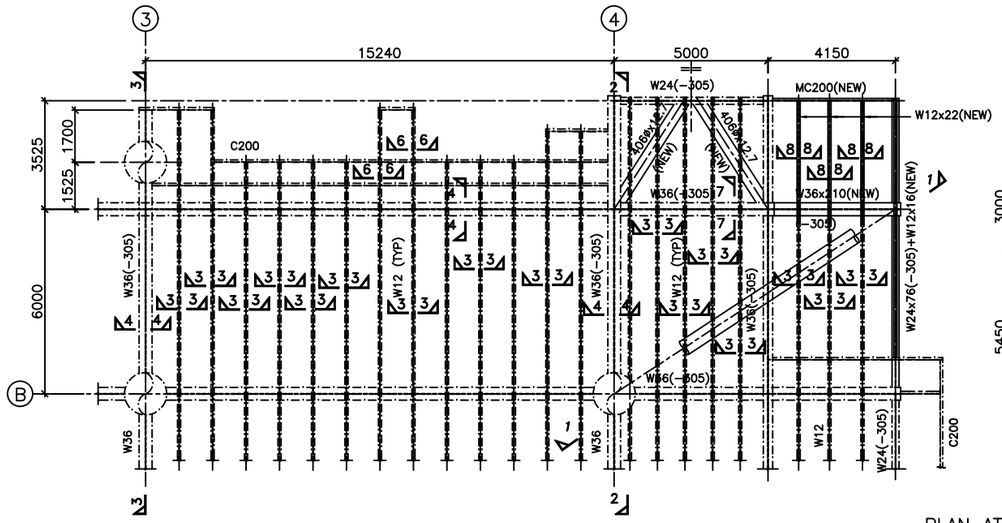
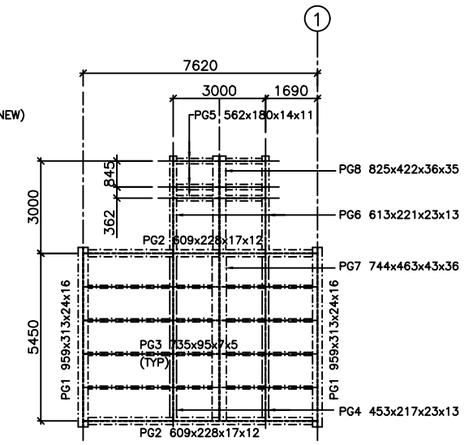
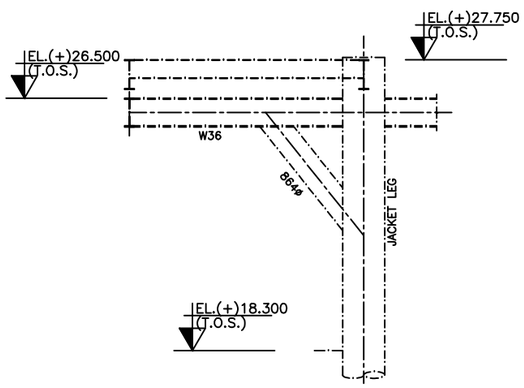
B774-101-83-43-0002



PLAN AT EL(+)18.300 (T.O.S.)



PLAN AT LQ LEVEL AT BRIDGE LANDING AREA EL(+)27.750 (T.O.S.)



PART ELEVATION OF ROW B

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICW–70–050 TO ICW–70–059	ICW PLATFORM DECK DRWAINGS

NOTES :-

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5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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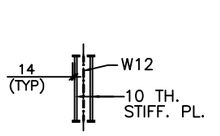
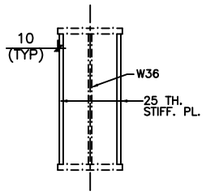
* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER



SECTION 3–3



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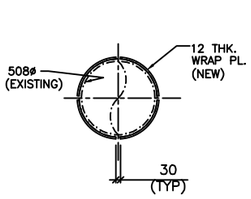

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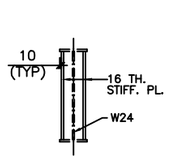

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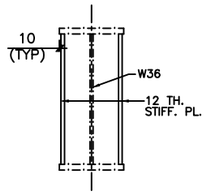

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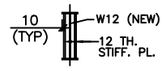
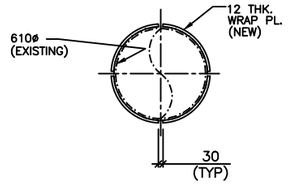


SECTION 8–8



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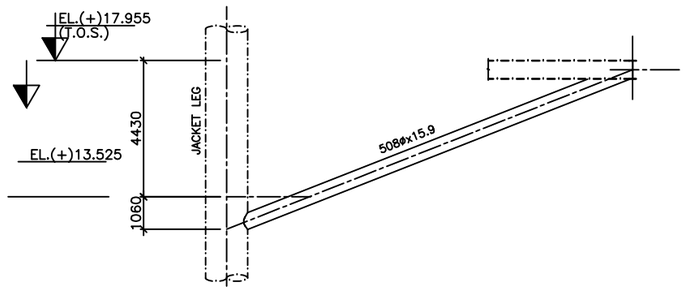
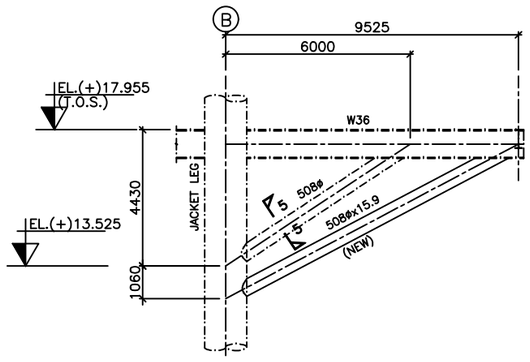


SECTION 1–1



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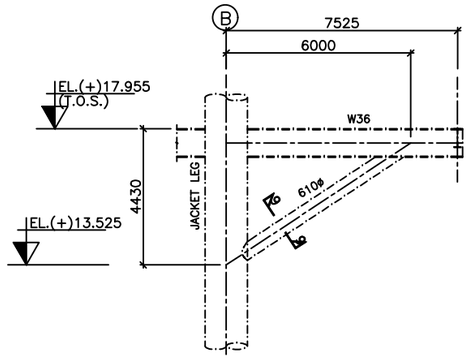

SECTION 3–3

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	01.12.25	ISSUED FOR TENDER	MUNISH	AMS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF ICW UPPER DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	101	83	43	0002	A

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए है और उसकारकर्ता ने यह स्पष्ट समझौता किया है कि न तो उन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही संसिद्ध और किसी प्रयोग के अलावा इसका कार्य अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उसकारकर्ता को लिखित रूप से दी गई सहमति से होगा।
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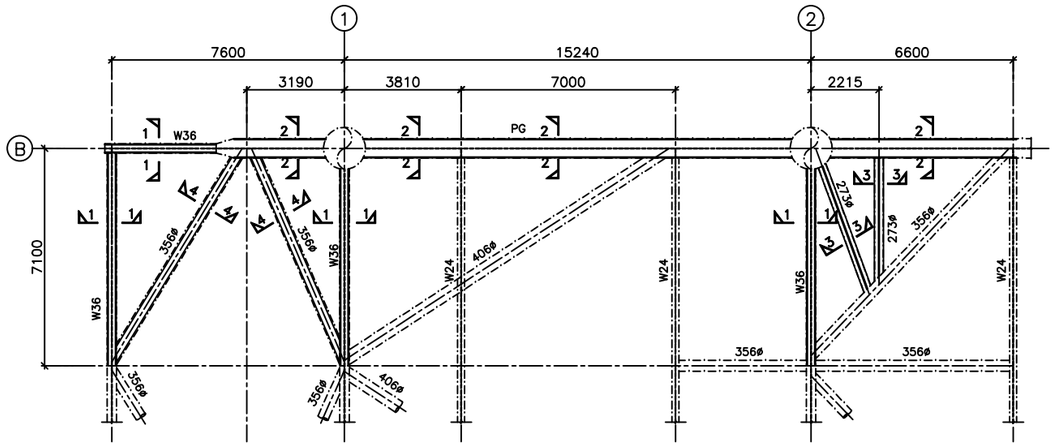
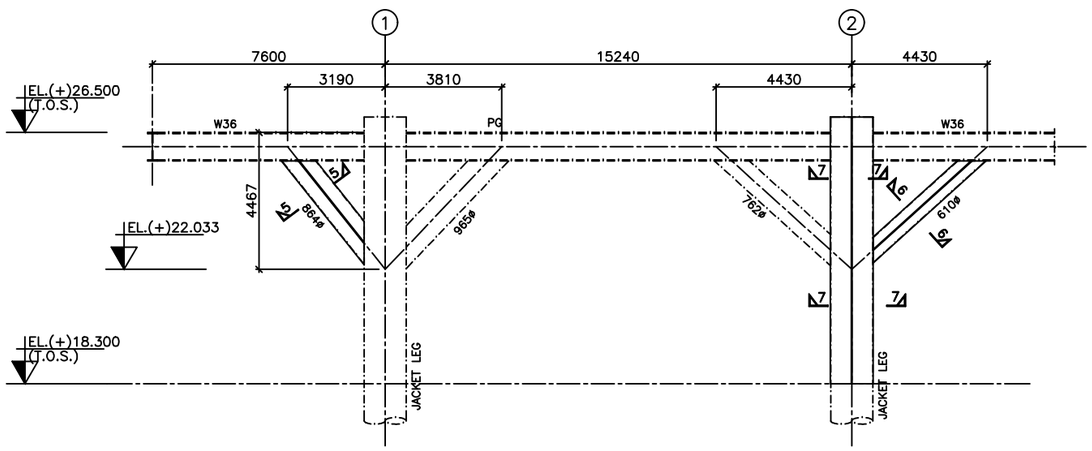
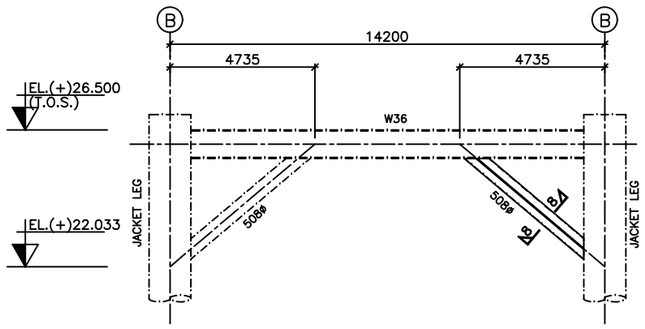
B774-101-83-43-0003



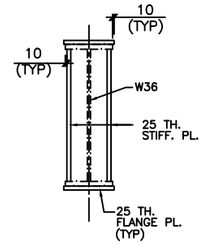
PART PLAN OF MAIN DECK AT EL(+)26.500 (T.O.S.)



PART ELEVATION OF ROW B



PART ELEVATION OF ROW 2



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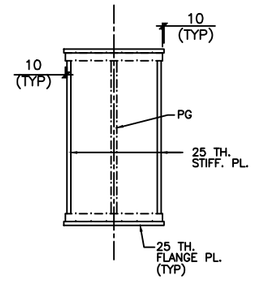

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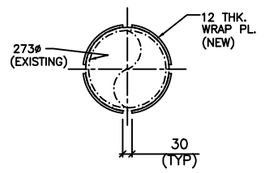
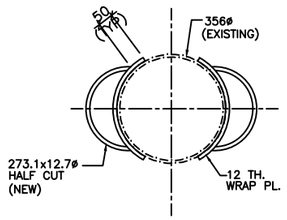


SECTION 3–3



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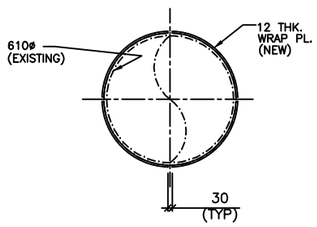

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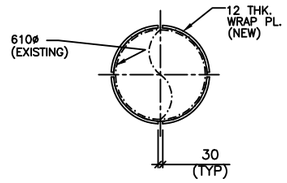

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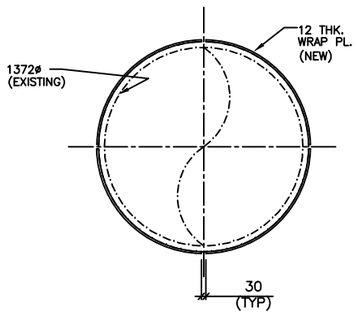

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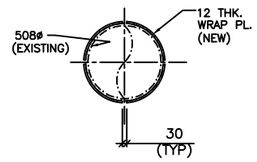

SECTION 8–8

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICW-70-050 TO ICW-70-059	ICW PLATFORM DECK DRWAINGS

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
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STRENGTHENING OF ICW UPPER DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	101	83	43	0003	A

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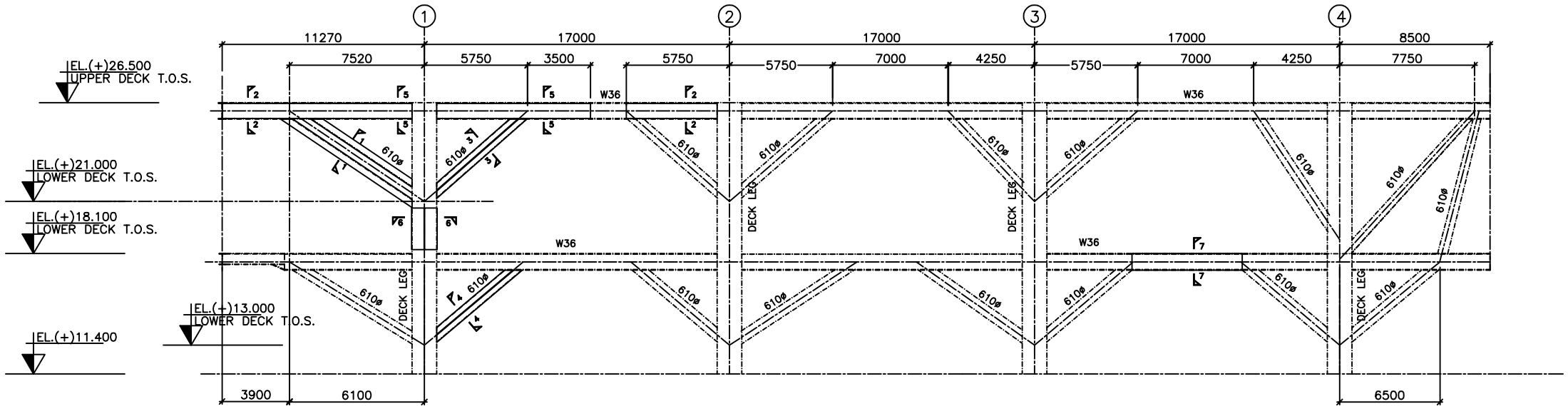
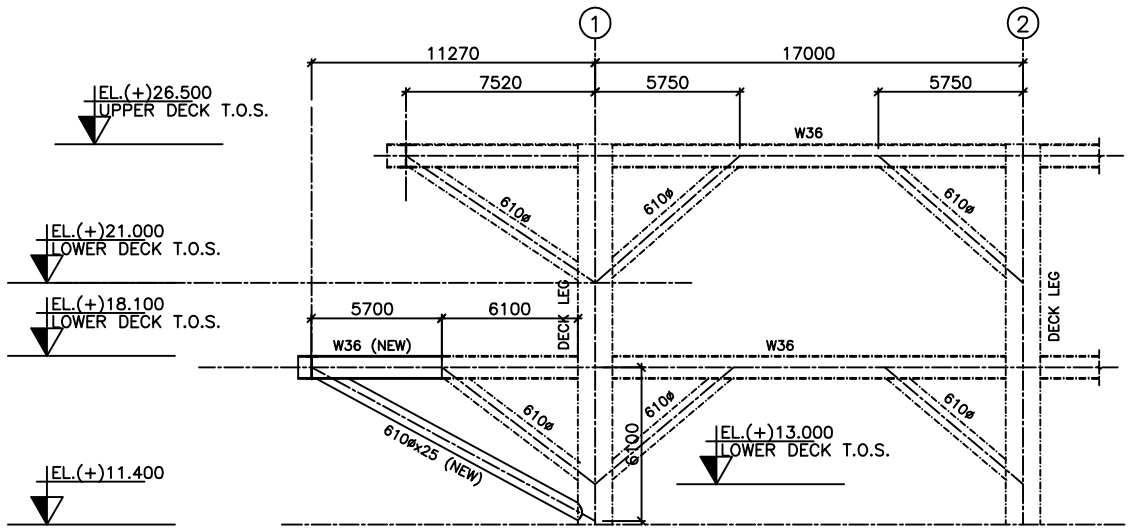
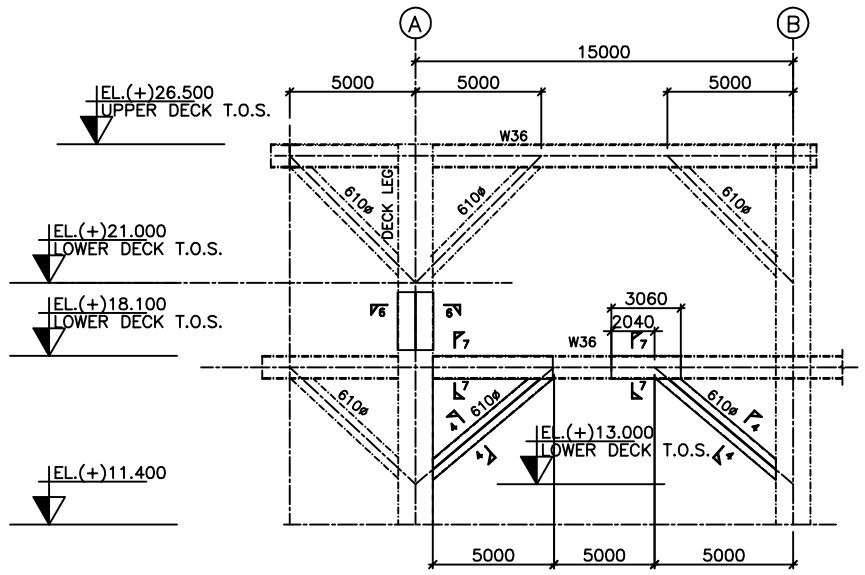
B774-101-83-43-0004



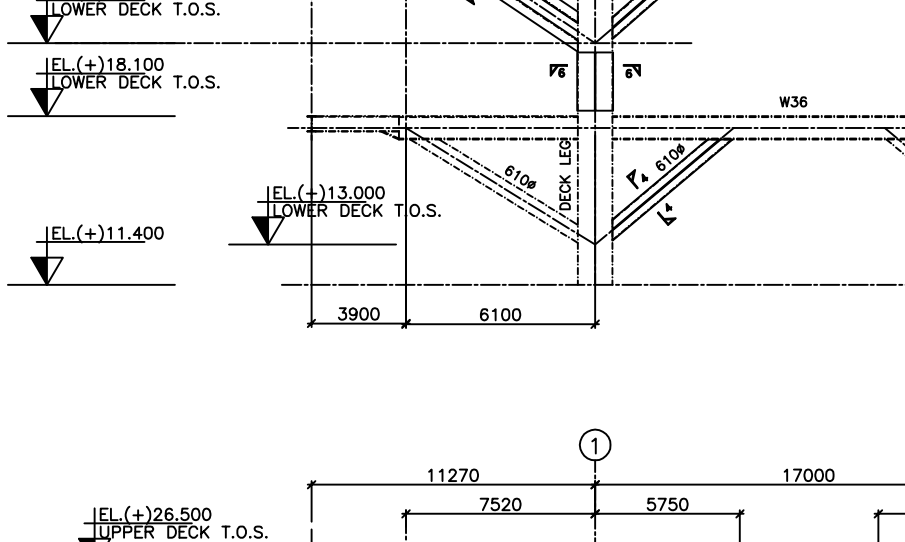
ELEVATION OF DECK ROW A



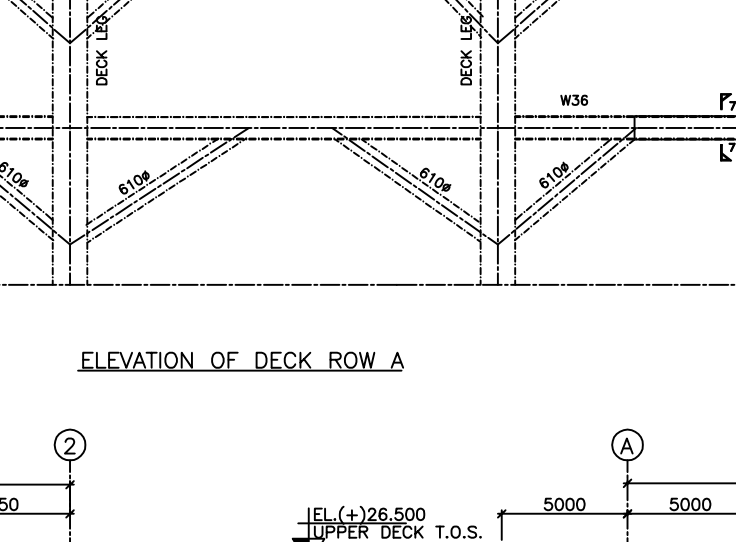
PART ELEVATION OF DECK ROW B



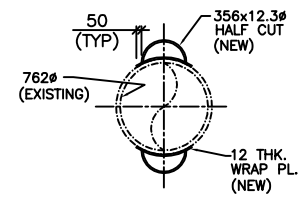
PART ELEVATION OF DECK ROW 1



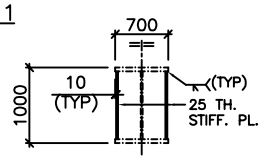
PART ELEVATION OF DECK ROW 3



PART ELEVATION OF DECK ROW 2



SECTION 1–1



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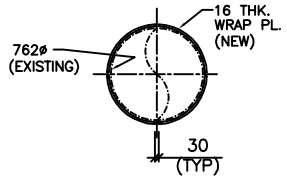
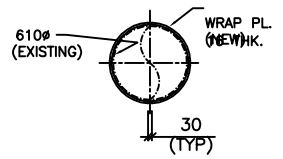
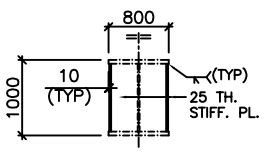


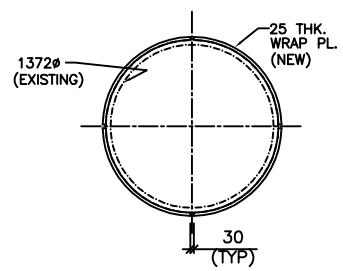
SECTION 3–3



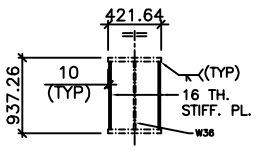
SECTION 4–4



SECTION 5–5



SECTION 6–6



SECTION 7–7

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICG-DD-MD-103 TO 108	ICG DECK DRAWINGS

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. ALL ELEVATIONS ARE W.R.T. CHART DATUM (CD) 0.0M CHART DATUM FOR INDIAN WATERS SHALL BE 2.5M BELOW MEAN SEA LEVEL (MSL).
4. MATERIAL SPECIFICATIONS
   a) ALL STIFFENERS, FLANGE EXTENSION PLATES ETC. SHALL BE OF SAME / HIGHER GRADE AS ELEMENT TO WHICH THEY ARE CONNECTED U.N.O.
5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	09.12.25	ISSUED FOR TENDER	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

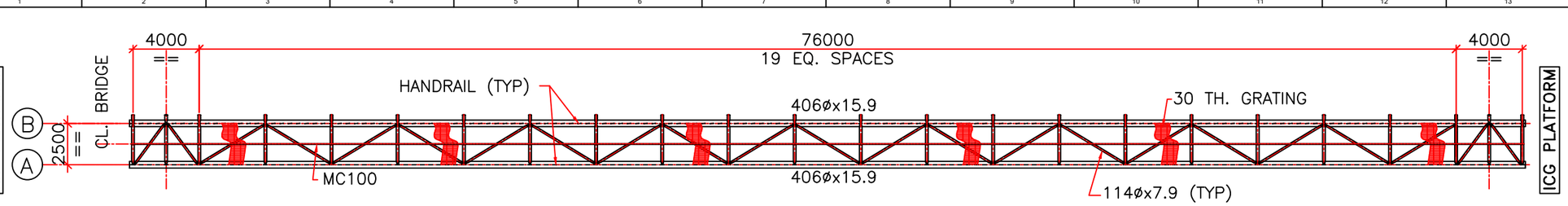
ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

ICG LANDING AREA-MODIFICATIONS

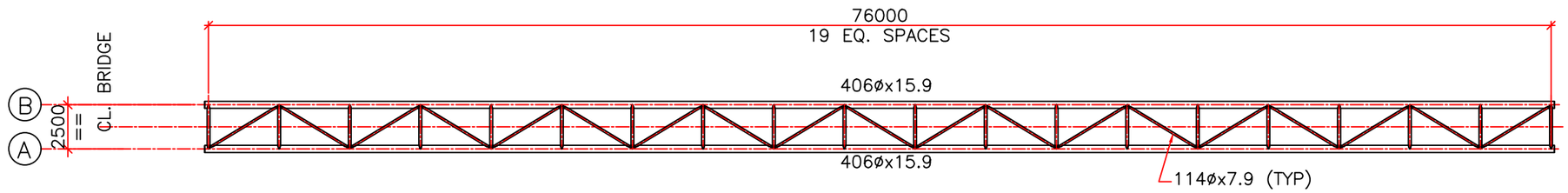
SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	101	83	43	0004	A

3-1641-0501 REV.2 A1-841x594



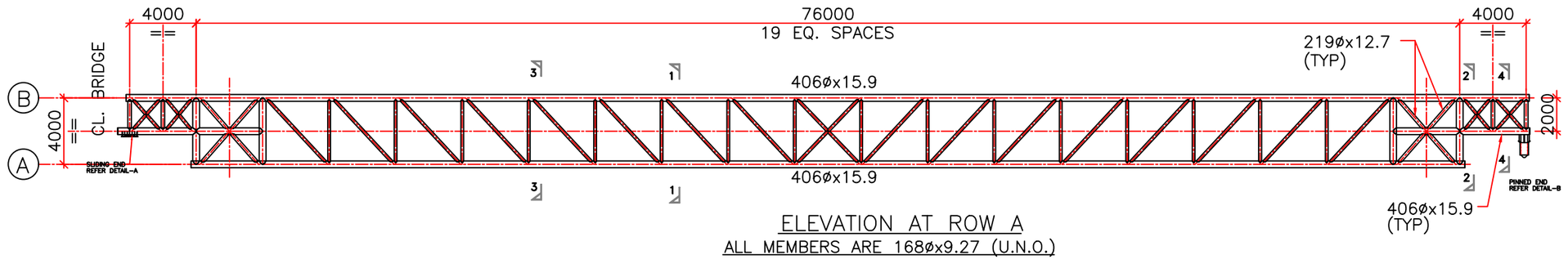
BRIDGE WALKWAY AND HANDRAIL LAYOUT AT TOP CHORD

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



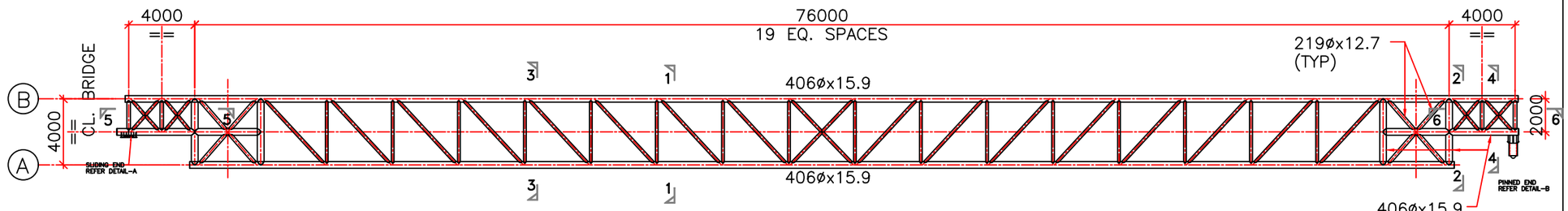
BRIDGE AT BOTTOM CHORD

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



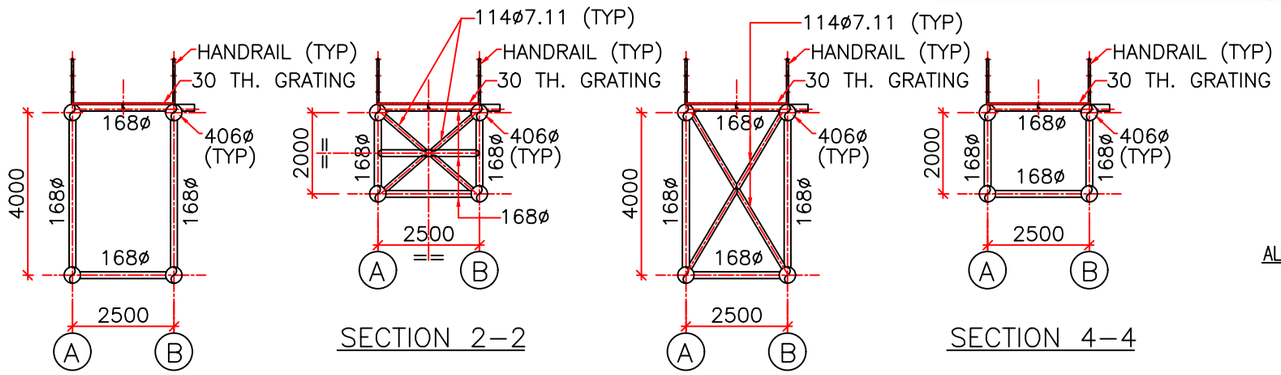
ELEVATION AT ROW A

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)

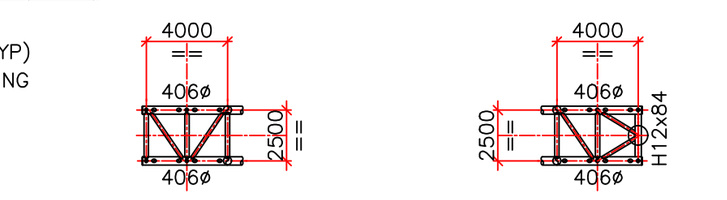


ELEVATION AT ROW B

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)



SECTION 1–1 | SECTION 2–2 | SECTION 3–3 | SECTION 4–4



SECTION 5–5 | SECTION 6–6

ALL MEMBERS ARE 168ø x9.27 (U.N.O.)

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	09.10.25	ISSUED FOR BIDS	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

एन एच संपत्तों की मौजूदा पुल का नवीकरण-बदलाव

REFURBISHMENT/REPLACEMENT OF EXISTING BRIDGE AT MH ASSET

आई सी पु से आई सी जी के निचली छत पुल का सामान्य विन्यास एवं ब्योरा

G.A. AND DETAILS OF ICP TO ICG LOWER DECK BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:225	B774	102	83	43	0001- sht 1 of 2	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है । ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो उन्हें पुन: मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही लिखित अंश में दी गई सहमति के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार दी गई इसी प्रणाली के अलावा इनका कोई अन्य प्रयोग किया जाएगा ।
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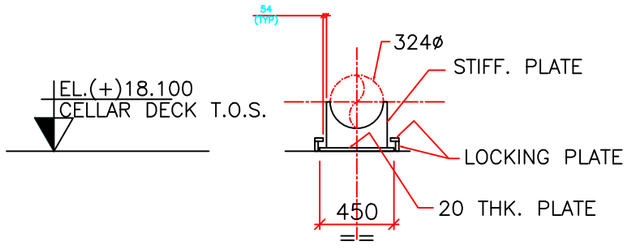
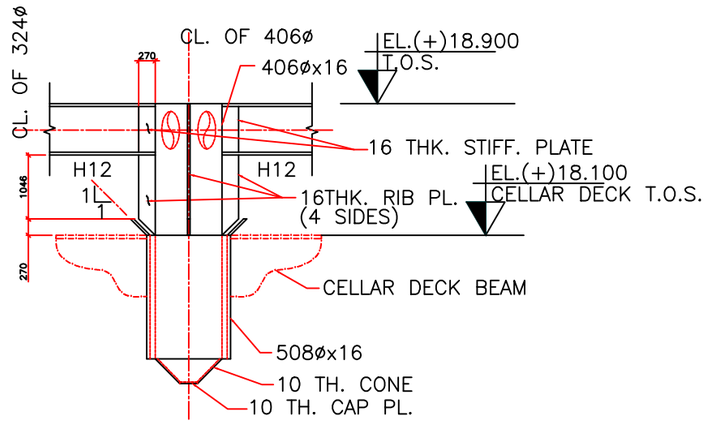
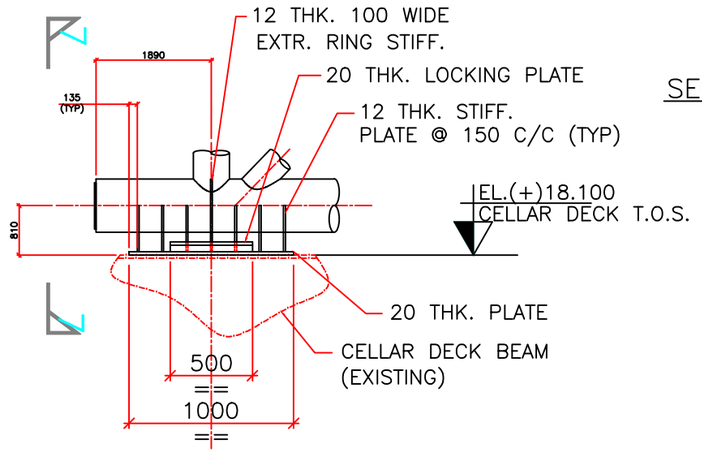
B774-102-83-43-0001



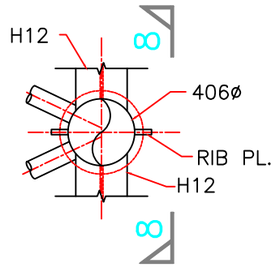
SECTION 7–7



SECTION 8–8



DETAIL–A



DETAIL–B

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. THE DRAWING SHALL BE READ IN CONJUNCTION WITH SHT 1 OF 2

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	09.10.25	ISSUED FOR BIDS	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

एम एच संपत्तियों की मौजूदा पुल का नवीकरण/बदलाव	REFURBISHMENT/REPLACEMENT OF EXISTING BRIDGE AT MH ASSET
आई सी पी से आई सी जी के निचली छत पुल का सामान्य विन्यास एवं ब्यौरा	G.A. AND DETAILS OF ICP TO ICG LOWER DECK BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:225	B774	102	83	43	0001- sht 2 of 2	A

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और इन्हें पुनः वापिस किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सीमित और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग किया जाएगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।

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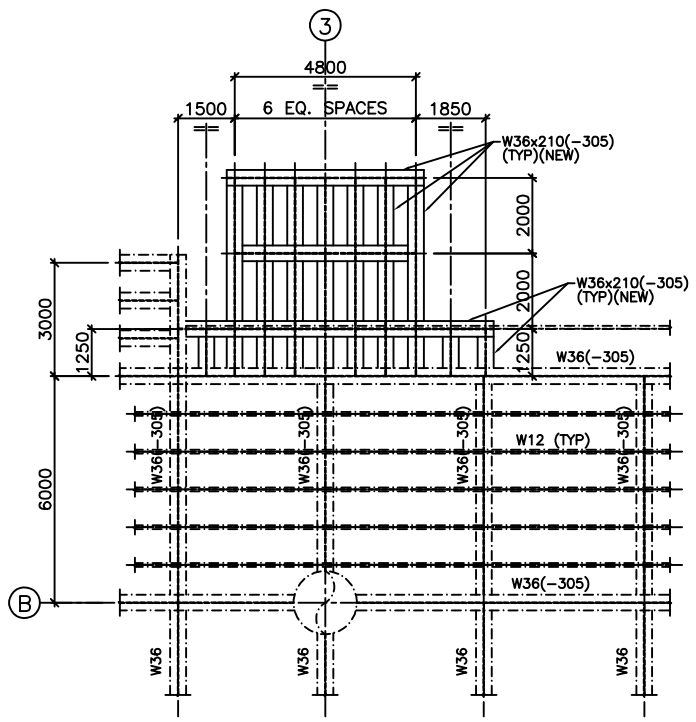
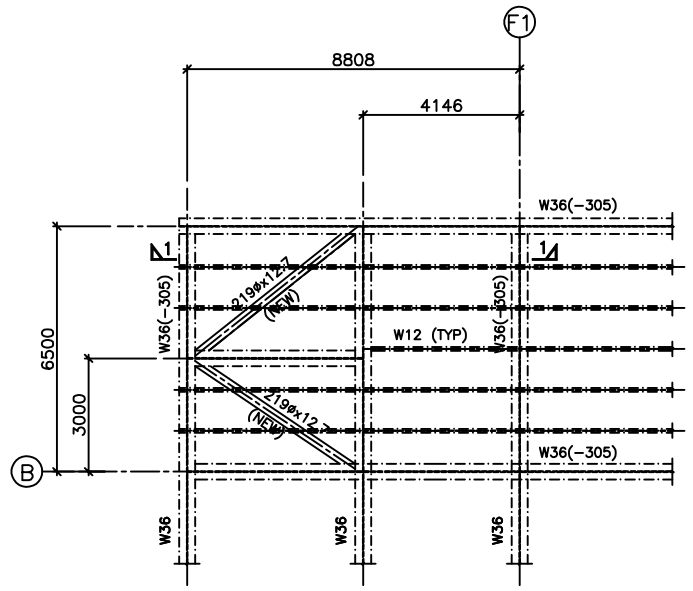
B774-102-83-43-0002



PART PLAN AT EL(+)18.500 (T.O.S.)



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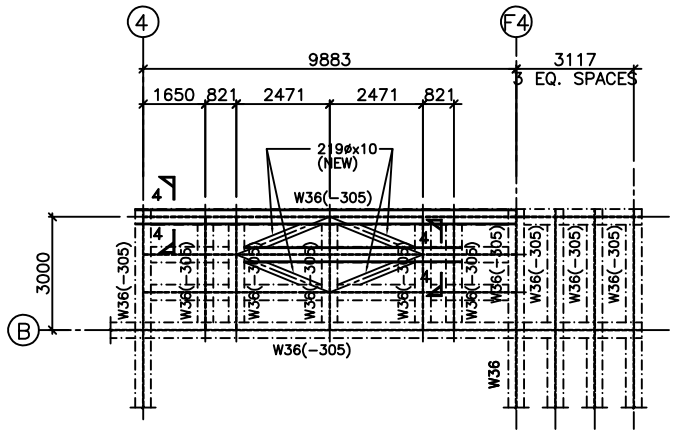

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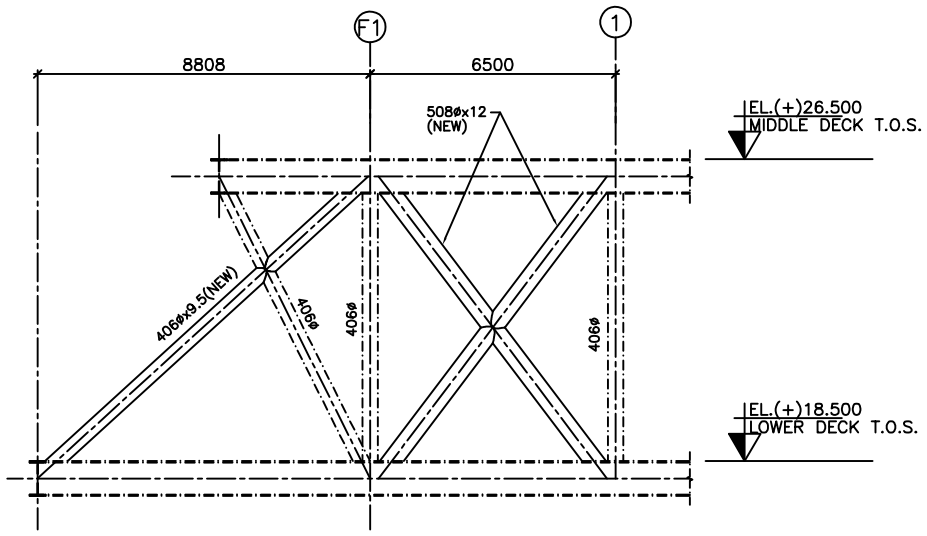
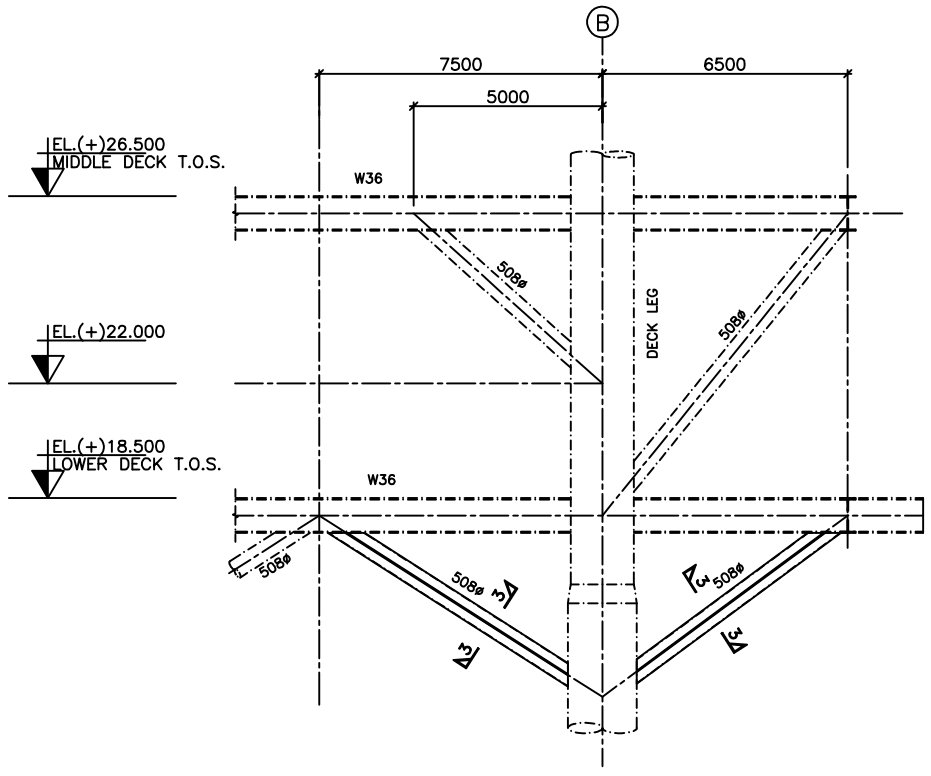
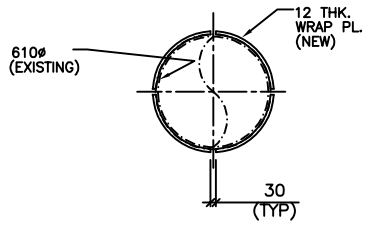


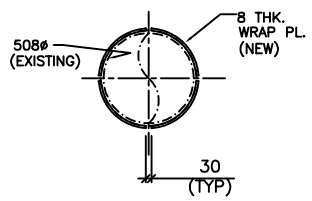
SECTION 1–1



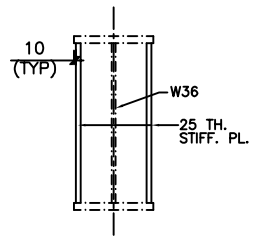
PART ELEVATION OF DECK ROW 1



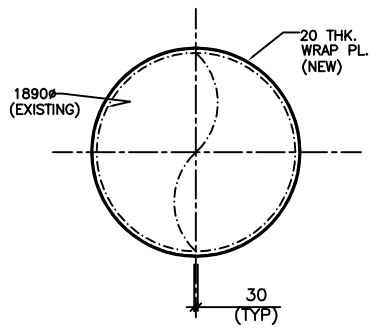
SECTION 2–2



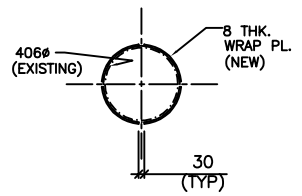
SECTION 3–3



SECTION 4–4



SECTION 5–5



SECTION 6–6

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICP–70–050 TO ICP–70–062	ICP PLATFORM DECK DRAWINGS

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
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9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	01.12.25	ISSUED FOR TENDER	MUNISH	AP	DP	

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ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

ICP LANDING AREA -MODIFICATIONS

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:100	B774	102	83	43	0002	A

3-1641-0501 REV.2 A1-841x594

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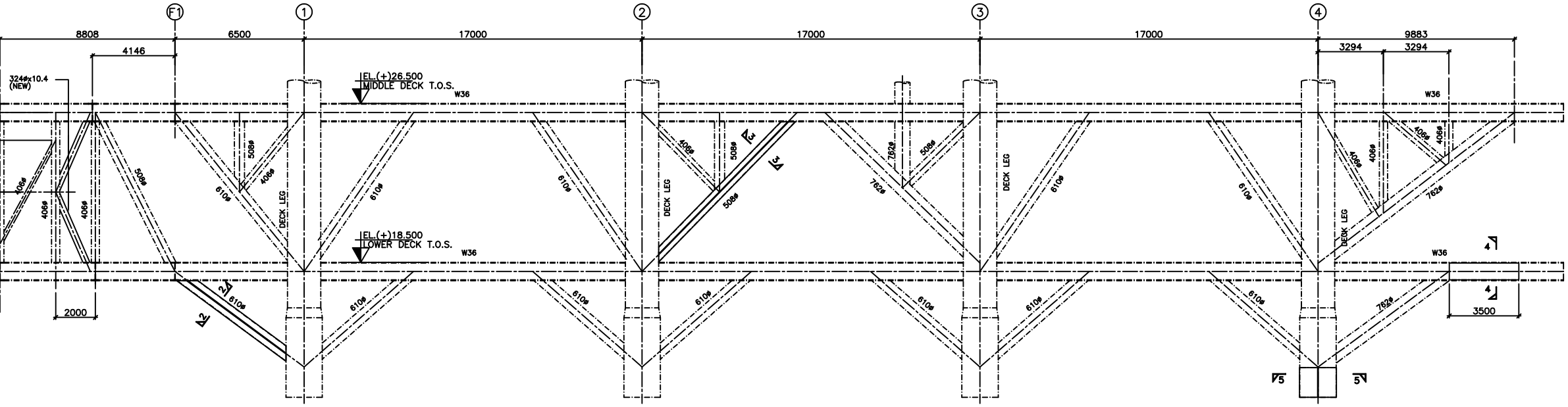
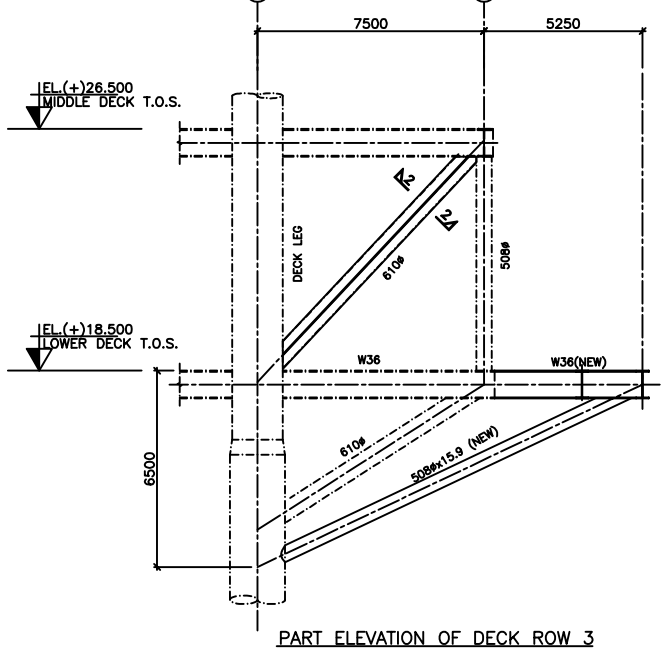
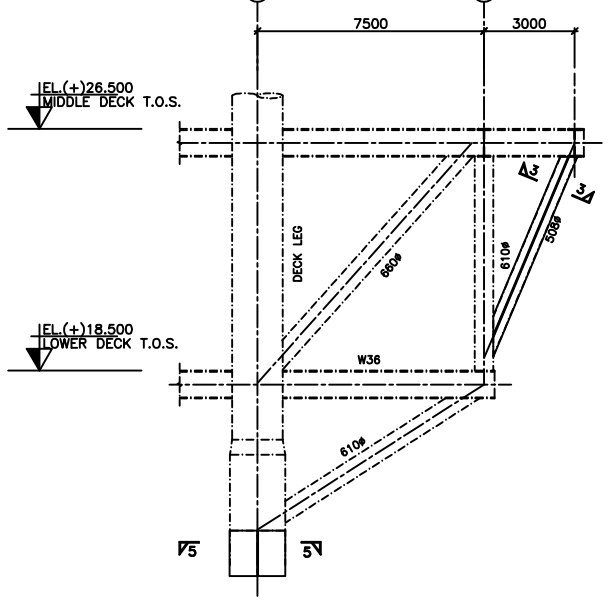


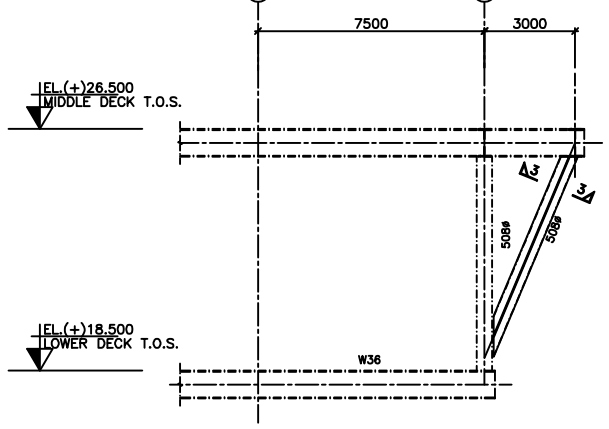
PART ELEVATION OF DECK ROW B



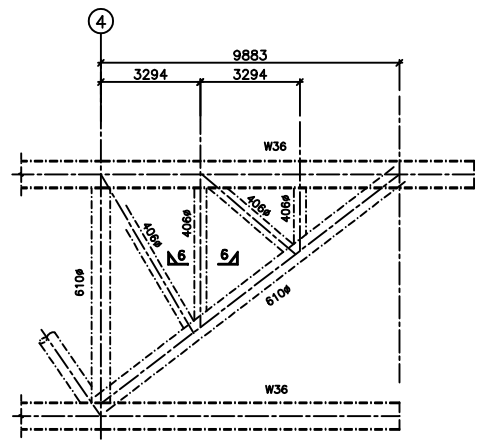
PART ELEVATION OF DECK ROW 3



PART ELEVATION OF DECK ROW 4



PART ELEVATION OF DECK ROW F4



PART ELEVATION OF DECK ROW FB

REF. DWG. NO.	REFERENCE DRAWING TITLE
ICP-70-050 TO ICP-70-062	ICP PLATFORM DECK DRAWINGS

NOTES :-

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5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7-60-0802.
9. FOR HANDRAIL REFER EIL STD. 7-60-0801.

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* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	01.12.25	ISSUED FOR TENDER	MUNISH	AP	DP	

ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

ICP LANDING AREA- MODIFICATIONS

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:125	B774	102	83	43	0003	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है कि न तो उन्हें पुनः तुलित किया जाएगा, न नकल की जाएगी, न उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो उन्हें पुनः तुलित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रयोगित किए जाएंगे और न ही वांछित और लिखित रूप से ऐसे द्वारा उधारकर्ता को लिखित रूप से ऐसे ही नहीं सकेंगे से होगा ।
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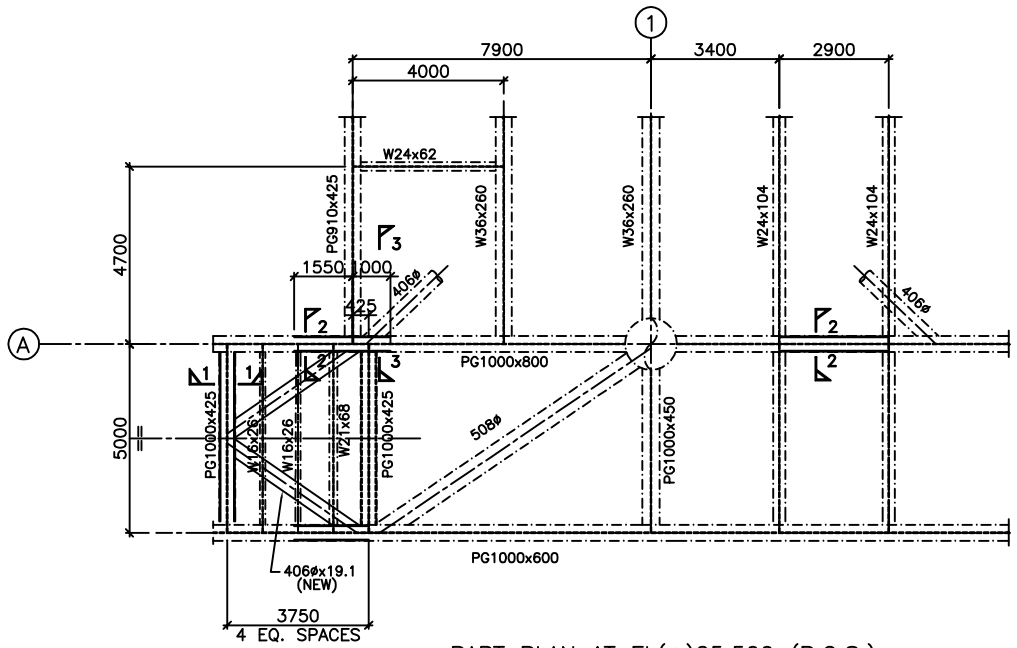
B774-102-83-43-0004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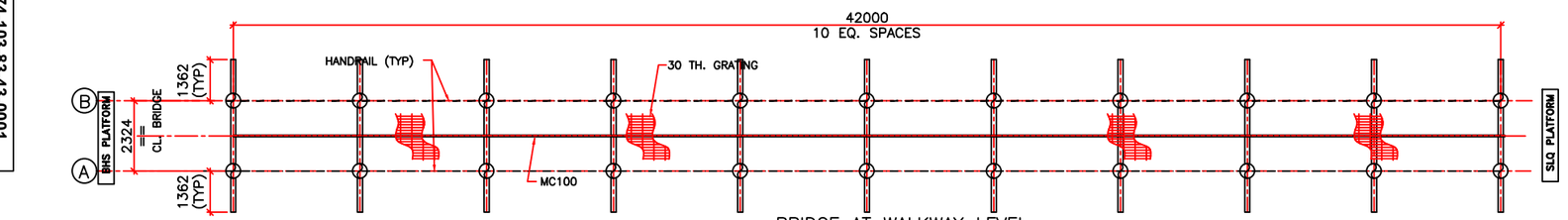
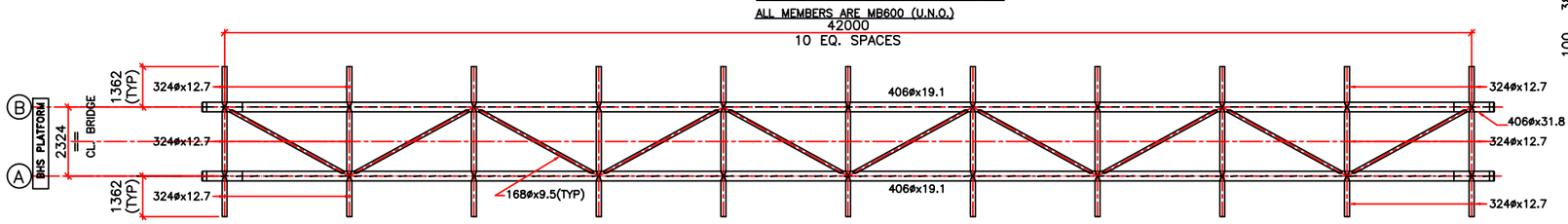
B774-103-83-43-0001



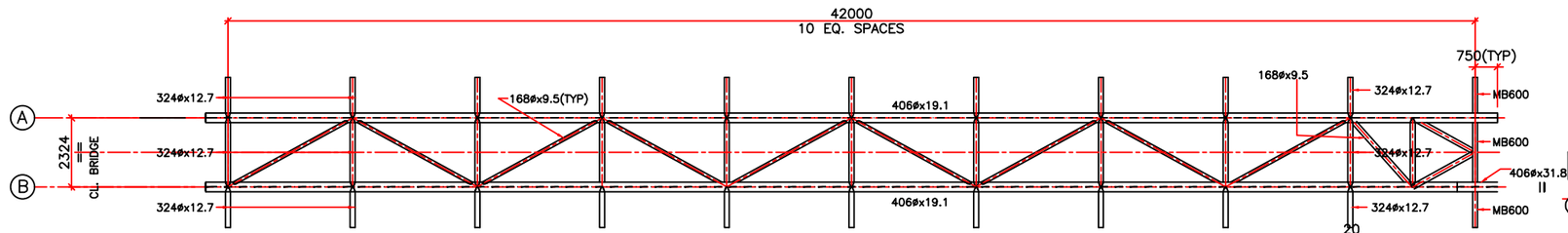
BRIDGE AT WALKWAY LEVEL

ALL MEMBERS ARE MB600 (U.N.O.)



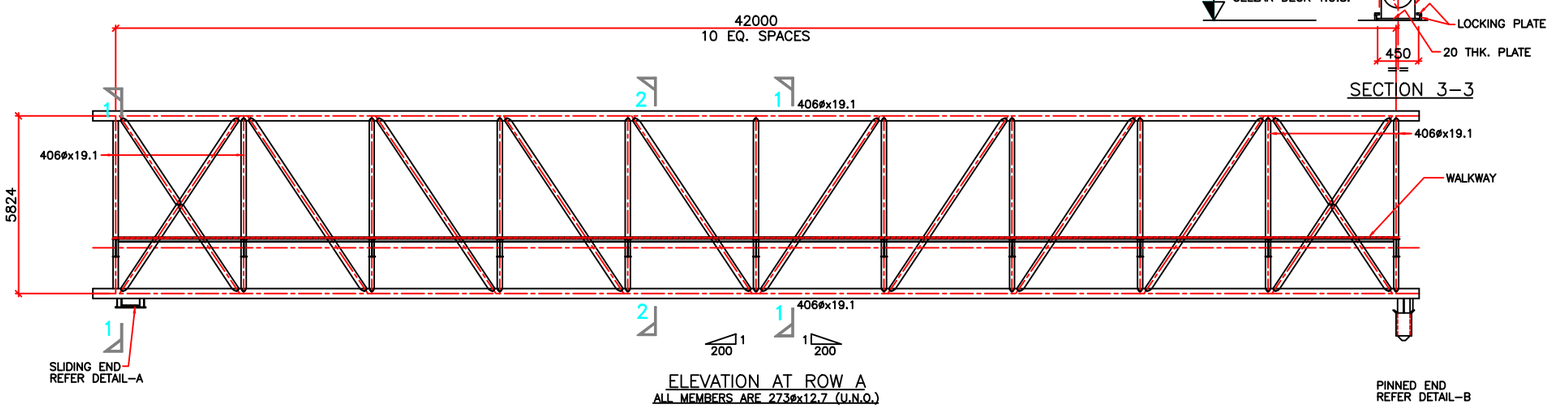
BRIDGE AT TOP CHORD

ALL MEMBERS ARE 273ø x12.7 (U.N.O.)



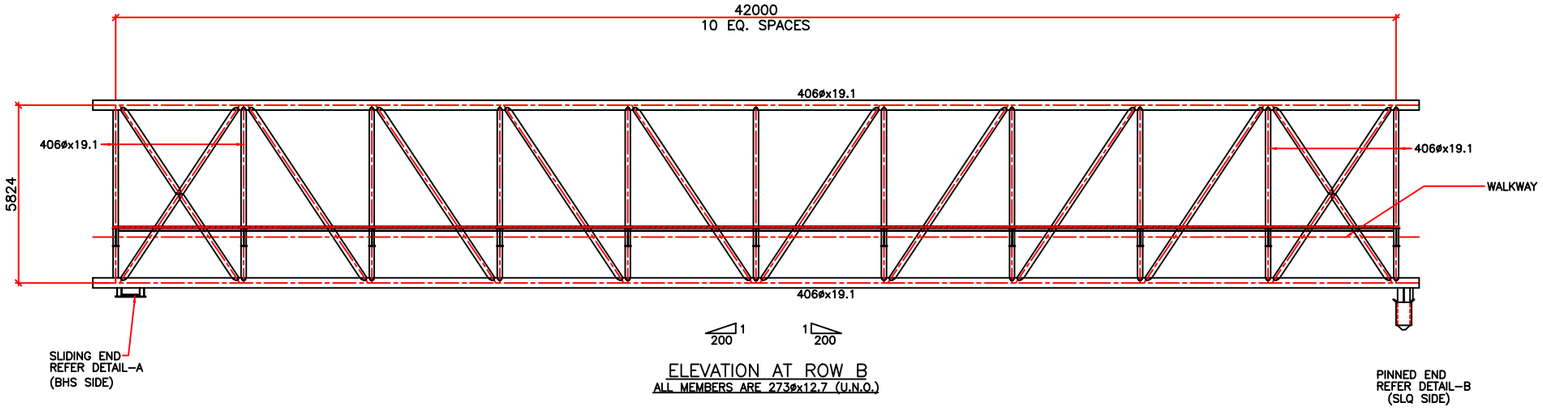
BRIDGE AT BOTTOM CHORD

ALL MEMBERS ARE 273ø x12.7 (U.N.O.)



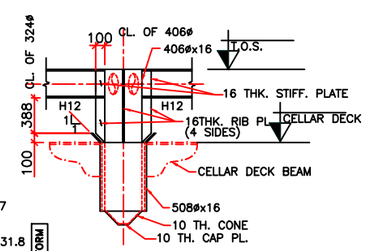
ELEVATION AT ROW A

ALL MEMBERS ARE 273ø x12.7 (U.N.O.)

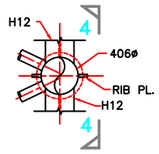


ELEVATION AT ROW B

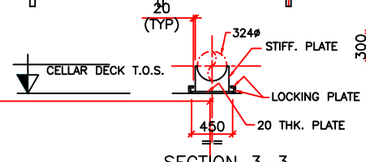
ALL MEMBERS ARE 273ø x12.7 (U.N.O.)



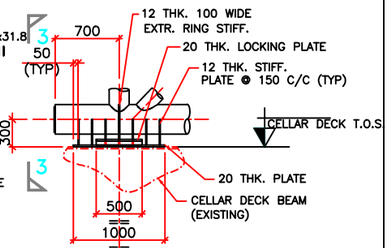
SECTION 4–4



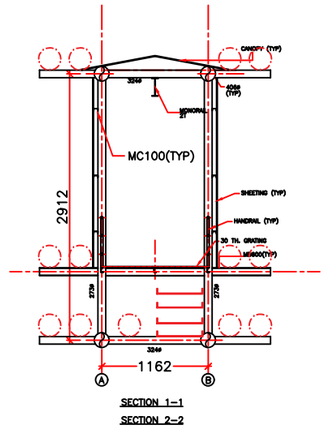
DETAIL–B



SECTION 3–3



DETAIL–A



SECTION 1–1

SECTION 2–2

REF. DWG. NO.	REFERENCE DRAWING TITLE

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FOR TENDER PURPOSE ONLY

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.08.25	ISSUED FOR BIDS	MUNISH	AR	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

एम एच संपर्कता की मौजूदा पुल का नवीकरण/बदलाव	REFURBISHMENT/REPLACEMENT OF EXISTING BRIDGE AT MH ASSET
आई सी डबलू से आई सी जी के निचली छत पुल का सामान्य विन्यास एवं व्यौरा	G.A. AND DETAILS OF BHS TO SLQ LOWER DECK BRIDGE

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
	B774	103	83	43	00001	A

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सार्वजनिक और निजी प्रयोग के अलावा इसका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा।

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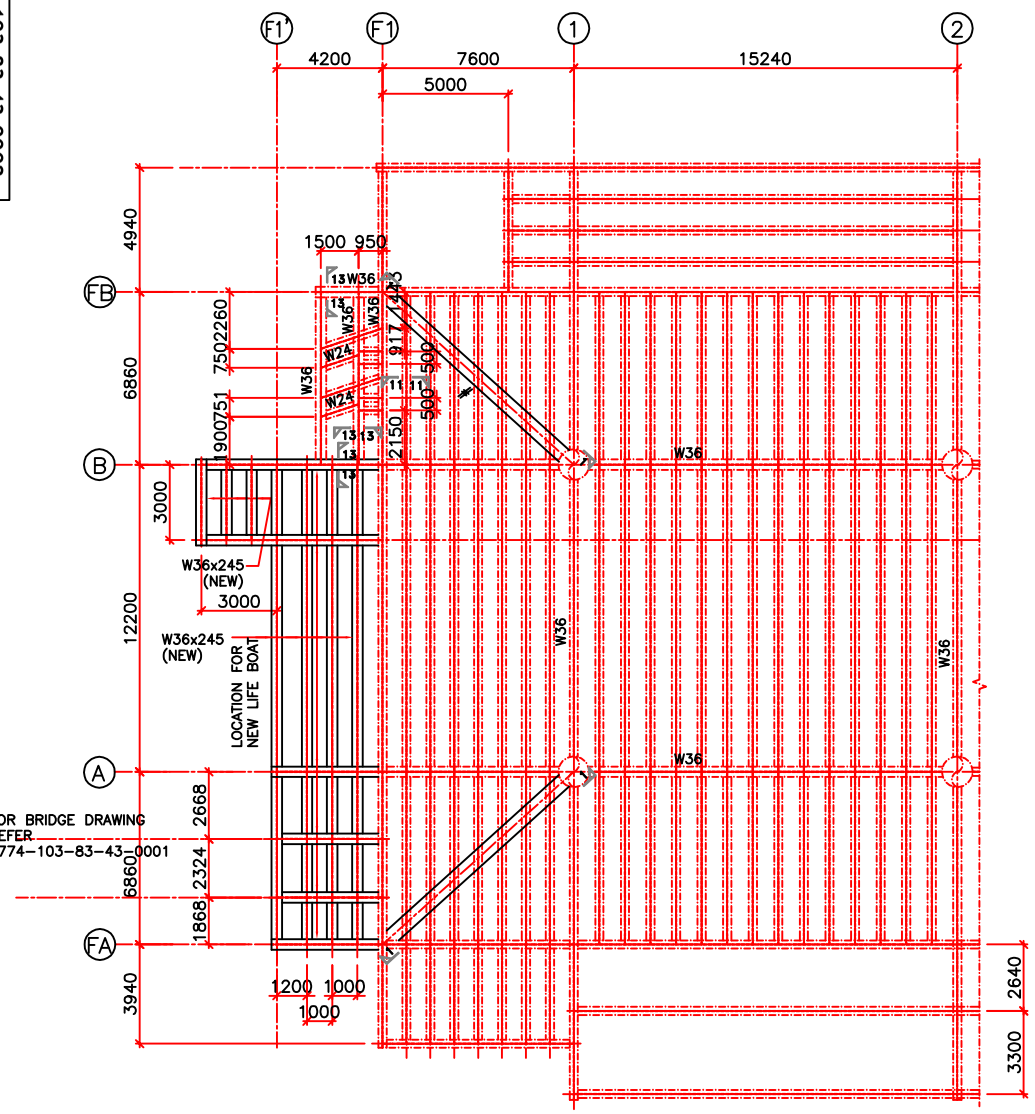
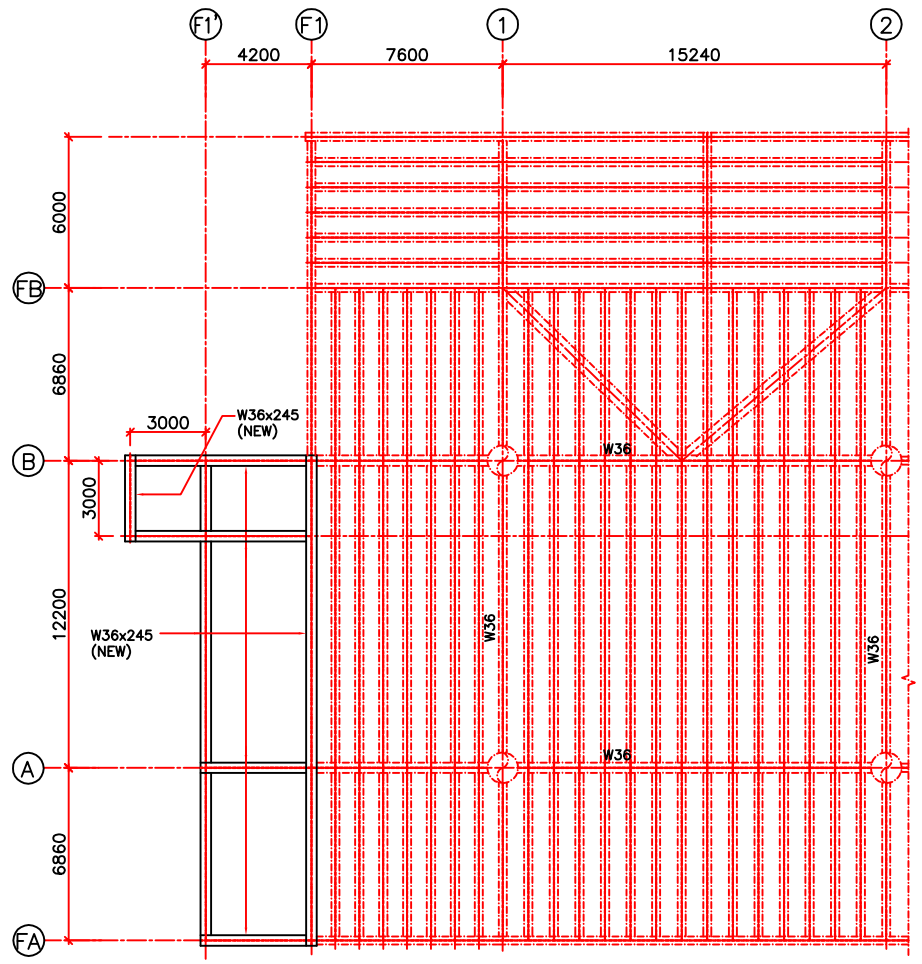
B774-103-83-43-0002

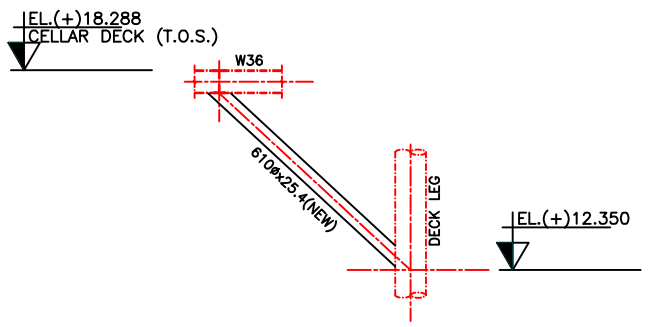


FOR BRIDGE DRAWING REFER B774-103-83-43-0001

PART PLAN OF CELLAR DECK EL(+)18.288(T.O.S.)
(ALL MEMBERS ARE W24 (U.N.O.)



PART PLAN OF MAIN DECK EL(+)25.292(T.O.S.)
(ALL MEMBERS ARE W24 (U.N.O.)



VIEW 1–1

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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FIELD NOTES

THE MEMBER MARKED SHALL BE ADJUSTED BY AVOIDING THE FOULING WITH EXISTING WALKWAY AS PER SITE DETAILS DURING DETAILED ENGINEERING.
* SINCE DRAWINGS OF BHS PLATFORM ARE NOT AVAILABLE, THE 3D MODEL AND NDT RESULTS ARE CONSIDERED FOR SIZING. THE MEMBER SIZE AND THICKNESS SHALL BE VERIFIED DURING DETAILED ENGINEERING FOR ALL MEMBERS BEFORE FINALIZING THE MODIFICATIONS.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	23.01.26	ISSUED FOR TENDER	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

MODIFICATION OF BHS DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	103	83	43	0002	A

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये न तो उन्हें मुफ़्त: सुरक्षित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सीमित और किसी प्रयोग के अलावा इसका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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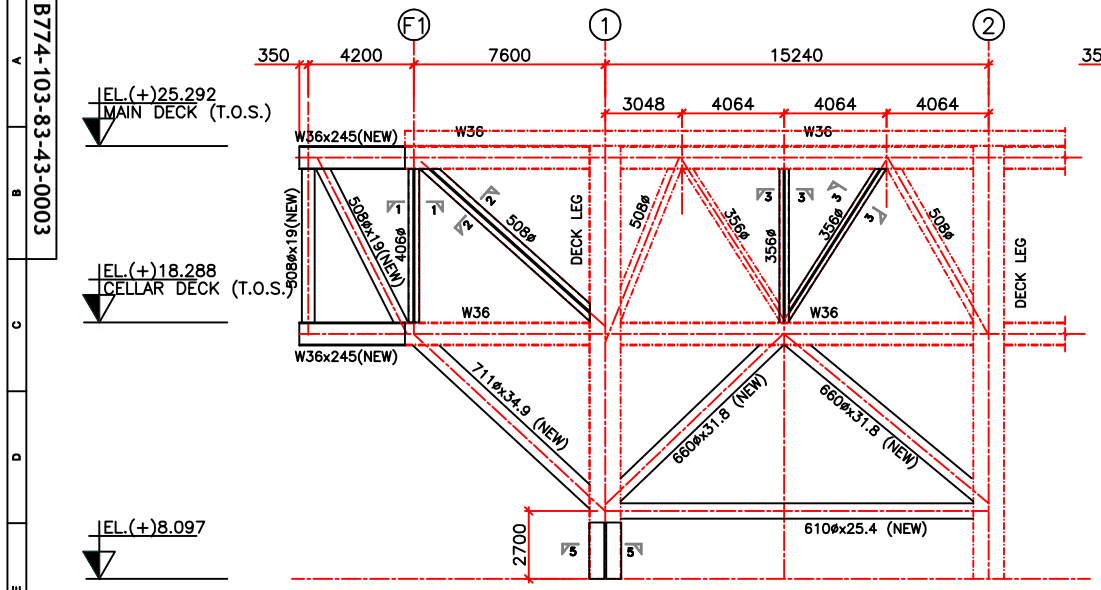
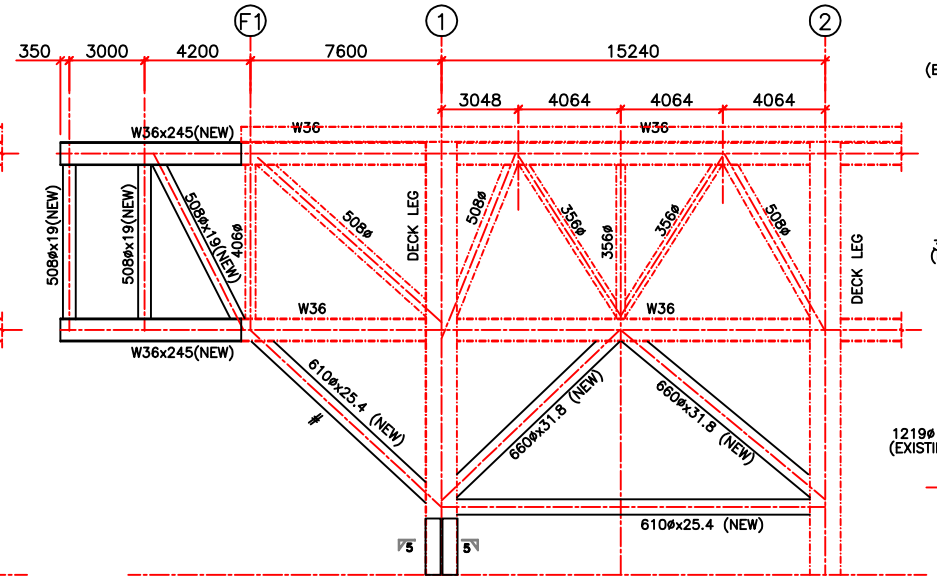
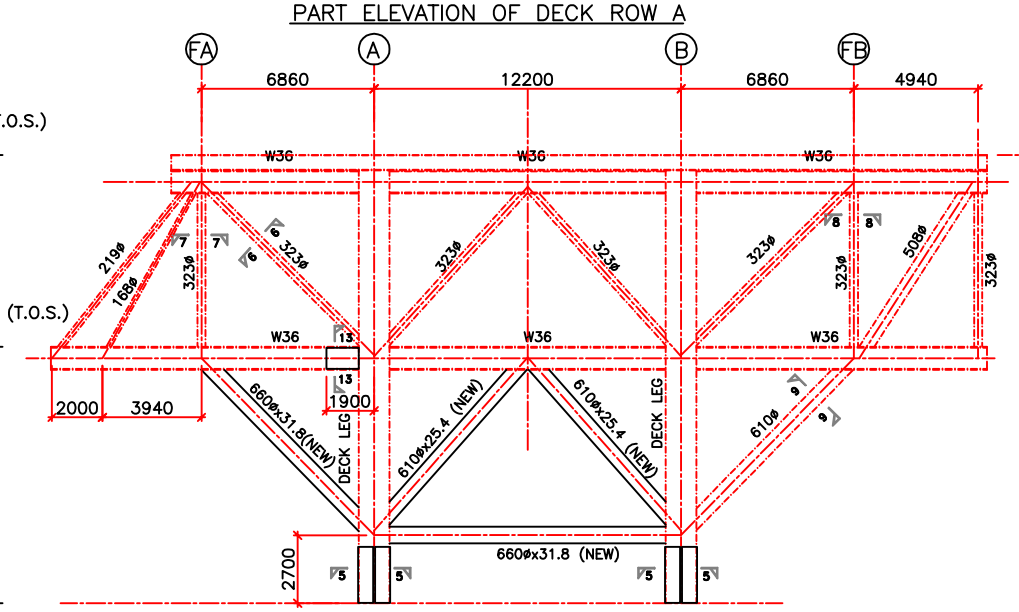
B774-103-83-43-0003



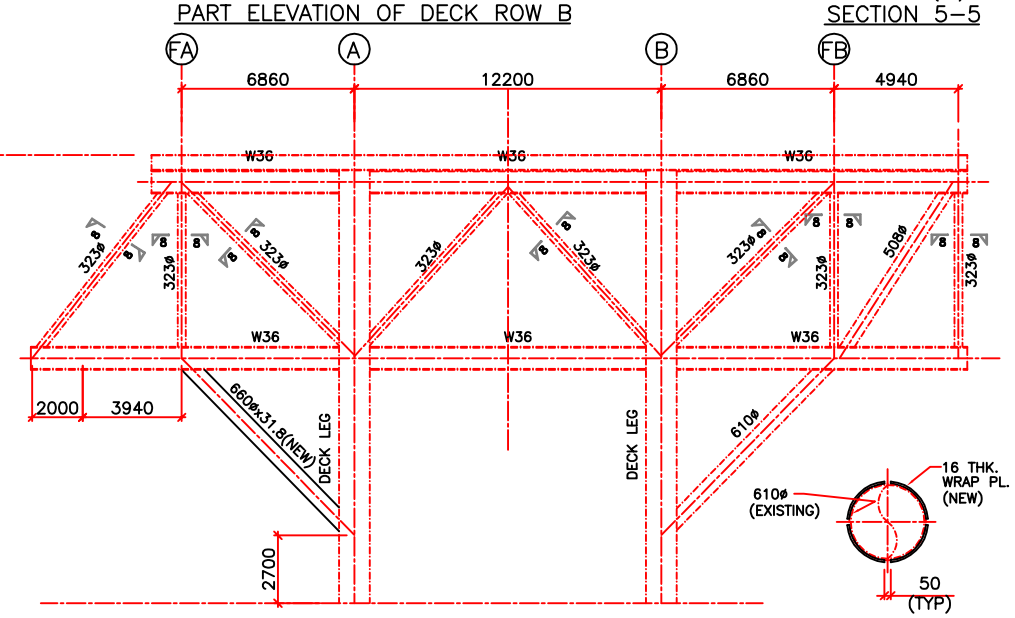
PART ELEVATION OF DECK ROW A



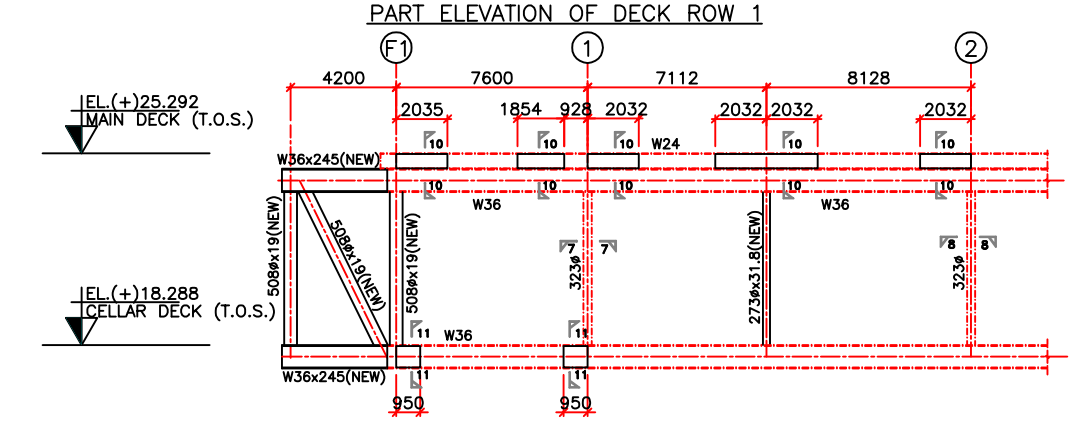
PART ELEVATION OF DECK ROW B



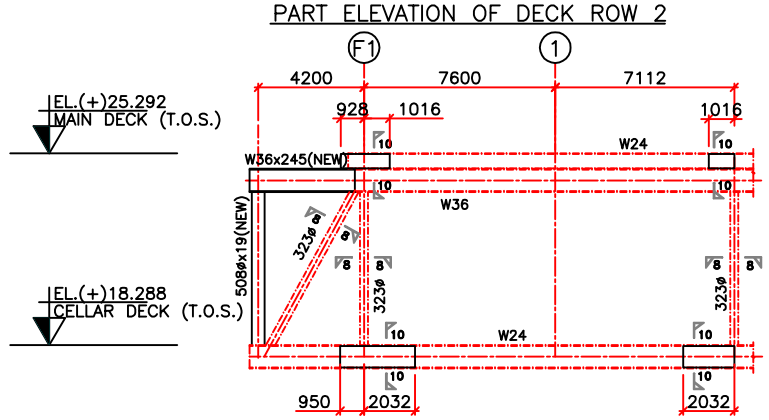
PART ELEVATION OF DECK ROW 1



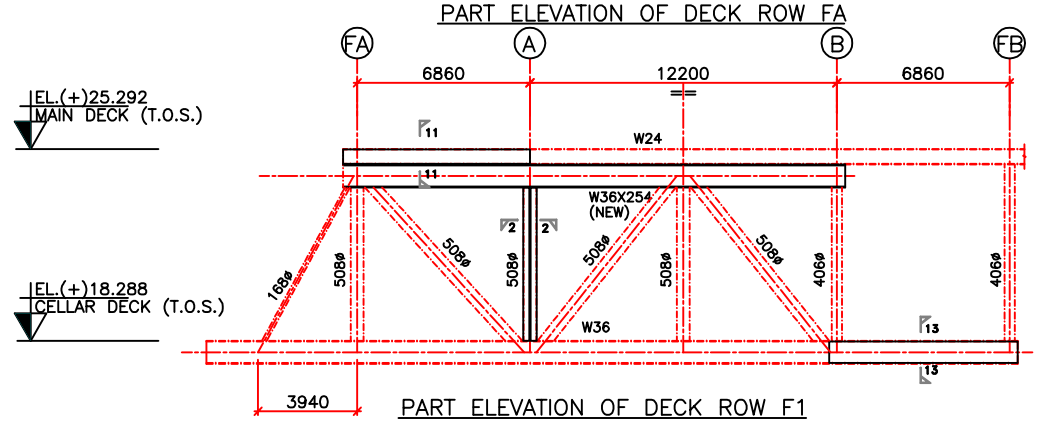
PART ELEVATION OF DECK ROW 2



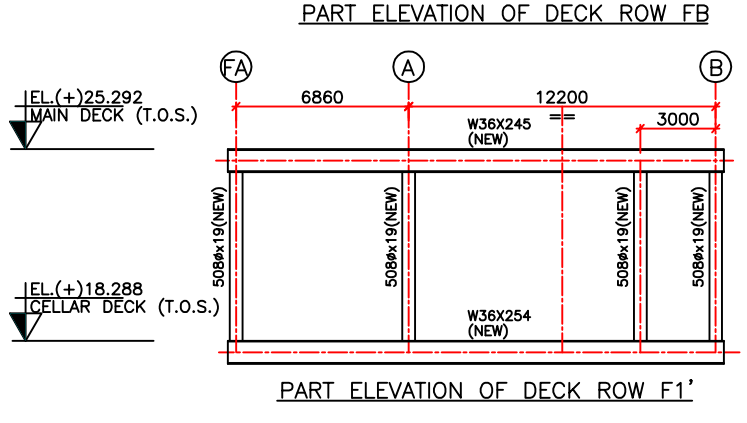
PART ELEVATION OF DECK ROW FA



PART ELEVATION OF DECK ROW FB



PART ELEVATION OF DECK ROW F1



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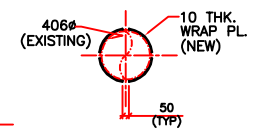
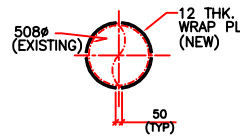


SECTION 1–1



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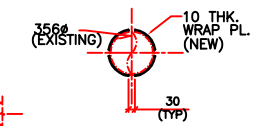

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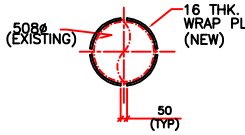

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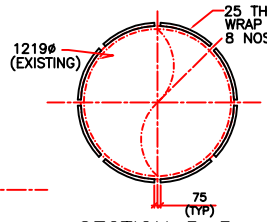

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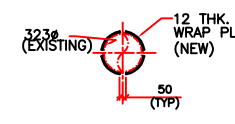

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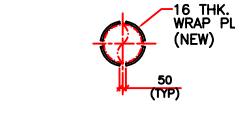

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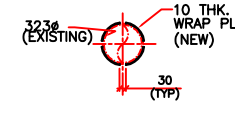
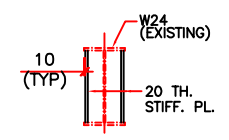
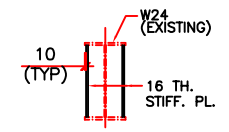


SECTION 8–8

SECTION 9–9



SECTION 10–10



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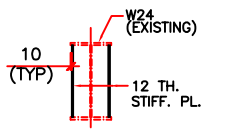
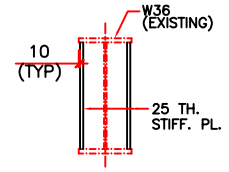


SECTION 12–12



SECTION 13–13

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

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5. MATERIAL TYPES LISTED IN SC NOS SHALL CONFORM TO ONGC SPEC 6001 F.
6. FABRICATOR TO ENSURE MIN 75MM CLEAR GAP B/W HE WELDS FOR CROSS BRACES AND CROSS JOINTS.
7. ALL WELDS SHALL BE FULL PENETRATION WELDS.
8. FOR GRATING REFER EIL STD. 7–60–0802.
9. FOR HANDRAIL REFER EIL STD. 7–60–0801.

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* ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

FIELD NOTES

\# THE MEMBER MARKED SHALL BE ADJUSTED BY AVOIDING THE FOULING WITH EXISTING WALKWAY AS PER SITE DETAILS DURING DETAILED ENGINEERING.
* SINCE DRAWINGS OF BHS PLATFORM ARE NOT AVAILABLE, THE 3D MODEL AND NDT RESULTS ARE CONSIDERED FOR SIZING. THE MEMBER SIZE AND THICKNESS SHALL BE VERIFIED DURING DETAILED ENGINEERING FOR ALL MEMBERS BEFORE FINALIZING THE MODIFICATIONS.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
A	23.01.26	ISSUED FOR TENDER	MUNISH	AP	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

MODIFICATION OF BHS DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	103	83	43	0003	A

3-1641-0501 REV.2 A1-841x594

अस्वीकृत अंतरण एवं इसमें निहित डिज़ाइन संबंधित विवरण इंजीनियर्स इंडिया लिमिटेड की संपत्ति है कि न तो उन्हें मुफ्त: सुरक्षित किया जाएगा, न नकल की जाएगी, न प्रकाशित किए जाएंगे, न प्रदर्शित किए जाएंगे और न ही लिखित और किसी प्रयोग के अलावा इसका कोई अन्य प्रयोग होगा और यह प्रयोग प्रकार से ऐसे पार्टी द्वारा उधारकर्ता को लिखित रूप में दी गई सहमति से होगा।
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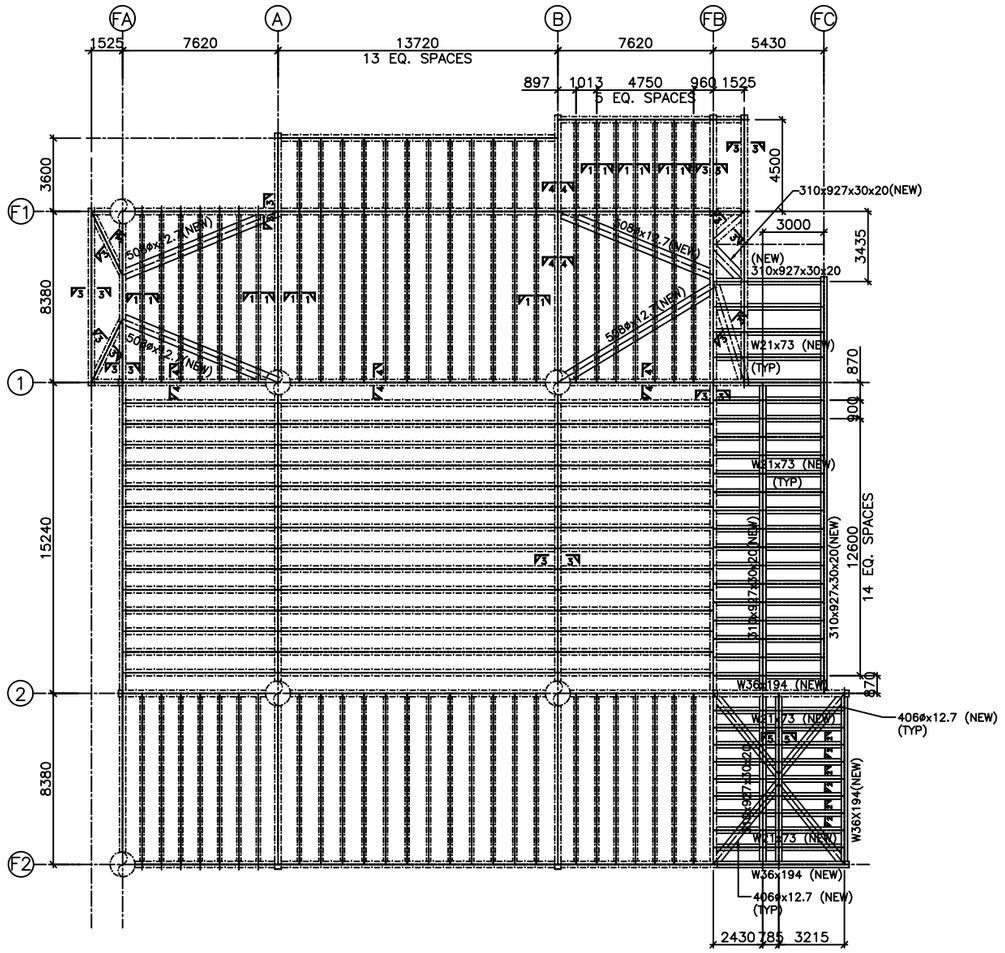
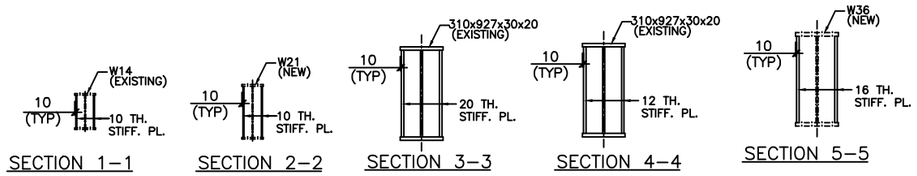
B774-103-83-43-0004



PART PLAN OF LOWER DECK EL(+)18.288(T.O.S.)



SECTION 1–1 SECTION 2–2 SECTION 3–3 SECTION 4–4 SECTION 5–5

REF. DWG. NO.	REFERENCE DRAWING TITLE
HTFC1-8-401-2101	LOWER DECK FRAMING PLAN (SLQ)
HTFC1-8-402-2102	LOWER DECK PLATING PLAN (SLQ)
HTFC1-8-403-2201	UPPER DECK FRAMING PLAN (SLQ)
HTFC1-8-404-2202	UPPER DECK BEAM PLAN (SLQ)
HTFC1-8-421-2401	TRUSS ELEVATION ROW 1&2 (SLQ)
HTFC1-8-404-2402	TRUSS ELEVATION ROW A&B (SLQ)

NOTES :-

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8. FOR GRATING REFER EIL STD. 7-60-0802.
9. FOR HANDRAIL REFER EIL STD. 7-60-0801.
10. THE MATERIAL AND AVAILABILITY OF INTERNAL STIFFENERS FOR LEG & CONE SHALL BE CHECKED DURING DETAILED ENGINEERING STAGE & ACCORDINGLY DESIGN SHALL BE UPDATED.

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- ALL THE DIMENSIONS ARE IN MM AND LEVELS ARE IN METER (U.N.O.)

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	29.01.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SLQ DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	103	83	43	0004	0

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड को संरक्षित है। ये मात्र उधार दिए गए है और उपकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उपकर्ता को लिखित रूप में दी गई सहमति से होगा ।
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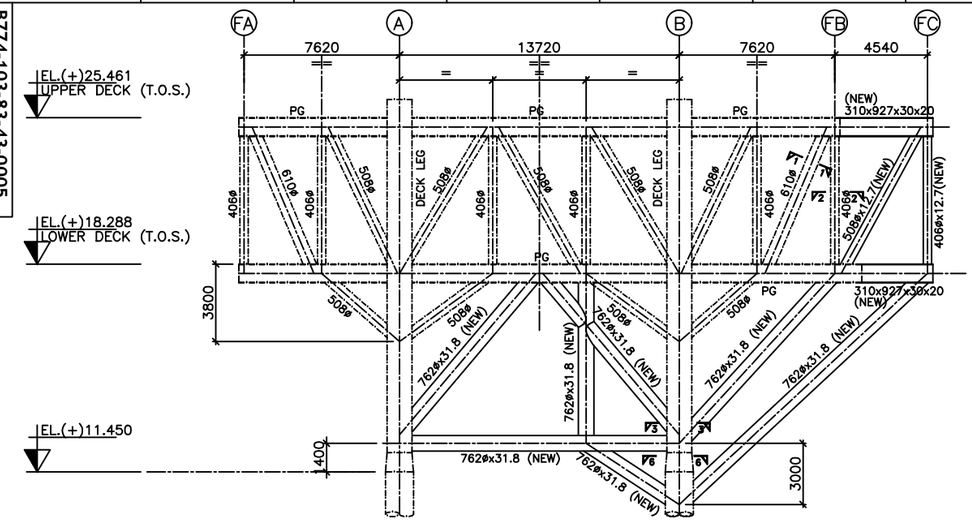
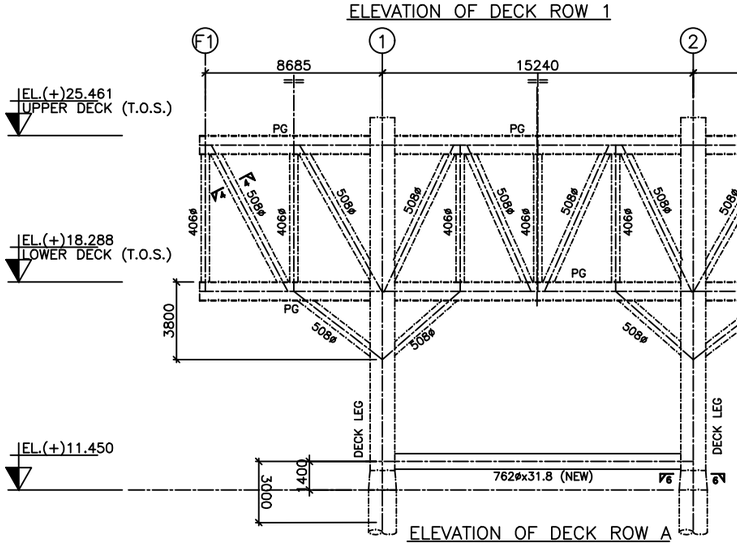
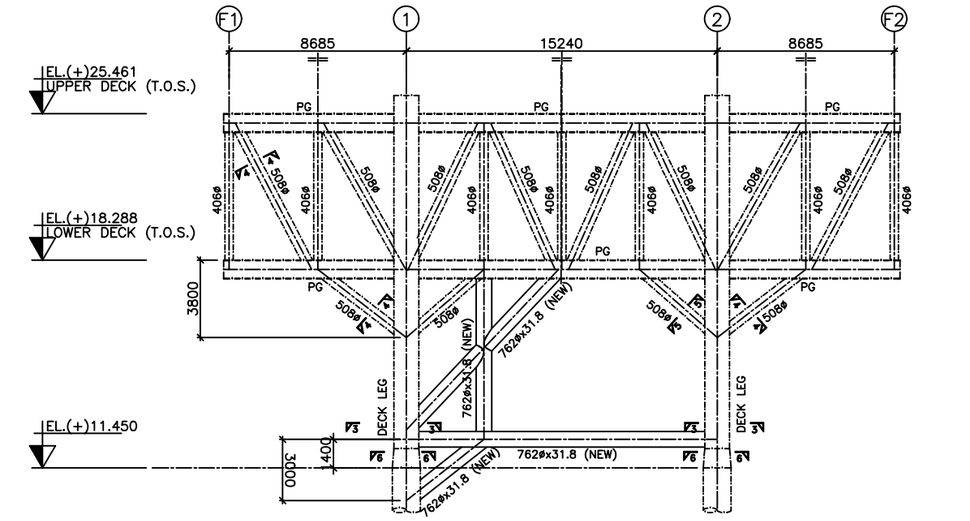
B774-103-83-43-0005



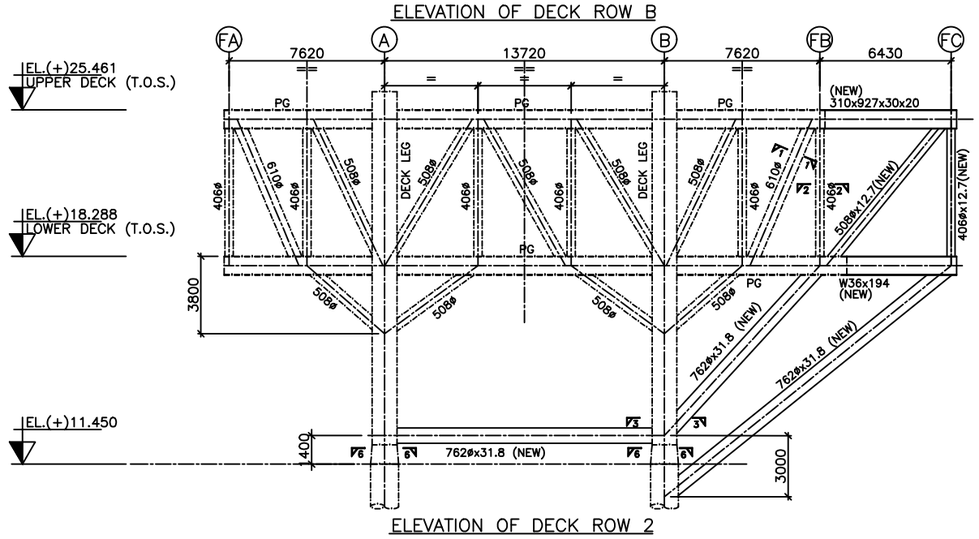
ELEVATION OF DECK ROW 1



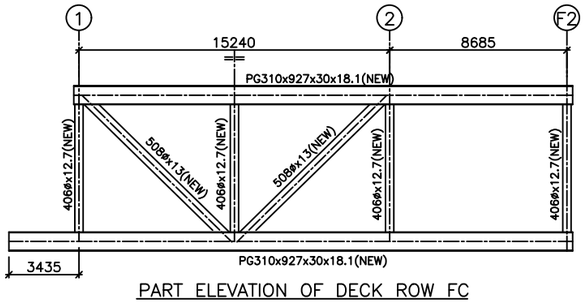
ELEVATION OF DECK ROW A



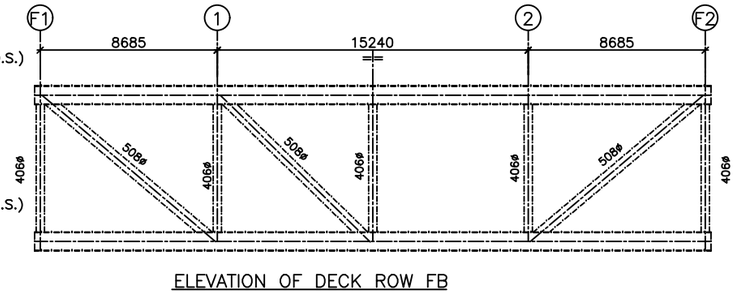
ELEVATION OF DECK ROW B



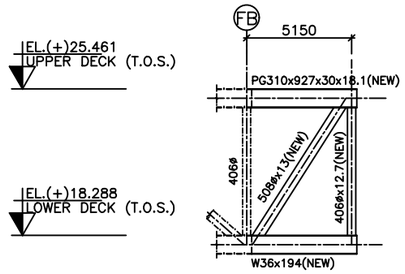
ELEVATION OF DECK ROW 2



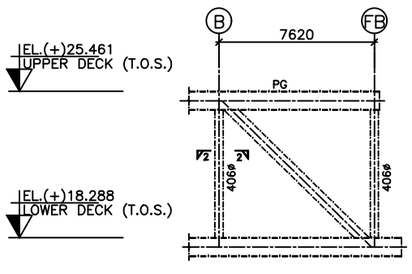
PART ELEVATION OF DECK ROW FC



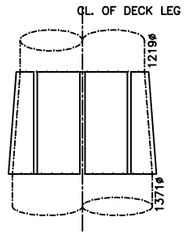
ELEVATION OF DECK ROW FB



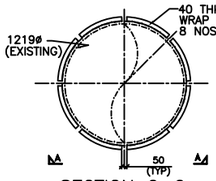
PART ELEVATION OF DECK ROW F2



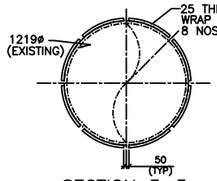
PART ELEVATION OF DECK ROW F1



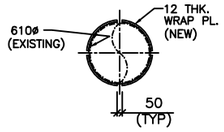
VIEW A–A
STRENGTHENING STABBING POINT AT ROW B



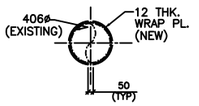
SECTION 6–6



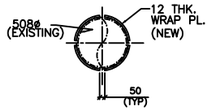
SECTION 3–3



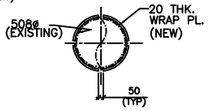
SECTION 1–1



SECTION 2–2



SECTION 4–4



SECTION 5–5

REF. DWG. NO.	REFERENCE DRAWING TITLE
HTFC1-8-401-2101	LOWER DECK FRAMING PLAN (SLQ)
HTFC1-8-402-2102	LOWER DECK PLATING PLAN (SLQ)
HTFC1-8-403-2201	UPPER DECK FRAMING PLAN (SLQ)
HTFC1-8-404-2202	UPPER DECK BEAM PLAN (SLQ)
HTFC1-8-421-2401	TRUSS ELEVATION ROW 1&2 (SLQ)
HTFC1-8-404-2202	TRUSS ELEVATION ROW A&B (SLQ)

NOTES :-

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ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	29.01.26	ISSUED FOR TENDER	MUNISH	AMS	DP	

ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

STRENGTHENING OF SLQ DECK

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	103	83	43	0005	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिज़ाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए है और उपकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न पुनर्निर्मित किए जाएंगे और न ही संलिप्त और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उपकर्ता को लिखित रूप में दी गई सहमति से होगा।
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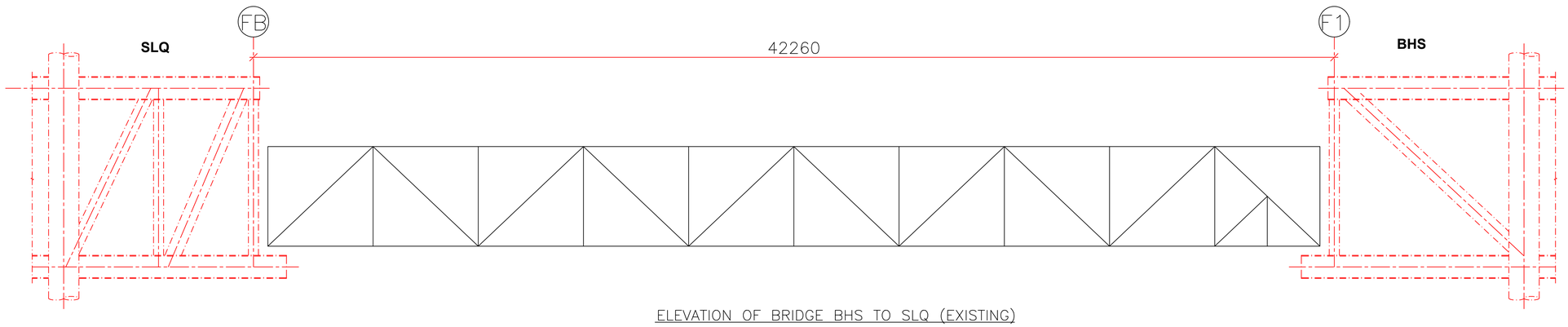
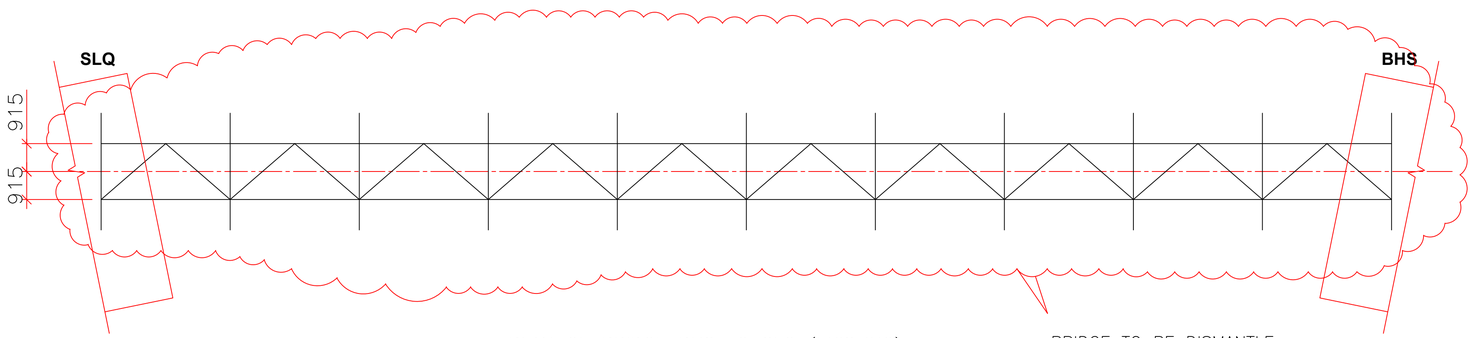
B774-103-83-43-0006



ELEVATION OF BRIDGE BHS TO SLQ (EXISTING)



PLAN OF BRIDGE BHS TO SLQ (EXISTING)

BRIDGE TO BE DISMANTLE

REF. DWG. NO.	REFERENCE DRAWING TITLE

NOTES :-

1. DO NOT SCALE THE DRAWING, WORK TO DIMENSIONS ONLY.
2. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN M U.N.O.
3. THE EXISTING BHS–SLQ BRIDGE SHALL BE DISMANTLED AFTER ERECTION AND COMMISSIONING OF NEW BHS–SLQ BRIDGE ON THE SOUTH SIDE.
4. REQUIRED STRENGTHENING OF EXISTING BHS–SLQ BRIDGE SHALL BE DESIGNED DURING DETAILED ENGINEERING STAGE FOR DISMANTLING THE BRIDGE AS A WHOLE (PREFERABLE).
5. CONTRACTOR SHALL SUBMIT THE DETAILED DISMANTLING SCHEME DURING DETAILED ENGINEERING STAGE.

ISSUED FOR TENDER

REV.	DATE	REVISIONS	BY	CHKD	APPD	PEM/PC
0	10.10.25	ISSUED FOR TENDER	MUNISH	NS	DP	

इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

ऑयल एण्ड नेचुरल गैस कारपोरेशन लिमिटेड
OIL AND NATURAL GAS CORPORATION LIMITED

REFURBISHMENT/REPLACEMENT OF BRIDGES IN MH ASSET

BHS SLQ BRIDGE (DISMANTLING)

SCALE	JOB NO.	UNIT	DIVN.	DEPT.	DWG. NO.	REV.
1:150	B774	103	83	43	0006	0

3-1641-0501 REV.2 A1-841x594

प्रस्तुत आरेख एवं इसमें निहित डिजाइन इंजीनियर्स इंडिया लिमिटेड की संपत्ति है। ये मात्र उधार दिए गए हैं और उधारकर्ता ने यह स्पष्ट समझौता किया है कि न तो इन्हें पुनः मुद्रित किया जाएगा, न नकल की जाएगी, न उधार दिए जाएंगे, न प्रदर्शित किए जाएंगे और न ही सहित और किसी प्रयोग के अलावा इनका कोई अन्य प्रयोग होगा और यह प्रयोग उधार देने वाले द्वारा उधारकर्ता को लिखित रूप से दी गई सहमति से होगा।
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अग्निरोधक लाइटिंग और पावर पैनलों के लिए विनिर्देश

SPECIFICATION FOR FLAMEPROOF LIGHTING AND POWER PANELS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
6	22.03.21	REVISED AND ISSUED AS STANDARD SPECIFICATION	RKS	ANPS	PG	SM
5	24.02.16	REVISED AND ISSUED AS STANDARD SPECIFICATION	NNB/AK	ANPS	BRB	SC
4	11.04.11	REVISED AND ISSUED AS STANDARD SPECIFICATION	PS	ANPS	UAP	DM
3	24.08.07	REVISED AND ISSUED AS STANDARD SPECIFICATION	ANPS	UAP	JMS	VC
2	30.01.02	REVISED AND ISSUED AS STANDARD SPECIFICATION	UAP	AAN	VPS	GRR

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Abbreviations:

AC	Alternating Current
BIS	Bureau of Indian Standards
CEA	Central Electricity Authority
DGMS	Directorate General of Mines Safety
EIL	Engineers India Limited
ELCB	Earth Leakage Circuit Breaker
FRLS	Flame Retardant Low Smoke
GI	Galvanized Iron
IEC	International Electro-technical Commission
IP	Ingress Protection
IS	Indian Standards
LV	Low Voltage
MCB	Miniature Circuit Breaker
MR	Material Requisition
NABL	National Accreditation Board for Testing and Calibration Laboratories
PESO	Petroleum and Explosives Safety Organisation
PO	Purchase Order
PVC	Poly Vinyl Chloride
SPN	Single Phase and Neutral
SS	Stainless Steel
TPN	Three Phases and Neutral
XLPE	Cross Linked Poly Ethylene

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1.0 SCOPE

This specification covers the requirements of design, manufacture, testing, packing and supply of flameproof Lighting and Power panels/ accessories suitable for installation in locations handling flammable liquids and gases / vapours.

2.0 CODES AND STANDARDS

2.1 The equipment shall comply with the requirements of latest revision of the following standards issued by BIS:

IS-5	:	Colours for ready mixed paints and enamels
IS-12640	:	Residual current operated circuit breakers
IS / IEC 60079-0	:	Electrical apparatus for explosive gas atmospheres (General requirements)
IS / IEC 60079-1	:	Electrical apparatus for explosive gas atmospheres (Equipment Protection by Flameproof Enclosures "d")
IS / IEC 60529	:	Degree of protection provided by enclosures (IP Code)
IS / IEC 60898	:	Electrical Accessories - circuit breakers for over protection for household and similar installations
IS / IEC 60947	:	LV switchgear and control gear.

2.2 In case of imported equipment, only IECEx (or equivalent, if more stringent than the applicable IEC) certified equipment shall be used.

2.3 The equipment shall also confirm to the provisions of CEA Regulations with latest amendments and other statutory regulations currently in force in the country.

2.4 In case Indian standards are not available for any equipment, standards issued by IEC or equivalent agency (if more stringent than IEC) shall be applicable.

2.5 In case of any conflict between requirements specified in various applicable documents for the project, the most stringent requirement shall govern. However, Owner's/ EIL's decision in this regard will be final and binding.

3.0 GENERAL REQUIREMENTS

3.1 The offered equipment shall be brand new with state of art technology and having proven field track record. No prototype equipment shall be offered.

3.2 Vendor shall ensure availability of spare parts and maintenance support services for the offered equipment for at least 10 years from the date of supply.

4.0 SITE CONDITIONS

The equipment shall be suitable for installation and satisfactory operation in classified hazardous locations in tropical, humid and corrosive atmosphere as prevalent in refineries, petrochemical and fertilizer plants. The lighting and power panels shall be designed to operate under site conditions as specified in the data sheet. If not specifically mentioned therein, a design ambient temperature of 40° C and an altitude not exceeding 1000m above mean sea level shall be considered.

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5.0 POWER SUPPLY

5.1 Lighting and power panels shall be suitable for 415V, 50Hz, TPN incoming power supply and outgoing circuits for 240V, SPN power supply.

5.2 For projects falling under the jurisdiction of DGMS, the following additional requirements shall apply:

5.2.1 The phase-to-phase voltage shall not exceed 125V (where electric lighting is used in underground mine)/ 250V (where electric lighting is used in open cast mine or surface of a mine), with the neutral or the mid-point of the lighting system connected with earth.

5.2.2 The panels shall have three phase, 3-wire system, without neutral.

5.2.3 Outgoing circuits shall be controlled by 2-pole MCBs. Earth leakage protection shall be provided for all outgoing circuits, which shall isolate both poles.

6.0 CERTIFICATION

The equipment shall have test certificates issued by NABL-approved/ Central Government labs in India/ IECEx-approved/ equivalent labs. All equipment (indigenous & imported) shall also have valid statutory approvals, as applicable, for the specified location, and marking as per IS/IEC 60079 and as required by statutory authorities. All indigenous flameproof equipment shall have valid BIS license.

7.0 TECHNICAL REQUIREMENTS

7.1 Construction

7.1.1 The enclosures of the lighting and power panels shall be made of cast light metal alloy, unless otherwise specified. The enclosures of the lighting and power panels intended for use in units handling sulphuric acid/ other corrosive materials shall be made of stainless steel grade SS-316L, if specified in MR/ Tender.

7.1.2 The panels shall be suitable for use in outdoor open locations and shall have minimum IP-55 degree of protection. They shall preferably be provided with integral canopy. However, where the enclosure has been certified without integral canopy, a separate canopy can be accepted. The separate canopy shall be made of at least 1.6 mm (16 gauge) galvanized sheet steel/ stainless steel minimum SS-304 (as specified in MR/ Tender). The canopy shall be suitable for providing protection against rain from top, back and two sides.

7.1.3 The lighting and power panels shall be provided with gaskets made of non-inflammable and self-extinguishing material.

7.1.4 All metal surfaces shall undergo manufacturer's standard cleaning/ painting/ powder coating cycle. After surface preparation, the panels shall be painted with two coats of epoxy based final paint or epoxy powder coated with minimum coating thickness of 80 microns, with colour shade as below:

- Flame proof Exd (Gas group IIA/IIB): Dark admiralty grey shade 632 of IS-5/ RAL 7031
- Flame proof Exd (Gas group IIC): Light yellow shade 355 of IS-5/ RAL 1012.

All unpainted steel parts shall be suitably treated to prevent rust formation/ corrosion. If these parts are moving elements, then these shall be greased with non-solidifying grease. However, in case of stainless steel lighting and power panels, finish shall be Electropolish/ Buffed/ equivalent finish.

7.1.5 A warning inscription "DO NOT OPEN WHEN ENERGIZED" shall be provided on each enclosure. The warning inscription shall be embossed on the enclosure or a separate warning plate with above inscription shall be fixed to the enclosure with screws. The warning plate shall be of nickel plated brass or stainless steel.

7.1.6 All accessories like nuts, bolts, washers etc. shall be made of stainless steel SS-304.

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7.1.7 All the non-current carrying metallic parts of the panel shall be inherently bonded together. Each lighting and power panel shall be provided with two earthing studs (bolt, nut with spring and plain washer) with lugs on the exterior of the panel enclosure suitable for termination of 10 mm dia GI wire rope. No screw type fixing arrangement shall be allowed.

7.1.8 Each outgoing feeder shall be provided with distinct terminals for phase, neutral and earth. The terminal block enclosures shall be adequately sized to properly terminate the cables by taking into account the required bending radii of cable cores and shall have the following minimum gland to terminal distances:

Conductor Size	Up to 2.5 mm^2	Above 2.5 mm^2 & Up to 10 mm^2	Above 10 mm^2 & Up to 35 mm^2	Above 35 mm^2 & Up to 95 mm^2
Gland to terminal distance	40 mm	60 mm	100 mm	150 mm

7.1.9 The panels shall be provided with suitably sized cable entries at the bottom/ sides, for incoming and outgoing cables. Panels shall be complete with double compression type nickel plated brass flameproof cable glands. Flameproof nickel plated brass sealing plugs shall be supplied, for plugging the unused cable entries. The quantity of sealing plugs shall be equal to 20% of the total number of outgoing cable entries.

7.1.10 The panels shall have external fixing lugs for mounting on wall or column. The holes provided on these lugs shall be of oblong type.

7.1.11 A tag plate indicating Tag Number, engraved, 3 ply laminate fixed with screws, shall be provided on each lighting and power panel. Nameplates shall also be provided for each incoming and outgoing feeder. A separate nameplate shall be provided to indicate the name of manufacturer, testing agency, test certificate number, serial number, BIS license number, applicable gas group as per IS/IEC 60079, statutory agency (PESO) approval number and any additional marking required by statutory authority. The nameplates shall be Al anodized or SS-304 engraved type or laser-marked and permanently fixed on the equipment. In case the standard details given above are embossed on the enclosures, the same need not be repeated on the name plate.

7.2 Component Specification

7.2.1 Bus bars in the lighting and power panels shall be made of high conductivity copper, and shall be supported by non-hygroscopic insulators. Bus bars shall be colour coded for identification of phases and neutral.

7.2.2 The incomer shall have one no. 4 pole MCB isolator (without overload and short-circuit release) and one no. 4 pole ELCB (if specified in the MR/ Tender). ELCBs shall have a maximum sensitivity of 30mA.

7.2.3 The outgoing feeders shall be provided with double pole MCBs having overload and short-circuit releases. Further, separate 2 pole ELCBs with maximum sensitivity of 30mA shall be provided for outgoing feeders if specified in MR/ Tender.

7.2.4 All MCBs (except isolators) and ELCBs shall be with 10kA interrupting capacity.

7.2.5 The ELCB shall be hand reset type. Door mounted reset push button shall be provided for ELCB.

7.2.6 The operating knobs (ON/ OFF/ RESET) shall be provided with a suitable rack and pinion arrangement or cam type operating mechanism for operating them smoothly from outside.

7.3 Terminals & Wiring

7.3.1 The panels shall be provided with sufficient number of terminals. More than 2 wires per terminal shall not be permitted. If required, additional terminal with shorting link may be

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used. Tinned Copper lugs shall be provided for cable termination. Incoming and outgoing terminals shall be suitably segregated.

7.3.2 All internal wiring in lighting and power panels shall employ adequately sized, 660V/ 1100V grade, FRLS type, XLPE/ PVC insulated, Copper conductor wires, colour coded for phase, neutral and earth, with minimum conductor sizes as below:

Incomer : 16 mm²

Outgoing : 2.5 mm²

All terminations shall be with suitable lugs.

8.0 INSPECTION, TESTING AND ACCEPTANCE

8.1 During fabrication, the equipment shall be subjected to inspection by EIL/ Owner or by an agency authorised by the Owner, as per agreed Inspection Test Plan. Manufacturer shall furnish all necessary information concerning the supply to EIL/ Owner's inspector. All routine/ acceptance tests shall be carried out at manufacturer's works under his care and expense.

8.2 Type test certificates from NABL-approved/ Central Government Labs in India/ IECEx-certified/ equivalent Labs, Manufacturer's works test reports, applicable PESO approval and BIS license shall be shown to the inspection agency on demand during inspection. The certificates, BIS license and PESO approval must be valid at the time of despatch.

8.3 Test certificates of bought out components shall be submitted to the inspection agency, as per inspection and Test Plan no. 6-81-1008.

8.4 All equipments shall be subjected to various routine/ acceptance tests as per Inspection and Test plan no. 6-81-1008.

9.0 PACKING AND DESPATCH

All the equipment shall be divided into several sections for protection and ease of handling during transportation. The equipment shall be properly packed for the selected mode of transportation, i.e. by ship/ rail or trailer and shall be wrapped in polythene sheets before being placed in crates/ cases to prevent damage to finish. The crates/ cases shall have skid bottom for handling. Special notations such as 'Fragile', 'This side up', 'Center of gravity', 'Weight', 'Owner's particulars', 'PO no.' etc., shall be clearly marked on the packages together with other details as per purchase order.

The equipment may be stored outdoors for long periods before installation. The packing should be suitable for outdoor storage in areas with heavy rains and high ambient temperature unless otherwise agreed. A set of instruction manuals for installation, testing and commissioning, a set of operation & maintenance manuals and a set of final drawing shall be enclosed in a waterproof cover along with the shipment.

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मध्यम एवं उच्च वोल्टेज केबल एवं असेसरिज के लिए विनिर्देश

SPECIFICATION FOR MEDIUM AND HIGH VOLTAGE CABLES & ACCESSORIES

Rev. No	Date	Purpose	Prepared by	Checked by	StandardsCommitteeConvenor Approved by	Standards Bureau Chairman
9	30/06/22	REVISED AND ISSUED AS STANDARD SPECIFICATION	RG	RSR	MKS	SM
8	20/01/17	REVISED AND ISSUED AS STANDARD SPECIFICATION	RG	MKS	BRB	RN
7	01/04/16	REVISED AND ISSUED AS STANDARD SPECIFICATION	RG	MKS	BRB	RN
6	11/04/11	REVISED AND ISSUED AS STANDARD SPECIFICATION	PS	ANPS	UAP	DM

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Abbreviations:

ASTM	:	American Society for Testing and Materials
BIS	:	Bureau of Indian Standards
BS	:	British Standard
CEA	:	Central Electricity Authority
DGMS	:	Director General of Mines and Safety
EIL	:	Engineers India Limited
EPR	:	Ethylene Propylene Rubber
FR	:	Flame Retardant
FRLS	:	Flame Retardant Low Smoke
FS	:	Fire Survival
ICEA	:	Insulated Cable Engineers Association
IEC	:	International Electro technical Commission
IEEE	:	The Institute of Electrical & Electronic Engineers
IS	:	Indian Standards
LF	:	Low Frequency
NEMA	:	National Electrical Manufacturers Association
PE	:	Polyethylene
PO	:	Purchase Order
PVC	:	Polyvinyl Chloride
UV	:	Ultra Violet
XLPE	:	Cross linked Polyethylene

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1.0 SCOPE

This specification along with data sheets covers requirements for design, manufacture, testing at works and supply of Fire survival Medium voltage cables and Flame Retardant Low Smoke Medium & High Voltage cables and cable jointing / terminating accessories for medium and high voltage systems.

2.0 STANDARDS

2.1 The cables and cables jointing & terminating accessories shall comply with the latest edition of the following standards as applicable:

IS: 209	:	Specification for zinc.
IS: 1554 Part-1	:	PVC insulated (heavy duty) electric cables for working voltages upto and including 1100V (including amendment no. 1, 2, 3, 4 & 5)
IS:1554 Part-2	:	Specification for PVC insulated (heavy duty) electric cables for working voltages from 3.3 kV up to and including 11 kV (including amendment no. 1, 2, 3 & 4)
IS: 3961(Pt-6)	:	Recommended current rating for cables: Part- 6 Cross-linked Polylene insulated PVC sheathed cables
IS: 3961(Pt-7)	:	Recommended current ratings for cables part 7 cross-linked insulated thermoplastic sheathed cables
IS: 3975	:	Mild steel wires, strips and tapes for armouring of cables.
IS: 5831	:	PVC insulation and sheath of electric cables.
IS: 6380	:	Specification for elastomeric insulation and sheath of electric cables.
IS: 7098 Part-1	:	Cross linked polyethylene insulated PVC sheathed cables for working voltage upto and including 1100V (including amendment no. 1,2,3,4 & 5)
IS: 7098 Part-2	:	Cross linked polyethylene insulated thermoplastics sheathed cables for working voltages from 3.3 kV up to and including 33 kV (including amendment no. 1 & 2)
IS: 8130	:	Conductors for insulated electric cables and flexible cords.
IS: 10418	:	Drums for electric cables.
IS:10462 (Pt-1)	:	Fictitious calculation method for determination of dimensions of protective coverings of cables: Part - I Elastomeric and thermo-plastic insulated cables.
IS: 10810	:	Methods of test for cables: - Part 41: Mass of zinc coating on steel armour - Part 58: Oxygen index test - Part-61: Flame retardant test - Part-62: Fire resistance test for bunched cables. - Part-63: Measurement of Smoke density of Electric cables under fire conditions
IS: 13573 (Pt-1):		Cable accessories for extruded power cables - part 1 for working voltages for 1.1 kv upto and including 3.3kV (E) - test methods and test requirements
IS: 13573 (Pt-2):		Cable accessories for extruded power cables - part 2 for working voltages for 3.3kV (UE) upto and including 33kV (E) - test requirements
IS: 13573 (Pt-3):		Cable accessories for extruded power cables - part 3 for working voltages for 3.3kV (UE) upto and including 33kV (E) - test methods
IS: 16269	:	Recommended short circuit ratings of electric cables with rated voltage from 1.1kV to 220kV

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IEC: 60331-21	:	Tests for electric cables under fire conditions circuit integrity – Procedures and requirements – Cables of rated voltage up and including 0.6/1.0kV.
IEC: 60332-1	:	Test of the fire behaviour on single core or single cable (flame retardancy)
IEC: 60332-3	:	Tests of the fire behaviour on bunched cables (reduced flame propagation)
IEC-60754-1	:	Test on gases evolved during combustion of materials from cables, Determination of the amount of halogen acid gas content
IEC: 61034	:	Measurement of smoke density of cables burning under defined condition
NEMA-WC53	:	Standard test methods for extruded dielectric power, control, instrumentation and portable cables for test.
IEEE-383	:	Standard for qualifying electrical cables and splices for nuclear facilities
ASTM-G-154	:	Standard practice for operating fluorescent light apparatus for UV exposure of non metallic materials.
ASTM-D-2863	:	Measurement of minimum oxygen concentration to support candle like combustion of plastics
BS: 7846	:	Electric cables- 600/1000V armoured fire resistant cables having thermosetting insulation and low emission of smoke and corrosive gases when affected by fire.

2.2 The cables and accessories shall also conform to the provisions of CEA Regulations and other statutory regulations, as applicable.

2.3 In case of any conflict between requirements specified in various applicable documents for the project, the most stringent one shall prevail. However, Owner's decision in this regard will be final and binding.

3.0 GENERAL CONSTRUCTION

3.1 The cables shall be suitable for laying in trays, trenches, ducts, and conduits and for underground-buried installation with uncontrolled backfill and possibility of flooding by water and chemicals.

3.2 Outer sheath of all cables shall be black in colour and the minimum value of oxygen index shall be 29% at 27 ± 2 ° C. In addition suitable chemicals shall be added into the PVC compound of the outer sheath to protect the cable against rodent and termite attack. However, for earthing cables; colour of outer sheath shall be green.

3.3 All cables covered in this specification shall be Flame Retardant Low Smoke (FRLS) or Fire Survival (FS). The outer sheath of FRLS cables shall possess flame propagation properties meeting requirements as per IS-10810 (Part-62) category AF. FRLS cable shall be identified by indenting, embossing or printing the appropriate legend i.e. 'FRLS, Category – C2' throughout the cable length. FRLS properties shall be as per IS:10810 Part 61 & 62, IEC-60332 Part 1 & Part 3, IEEE-383, IEC-61034, IEC-60754 Part 1, ASTM-D-2863.

3.4 Sequential marking of the length of the cable in metres shall be provided on the outer sheath at every one metre. The embossing/engraving shall be legible and indelible.

3.5 The overall diameter of the cables shall be strictly as per the values declared by the manufacturer in the technical information subject to a maximum tolerance of ±2 mm up to overall diameter of 60mm and ±3mm for beyond 60mm.

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3.6 PVC/ Rubber end caps shall be supplied free of cost for each drum with a minimum of eight per thousand metre length. In addition, ends of the cables shall be properly sealed with caps to avoid ingress of water during transportation and storage.

3.7 The joints in armour wire/strips shall be made by brazing or welding and any surface irregularities shall be removed. A joint in any wire/strip shall be at least 300mm. from the nearest joint in any wire/strip in the complete cable.

3.8 The cables meant to be used in mining area under the jurisdiction of DGMS shall satisfy the following

(a) "DGMS CERTIFIED" to be embossed at outer sheath at regular interval.

(b) Cables to be used in Mines shall have copper conductor only.

(c) The resistance of armour shall not exceed that of the conductor as specified in IS-8130 by more than 33%. To satisfy this, substitution of galvanised steel wire/strip in armouring by the required number of tinned copper wires/strips is permissible.

(d) For mining cables, the size and type of armour shall be such that the combined conductance of armour shall be equivalent to 75 percent of the conductance of the largest conductor of the cable.

3.9 Medium Voltage Cables

3.9.1 All power/control cables for use on medium voltage systems shall be heavy-duty type, 650/1100V grade with aluminium/ copper conductor (as mentioned in datasheet / Material requisition), XLPE insulated, inner-sheathed, armoured/ unarmoured and overall PVC sheathed. XLPE insulated cables shall meet the requirement specified in IS-7098 (Part-1).

3.9.2 The conductors shall be stranded for all sizes of MV power cable. Conductors of nominal area less than 16 mm^2 shall be circular only. Conductors of nominal area 16 mm^2 and above may be circular or shaped as per IS 8130. Cables with reduced neutral conductor shall have sizes as per Table- 2 of IS 7098 (Part-1).

3.9.3 The core insulation shall be with XLPE compound applied over the conductor by extrusion. It shall conform to the properties given in Table-1 of IS: 7098 (Part-1).

3.9.4 The inner sheath shall be applied over the laid-up cores by extrusion and shall be of PVC conforming to the requirements of Type ST-2 PVC compound as per IS: 5831. The minimum thickness of inner sheath shall be as per Table-5 of IS: 7098 (Part-1). Single core cables shall have no inner sheath.

3.9.5 All cables shall be provided with armour except those specifically specified as unarmoured. For single core cables intended for use on AC system, the armouring shall be of non-magnetic material. For multicore cables, the armour shall be by single round galvanised steel wires where the calculated diameter below armouring does not exceed 13 mm and by galvanised steel strips where this dimension is greater than 13 mm. Requirement and methods of tests for armour material and uniformity of galvanisation shall be considered as per IS - 3975 and IS - 10810 (Part 41). The dimensions of armour shall be considered as per method (b) of Table-6 of IS - 7098 (Part -1). For single core cables, the armour shall be with H4 grade hard drawn aluminium round wire of 2.5 mm diameter for cable having calculated diameter under armour upto 55 mm. In case of calculated diameter under armour is more than 55 mm. Table-6 of IS: 7098 (Part-1) shall be followed for dimension of armour.

3.9.6 The outer sheath for the cables shall be applied by extrusion and shall be formulated for lower smoke and shall be of PVC compound conforming to the requirements of type ST-2

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compound as per IS: 5831. The minimum and average thickness of outer sheath for unarmoured cables and minimum thickness of outer sheath for armoured cables shall be as per Table-8 of IS: 7098 (Part -1).

3.9.7 For XLPE insulated cables, it shall be possible to continuously operate the cable at a maximum conductor temperature of 90°C under full load condition and 250°C under short-circuit condition.

3.9.8 The fire survival cables shall meet the following additional requirements :

i) The insulation shall be of EPR or equivalent material with glass mica tape below or above insulation.

ii) The cables shall meet requirement of circuit integrity test for a minimum period of 3 hours at maximum temperature of 950° C.

3.10 High Voltage Cables

3.10.1 Power cables from 3.3kV and up to and including 33kV systems shall be Aluminium/ Copper conductor, XLPE insulated, sheathed, armoured/ unarmoured and overall PVC sheathed.

3.10.2 The conductors shall be stranded and compacted circular for all cables.

3.10.3 All cables rated 3.8/ 6.6kV and above shall be provided with both conductor screening and insulation screening. The conductors shall be provided with non-metallic extruded semi conducting screen.

3.10.4 The core insulation shall be with cross linked polyethylene insulating compound dry cured, applied by extrusion. It shall be free from voids and shall withstand all mechanical and thermal stresses under steady state and transient operating conditions. It shall conform to the properties given in Table-1 of IS: 7098 (Part-2).

3.10.5 The insulation screen shall consist of non-metallic extruded semi-conducting compound in combination with a non-magnetic metallic copper screen. The copper screen for all the three cores together shall be capable of carrying the single line to ground fault current value and the duration specified in the data sheet. For single core cable, copper screen provided shall be capable of carrying the single line to ground fault current value and the duration specified in the data sheet. For screen sizing, final temperature shall be considered as 200°C under adiabatic condition as per IS: 16269.

3.10.6 The conductor screen, XLPE insulation and insulation screen shall all be extruded in one operation by 'Triple Extrusion' process to ensure perfect bonding between the layers. The core identification shall be by coloured strips or by printed numerals.

3.10.7 The inner sheath shall be applied over the laid up cores by extrusion and shall conform to the requirements of type ST 2 compound of IS: 5831. The extruded inner sheath shall be of uniform thickness. The minimum thickness of inner sheath shall be as per Table-5 of IS: 7098 (Part-2). In case of single core cables, there shall be extruded inner sheath between insulation metallic screen and armouring.

3.10.8 All cables shall be provided with armour except those specifically specified as unarmoured. For single core cables intended for use on AC system, the armouring shall be of non-magnetic material. For multicore cables, the armour shall be by galvanised steel strips. Requirement and methods of tests for armour material and uniformity of galvanisation shall be as per IS-3975 and IS -10810 (Part 41). The dimensions of armour shall be as per method (ii) of Table-6 of IS -7098 (Part -2). For single core cables, the armour shall be with H4 grade hard drawn aluminium round wire of 2.5 mm diameter for cable having calculated diameter under armour

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upto 55 mm. In case of calculated diameter under armour is more than 55 mm. Table-6 of IS: 7098 (Part-2) shall be followed.

3.10.9 The outer sheath of the cables shall be applied by extrusion over the armouring and shall be of PVC compound conforming to the requirements of Type ST 2 compound of IS: 5831. The minimum and average thickness of outer sheath for unarmoured cables and minimum thickness of outer sheath for armoured cables shall be as per Table-7 of IS: 7098 (Part-2).

3.10.10 The thickness of the insulation, inner sheath shall be governed by values given in Table-4 and Table-5 of IS: 7098 (Part -2).

4.0 CABLE ACCESSORIES

4.1 The termination and straight through jointing kits for use on the systems shall be suitable for the type of cables offered as per this specification and shall meet requirements of IS 13573.

4.2 The accessories shall be supplied in kit form. Each component of the kit shall carry the manufacturer's mark of origin.

4.3 The kit shall include all stress grading, insulating and sealing materials apart from conductor fittings and consumable items. An installation instruction sheet shall also be included in each kit.

4.4 The contents of the accessories kit including all consumable shall be suitable for storage without deterioration at a temperature of 45° C, with shelf life extending to more than 5 years.

4.5 Terminating Kits

The terminating kits shall be suitable for termination of the cables to an indoor switchgear or to a weatherproof cable box of an outdoor mounted transformer/ motor. For outdoor terminations, weather shields/ sealing ends and any other accessories required shall also form part of the kit. The terminating kits shall be from one of the makes/ types mentioned in the data sheet.

4.6 Jointing Kits

The straight through jointing kits shall be suitable for installation on overhead trays, concrete lined trenches, and ducts and for underground burial with uncontrolled backfill and possibility of flooding by water and chemicals. These shall have protection against any mechanical damage and suitably designed to be protected against rodent and termite attack. The inner sheath similar to that provided for cables shall be provided as part of straight through joint. The jointing kits shall be from one of the makes/ types mentioned in the data sheet.

5.0 INSPECTION, TESTING AND ACCEPTANCE

The cables shall be tested and inspected at the manufacturer's works. Manufacturer shall furnish all necessary information concerning the raw material supply to EIL/ Owner's inspectors. The inspector shall have free access to the manufacturer's works for the purpose of inspecting the process of manufacture in all its stages and will have the power to reject any material, which appears to be of unsuitable description or of unsatisfactory quality. For HV cables, the vendor shall give at least 2 weeks advance notice to the purchaser, regarding the date of testing to enable purchaser's representative to witness the tests.

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5.1 Cables

5.1.1 After completion of manufacture of cables and prior to despatch, the cables shall be subjected to type, routine, acceptance and special tests as detailed below. The test reports for all cables shall be got approved from the Engineer before despatch of the cables.

5.1.2 All routine tests, acceptance tests, type tests and additional type tests for improved fire performance shall be carried out as listed in IS: 7098 (Part-1 & Part-2) on XLPE insulated cables.

5.1.3 The test requirements for PVC sheath of cables shall be as per latest revision of IS: 5831.

5.1.4 Test for Resistance to Ultra Violet Radiation: This test shall be carried out as per ASTM-G-53 or ASTM-G-154 on outer sheath. The retention value of tensile strength and ultimate elongation after the test shall be minimum 60 % of tensile strength and ultimate elongation before the test. Test certificates with respect to this test (not older than one year) from recognised testing laboratory to be furnished for review by EIL before despatch clearance of cables. In case test certificates are not available, test is to be conducted by vendor at his own cost in any recognised test laboratory or in house testing laboratory, before despatch clearance of cables. Sampling for this test is to be done randomly once for each order, provided outer sheath remains same.

5.1.5 Acceptance tests as per IS-7098 (Part-1 & Part-2) and the following special tests to be performed on the cables as per sampling plan for all cables. However these tests are required to be witnessed by EIL/ Owner for HV cables.

a. Accelerated water absorption test for insulation as per NEMA-WC-53 (for XLPE/ EPR insulated cables). Test certificates with respect to this test (not older than one year) from recognised testing laboratory to be furnished for review by EIL before despatch clearance of cables. In case test certificates are not available, test is to be conducted by vendor at his own cost in any recognized test laboratory or in house testing laboratory, before despatch clearance of cables. Sampling for this test is to be done randomly once for each order, provided type of insulation remains same.

b. Oxygen Index Test: The test shall be carried out as per IS-10810 (Part 58). Sampling to be done for every offered lot/size as per sampling plan.

c. Flammability Test: The test shall be carried out on finished cable as per IS-10810 (Part 61 & 62). Sampling for these tests is to be done randomly once for each order, provided outer sheath remains same. The acceptance criteria for tests conducted shall be as under:

Part-61-The cable meets the requirement if there is no visible damage on the test specimen within 300 mm from its upper end

Part-62-The maximum extent of the charred portion measured on the test sample should not have reached a height exceeding 2.5 m above the bottom edge of the burner at the front of the ladder.

d. Test for rodent and termite repulsion property shall be done by analysing the property by chemical method.

5.1.6 Following tests shall be carried out to prove FRLS property of the cable.

a. Critical oxygen index as per ASTM-D-2863 i.e. Determination of % of oxygen required for combustion at room temperature of FRLS sheath which shall remain as 29% (min.)

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b. Temperature index as per ASTM-D-2863 i.e. to determine at what temperature normal oxygen content of 21% in air will support combustion of FRLS sheath which shall remain as 250°C.

c. Halogen acid gas emission as per IEC-60754 Part 1 i.e. to determine the % of release of hydrochloric acid gas from the FRLS sheath under fire which shall be 20% (max.)

d. Smoke Density Test shall be as per ASTM D - 2843 and Smoke Density Rating of FRLS Sheath shall be 60% (max.).

5.1.7 The test for circuit integrity for fire survival cables shall be carried out as per IEC-60331 (Part-21).

5.2 Cable Accessories

Type tests should have been carried out to prove the general qualities and design of a given type of termination/ jointing system as per IS-13573. The type test certificates from independent testing laboratory shall be submitted before despatch.

6.0 PACKING AND DESPATCH

6.1 Cables shall be despatched in non-returnable steel drums of suitable barrel diameter, securely packed, with the take-off end fully protected against mechanical damage. Ferrous parts used shall be treated with a suitable rust preventive finish or coating to avoid rusting during transit or storage.

6.2 On the flange of the drum, necessary information such as project title, manufacturer's name, type, size, voltage grade of cable, length of cable in metres, drum no., cable code, BIS certification mark, gross weight etc. shall be printed. An arrow shall be printed on the drum with suitable instructions to show the direction of rotation of the drum.

6.3 Cables shall be supplied in drum lengths as follows:

6.3.1 **MV Cables**

-	Multicore Power cables upto 50 mm^2	:	2000 m
-	Multicore Power cables from 70 mm^2 up to 500 mm^2	:	1000 m
-	Single Core Power cables upto 630 mm^2	:	1000 m

6.3.2 **HV Power Cables Upto 11kV Grade**

-	Three Core cables upto 400 mm^2	:	750 m
-	Single Core cables upto 400 mm^2	:	1000 m
-	Single Core cables above 400 mm^2 and upto 1000 mm^2	:	750 m

6.3.3 **HV Power Cables Above 11kV Grade and upto 33kV Grade**

-	Three Core cables upto 300 mm^2 grade	:	500 m
-	Single Core cables upto 400 mm^2	:	1000 m
-	Single Core cables above 400 mm^2 and upto 1000 mm^2	:	500 m

However exact drum lengths shall be finalised during order execution. A tolerance of ± 3 % shall be permissible for each drum. However overall tolerance on each size of cable shall be limited to (-2%) to (+0%).

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संकटमय क्षेत्रों में लाइटिंग फिक्सचरों एवं जंक्शन बक्सों के लिए विनिर्देश

SPECIFICATION FOR HAZARDOUS AREA LIGHT FIXTURES AND JUNCTION BOXES

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
6	23.12.20	REVISED AND ISSUED AS STANDARD SPECIFICATION	NNB/RKS	ANPS	PG	SM
5	24.02.16	REVISED AND ISSUED AS STANDARD SPECIFICATION	NNB/AK	ANPS	BRB	SC
4	11.04.11	REVISED AND ISSUED AS STANDARD SPECIFICATION	PS	ANPS	UAP	DM
3	27.08.07	REVISED AND ISSUED AS STANDARD SPECIFICATION	ANPS	UAP	JMS	VC
2	30.01.02	REVISED AND ISSUED AS STANDARD SPECIFICATION	UAP	AAN	VPS	GRR

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Abbreviations:

AC	Alternating Current
BIS	Bureau of Indian Standards
CEA	Central Electricity Authority
CRI	Colour Rendering Index
DC	Direct Current
DGCA	Director General Civil Aviation
EIL	Engineers India Limited
FRLS	Flame Retardant Low Smoke
IEC	International Electrotechnical Commission
IES	Illuminating Engineering Society
IP	Ingress Protection
IS	Indian Standard
LED	Light Emitting Diode
MR	Material Requisition
NABL	National Accreditation Board for Testing and Calibration Laboratories
PESO	Petroleum and Explosives Safety Organisation
PO	Purchase Order
PVC	Poly Vinyl Chloride
SPD	Surge Protection Device
SPN	Single Phase Neutral
SS	Stainless Steel
XLPE	Cross Linked Poly Ethylene

Electrical Standards Committee

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1.0 SCOPE

This specification covers the requirements of design, manufacture, testing, packing and supply of LED lighting fixtures, complete with LED lamp and driver, junction boxes and accessories suitable for installation in locations handling flammable liquids and gases/ vapours.

2.0 CODES AND STANDARDS

2.1 The equipment shall comply with the requirements of latest revision of the following standards issued by BIS:

IS-5	:	Colours for ready mixed paints and enamels
IS-10322	:	Luminaires
IS-13383	:	Photometry of luminaires - Method of measurement
IS 15885 (Part 2/ Sec 13)	:	Safety of lamp control gear-Part-2 Particular requirements-Section 13 d.c. or a.c. supplied electronic control gear for LED modules
IS-16101	:	General lighting – LEDs and LED modules – Terms and Definitions
IS-16102	:	Self-ballasted LED lamps for general lighting services
IS-16107	:	Luminaires performance
IS / IEC 60079-0	:	Electrical apparatus for explosive gas atmospheres (General requirements)
IS / IEC 60079-1	:	Electrical apparatus for explosive gas atmospheres (Equipment Protection by Flameproof Enclosures "d")
IS / IEC 60079-7	:	Electrical apparatus for explosive gas atmospheres (Equipment Protection by Increased Safety "e")
IS / IEC 60079-15	:	Electrical apparatus for explosive gas atmospheres Construction, test & marking of type of protection "n"
IS / IEC 60529	:	Degrees of protection provided by enclosures (IP Code).

2.2 In case of imported equipment, only IECEx (or equivalent, if more stringent than the applicable IEC) certified equipment shall be used.

2.3 The equipment shall also confirm to the provisions of CEA Regulations with latest amendments and statutory regulations currently in force in the country.

2.4 In case Indian standards are not available for any equipment, standards issued by IEC or equivalent agency (if more stringent than IEC) shall be applicable.

2.5 In case of any conflict between requirements specified in various applicable documents for the project, the most stringent requirement shall govern. However, Owner's/ EIL's decision in this regard will be final and binding.

3.0 GENERAL REQUIREMENTS

3.1 The offered equipment shall be brand new with state of art technology and having proven field track record. No prototype equipment shall be offered.

3.2 Vendor shall ensure availability of spare parts and maintenance support services for the offered equipment for at least 10 years from the date of supply.

4.0 SITE CONDITIONS

The equipment shall be suitable for installation and satisfactory operation in classified hazardous locations in tropical, humid and corrosive atmosphere as prevalent in refineries,

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petrochemical and fertilizer plants. Unless otherwise specified, a design ambient temperature of 40° C and an altitude not exceeding 1000 m above mean sea level shall be considered.

5.0 POWER SUPPLY

All AC lighting fixtures/ junction boxes shall be suitable for 240V ± 10%, SPN, 50Hz ± 3% power supply and all DC lighting fixtures/ junction boxes shall be suitable for 110/ 220V ± 10% power supply.

6.0 CERTIFICATION

The equipment shall have test certificates issued by NABL-approved/ Central Government labs in India/ IECEx-approved/ equivalent labs. All equipment (indigenous & imported) shall also have valid statutory approvals, as applicable, for the specified location, and marking as per IS/IEC 60079 and as required by statutory authorities. All indigenous flameproof equipment shall have valid BIS license.

7.0 TECHNICAL REQUIREMENTS

7.1 Construction

7.1.1 The enclosures of the lighting fixtures and junction boxes shall be of cast light metal alloy.

7.1.2 All equipment shall be suitable for use in outdoor open locations and shall have minimum IP-55 degree of protection.

7.1.3 The enclosures shall be provided with gaskets made of non-inflammable and self-extinguishing material.

7.1.4 All metal surfaces shall undergo manufacturer's standard cleaning/ painting/ powder coating cycle. After surface preparation, the equipment shall be painted with two coats of epoxy based final paint or epoxy powder coated with minimum coating thickness of 80 microns, with colour shade as below:

- Flameproof Exd (Gas group IIA/IIB) and AC supply: Dark admiralty grey shade 632 of IS-5/ RAL 7031
- Flameproof Exd (Gas group IIC) and AC Supply : Light yellow shade 355 of IS-5/ RAL 1012
- Non-sparking ExnR/ Increased safety Exec and AC Supply : Dark admiralty grey shade 632 of IS-5/ RAL 7031
- DC lighting fixtures : Bright red shade 538 of IS-5/ RAL 3003

All unpainted parts shall be suitably treated to prevent rust formation/ corrosion. If these parts are moving elements then these shall be greased with non-solidifying grease.

7.1.5 A warning inscription "DO NOT OPEN WHEN ENERGIZED" shall be provided on each enclosure. The warning inscription shall be embossed on the enclosure or a separate warning plate with above inscription shall be fixed to the enclosure with screws. The warning plate shall be of nickel plated brass or stainless steel.

7.1.6 All accessories like nuts, bolts, washers etc. shall be made of stainless steel SS-304.

7.1.7 All equipments on single phase supply shall be provided with an independent earth terminal with lug inside the enclosure for terminating the third (earth) core of the cable and shall have facility for looping.

7.1.8 A nameplate shall be provided to indicate the details of testing agency, name of manufacturer, test certificate number, statutory approval agency (PESO), serial number, BIS license number, applicable gas group etc. and any additional marking required by statutory authority like approval no. etc. The nameplates shall be Al anodized or SS-304 engraved type or laser-

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marked and permanently fixed/ fixed with rivets on the equipment. In case the standard details given above are embossed on the enclosures, the same need not be repeated on the name plate.

7.2 Lighting Fixtures

7.2.1 Glass used for all types of lighting fixtures shall be clear and suitable for use under conditions involving exceptional risk of mechanical damage. Glass shall be toughened type.

7.2.2 All well/ bowl glass fixtures shall be provided with Stainless Steel SS-304 protective wire cage using minimum 5mm welded steel construction and having mesh dimension not exceeding 50mm x 50 mm. The mounting bolts of SS-304 required for physically connecting the hazardous area well/ bowl glass lighting fixtures (bracket/ ceiling mounted type) to the respective mounting bracket shall be supplied by Vendor along with the fixtures.

7.2.3 The fixing parts of the enclosure, which are to be opened for replacement of lamp, shall be so fastened that they can be unfastened only with special tools.

7.2.4 The lighting fixtures in Zone-2 area shall be Ex nR type, unless otherwise specified in the MR/ Tender.

7.2.5 For LED fixture, integral junction box shall be provided with 3 nos. cable entries for loop-in, loop-out and associated wiring to fixture. Alternatively, lighting fixtures shall be provided with 2 nos. threaded entries and 2 nos. hazardous area double compression nickel plated brass cable glands, and internal wiring shall be provided for driver and lamp. The glands shall be suitable for the specified cable size. 20% of the fixtures shall be supplied with 1 no. nickel plated brass plug for sealing the unused entry.

7.2.6 The top of all well/ bowl glass lighting fixtures shall be identically drilled/ threaded to facilitate installation on pole/ column or ceiling.

7.2.7 The flood lighting fixtures shall be supplied with adjustable mounting arrangement both in horizontal and vertical plane.

7.2.8 The aviation warning lighting fixtures shall have cluster LED lamps of Type C, medium intensity (as per DGCA civil aviation statutory regulations) and of red colour, fixed (not flashing), 2000cd, unless otherwise specified.

7.2.9 All lighting fixtures shall be provided with suitable mirror-polished aluminum internal reflector of suitable thickness to ensure adequate mechanical strength.

7.2.10 Lighting fixtures shall be tested for resistance to impact in line with IS/ IEC 60079-0. Impact test certificates shall be furnished by vendor.

7.2.11 All LED fixtures shall be supplied complete with lamp and driver. LED driver shall be suitable for surge protection of minimum 4 kV surge. However, for flood lights, aviation warning lights, and lighting fixtures with wattage higher than 90W, LED driver shall be suitable for surge protection of minimum 10 kV surge. An additional fail safe SPD may be provided, as required, to achieve 10 kV surge protection. LED driver shall comply with the requirement of IS 15885-2-13. Fixture lifetime (L70) shall be greater than 50000 hours at ambient temperature of 40°C.

7.2.12 LED type lighting fixtures shall fulfill the following requirements:

a. Minimum power factor shall be 0.95.

b. LED lighting fixture luminous efficacy shall be minimum 100 lumen per watt.

c. LED lighting fixture beam angle shall be 120°/ 90° depending upon fixture location and mounting height.

d. The CRI for LED lighting fixtures shall be greater than 80 for indoor fixture and greater than 70 for outdoor fixture.

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e. The colour temperature of white colour LED shall be from 5700K to 6500K, unless otherwise specified.

f. Polar curves/ photometric drawings shall be submitted by vendor for the offered lighting fixtures.

g. LED lamps and drivers shall be provided with minimum 5 years warranty period, unless specified otherwise in the MR/ Tender. Driver shall have valid BIS certification and the same shall be furnished during drawing review stage.

h. The LED light fixtures, should be tested for luminous lux level as per IES-LM-79. Wherever wire guard is provided, testing shall be done with wire guard. Type test certificates for LM-79 for all fixtures shall be from an NABL-accredited/ IECEx-certified/ equivalent lab.

i. The LEDs used in the lighting fixtures should be tested for the service life as per IES-LM-80. Type test certificates shall be furnished in compliance to same.

7.3 Junction Boxes

The junction boxes shall be of flameproof or non-sparking type construction as mentioned in the MR/ Datasheet. All junction boxes for looping of single phase lighting circuits using cables up to 6mm^2 shall be minimum 125mm diameter in size. Each junction box shall be complete with requisite number of cable entries and double compression flameproof nickel plated brass cable glands to suit the specified cable size.

7.4 Terminals & Wiring

7.4.1 The fixtures and junction boxes shall be provided with sufficient number of terminals. More than 2 wires per terminal shall not be permitted. If required, additional terminal with shorting link may be used. The terminals for external cable termination shall be suitable for 2.5mm^2 solid Copper conductor. Tinned copper lugs shall be provided for cable termination wherever applicable.

7.4.2 All internal wiring shall employ 1.5mm^2, 660/ 1100 V grade, FRLS type, XLPE/ PVC insulated Copper conductor wires, colour coded for phase, neutral and earth. All termination shall be with suitable lugs.

8.0 INSPECTION, TESTING AND ACCEPTANCE

8.1 During fabrication, the equipment shall be subjected to inspection by EIL/ Owner or by an agency authorised by the Owner, as per agreed Inspection Test Plan. Manufacturer shall furnish all necessary information concerning the supply to EIL/ Owner's inspector. All routine/ acceptance tests shall be carried out at manufacturer's works under his care and expense.

8.2 Type test certificates from NABL-approved/ Central Government Labs in India/ IECEx-certified/ equivalent Labs, Manufacturer's works test reports, applicable PESO approval and BIS license shall be shown to the inspection agency on demand during inspection. The certificates, BIS license and PESO approval must be valid at the time of despatch.

8.3 Test certificates of bought out components shall be submitted to the inspection agency, as per Inspection and Test Plan no. 6-81-1061.

8.4 All equipments shall be subjected to various routine/ acceptance tests as per Inspection and Test Plan no. 6-81-1061.

9.0 PACKING AND DESPATCH

All the equipment shall be divided into several sections for protection and ease of handling during transportation. The equipment shall be properly packed for the selected mode of transportation, i.e. by ship/ rail or trailer, and shall be wrapped in polythene sheets before

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being placed in crates/ cases to prevent damage to finish. The crates/ cases shall have skid bottom for handling. Special notations such as 'Fragile', 'This side up', 'Center of gravity', 'Weight',' Owner's particulars', 'PO nos.' etc., shall be clearly marked on the packages together with other details as per purchase order.

The equipment may be stored outdoors for long periods before installation. The packing should be suitable for outdoor storage in areas with heavy rains and high ambient temperature unless otherwise agreed. A set of instruction manuals for installation, testing and commissioning, a set of operation & maintenance manuals and a set of final drawing shall be enclosed in a waterproof cover along with the shipment.

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अग्निरोधक लाइटिंग एवं पावर पैनलों के लिए निरीक्षण एवं परीक्षण योजना

INSPECTION AND TEST PLAN FOR FLAMEPROOF LIGHTING AND POWER PANELS

Rev. No.	Date	Purpose	Prepared by	Checked by	Standards Committee Convener	Standards Bureau Chairman
					Approved by	
4.	28.08.2024	Revised and Reissued	RP	GS	RK	MN
3.	14.01.2020	Revised and Reissued	MM	RS	RKS	RKT
2.	21.11.2013	Revised and Reissued	MJ	RS	SCG	SC
1	15.07.2011	Revised and Reissued	GS	SA	AKC	DM
0	07.10.2008	Issued for implementation	SA	SKD	SKP	VC

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Abbreviations

BIS	:	Bureau of Indian Standard	MTC	:	Material Test Certificate
CIMFR	:	Central Institute of Mining & Fuel Research	NEMA	:	National Electrical Manufacturers Association
DFT	:	Dry Film Thickness	PO	:	Purchase Order
ERTL	:	Electronics Regional Test Laboratory	PESO	:	Petroleum Explosive Safety Organization
FM	:	Factory Mutual	PQR	:	Procedure Qualification Record
FLP	:	Flame Proof	PR	:	Purchase Requisition
FRLS	:	Flame retardant low smoke	PVC	:	Poly Vinyl Chloride
ITP	:	Inspection and Test Plan	QC	:	Quality Control
IP	:	Ingress Protection	TC	:	Test Certificate
IS	:	Indian Standard	TPI or TPIA	:	Third Party Inspection Agency
IC	:	Inspection Certification	VDR	:	Vendor Data Requirement
IEC	:	International Electro technical Commission	XLPE	:	Cross Linked Poly Ethylene
JEC	:	Japanese Electro technical Committee			

Inspection Standards Committee

Convenor : Mr. Rajeev Kumar

Members:

Mr. G Suresh	Mr. Himangshu Pal	Mr. Chandrashekhar	Mr. R Muthu Ramalingam
Mr. Mahendra Mittal	Mr. Avdesh Agrawal	Mr. Anil Kumar	Mr. Arvind NP Singh (Engg)

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1.0 SCOPE

This Inspection and Test Plan covers the minimum inspection and testing requirements for Flameproof Lighting And Power panels.

2.0 REFERENCE DOCUMENTS

PO/PR / Standards referred there in /Job specifications / Approved documents/Relevant IS/IEC Standards.

3.0 INSPECTION AND TEST REQUIREMENT:

SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
1.0	**Procedures**	---	---	---	---	---	---
2.0	**Material Inspection**						
2.1	Incoming Materials like Castings, Glass, Meters, Wires/ Cables, Bus-Bars, Gaskets, Isolators, MCBs, MCCBs, RCCBs, CTs/ PTs etc (As applicable)	Visual, Dimensional, Physical, chemical & functional checks (as applicable)	100%	Material test certificates / Manufacturer's Test records/	H	H	R
3.0	**In process Inspection**						
3.1	Machining, Joints, Sealing methods	As per Test Lab Certified Drawings for Hazardous Area Applications	100%	Test records	-	H	---
3.2	Panel assembly	Visual, Dimensional, Bill of Material, Rating of Components.	100%	Test Records	-	H	---
3.3	Wiring Checks	Size, type (FR/FRLS), make, termination tightness, ferruling, continuity, color coding	100%	Test Records	-	H	----
3.4	Pressure Check	Over Pressure test on individual enclosures as per relevant standard	100%	Test Records	-	H	R

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.0	**Final Inspection**						
4.1	Final Inspection	• Visual and Dimensional check • Adherence of panels to Hazardous Area Certification conditions including Flame Path, Flame Gap, Thread Check, Thickness of Enclosure, No. of Cable Entries, Rating of mounted components, Maximum number of door/ cover mounted components, Joints, Sealing etc. • Bill of material. • Wiring checks • Paint shade, adhesion and thickness. • Verification of mechanical and electrical operations • Dielectric test (High Voltage) • Insulation check before & after High Voltage • Correctness of nameplate • Annunciation/ Indication Checks	100%	Test Records	-	H	H
4.2	Submission of Certificates & Documents	• Test Certificate of testing agency (CIMFR/ KLPL etc) for suitability of model to intended Hazardous area classification as per latest IS/IEC standards. • Certificate of Statutory authority (PESO). • Valid BIS License.	For each Type/Model	Certificates from Test Labs/ Agencies	-	H	R

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
5.0	**Painting**						
5.1	Painting and Packing	• Visual checks for compliance to Painting requirements as per contract. • Suitable protection to prevent entry of foreign material. • Proper packing to prevent any damage during transportation and storage as per contract.	100%	Packing list / Supplier's Records	-	H	-
6.0	**Documentation and IC**						
6.1	Documentation and IC	• Review of Internal Test Reports & MTC • IC Issuance	100%	Supplier's Test Records / Inspection Certificate	-	H	H

Legends: H- Hold (Do not proceed without approval), Random -10%, R-Review, RW-Random witness, W- Witness (Give due notice, work may proceed after scheduled date).

NOTES :-

1. This document describes the generic test requirements. Any additional test or inspection scope if specified in contract documents shall also be applicable. (Unless otherwise agreed upon).
2. Acceptance Norms for all the activities shall be as per PO/PR/ Standards referred there in/ Job specifications /Approved documents/ Hazardous Area certifications.
3. Supplier has to furnish internal test reports before scheduled inspection date.

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मध्यम एवं उच्च वोल्टेज केबल तथा सहायक सामग्री के लिए निरीक्षण व परीक्षण योजना

INSPECTION AND TEST PLAN FOR MEDIUM AND HIGH VOLTAGE CABLES & ACCESSORIES

Rev. No.	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
4	26.05.2020	Revised and Reissued	MM	RS	RKS	SKS
3	30.01.2014	Revised and Reissued	MJ	RS	SCG	SC
2	15.07.2011	Revised and Re-issued	AB	RJD	AKC	DM
1	28.09.07	Revised and Re-issued	AKG	PPM	MVKK	VC
0	30.04.02	Issued for implementation	RG	AKC	AKB	GRR

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Abbreviations:

BIS	:	Bureau of Indian Standard
CEIL	:	Certification Engineers International Limited
Cu	:	Copper
EPC	:	Engineering Procurement & Construction
ERTL	:	Electronics Regional Test Laboratory
FM	:	Factory Mutual
FRLS	:	Flame Retardant Low Smoke
HT	:	High Tension
ITP	:	Inspection and Test Plan
IP	:	Ingress Protection
IC	:	Inspection Certification
IS	:	Indian Standard
IEC	:	International Electro technical Commission
JEC	:	Japanese Electro technical Committee
Min	:	Minimum
MTC	:	Material Test Certificate
NEMA	:	National Electrical Manufacturers Association
No	:	Number
PD	:	Partial Discharge
PO	:	Purchase Order
PR	:	Purchase Requisition
PVC	:	Poly Vinyl Chloride
QC	:	Quality Control
TC	:	Test Certificate
TPI or TPIA	:	Third Party Inspection Agency
VDR	:	Vendor Data Requirement
XLPE	:	Cross Linked Poly Ethylene

Inspection Standards Committee

Convenor : Mr. R.K. Singh

Members: Mr. Rajesh Sinha, Mr. Himangshu Pal, Mr. Chandrasekhar, Mrs. Arvind NP Singh (Electrical)

Mr. R Muthuramalingam, Mr Avdhesh Agrawal, Mr. Mahendra Mittal

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1.0 SCOPE

This Inspection and Test Plan covers the minimum testing requirements of PVC/XLPE insulated cables and cable jointing/terminating accessories for high and medium voltage systems.

2.0 REFERENCE DOCUMENTS

PO/PR / Standards referred there in /Job specifications / Approved documents.

3.0 INSPECTION AND TEST REQUIREMENTS

SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
1.0	**Procedures**	--	--	--	--	--	--
2.0	**Material Inspection**						
2.1	Incoming raw material like Copper, Aluminum, Semi conducting compound PVC, XLPE, Galvanized Armour, Cu tape, as applicable	Purity, Dimensions, Physical and Chemical Properties, as applicable	100%	Material Test Certificates/ Test Records/ Third Party Lab certificates	H	H	R
3.0	**In process Inspection**						
3.1	Inspection during insulation, sheathing	Spark test, thickness of insulation, Conductor Resistance	100%	Test Records	--	H	R
4.0	**Final Inspection**						

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.1	Finished Cable	Visual, Drum details, Size, Ends Capping, Marking on outer sheath (such as Incremental Length, Voltage grade, manufacturer's name), Colour coding, etc.	100% for Drum details and Sample drums for balance details	Test Records	-	H	RW/R**
4.2	Finished Cable (Routine Tests)	• Conductor Resistance. • High Voltage • Partial Discharge (for HT cable only)	100%	Supplier's Test Records	-	H	R
4.3	Finished Cable (Acceptance tests)	• Check calibration status of all testing equipment. • Dimension checks – overall dia, over/under armour, thickness of insulation, sheath and semi conducting screen, copper screen, dimension, etc as applicable • Conductor Resistance • Volume resistivity at room temperature and elevated temperature as per IS • Thermal stability test for Insulation (for PVC only) and sheath. • High Voltage tests • Partial Discharge(for HT Cable) • Hot Set (for XLPE insulation) • Tests on Conductor as applicable • Tests on PVC/XLPE Insulation & Sheath * • Armour Galvanization - dip test & coverage • Tensile and elongation (without ageing) • Silicon oil and wafer boil test for insulation and screen (for HT cable).	As per IS sampling plan	Inspection Test Records	-	H	H/R**

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.4	Finished Cable (Special tests)	• Flammability test as per IS 10810 part 61 & part 62 / IEC 60332 Part 1 & part 3 • Smoke density test (for FRLS outer sheet) • Acid Gas generation (for FRLS outer sheet) • Accelerated water absorption test as per NEMA(Electrical method) as applicable # • Test for resistance to Ultra Violet Radiation # • Oxygen Index and temperature index test as per ASTM D2863 • Rodent & Termite repulsion check • Test for circuit integrity for fire survival cables as per IEC • Drum length /surface finish/color/marking check by re-winding (1 or 2 drums)	For ladder test (1 sample/order) 1 Sample/Lot	Inspection test records		H	H/R**
4.5.	Finished Cable (Type tests **)	• Dielectric Retention. • Type tests on PVC insulation and Sheath High Voltage test (Water Immersion) for HT cable • Dielectric power factor tests as function of voltage, temperature. (HT cables) • Heating cycle test (HT Cables) • Impulse withstand test (HT Cables) • Short circuit test on conductor and screen (HT Cable)	On Similar type cable	Test Records		H	R

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.6	Cable accessories	• Visual • Dimensions • Type tests** as per IS/IEC	Sample	Test Records		H	R
5.0	**Painting**	--	--	--	--	--	--
6.0	**Documentation and IC**						
6.1	Documentation and IC	• Review of Routine, Acceptance, Type test reports. • IC issuance	100%	Supplier TC & IC	-	-	H
6.2	Final Document submission	Compilation of Inspection reports, drawings, etc. as per VDR / PR	100%	Final data folder / Completeness certificate	-	H	H

\# In case TC (not older than one year) from independent test lab is available & type of compound used is same then the same may be accepted.

* Hot deformation, Heat shock and Shrinkage tests on insulation & sheath (One sample/Lot- provided sheath material is from same compound/batch) shall be witnessed by EIL/TPIA, all other tests as per IS/IEC shall be carried out by supplier and test reports shall be reviewed by EIL/TPIA.

** EIL witness for HT cables and EIL surveillance checks on MV Power cable.

Legends: H- Hold (Do not proceed without approval, R-Review, RW-Random witness (As specified or 10 % - Samples must include min 1 No of each type), W- Witness (Give due notice, work may proceed after scheduled date).

NOTES :-

1. This document describes the generic test requirements. Any additional test or inspection scope if specified in contract documents shall also be applicable. (Unless otherwise agreed upon).
2. Acceptance Norms for all the activities shall be as per PO/PR/ Standards referred there in/ Job specifications /Approved documents.
3. For EPC jobs, all inspection shall be carried out by TPIA, unless notified otherwise.

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संकटमय क्षेत्रों के लिए लाईटिंग फिक्सचरों एवं जंक्शन बक्सों के लिए निरीक्षण व परीक्षण योजना

INSPECTION AND TEST PLAN FOR HAZARDOUS AREA LIGHTING FIXTURES AND JUNCTION BOXES

Rev. No.	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
4.	20.05.2020	Revised and Reissued	JA	RS	RKS	SKS
3.	21.11.2013	Revised and Reissued	MJ	RS	SCG	SC
2	29.06.2012	Revised and Reissued	AB	HVJ	AKC	DM
1	28.09.2007	Revised and Re-issued	AKG	PPM	MVKK	VC
0	12.08.2002	Issued for implementation	RG	AKC	AKB	GRR

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Abbreviations

BIS	:	Bureau of Indian Standard	MTC	:	Material Test Certificate
BOM	:	Bill Of Materials	IR	:	Insulation Resistance
CIMFR	:	Central Institute of Mining & Fuel Research	No.	:	Number.
CE	:	Carbon Equivalent	NEMA	:	National Electrical Manufacturers Association
DFT	:	Dry Film Thickness	PD	:	Partial Discharge
DIV	:	Division	PO	:	Purchase Order
DPT	:	Dye Penetrate Testing	PESO	:	Petroleum Explosive Safety Organization
ERTL	:	Electronics Regional Test Laboratory	EPC	:	Engineering Procurement Construction
FCRI	:	Fluid Control Research Institute	PR	:	Purchase Requisition
FLP	:	Flame Proof	PVC	:	Poly Vinyl Chloride
HV	:	High Voltage	IS	:	Indian Standars
ITP	:	Inspection and Test Plan	TPI or TPIA	:	Third Party Inspection Agency
IP	:	Ingress Protection	VDR	:	Vendor Data Requirement.
IC	:	Inspection Certification	UL	:	Under writer Laboratories
IEC	:	International Electrotechnical Commission			

Inspection Standards Committee

Convenor : Mr. R.K.Singh

Members:

Mr. Rajesh Sinha
Mr. R. Muthuramalingam
Ms. Arvind N P Singh (Electrical)
Mr. Himangshu Pal
Mr. Avdhesh Agrawal
Mr. Chandrashekhar
Mr. Mahendra Mittal

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1.0 SCOPE

This Inspection and Test Plan covers the minimum testing requirements of Lighting fixtures, control gearboxes, junction boxes and accessories

2.0 REFERENCE DOCUMENTS

PO/PR/ Standards referred there in/Job specifications/Approved documents.

3.0 INSPECTION AND TEST REQUIREMENTS

SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
1.0	**Procedures**	--	--	--	--	--	--
2.0	**Material Inspection**						
2.1	Incoming materials like castings, terminals, Glass, Rubber gaskets, reflectors, Guards, Chokes, Capacitors, etc.	Physical & Chemical Properties, Dimensions, Surface finish, Light transmission thro' glass, tests on glass, Material Certificate, Make, IR for chokes, etc.	Samples as per relevant standard/ supplier's standard	Supplier's Test Records	H	H	R
3.0	**In process Inspection**						
3.1	Well glasses	Static Hydraulic test	100% by supplier and sample basis by EIL	Supplier's Test Records	-	H	R
3.2	Control gear boxes and junction Boxes	Routine Pressure test	100% by supplier and sample basis by EIL	Supplier's Test Records	-	H	R
4.0	**Final Inspection**						

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.1	Final Inspection	Acceptance Tests:- • Physical, Dimensional and BOM Verification • Flame paths, Clearances • Marking on each piece for suitability to area classification, Tag no. and other details as per relevant IS/IEC. • Marking on glass for type and other details as per IS/IEC • Test for restricted breathing enclosure for Div-2 • Impact test (on samples) • Thermal shock test (on samples) • Electrical operation check • Power factor test on choke • HV test • IR test before & after HV	100%	Inspection Witness Test Records	-	H	R
4.2	Type tests:-	• Temperature rise test • Test for suspension for Div 2 fittings • Test for enclosed Break lamp holder for Div 2 fittings. • Pressure Test (As Applicable)	Samples	Certificates from test agencies	-	R	R

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SL NO.	STAGE/ ACTIVITY	CHARACTERISTICS	QUANTUM OF CHECK	RECORD	SCOPE OF INSPECTION		
					SUB SUPPLIER	SUPPLIER	EIL/TPIA
4.3	Submission of Certificate & Documents	• Certificate of Statutory testing agency for suitability of area classification • Certificate of Statutory approval authority like PESO • License for Indigenous fittings from BIS • Degree of Ingress protection certificate.	Samples	Certificates from test agencies	-	H	R
5.0	**Painting**						
5.1	Painting and Packing	• Visual • Suitable protection to prevent entry of foreign material. • Proper packing to prevent any damage during transportation	100%	Packing list / Supplier's Records	-	H	R
6.0	**Documentation and IC**						
6.1	Documentation and IC	• Review of suppliers internal test reports • Issuance of IC	100%	Internal test reports & IC	-	H	H
6.2	Final Document submission	• Compilation of Inspection reports, drawings etc as per VDR / PR	100%	Final data folder /Completeness certificate	-	H	H

Legends: H- Hold (Do not proceed without approval, R-Review, RW-Random witness (As specified or 10 % - Samples must include minimum 1 No of each type), W- Witness (Give due notice, work may proceed after scheduled date).

NOTES :-

1. This document describes the generic test requirements. Any additional test or inspection scope if specified in contract documents shall also be applicable. (Unless otherwise agreed upon).
2. Acceptance Norms for all the activities shall be as per PO/PR/ Standards referred there in/ Job specifications /Approved documents.
3. For EPC jobs, Scope of Inspection shall be under TPIA only, unless specified otherwise.

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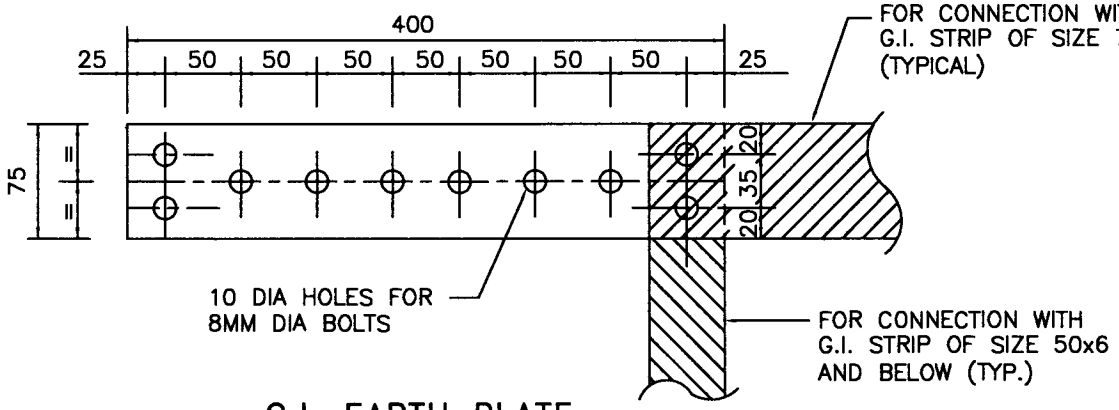
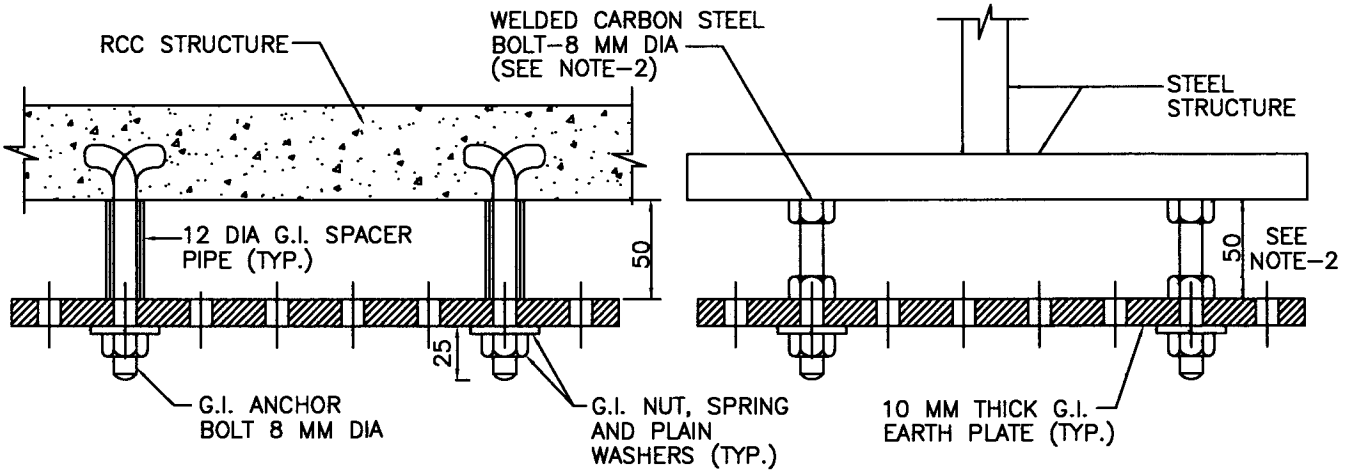
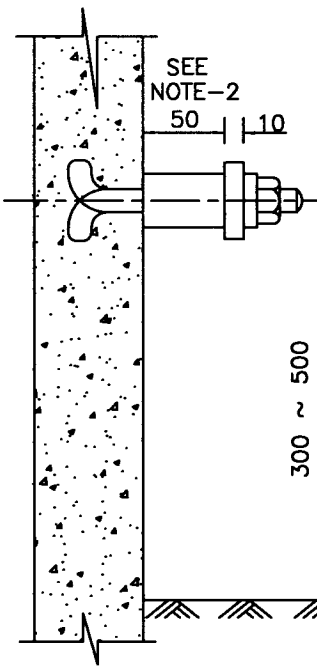
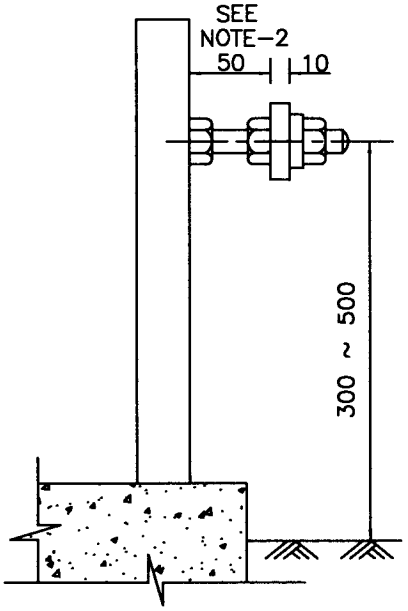


G.I. EARTH PLATE

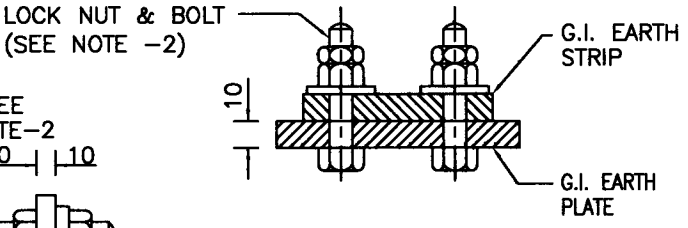




TYPICAL INSTALLATION OF EARTH PLATE ON R.C.C. STRUCTURES



TYPICAL INSTALLATION OF EARTH PLATE ON STEEL STRUCTURES



TYPICAL FIXING DETAIL OF G.I. EARTH STRIP TO EARTH PLATE

NOTES:–

1. ALL DIMENSIONS ARE IN MM.
2. WHERE FIRE PROOFING OF STEEL STRUCTURES ARE ENCOUNTERED LENGTH OF CARBON STEEL BOLTS SHALL BE SUITABLY INCREASED FOR FIXING OF EARTH PLATE.
3. ALL CONNECTIONS WITH EARTH PLATE SHALL BE MADE WITH 8MM DIA. G.I. BOLT, NUT, SPRING AND PLAIN WASHERS.
4. CRIMP TYPE CABLE LUGS SHALL BE USED FOR CONNECTION WITH G.I. WIRE ROPE/CU/AL CABLES.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
7	01.11.21	UPDATED & ISSUED AS STANDARD	JSK	VKS/HK	PG	SM
6	08.11.16	UPDATED & ISSUED AS STANDARD	BP	FA/HK	BRB	RN
5	27.06.11	REAFFIRMED & ISSUED	BP	RKS/RSC	UAP	DM

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TYPE OF EQUIPMENT	EARTH CONDUCTOR SIZE (SEE NOTE-1)
LV MOTORS UPTO 3.7KW	8 SWG SOLID G.I. WIRE
LV MOTORS FROM 5.5KW TO 30KW & WELDING RECEPTACLES	10 MM (3/8") DIA G.I. WIRE ROPE
LV MOTORS ABOVE 30KW INCLUDING MV MOTORS	16 MM (5/8") DIA G.I. WIRE ROPE/ 40x5 MM G.I. STRIP
BUILDING COLUMNS	40 x 5 G.I. STRIP
STORAGE TANKS (VERTICAL & HORIZONTAL)	40 x 5 G.I. STRIP
LOADING RACKS	40 x 5 G.I. STRIP
VESSELS & HEAT EXCHANGERS	40 x 5 G.I. STRIP
SMALL EQUIPMENT & INSTRUMENTS	8 SWG SOLID G.I. WIRE
LIGHTING, POWER & INSTRUMENT PANELS	10 MM (3/8") DIA G.I. WIRE ROPE
MAIN EARTH BUS / LV, MV, HV & EHV SWITCH GEAR INTERCONNECTIONS/ POWER TRANSFORMER	AS SPECIFIED
EHV, HV & MV SUB-STATIONS	AS SPECIFIED
PUSH BUTTON STATIONS	8 SWG SOLID G.I. WIRE
STREET LIGHT POLES	10 MM (3/8") DIA G.I. WIRE ROPE
LIGHTING TRANSFORMER	16 MM (5/8") DIA G.I. WIRE ROPE
PIPE RACK	40 X 5 G.I. STRIP
BONDING OF PIPE	25 SQ. MM INSULATED FLEXIBLE CU. CABLE
LIGHTNING PROTECTION	25 X 3 MM G.I. EARTH STRIP
CABLE TRAYS	40 X 5 G.I. STRIP

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
9	31.07.24	UPDATED AND ISSUED AS STANDARD	SKC	SKS/HK	MKS	MN
8	01.11.21	UPDATED AND ISSUED AS STANDARD	JSK	VKS/HK	PG	SM
7	08.11.16	REVISED AND ISSUED AS STANDARD	BP	FA/HK	BRB	RN

Approved by

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TYPE OF EQUIPMENT	ALUMINUM CABLE (SEE NOTE-1&2)	COPPER CABLE (SEE NOTE-1&2)
LV MOTORS UPTO 3.7KW	16 SQ MM	10 SQ MM
LV MOTORS FROM 5.5KW TO 30KW	16 SQ MM	10 SQ MM
LV MOTORS ABOVE 30KW TO 55KW	25 SQ MM	16 SQ MM
LV MOTORS ABOVE 55KW TO 160KW	70 SQ MM	50 SQ MM
MV MOTORS	120 SQ MM	95 SQ MM
STORAGE TANKS (VERTICAL & HORIZONTAL)	70 SQ MM	50 SQ MM
VESSELS & HEAT EXCHANGERS	25 SQ MM	16 SQ MM
LIGHTING, POWER & INSTRUMENT PANELS	10 SQ MM	10 SQ MM
PUSH BUTTON STATIONS	10 SQ MM	10 SQ MM
STREET LIGHT POLES & FLOOD LIGHT MAST	16 SQ MM	10 SQ MM
PIPE RACK	25 SQ MM	16 SQ MM
WELDING RECEPTACLES	25 SQ MM	16 SQ MM

NOTE:-

1. TYPE OF EARTH CONNECTIONS TO INDIVIDUAL EQUIPMENT (I.E. WIRE ROPE / STRIP OR CABLE) FROM NEAREST EARTH PLATE / GRID SHALL BE SELECTED AS SPECIFIED ELSEWHERE. CABLE/WIRE ROPE CONNECTIONS SHALL BE MADE USING CRIMP TYPE OF LUGS.

2. EARTHING CABLE SHALL BE SINGLE CORE PVC INSULATED GREEN COLOR/GREEN WITH YELLOW COLOR STRIPE OUTER SHEATH.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
9	31.07.24	UPDATED AND ISSUED AS STANDARD	SKC	SKS/HK	MKS	MN
8	01.11.21	UPDATED AND ISSUED AS STANDARD	JSK	VKS/HK	PG	SM
7	08.11.16	REVISED AND ISSUED AS STANDARD	BP	FA/HK	BRB	RN

Approved by

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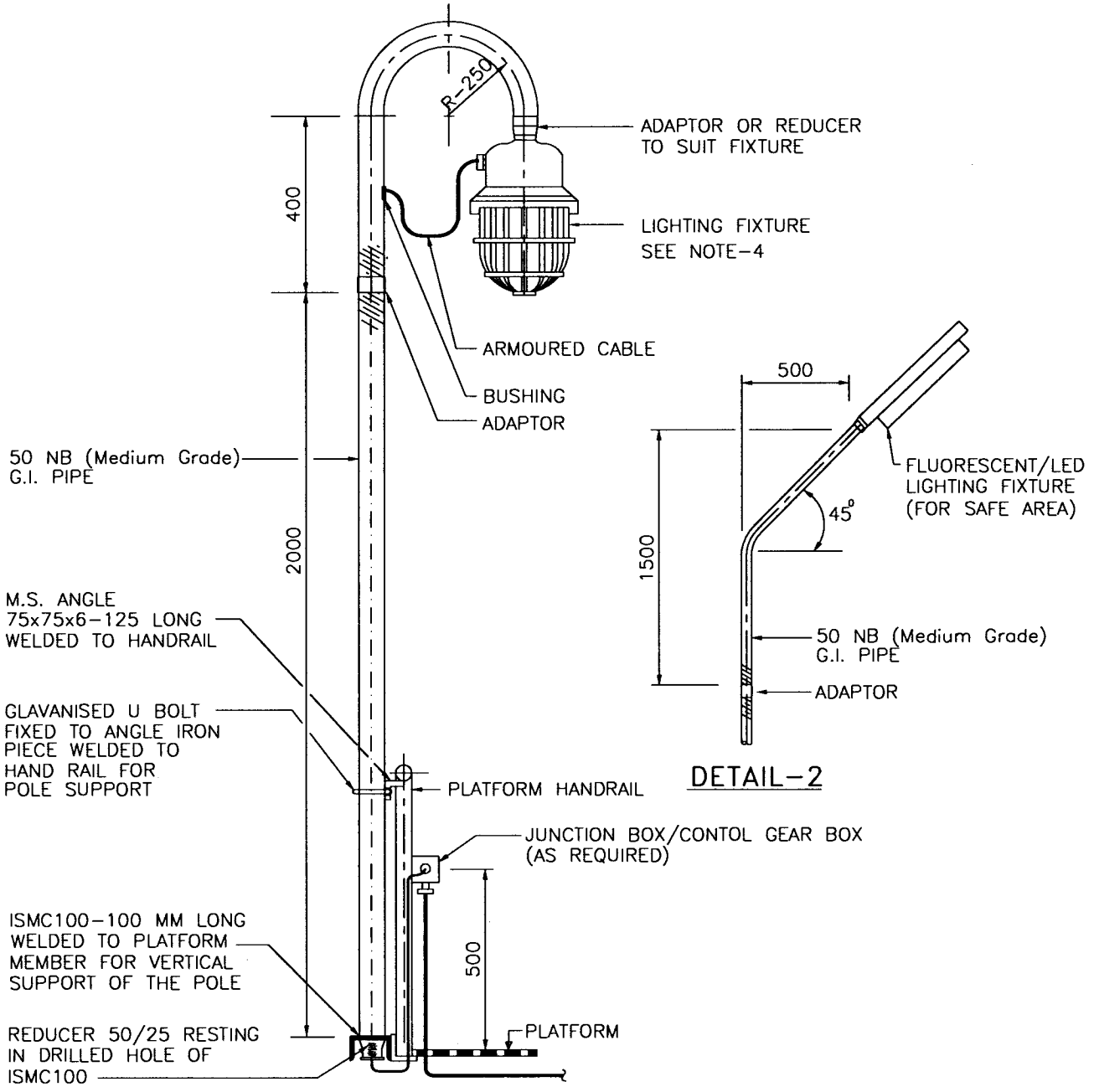

DETAIL-2

DETAIL-1

NOTES:-

1. ALL DIMENSIONS ARE IN MM.
2. FOR TYPE OF FIXTURE REFER LAYOUT DRAWING / LIGHTING FIXTURE SCHEDULE.
3. ALL G.I. PIPES SHALL BE MEDIUM GRADE.
4. LIGHTING FIXTURE CAN BE WELL GLASS/DOME GLASS/FLAT GLASS TYPE BASED ON TYPE OF LAMP.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	29.09.20	UPDATED & ISSUED AS STANDARD	JSK	VKS/HK	SA	SM
5	10.07.17	UPDATED & ISSUED AS STANDARD	BP	FA/HK	BRB	RN

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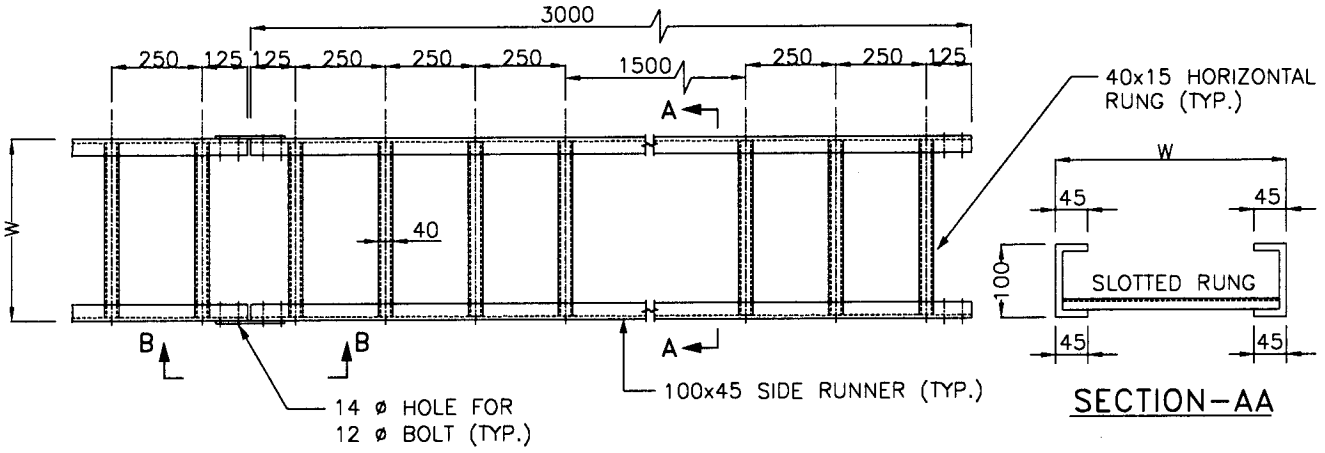
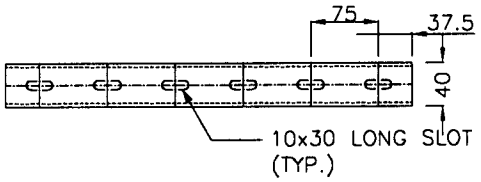
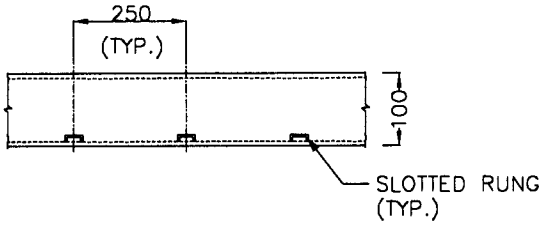


LADDER TYPE CABLE TRAY
(W=300/450/600/750/1000)

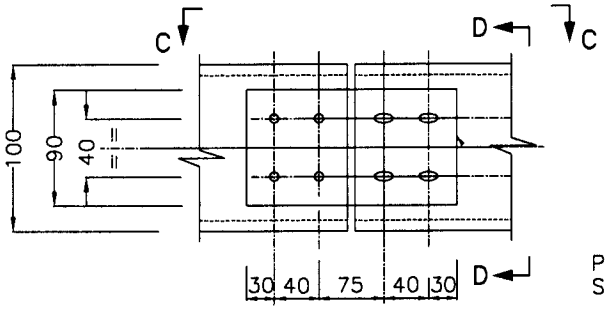
SECTION–AA



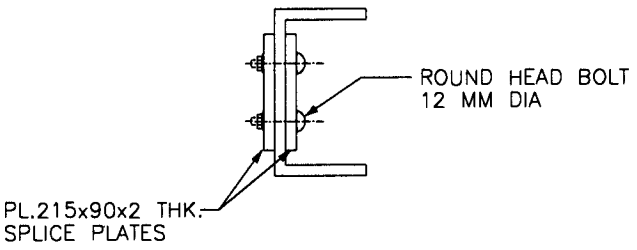
SLOTTED RUNG



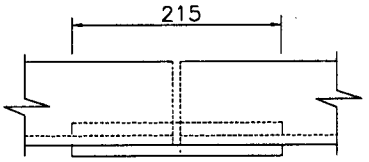
SIDE RUNNER



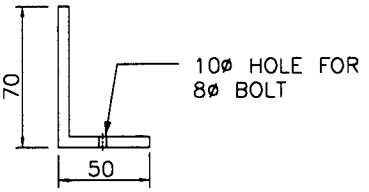
VIEW B–B



SECTION D–D



VIEW C–C



BARRIER PLATE
(100 MM LONG)

NOTES:–

1. 12 DIA ROUND HEAD BOLT WITH WASHERS & NUTS CONFORMING TO IS 1367 SHALL BE USED FOR FIXING OF SPLICE PLATE.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
4	29.09.20	UPDATED & ISSUED AS STANDARD	JSK	VKS/HK	SA	SM
3	10.07.17	REAFFIRMED & ISSUED AS STANDARD	BP	FA/HK	BRB	RN

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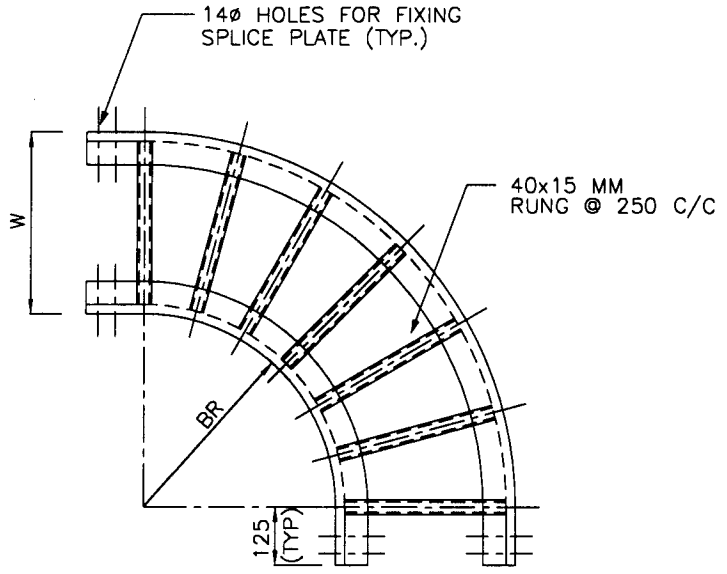
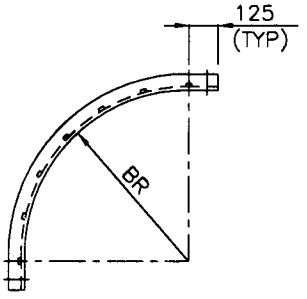
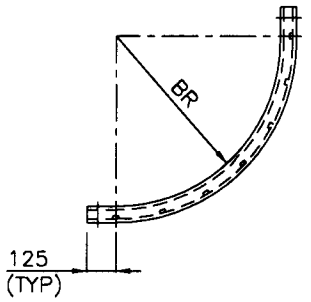


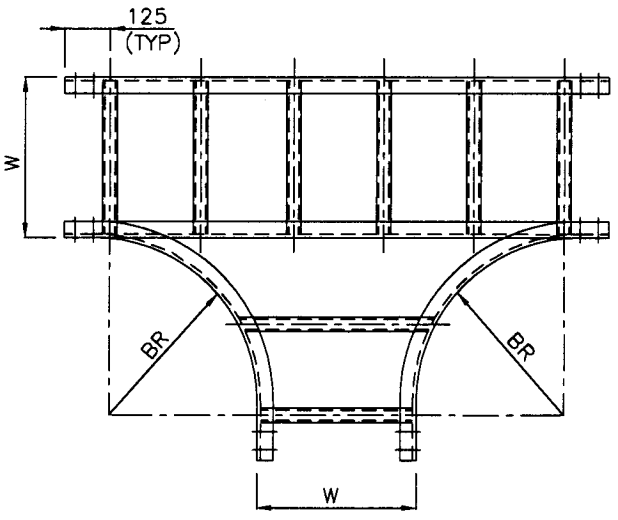
HORIZONTAL BEND



VERTICAL OUTSIDE BEND



VERTICAL INSIDE BEND



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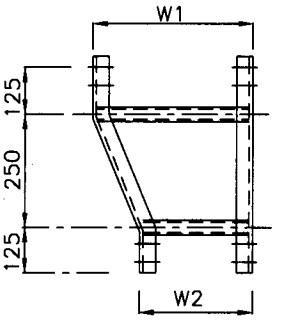
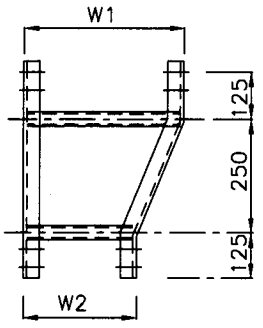


LEFT HAND REDUCER



RIGHT HAND REDUCER

W – WIDTH OF CABLE TRAY (300/450/600/750/1000 MM)
W1 – WIDTH OF CABLE TRAY ON ONE SIDE
W2 – WIDTH OF CABLE TRAY ON OTHER SIDE
BR – BENDING RADIUS (750/1000/1250/1600/2000 MM)

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4	29.09.20	UPDATED & ISSUED AS STANDARD	JSK	VKS/HK	SA	SM
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GENERAL NOTES:-

1. THE CABLE TRAYS AND ACCESSORIES SHALL COMPLY WITH LATEST EDITION OF INDIAN STANDARDS IS: 1367(PART-13), IS:2629, IS:5986.
2. THE CABLE TRAYS SHALL BE LADDER TYPE PREFABRICATED, HOT DIP GALVANIZED. THE LADDER TYPE TRAYS SHALL CONSIST OF SIDE RUNNERS & HORIZONTAL RUNGS AS PER DETAILS SHOWN IN PAGE 1 & 2.
3. THE LADDER TYPE TRAYS AND ACCESSORIES SHALL HAVE RIGID WELDED CONSTRUCTION AND SHALL BE FABRICATED OUT OF 2 mm THICK HOT ROLLED SHEET STEEL. THE RUNGS SHALL BE WELDED TO THE SIDE RUNNERS.
4. SIDE RUNNERS SHALL BE 100 x 45 mm CHANNEL INCLUDING FOR HOIZONTAL & VERTICAL BENDS WITH THE FLANGES FACING INSIDE. RUNGS SHALL BE 40 x 15 mm SLOTTED CHANNEL TYPE.
5. CABLE TRAYS SHALL BE SUITABLE FOR A CABLE WEIGHT OF 75 kg PER METRE OF RUNNING LENGTH OF TRAY AND SHALL BE SUPPORTED AT 3 m INTERVALS. IN ADDITION, TRAYS SHALL BE SUITABLE FOR A POINT LOAD OF 70 kg (EQUIVALENT WEIGHT OF A MAN WORKING ON THE CABLE TRAY).
6. EACH SIDE RUNNER CHANNELS SHALL BE CONNECTED AT ENDS USING 2 SPLICE PLATES ONE ON EACH SIDE AS SHOWN IN PAGE 1.
7. THE SIDE RUNNER WILL ALSO HAVE SUITABLE HOLES AT EVERY METRE FOR CLEATING EARTHING STRIP. SUITABLE THREADED HOLES SHALL BE PROVIDED ON THE RUNNER TOP AND BOTTOM FOR SUPPORTING AND FIXING TRAY COVERS WITH SCREWS AT EVERY METRE.
8. HOT DIP GALVANIZING SHALL BE DONE AFTER FABRICATION AS PER THE RELEVANT INDIAN STANDARD SPECIFICATIONS. THE AMOUNT OF GALVANIZING SHALL BE MINIMUM 610 g/m^2.
9. THE TYPE OF CONSTRUCTION SHALL BE SUCH AS TO FACILITATE EASY HANDLING, ASSEMBLY AND INSTALLATION AT SITE. FOR THE CONSTRUCTION OF CABLE TRAYS THE STANDARD STRAIGHT LENGTH OF CABLE TRAY SHALL BE 3 METERS UNLESS OTHERWISE SPECIFIED.
10. SPLICE PLATE OF CABLE TRAYS SHALL BE PLACED NOT MORE THAN 750 MM FROM SUPPORT ANGLE.
11. THE WORKMANSHIP SHALL BE SUCH AS TO ENSURE EASY LAYING OF CABLES WITHOUT CAUSING DAMAGE TO CABLES. ALL SURFACES SHALL BE FREE FROM DEFECTS SUCH AS BURRS, SHARP EDGES ETC.
12. THE HARDWARE SHALL CONFORM TO THE RELEVANT INDIAN STANDARD SPECIFICATIONS AND SHALL BE ABLE TO WITHSTAND THE MAXIMUM LOADING CONDITIONS AS REQUIRED. ALL HARDWARE AND FITTINGS SHALL BE STAINLESS STEEL SS-304. HARDWARE SHALL INCLUDE BOLTS, NUTS, WASHERS ETC.
13. THE BENDS, TEES, REDUCERS AND DROPPERS SHALL HAVE A BENDING RADIUS AS DESCRIBED IN THE SCHEDULE OF RATES.
14. THE FOLLOWING TESTS FOR THE AMOUNT OF ZINC COATING SHALL BE CARRIED OUT:
    - THICKNESS OF GALVANIZED COATING BY ELCOMETER.
    - MASS OF GALVANIZED COATING BY STRIPPING TEST.
    - DETERMINATION OF UNIFORMITY OF GALVANIZED COATING.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
4	29.09.20	UPDATED & ISSUED AS STANDARD	JSK	VKS/HK	SA	SM
3	10.07.17	REAFFIRMED & ISSUED AS STANDARD	BP	FA/HK	BRB	RN

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JOB SPECIFICATION (ELECTRICAL)

PROJECT : **ENGINEERING / TECHNICAL CONSULTANCY SERVICES FOR PREPARATION OF EXECUTION METHODOLOGY, SOW AND COST ESTIMATES FOR REPLACEMENT / REFURBISHMENT OF BRIDGES IN MH ASSET**

OWNER : **M/s ONGC**

LOCATION : **MH ASSET (NORTH AND SOUTH FIELD)**

JOB NO. : **B774**

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
A	09.09.2025	ISSUED FOR TENDER	PM	RKS	ANPS

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1.0 PURPOSE

Intent of this job specification is to define the electrical system and the specific requirement applicable to the scope covered under this contract in addition to the equipment specifications, design philosophy, drawings etc. as enclosed with the bid document and shall be complied for engineering, supply, installation, testing and commissioning. The designed system shall also comply with the requirement of applicable Indian/ international standards and must comply with the Indian statutory regulations. Wherever owner specification is not available, specification prepared by contractor subject to owner approval shall be followed. Critical job specific requirement is also covered in this document as conflict avoidance.

Codes and standards as prevailing in the country of origin may be followed for equipment purchased abroad provided that the equivalent standards are more stringent than Indian / International standards and meet the statutory regulations of the approving Indian statutory authorities.

It is responsibility of the Contractor to highlight any contradiction amongst various documents at the bid stage itself. In case of conflict between owner specifications, the most stringent requirements will apply.

The following order of priority shall be followed. In case of conflict between requirements specified in various documents, the more stringent one shall be followed. Owner's concurrence shall, however, be obtained before taking a decision in such matters.

- Statutory Regulations

  In particular, the latest version of the following statutory regulations shall be followed for design of electrical system:

  - The Petroleum Rules
  - OISD standards
  - Requirements of other statutory bodies as applicable, e.g. PESO.
- Data sheets & drawings
- Job Scope of Supply & Works
- Job specifications for electrical facilities
- EIL Standard Specifications
- Codes & standards

Contractor shall furnish list of conflicts, if any, along with their offer and also furnish the basis that is considered for submitting offer. Owner/EIL will resolve listed conflicts prior to award. In case of ambiguity, bidder shall inform owner/EIL of their interpretation. In case bidder fails to convey the same prior to award, the Owner's decision on interpretation shall be considered final if need arises during the execution. No additional cost or extra time on account of conflicts/ambiguities shall be admissible.

2.0 SITE CONDITIONS

All electrical equipment and accessories / material shall be suitable for installation and operation under extremely saline, humid, corrosive and hostile marine environment with specified degree of hazards including H2S (wherever required). All outdoor electrical equipment shall be designed for 45°C temperature and maximum 95% relative humidity.

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3.0 ADDITIONS/ DELETIONS/ MODIFICATIONS TO STANDARD SPECIFICATION

4.1 Specification for flameproof lighting & power panel (6-51-0008 Rev 6)

a. Clause 7.1.2 shall be modified as under

The enclosures of the lighting panels intended for use in offshore platform shall be made of stainless-steel grade GRP /COPPER FREE ALUMINIUM.

b. Clause 7.1.11 shall be modified as under

The nameplates shall be SS-304 engraved type or laser-marked and permanently fixed on the equipment.

4.2 Specification for hazardous area light fixtures and junctions' boxes (6-51-0061 Rev.6)

a. Clause 7.1.1 shall be modified as under

The enclosures of the lighting fixtures and junction boxes shall be made of stainless-steel grade GRP /COPPER FREE ALUMINIUM.

b. Clause 7.2.4 shall be modified as under

The lighting fixtures in Zone-2 area shall be Flameproof type, and suitable for Gas Group IIC & Temperature Class T4.

c. Clause 7.3 shall be modified as under

The junction boxes shall be of flameproof type and suitable for Gas Group IIC & Temperature Class T6. All junction boxes for looping of single-phase lighting circuits using cables up to 6mm2 shall be minimum 125mm diameter in size. Each junction box shall be complete with requisite number of cable entries and double compression flameproof nickel-plated brass cable glands to suit the specified cable size. The junction box shall be provided with threaded type cover.

4.0 SPECIFIC JOB REQUIREMENTS

Various equipment covered in the bid package to be supplied by the contractor shall in general be, in accordance with the requirements defined in standard specifications, respective equipment datasheet and. Besides these, the following requirements as detailed out in respective clauses for the equipments shall also be considered:

(i) Contractor shall ensure that all electrical equipment comply to site and system conditions, rating and other technical requirement as specified.

(ii) CONTRACTOR shall use this specification, datasheet, and OWNER standard specifications for various equipments. However, for equipment where OWNER standard specifications have not been attached, the CONTRACTOR shall use their own specification developed based on project specific requirement and good engineering practices prevalent in offshore petroleum industry so as to ensure satisfactory operational and maintenance requirements subject to approval by owner/EIL.

(iii) CONTRACTOR shall note that data sheets for some of the equipment are not enclosed separately. Equipment data sheets for all equipment shall be prepared by the CONTRACTOR based on relevant codes and specifications and data sheet shall contain all technical data and information which are essential for review and technical acceptability, detailed engineering, installation, testing, repair and maintenance, replacement etc.

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(iv) All equipment shall be subjected to routine and acceptance tests as per applicable specifications. In addition, equipment shall be subjected to type tests if specified elsewhere.

(v) CONTRACTOR shall clearly specify in their purchase specifications, the requirement of conducting other special tests/ type tests, which are envisaged for various electrical equipment, which shall have no impact on cost and time.

5.0 All lighting fixtures shall be LED type only & Lighting system shall be designed in accordance with recommendations of illumination engineering society (IES) and to achieve minimum level of illumination as per API RP 14F. LED lamps selected shall provide min. 110 lumens per watt for flood lights and 100 lumens per watt for all other lights. All power and control cable shall be XLPE insulated and copper conductor.

6.0 AREA CLASSIFICATION

Irrespective of hazardous area classification all items outdoor items shall be flameproof type and suitable for Zone-1, gas group IIC.

7.0 MISCELLANEOUS

a. Three number support shall be provided for pole mounted linear lighting fixtures with length more than 1200mm.

b. All light fittings to be secured and fastened using 316 SS fasteners and accessories.

c. All lighting fixtures shall be easily maintainable with separate detachable assemblies.

d. All nuts, bolts, clamps required under this project shall be made of SS316.

8.0 VOLTAGE LEVEL

a. NORMAL LIGHTING

Linear LED Lighting Fixture	240 V AC, 50 Hz
LED Flood Light	

b. EMERGENCY LIGHTING

LED Lighting Fixture With self-contained Battery / Inverter having back up duration of 90 minutes	240 V AC, 50 Hz

c. CRITICAL LIGHTING

LED Lighting Fixture with Inverter circuit	110 V DC

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9.0 INSPECTION AND TESTING AT MANUFACTURER'S WORKS

(i) All major electrical equipment and material shall be subject to inspection by owner/ PMC or authorized representative at manufacturers' works. Contractor/Vendor/ Sub-Vendor shall furnish all necessary information concerning the supply to owner/PMC.

(ii) Contractor shall ensure that the hazardous area electrical equipment procured have valid hazard area type test certificates and PESO approval.

(iii) During fabrication, all the equipment shall be subject to inspection by owner/PMC or by an agency authorized by the owner to assess the progress of work as well as to ascertain that only quality raw material is used.

(iv) Routine and acceptance tests as listed in relevant Indian standard and equipment specifications shall be conducted. Type test if listed in ITPs and in the relevant data sheet shall be conducted. Type test, if specified, shall be conducted only on one of equipment of each type and rating. These tests shall be carried out by the Contractor/vendor/sub-vendor and shall be witnessed by owner/PMC or an agency authorized by owner/PMC.

10.0 FIELD TESTING & COMMISSIONING

(i) Contractor shall carry out the installation, field testing and commissioning of all items. Further appropriate test and commissioning reports and as-built documentation as necessary for all electrical equipment shall be provided.

(ii) All electrical equipment shall be said to be installed and mechanically complete after circuit testing. Due care and consideration shall be given to the installation of all equipment, materials and facilities.

11.0 TWO YEARS SPARES

Bidder shall furnish the required spares for two years of normal operation and maintenance for all electrical equipment with itemized quantity and unit rate. However, owner will decide ordering as per their requirement.

12.0 MAKES OF EQUIPMENT/COMPONENTS

(i) Refer elsewhere in the bid document for acceptable makes of major electrical equipment and electrical components.

(ii) Contractor/ vendor while ordering shall ensure the availability of spare parts and maintenance support services for the offered equipment at least for 10 years from the date of supply. Contractor/ vendor shall give a notice of at least one year to the Owner and PMC before phasing out the products/spares to enable the owner for placement of order for spares and services.

(iii) Contractor/ Vendor can offer equipment/components of makes other than specified in the tender during order execution. The alternate make of equipment/components will be evaluated post order, based on the satisfactory track record and test certificates to be furnished by the Vendor/ Contractor. In case the alternate makes are not found acceptable, equipment/components shall be strictly as per vendor list enclosed with the tender.

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FUNCTIONAL SPECIFICATION FOR FRP CABLE TRAY SYSTEMS

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES	DATE	REV.
B.R	**K.R**	**A.S**	**5**	**02-03-2017**	**3**

1. SCOPE :

The intent of this specification is to define the requirements of Fibre Glass reinforced plastic cable tray systems, including ladder type cable trays, ventilated and solid bottom cable trays, tray cover, necessary hardware, accessories and fittings etc. required for laying of power and control cables on offshore platforms.

2. CODES AND STANDARDS:

2.1. The following codes and standards shall form part of this specification.

ANSI/NFPA	American National Standards Institute/National Fire Protection Association.
NEC	U.S. National Electrical Code
NEMA FGI-1	U.S. National Electrical Manufacturer's Association.
ASTM	American Society for Testing and Material
BSI	British Standards Institution.
IEC	International Electrotechnical Commission
BIS	Bureau of Indian Standards

3. SITE CONDITIONS :

The cable trays shall be designed for installation on an offshore location. Site conditions are specified in General Design Criteria enclosed in the bid package.

4. SCOPE OF SUPPLY :

The scope of supply through this specification includes:

4.1. Fibre glass reinforced, ladder type cable tray, ventilated and solid bottom cable trays, tray covers.

4.2. Accessories such as bends, tees, crosses, reducers, splicers and necessary hardware etc.

5. TECHNICAL REQUIREMENTS :

Hereafter the term "cable tray" shall refer to ladder type cable trays, ventilated and solid bottom cable trays unless specifically mentioned.

5.1. Fibre Glass Cable Trays :

5.1.1. All FRP products shall be halogen free. Fibre glass cable trays shall be manufactured in accordance with NEMA FGI 1993.

5.1.2. All cable trays and accessories shall be manufactured of the fibre glass by Pultrusion process and shall be corrosion resistant and fire retardant (Flame spread index <=25, Smoke developed index <=450) in accordance with the latest ASTM E-84. An additive material shall be mixed with the fibre glass to make them resistant to ultraviolet light.

5.1.3. The minimum glass content in FRP material shall be 55%.

5.1.4. The ladder type trays shall have side rails and horizontal rungs. Both of these shall be of same material i.e. the Fibre glass with additive. The rungs shall be placed 300 mm apart from each other.

5.1.5. The inside depth of the cable trays shall not exceed 150 mm.

5.1.6. The tray construction shall have high degree of finish and shall be such as to facilitate easy handling as also to ensure easy laying of cables without causing damage to cables. The inside surface shall be perfectly smooth and free from sharp edges, burrs or projections, etc. The edges of the trays which have been cut should always be protected with the help of suitable coating system so as to protect the edges.

5.1.7. Cable ladder should follow NEMA FG1 standard with minimum of 12C class and the cable trays should follow IEC 61537 standard for loading. Span for ventilated/ solid bottom trays shall be 3m. Electrical performance (Resistivity) shall conform to ASTM D257 (less than 1.5x10^9 ohm). All cable trays shall have chemical additives to protect

against ultra violet (UV) ray (Tensile strength reduction and flexural strength reduction should be less than 7% when tested for 1000 hours as per ASTM G 154 and flexural strength reduction).

5.1.8. For perforated and solid bottom tray the following load list shall be considered in addition to self load

50 mm width – 10 kg/m
100 mm width – 20 kg/m
150 mm width – 30 kg/m
300 mm width – 60 kg/m
600 mm width – 90 kg/m

5.1.9. Deflection of cable trays and ladders shall be as per IEC 61537.

5.1.10. Cable Trays / Ladders and Fittings shall be supplied with pre-drilled resin finished holes to accept manufacturer standard coupler plates.

6. TESTING AND INSPECTION :

Testing and inspection of cables trays shall include but not necessarily be limited to the following:

6.1. A quantitative check shall be made to ensure that all items with accessories are present according to order and the packing list.

6.2. A visual inspection shall be made to ensure that good workmanship and practice were maintained throughout the fabrication.

6.3. Acceptance test for load deflection test shall be carried out for one sample of each type.

6.4. Following type test certificates shall be submitted for each type of tray system.

6.4.1. Flame spread test and smoke index test
6.4.2. Anti static property test (ASTM D257)
6.4.3. UV resistance property test (ASTM G154)

6.5. Company reserves the right to witness the final test and inspection. Prior notice of at least 20 days shall be given to enable the Company to depute its representative to witness the test.

7. SPARES :

Vendor shall supply all accessories for cable trays and cable ladders such as splice connector complete with nuts, bolts, cover clamps, hold on clamps and cover. Additionally, 20% spares of accessories such as splice connector complete with nuts, bolts, cover clamps, hold on clamps shall be provided.
These shall be properly wrapped and packed separately, clearly marked as “spare hardware” for cable trays and shipped along with trays. Packing list shall be furnished so that the parts can be handled without uncrating, if required.

8. CERTIFICATION :

All offered cable trays or those of similar designs manufactured by the same supplier:

- shall have been type tested by an authority approved by the Company.
- shall be having current certification/approval/listing by an approved agency or UL or FM

9. PACKING AND TRANSPORT :

The offered cable trays along with their accessories shall be shipped to site packed in wooden crates. They shall be wrapped with polythene sheets, before being placed in the crates to prevent damage to finish. Crates shall have skid bottom for handling.

FUNCTIONAL SPECIFICATION

FOR

Lighting Transformer

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
KL	AK	PK	9	21.01.2022	3
KL	NNJ	PK	9	31.03.2021	2

1. SCOPE

Scope of the specification describe the minimum requirements for the design, manufacture, inspection and testing, installation and commissioning of lighting power transformer (440/240 volt, 3 phase, 50 hz) along with accessories and protective enclosures suitable for indoor operation at offshore Platform in safe area.

2. CODES AND STANDARDS

2.1 Codes, Standard & Regulations: - The contractor shall ensure that the scope of work is executed in accordance with all mandatory Indian Statutory requirement and the transformer supplied shall comply with the latest revision of the applicable codes &standards.

IEC	International Electro-Technical Commission
BSI	British Standards Institution
NEMA	National Manufacturer's Association
NEC	National Electrical Code
IEEE	Institute of Electrical & Electronics Engineers
NFPA	National Fire Protection Association
API	American Petroleum Institute.
BIS	Bureau of Indian Standards

Any other international codes or standards may be used in lieu of above only with the approval of ONGC.

3. GENERAL REQUIREMENTS

3.1 Scope of Supply

Lighting Power transformer to be provided with all equipment's necessary to provide transformer that are operable, safe, reliable and efficient in accordance with this specifications to make a complete and operational package.

This shall include but is not limited to the following.

- Cast resin dry type Transformer with protective enclosures and all the accessories required.
- Terminal box.

Any other item(s) specifically not mentioned here but required for making the system complete, efficient and operable shall be under the scope of supply by contractor at no extra cost and time impact. All components should be appropriately rated, properly wired, tested & painted for marine condition.

General Requirement consideration for contractor: -

- The transformers shall be a product of a manufacturer regularly engaged in the manufacture of transformers preferably of repetitive manufacture.
- All components utilized in the construction of the transformers and the ancillary equipment shall be of the latest design, new and in current production.
- The contractor shall submit documentary evidence (copies of test certificates, type test report, etc.) from recognized testing authorities.
- Contractor shall not proceed with the deviation(s) without the company's written approval. Moreover any omission of items essential for the proper functioning of the transformers need to be brought to the notice of the company.
- All equipment shall be designed to minimize any risk due to internal short circuit and to ensure safety to personnel during all operating conditions, inspections, commissioning and maintenance.
- The equipment shall meet approved limits for electromagnetic compatibility for emission and immunity.

3.2 SAFETY:

It is the intent of the company that operational hazards be reduced to a minimum. Contractor shall not use toxic material. Due consideration shall be made towards fire hazards involved. Vendor shall use sound engineering judgment to complete an installation that will perform the required service function without compromising this aim.

3.3 MATERIALS WORKMANSHIP & SUITABILITY

All metal surfaces shall be thoroughly cleaned and degreased to remove mill scale, rust, grease and dirt. Fabricated structures shall be pickled and then rinsed to remove any trace of acid. The under surface shall be prepared by applying a coat of phosphate paint and a coat of yellow zinc chromate primer. Finally the metal surface shall be spray painted with two coat of epoxy paint

All unpainted steel parts shall be cadmium plated.

4. TECHNICAL REQUIREMENTS

Transformers required are of cast resin encapsulated-winding (dry type insulation) natural air-cooled, indoor type and suitable for operation at offshore platform in safe area. The room accommodating the transformer shall be located in non-hazardous area.

4.1 TRANSFORMER PERFORMANCE REQUIREMENT

Transformer shall be of continuous rated type at the maximum specified indoor ambient temperature.

Transformer shall be capable of withstanding short circuit.
Over load capacity need to be as per IEC 60076-12.

5. CONSTRUCTION REQUIREMENTS

The power transformer core shall be made of low loss rolled grain oriented steel lamination. The limb and yoke shall be preferably of same cross section having stepped design to give a circular shape.

The core assembly shall be treated with a resin base lacquer for corrosion protection.

5.1 COILS &INSULATION

The primary and secondary windings shall be made from high conductivity electrolytic copper, without brazed joints with extra-reinforced insulation applied to the points where mechanical & electrical stress may be more.

Conductor insulation system shall be of class H or better with temperature rise limited to class B as per IEC 60076-11. When subject to a continuous bolted short circuit on outgoing terminals, the transformer shall not explode or expel molted materials. Authorized test certificate to be submitted in this regard.

Coil assemblies shall be acoustically insulated from each other and from core base frame to reduce noise.

5.2 ENCLOSURE

Painted sheet steel enclosure-having degree of protection IP23 shall be provided. Housing shall be fabricated out of 2mm minimum thickness cold roll sheet steel with cover, bottom, and sides adequately reinforced with stiffeners to ensure rigidity & hoisting of complete transformer through suitable and rigid lifting provision. Steel fabrication & painting method shall be in accordance with relevant specifications prescribed under project.

5.3 TERMINALS BOXES

Winding shall be brought out on suitable terminals for cable terminations. Incoming and outgoing cable box shall be with bolted type cable gland plates. The cable box shall have sufficient space for ease in the cable termination.

An extra neutral bushing terminal shall be provided for grounding of transformer winding neutral on the secondary side.

5.4 COOLING

The transformer shall be of natural air- cooled type.

5.5 ACCESSORIES

All accessories shall be included in the contractor scope of supply.

5.6 NAMEPLATE

Nameplate of suitable thickness 316 stainless steel screwed type shall be fixed to the transformer frame also screwed to the enclosure suitably seen.

Following minimum information need to be indelibly marked in English.

- Type of transformer
- Number of standard applicable to the design
- Manufacture' name/trade mark
- Country of origin
- Year of manufacture
- Company' identification number

- Temperature class of insulation of each winding
- Voltage rating
- Power rating
- Number of phases
- Rated frequency
- Connection symbols
- Impedance voltage at rated current
- Type of cooling
- Total weight

5.6 NOISE LEVEL

The noise level of the transformer, measured at 1 meter from the transformer housing shall not be more than 65 dB.

6. TESTING AND INSPECTION

Vendor shall carryout various tests as per applicable codes/standards. Test certificate shall be submitted for company's approval.

ONGC or its authorized representatives shall witness all transformer testing. Shop inspection may be required at regular intervals.
Transformer testing shall include, but not necessarily be limited to the following

- Winding resistance measurement
- Voltage ratio test
- Polarity & phase relation test
- No load loss & no load current measurement
- Load loss measurement
- Impedance voltage measurement

- Separate-source voltage withstand test
- Induced over voltage withstand test
- Temperature rise test (for one only)
- Insulation resistance test
- Noise level test.

7. SPARES

Vendor shall recommend and provide spare parts needed for startup and commissioning. Vendor shall furnish recommended spares for two year of normal operation & maintenance.

8. TOOLS:

Two set of special tools needed for repair and maintenance shall be provided.

9. PACKING AND TRANSPORT:

The offered equipment along with their accessories shall be shipped to site packed in wooden crates they shall be wrapped with polythene sheets, before being placed in the crates to damage to finish. Crates shall have skid bottom for handling.

10. VENDOR DATA REQUIREMENTS

Following information shall be furnished:-

i. Type test certificate from a recognized independent testing authority for the specified short circuit withstand test on a similar offered rating of transformer.

ii. Manufacturing, Painting & testing procedure for the transformer offered.

iii. Drawing for Cable termination box and terminal.

iv. Brochure and catalogues.

v. Installation, operation and maintenance manual.

11. CERTIFICATION

All offered Transformer and accessories of similar design and construction feature manufactured by the same vendor.

- Shall have been type tested by an authorized testing agency/lab viz. CPRI, ERDA etc.
- Shall be having current certification/approval/listing by UL/FM/ATEX/IECEx etc.

FUNCTIONAL SPECIFICATION FOR SELF CONTAINED EMERGENCY LUMINAIRE

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES	DATE	REV.
BR	**KR**	**AS**	**4**	**06-09-2016**	**04**

1. SCOPE

This specification covers the minimum requirements for self-contained emergency light fixtures to be used on offshore platforms.

2. CODES & STANDARDS:

2.1. The self contained emergency luminary assembly shall comply with the requirements of the latest revisions of relevant standards of:

BIS : Bureau of Indian Standards.
NEC : U.S. National Electrical Code.
NEMA: National Electrical Manufacturer's Association
BSI : British Standards Institution.
ANSI : American National Standards Institute
IEC : International Electro-technical Commission.
ATEX :

2.2. Any other recognised international codes or standards may be followed in lieu of above with prior approval of the Company.

3. SITE CONDITIONS:

All the luminaries shall be designed and tropicalised to withstand the site conditions specified in the General Design Criteria enclosed in the Bid Package.

4. SCOPE OF SUPPLY:

The scope of supply through this specification includes:

4.1. Self-contained emergency luminaire assembly suitable for use in Class I Div-1 or Zone 1 (Temp. class T3).

4.2. All hardware required for mounting and fastening of the assembly.

5. TECHNICAL REQUIREMENTS:

5.1. General:

The emergency luminaire shall be fluorescent or LED type. The fixture shall provide normal area lighting on offshore platforms and also provide emergency lighting in case of Loss of AC power. In the event of a power

failure the luminaire will automatically switch over to emergency light operation.

5.2. Included Equipment:

5.2.1. The fixture shall be weatherproof (IP-56/NEMA-4) and shall contain two 18/20 watt fluorescent tubes or equivalent LED along with necessary batteries, inverter/LED driver and rectifier for emergency operation.
Necessary Indication shall be provided to indicate the status of battery charge.

5.2.2. The batteries provided shall be gas-tight nickel-cadmium type with a capacity to power the luminaire for at least 90 minutes continuously. The recharge period shall not exceed 24 hours.

6. UTILITIES AVAILABLE:

Electric power: 240 volts $\pm$ 5%, 1 phase 50 Hertz $\pm$ 3%.

7. SAFETY:

Each fixture shall be provided with as a minimum two cable entries, two glands & one loose blocking plug.

8. TESTING AND INSPECTION:

Testing & inspection of the luminaire assembly shall include but not necessarily be limited to the following:

8.1. A visual inspection shall be made to ensure that the finishing & workmanship are up to the mark.

8.2. Company reserves the right to witness the final testing & inspection. Advance intimation of at least 20 days shall be given to enable the Company to depute its representative to witness the test.

9. SPARES:

9.1. Vendor shall furnish a separate priced list of recommended spares for two(2) years of normal operation and maintenance. Recommended spares shall take into account the related factors of system reliability, effect of down- time

upon production, safety, and cost of parts and availability of equipment service facilities.

9.2. All spares furnished by the vendor shall be wrapped and packaged so that they could be preserved in original as new condition under the normal conditions of storage prevailing in India and shall be properly tagged and coded to facilitate future identification. They shall be packaged separately, clearly marked as "spare parts" and shipped along with the main equipment i.e. luminaire assembly. Packing lists shall be furnished so that the parts can be handled without uncrating if required.

10. WARRANTY:

Vendor shall have final and total responsibility for the design and performance of all equipments supplied under this specification. Warranty/Guarantee shall be as covered in Vol-I of Bid documents.

11. VENDOR DATA REQUIREMENTS:

11.1. Technical literature and drawings consisting of information regarding overall dimensions, weight, material of construction, mounting details etc. shall be provided. Details regarding batteries like make, type, capacity, year and month or year and week of manufacture etc shall also be provided.
11.2. Copies of type test certificates shall be provided.
11.3. Certificates showing the suitability to use in the hazardous area in which the luminaries are to be installed shall also be provided.

12. CERTIFICATION

All emergency luminaire assemblies offered or those of similar design and construction features, manufactured by the same vendor:

12.1. Shall have been type tested by an authority approved by the Company.
12.2. Shall be having current certification/approval/listing by an agency approved by the Company or UL or FM.

13. PACKING & TRANSPORT:

The emergency luminary assemblies along with their accessories shall be shipped to site packed in wooden crates. They shall be wrapped with polythene sheets, before being placed in the crates to prevent damage to finish. Crates shall have skid bottom for handling.

FUNCTIONAL SPECIFICATION

FOR

ELECTRICAL POWER AND CONTROL CABLES

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES INCLUDING THIS PAGE	DATE	REV.
LNVR	HSM	RK	16	31-01-05	2
SRS	AS	HMG	15	19 / 02 / 08	3
PK	KR	AS	11	10 / 08 / 15	4
KL	AK	PL	11	24/01/2022	5

1.0 SCOPE

This Specification defines the minimum requirements for design, manufacture, supply and testing of electrical power and control cables for high voltage and low voltage systems and cable accessories for installation on the offshore process / well platforms.

2.0 CODES AND STANDARDS

2.1 Codes Standards and Regulations:

All cables, accessories and other components supplied, shall comply with the latest revisions of the applicable codes and standards.

IEC	International Electro technical Commission.
NEMA	National Electrical Manufacturers Association.
NEC	U.S. National Electrical Code
IEEE	Institute of Electrical and Electronics Engineers.
API	American Petroleum Institute.
B I S	Bureau of Indian Standards

Further the Contractor shall ensure all mandatory Indian statutory requirements are complied with.

2.2 Any other recognized International code or standard may be used in lieu of above, subject to approval of company.

3.0 SITE CODITION

All the cables and accessories shall be designed to withstand the conditions specified in the General Design Criteria enclosed elsewhere in the bid package.

4.0 SCOPE OF SUPPLY

The Scope of this specification includes the following items:

4.1 Flame / Fire retardant, low smoke, halogen free, heat and oil resistant, high voltage cables.

4.2 Flame / Fire retardant, low smoke, halogen free, heat and oil resistant, low voltage Power and Control cables.

4.3 Flame / Fire retardant and Fire resistant, low smoke, halogen free, mica glass tape insulated, heat and oil resistant, low voltage cables.

4.4 High voltage cable terminations and accessories.

5.0 GENERAL REQUIREMENT

5.1 Safety

5.1.1 Non-metallic materials, such as conductor insulations shall be non-hygroscopic, and nontoxic.

5.1.2 Materials that are highly persistent or toxic to the environment or potential Carcinogen such as asbestos, poly chlorinated bi-phenyls (PCBs), Mercury and the like shall not be used.

5.2 Materials, Workmanship and suitability:

5.2.1 All materials utilized in the construction of the cables shall be to the latest specifications, new and in current production. Should any material prove unsatisfactory, it will be rejected not withstanding any previous satisfactory examination or tests of similar material or of completed cables.

5.3 Cables and cable accessory shall be suitable for operations without periodic maintenance applied throughout the life span of the cable.

5.4 Units and information.

All quantities and dimensions shall be expressed in metric units. All information, manual, certificates, data and inscriptions shall be in the English language.

6.0 TECHNICAL REQUIREMENTS

6.1 General

6.1.1 The cable shall be new and in good condition and shall be suitable for layering and bunching in ladders, trays and ducts and flexible enough for easy termination using standard terminals.

6.1.2 All power and control cables shall be flame retardant, low smoke, halogen free type and oil resistant outer sheath suitable for installation on offshore oil and gas processing platforms.

6.1.3 For sheathing materials of cables, Oxygen Index shall not be less than 30 at 27 ± 2^0 C and temperature index shall be 250^0 C at 21% Oxygen, tested as per ASTM D2863.

6.1.4 For Flame Retardant cables, Flame spread (retardant) characteristics of cables shall be as per recommendations mentioned in IEC-60332-3 (category A).

6.1.5 Halogen free cables shall be compliant to properties as defined in IEC 60092 – 350.

6.1.6 Low smoke properties shall conform to recommendation in IEC 61034 – 2.

6.1.7 Sheathing material shall be halogen free elastomeric/ thermosetting SHF2.

6.1.8 Cables for safety and emergency systems as defined in electrical design criteria shall be fire resistant type in addition to the specifications at 6.1.2 above. Fire resistance characteristics shall be as per IEC 60331 – 11 & 60331 – 21 (withstand capacity of at least for 750^0 C for 90 mins).

6.2 Voltage Designation

Voltage designation for cables shall be as following in line with IEC – 60183 & 60092 – 353. However cables need to be supplied as per following rating

System Voltage (kV)	Voltage grading for cables and accessories [U_0 / U (U_m)] (as per IEC – 60183) (kV)
0.440	0.6 / 1.0 (1.2)
3.3	**3.6 / 6 (7.2)**
6.6	**6 / 10 (12)**
11	8.7 / 15 (17.5)

7.0 CABLE CONSTRUCTION

In addition to the technical requirements stated in the previous section, the constructional requirements specific to all the power and control cables as described below shall be followed. Sections applicable for particular voltage ranges are mentioned against those. Other sections are common for all voltage ranges.

7.1 Conductor

Conductor shall be annealed, circular, stranded tinned / metal coated copper, smoothly drawn, free from defect, complying with the requirement of IEC-60228.

The conductor class (min. class 2) shall be selected to obtain the desired bending radius as per clause no 7.9.

7.2 Conductor Insulation

Conductor insulation shall be extruded insulation made out of Cross linked compounds (EPR / HEPR / XLPE only) complying with the requirements of IEC–60502.

7.3 Internal Construction (for High Voltage Cables - 3.6 / 6 (7.2) kV and above)

7.3.1 Cables rated 3.6 / 6 (7.2) kV and above shall be provided with non-metallic extruded semi conducting compound screening, to be applied on top of a semi-conducting tape for conductor and extruded nonmetallic semiconducting layer in combination with a metal layer as insulation screening over each core as per IEC 60092-354. The dimensional, physical and electrical requirements of the metallic layer (metallic screen) shall be determined as per IEC 60092-354.

7.3.2 The fillers, binders and extruded inner covering shall be of non-hygroscopic material, shall be suitable for the operating temperature of the cable and compatible with the insulating material and shall be provided as per IEC 60092 – 354; the thickness of inner covering shall be according to IEC 60092 – 354 (for extruded). The inner covering shall be extruded cross-linked elastomeric halogen free compound (SHF 2).

7.4 Internal Construction (for Low Voltage Cables - 0.6 / 1.0 (1.2), 1.8 / 3 (3.6) kV).

For above mentioned voltages, cores of a multicore cable shall be laid up and the interstices filled up with non-hygroscopic fillers & extruded non-hygroscopic inner covering shall be provided as per IEC 60092 – 353. The material selected shall be compatible with cable components with which it is in contact and compatible with the operating temperature of the cable. A suitable binder is to be to be applied before application of the extruded inner covering. The thickness of extruded inner covering shall be according to IEC 60092–353. The inner covering shall be extruded cross-linked elastomeric halogen free compound (SHF 2).

7.5 Armouring

Armouring shall be selected to as to achieve the bending radius requirement of clause7.9.Multicore cables shall be mechanically protected with hot dipped galvanized steel wire braiding (GSWB) or galvanized round steel wire armouring (GSWA).Single core low voltage and high voltage cables shall be mechanically protected with tinned phosphor bronze wire braiding (PBWB) or hard drawn aluminium round wire armouring which shall be applied over the inner covering to confirm to the requirements of international standards. In cases where the steel armour layer is required to comply with a minimum conductance, it is permissible to include sufficient copper or tinned copper wires in the armour layer to ensure compliance. The armour of single-core cables for use on a.c. circuits shall consist of non-magnetic material. In special cases, for instance when the cables are used on d.c. circuits, magnetic materials can also be used.

7.6 Outer Sheath

The outer / over sheath shall be heat and oil resistant, suitable for installation in an offshore oil and gas processing platform. Material for outer sheath shall be cross-linked elastomeric halogen free compound (SHF 2) extruded in one or more closely adherent layers as per IEC 60092-350.

Minimum thickness of outer sheath is to be as per IEC 60092-353/ 354 as per voltage rating of cable. The thickness at any point shall satisfy the prescriptions given in 4.9.3 of IEC 60092- 350.

Outer sheath marking shall be printed type as per sec 4.2, IEC60092-354 & section 4.1.3, IEC 60092-350.

7.7 Fire Resistant Cables

7.7.1 In addition to the technical requirements stated in the previous sections, the following requirements are specific to Fire Resistant cables.

7.7.2 Fire resistance cables shall be in compliance with IEC 60331-11 & 60331 - 21 and shall have a Heat barrier of glass mica tape lapped in double layer shall be provided over the conductor, before the insulation.

7.8 Colour Coding

7.8.1 Cable outer sheaths shall be permanently colour coded such that colours do not deteriorate with age or fade with exposure to direct sunlight.

7.8.2 Cable shall be coded as per the colours indicated below. Any alternate colour coding shall need prior approval by the company.

7.8.3 Cable Colour Coding

1	High Voltage Power (greater than 1 KV)	:	Red / Black (as per IEC 60992 – 353 / 354 respectively)
2	Low Voltage Power & Control (equal to or less than 1 KV)	:	Black
3	Earth	:	Green/Yellow

7.8.4 Cable cores shall be identified either by colors or by numbers. Color coding shall be as following:

1	Number of cores (1 core)	:	Black
2	Number of cores (2cores + Earth)	:	Black & Red + Green/Yellow
3	Number of cores (3 cores + Earth)	:	Red, Yellow & Blue + Green/Yellow
4	Number of cores (4 cores + Earth)	:	Red, Yellow, Blue & Black + Green/Yellow
5	Control cables: Multi-core + Earth	:	White with Black numbers + Green/Yellow

Individual cores of multi-core power, control cables, if identified by numbers, shall be done as per sec. 4.2.3.3 and Annex. B, IEC 60092 – 353.

7.9 Bending Radius

Cables shall be flexible and suitable for installation with the minimum bending radius, appropriate for the type of cable supplied and the bending radius shall be finalized based on installation contractor's requirement.

8. Cable Drums

8.1 Cables shall be dispatched on suitable barrel diameter, securely battened, with takeoff ends of the cable brought to the outside of the drum (either wooden or metallic) and fully protected against mechanical damage.

8.2 Besides customary markings as per any prevalent standard / code, each cable drum shall be fitted with a stainless steel or equivalent corrosion resistant label. The label shall be fixed to the outside of each flange, engraved with the following information:

i) Cable Manufacturer's name
ii) Cable type number
iii) No. of cores and cable cross sectional area in mm and core construction/stranding e.g. 3C+E 4mm2, (7/0.85)
iv) Voltage grade
v) Standard(s)
vi) Purchase order number/item number
vii) Cable length (in metres)
viii) Net and gross weights (kg)

ix) Date of manufacture.

9. Cable Accessories

9.1 Cable termination kits for use on high voltage systems shall be suitable for the type of cables offered as per this specification. The terminations shall be supplied in kit form. The terminations kit shall include all insulating and sealing materials apart from conductor fittings and consumable items. The Contractor shall ensure that installation instructions are included in each kit.

9.2 Cable termination kits shall be provided with semi conductive de-stressing appropriate for the working voltage.

9.3 The cable terminations kits shall be suitable for terminations of the cables to an indoor-switchgear, weatherproof enclosure, outdoor transformers or hazardous area motor terminal box, as the case may be. The terminating kits shall be heat/shrinkable sleeve type.

9.4 The Contractor shall provide type test certificates for cable termination kits as per IS 13573 from the manufacturers of these kits for review by company.

10. INSPECTION AND TESTING

10.1 General

10.1.1 The cable shall be tested and inspected at the manufacturer's works, or at a recognized testing facility. The Contractor shall ensure all the test equipment necessary to test the cables is provided by the manufacturer. Test certificates for all tests shall be provided to the Company, immediately on completion of tests performed. All tests shall be witnessed by the Company or its authorized representative.

10.1.2 Company shall be advised at least (4) weeks in advance of the scheduled date of each test and again 20 days prior to the actual test.

10.1.3 The Contractor shall ensure detailed records and log sheets regarding the results of the tests and certification are furnished for review.

10.1.4 The Contractor shall furnish all necessary information concerning the supply to the Company's inspector or representative. The inspector shall have free access to the manufacturer's works for the purpose of inspecting the process of manufacture in all the stages and will have the authority to reject any material which appears to be unsuitable or of unsatisfactory quality.

10.2 Testing

10.2.1 All LV Electrical cables shall be tested in compliance with the IEC standard 60092-353 and HV cables shall be tested in compliance with the IEC standard 60092-354. In case cable is tested as per other international standards (IEEE, BIS, NEMA or NEC), the tests shall be either equivalent to IEC or more stringent. After installation, tests will be carried out for cables with lower voltage rating in accordance with IEC 60502-1 and for cables with higher voltage rating, in accordance with IEC 60502-2.

10.2.2 For type tests, valid type test certificates certified by any reputed international agency are acceptable in lieu of tests. Type test certificates must be submitted to company representative / inspector during inspection.

10.3 Sealing

Immediately after electrical tests, the ends of the cable shall be sealed with heat shrink caps to prevent the ingress of moisture.

11.0 CERTIFICATION

All cables and cable accessories of similar design and construction features manufactured by the same Vendor:

a) Shall have been type tested by an authorized testing agency/lab viz. CPRI, ERDA etc.

b) Shall be having current certification/approval/listing by UL/FM/ATEX/IECEx etc.

FUNCTIONAL SPECIFICATION FOR EXPLOSION PROOF CONTROL STATION

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
			07		1
AKS	NNJ	PK	07	16.02.2021	2

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1.0 SCOPE:

The intent of this specification is to define the minimum requirements for explosion proof control stations / push button stations to be located near the motors on the offshore platforms.

2.0 CODES AND STANDARDS:

2.1 The following codes and standards shall form a part of this specification:

NEC	-	National Electrical Code.
NEMA	-	National Electrical Manufacturer's Association.
BSI	-	British Standards Institution.
API-RP-14F	-	Recommended practice for design and installation of Electrical system for offshore production platform.
API-RP-500	-	Classification of areas for electrical installations at drilling rigs and production facilities.
BIS	-	Bureau of Indian Standard

2.2 Any other recognized international code or standard may be used in lieu of above with prior approval of the Company.

3.0 SITE CONDITIONS:

All materials shall be selected and finished to resist/withstand the site conditions Specified in General Design Criteria enclosed in the Bid Package.

4.0 SCOPE OF SUPPLY:

The scope of supply includes.

4.1 Explosion proof control stations / push button stations of various types.

4.2 Cable lugs & explosion proof double compression type nickel chromium plated brass cable glands.

5.0 TECHNICAL REQUIREMENTS:

5.1 The control stations / push button stations shall be explosion proof and weather-proof suitable for operation in class-1, Division-2, Group-D or Zone 1, Group II A/II B, and temperature Class T-3 (200ºC). Their enclosures shall be explosion-proof to NEMA-7 and weather-proof to NEMA- 4 (IP-56) or equivalent.

5.2 They shall be made up of cast metal aluminum alloy with offshore painting as per ISO 12944-6 / high impact strength glass polyester material housing complete with requisite terminals and explosion-proof cable termination accessories suitable for cable as per FS 4011. If aluminum is used, the same shall be copper-free. The glass polyester material, if used, shall not get damaged due to ultra-violet rays over its life span.

5.3 Contact for push buttons shall be rated for 6 Amps. At 240VAC. AC-11 duty and 4.0 Amps at 11OV D.C., DC-13 duty.

5.4 Wherever start-stop push buttons are used, the start push button shall be Momentary close type while the 'Stop' push button shall be stay put type and lockable in pressed condition. The colours of push buttons shall be as follows:

Start: Green (Shrouded)

Stop: Red (Mushroom head type)

5.5 Auto - Manual / local - Remote Selector switches shall be knob type having Stay-put type contacts of rating same as that of push buttons. Auto - Manual / Local - Remote switch and start - stop pushbuttons, wherever required, shall be housed in a common enclosure.

5.6 The control stations shall have four lugs for mounting on the pre-fabricated available supports. The lugs shall be off-set so that the rear surface of the Unit stays at least 5 mm away from the surfaces. On lugs, two holes shall be round and two oblong to facilitate correct positioning of the control station.

5.7 Desired number of terminals plus 10% spare shall be provided for connection of external cable leads of copper conductor. Necessary cable termination

accessories shall be supplied with each control station. The cable entry shall be from the bottom. These entries shall be properly plugged to protect the equipment during transit.

5.8 1 No. external Earth terminal shall be provided on the body.

5.9 A nameplate indicating the equipment to be operated shall be provided for each control station. The particulars of the nameplate will be furnished later. Nameplates shall be of stainless-steel having letters of a minimum height of 5 mm.

5.10 Warning inscription "ISOLATE ELSEWHERE BEFORE OPENING" shall be prominently and indelibly displayed on the top cover of control stations.

5.11 Ammeters shall invariably be incorporated in the control station for motors irrespective of their KW ratings and sizes. Ammeter scale shall be suitable to read both rated current and starting current.

5.12 A red mark shall be provided at the full load current of the motor, the values of which will be furnished along with the Purchase Specification to the successful vendor.

5.13 The control stations shall be tested and certified as defined in Electrical Design Criteria enclosed in the Bid Package. Each equipment shall have certification mark on its nameplate.

5.14 All control stations shall be provided with canopy of suitable material and proper size for protections against rain, fog etc.

5.15 All hardware such as nuts, bolts, washers etc. shall be Cadmium / stainless steel plated.

6 TESTING AND INSPECTION:

6.1 Testing and inspection of Control Station / Push Button station shall include, but not necessarily be limited to the following:

6.1.1 A quantitative check shall be made to ensure that all items in specified quantity are present according to the order and the packing list.

6.1.2 A visual inspection shall be made to ensure that good workmanship and practices were adopted during fabrication and assembly.

6.2 DELETED

6.3 All type test certificates including explosion-proof certificate shall be furnished along with the offer.

7 SPARES:

7.1 Vendor shall recommend and provide spare parts needed for start-up. In addition, vendor shall also furnish a separate price list of recommended spares for two (2) years of normal operation and maintenance.

7.2 Recommended spares shall take into account related factors for system reliability, effect of downtime upon production, safety, cost of parts and availability of equipment service facilities.

7.3 All spares furnished by vendor shall be wrapped and packaged so that they could be preserve in original as new condition under the normal conditions of storage prevailing in India and shall be properly tagged and coded to facilitate future identification. They shall be packaged separately, clearly marked as "Spare Parts", and shipped along with the control / push button station. Packing lists shall be furnished so that the parts can be handled without uncrating, if required.

8 WARRANTY:

Vendor shall have final and total responsibility for the design and performance of All the Control Stations / Push Button Stations supplied under this specification. Vendor shall warrant them for their performance in accordance with this Specification.

9 VENDOR DATA REQUIREMENT:

Vendor shall furnish data for the supplied item-

1. Interconnection Drawing.
2. Test Certificates.
3. Support Drawing.
4. Other Relevant Data.

10 DELETED

11 CERTIFICATION:

All Control Stations / Push Button Stations offered or those of similar design and Construction features manufactured by the same vendor:

- Shall have been type tested by accredited laboratory / Institution.
- Shall have been in continuous satisfactory service in offshore for a minimum period two years,
- Shall be having current certification / approval / listing UL / FM /ATEX / IECEX / CPRI OR ERDA.

12 PACKING AND TRANSPORT:

The control station / push button stations along with their accessories shall be shipped to Site packed in wooden crates. They shall be wrapped with polythene Sheets, before being placed in the crates to prevent damage to finish. Crates shall have skid bottom for handling.

13 DELETED

FUNCTIONAL SPECIFICATION FOR **CABLE TRANSIT SYSTEM**

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES	DATE	REV.
BR	KR	AS	3	19-09-2016	0

1. SCOPE

This specification covers the minimum requirements for cable transit system to be used on offshore platforms.

2. CODES & STANDARDS:

2.1. The cable transit system shall comply to the requirements of the latest revisions of relevant standards/ directives of:

DNV : Det Norske Veritus.
API : American Petroleum Institute
IEC : International Electro-technical Commission.
ISO : International Standard Organization

2.2. Any other recognised international codes or standards may be followed in lieu of above with prior approval of the Company.

3. SITE CONDITIONS:

All cable transit systems shall be designed to withstand the site conditions specified elsewhere in the Bid Package.

4. SCOPE OF SUPPLY:

The scope of supply through this specification includes:

(a) Insert blocks or modules.

(b) Frames.

(c) Any required accessories such as gaskets, lubricants, stay plates etc required for proper installation of cable transit system.

5. TECHNICAL REQUIREMENTS:

5.1. General:

The cable transit system shall be tested and certified for fire integrity and gas tightness.

5.2. Material of construction**:** Cable transit frames shall be made of stainless steel.

5.3. Fire Integrity:

The cable transit frames shall have minimum class of A60 (SOLAS) or EI 60(BS EN 13501-2). Transit system modules or insert blocks shall have certified for fire ignitability as per ISO 11925-2 or equivalent.

5.4. Gas tightness: The cable transit system shall be rated for minimum gas tightness as stipulated in "DNV: Type Approval Programme for Cable and Pipe Penetration" or equivalent.

6. TESTING AND INSPECTION:

Testing & inspection of the cable transit system shall include but not necessarily be limited to the following:

6.1. Visual inspection.
6.2. Verification of Type Test reports

6.3. Company reserves the right to witness the final testing & inspection. Advance intimation of at least 20 days shall be given to enable the Company to depute its representative to witness the test.

7. WARRANTY:

Vendor shall have final and total responsibility for the design and performance of all the cable transit system supplied under this specification and provide necessary warranty for them in accordance with Volume 1 of bid documents.

8. VENDOR DATA REQUIREMENTS:

8.1. Technical literature and drawings consisting of information regarding overall dimensions, weight, material of construction, installation details etc. shall be provided. Details like make, type, capacity, year and month or year and week of manufacture etc shall also be provided.
8.2. Copies of type test certificates shall be provided.

9. CERTIFICATION

All cable transit system offered shall be having current certification/approval/listing from UL or FM or-by an agency approved by the Company.

FUNCTIONAL SPECIFICATION FOR LIGHT FITTINGS & JUNCTION BOXES FOR UNCLASSIFIED AREA

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES INCLUDING THIS	DATE	REV.
Ashish Kumar	**Pushkar Singh**	**Pradeep Kumar**	**05**	**13-02-23**	**5**
Ashish Kumar	**Pushkar Singh**	**Pradeep Kumar**	**05**	**20-12-22**	**4**
B.Rajkumar	**K.Rajamouly**	**A.Sezhian**	**05**	**15-03-17**	**3**

1. SCOPE

The scope of this specification is to define the requirements of LED light fixtures and junction boxes suitable for installation in unclassified areas on offshore platforms.

2. CODES & STANDARDS

2.1. The items shall be in accordance with the latest applicable Codes and Standards of :

NEC - National Electrical Code

NEMA - National Electrical Manufacturer's Association

BSI - British Standards Institution.

ANSI - American National Standards Institute

BIS - Bureau of Indian Standards

IEC - International Electrotechnical Commission

ATEX - Atmosphere Explosible

IESNA - Illuminating Engineering Society of North America

3. SITE CONDITIONS

All light fittings, junction boxes and their accessories shall be designed to withstand the site conditions specified in the General Design Criteria enclosed in the Bid Package.

4. SCOPE OF SUPPLY

The scope of supply through this specification includes:

- LED Light Fixtures complete with driver circuit.
- Junction Boxes
- Cable glands, plugs etc.
- Clamps, Brackets etc. for fixture supports.

5. TECHNICAL REQUIREMENTS

5.1. General :

5.1.1. All indoor LED light fittings and junction boxes (located in enclosed rooms) shall be industrial/decorative type with enclosure protection to IP-22, as minimum. They shall be provided with shrouding type terminals/live parts to make them shockproof, while cleaning, replacing etc.

5.1.2. The body of junction boxes shall be Stainless steel (SS) and shall be free from frictional sparking hazard. Copper or copper alloy shall not be used for manufacturing of any exposed parts.

5.1.3. The LED light fittings and junction boxes shall be suitable for mounting on poles/ceiling/columns. Suitable Clamps/brackets required for installation shall also be supplied.

5.1.4. All the lighting fixtures/junction boxes shall be provided with internal earthing terminal and external terminals as per IS/IEC 60079-0.

5.1.5. Double compression type nickel/chromium plated brass cable gland and tinned copper crimping type lugs shall be provided for all LED light fittings and junction boxes.

5.1.6. All LED light fittings and junction boxes nameplates shall bear the stamp of certifying agency.

5.1.7. The LED fixtures shall be high efficiency, durable, rugged, with minimum lumen depreciation and designed for ambient temperature of 45 deg C. The fixtures shall have been type tested in a NVLAP (National Voluntary Laboratory Accreditation Program, USA) accredited laboratory.

(1) As per LM-79 (IESNA) for
- Total Luminous Flux
- Luminous Intensity Distribution
- Electrical Power Characteristics
- Luminous Efficacy (calculated)
- Color Characteristics

(2) As per LM-80 (IESNA) for lumen maintenance and LED lifetime projections shall be made using TM-21 method.

In Situ Temperature Measurement Test (ISTMT) temperature shall be used while generating LM-80 report and TM-21 projection. The LEDs shall have

70% lumen maintenance at minimum 50,000 hours of operation. The LED fixture driver shall comply with IEC/FCC (Federal Communications Commission, USA) rules and regulations regarding electromagnetic radiation. Minimum power factor of LED fixtures shall be 0.9. Light colour shall be white and CRI shall be minimum 75. Fixture enclosure material shall be GRP or Copper free cast aluminum or Stainless steel. Protective cover shall be made of polycarbonate or toughened glass.
Offered LED fixtures should have been supplied to offshore installations.

5.2. Light Fittings

5.2.1. All fittings shall be suitable for operation on 240+5% V, Single phase, 50+3% Hz power supply.

5.2.2. The glass used for outdoor fittings shall be clear and toughened type suitable for conditions involving exceptional risk of mechanical damages. The light fittings shall have glass sealed into retaining ring, which in turn shall be secured to the body.

5.3. Junction Boxes

5.3.1. The junction boxes shall be provided with cable termination accessories for connecting, armoured, copper conductor, FRLS Cables. The Minimum number of side entries shall be three. Spare plugs shall be provided to block the unused entries.

5.3.2. Adequate no. of the terminals shall be provided to terminate three cables. Sufficient space for accommodating specified No. of cables and easy access to their terminals for termination purposes, shall be provided.

5.3.3. Four plugs shall be provided for mounting of junction box on wall/steel structure with the help of bolts.

6. TESTING AND INSPECTION

6.1. Testing and inspection of light fittings and junction boxes shall include but not necessarily be limited to the following:

- A visual inspection shall be made to ensure that the finishing and workmanship are up to the mark.

- Company reserves the right to witness the final testing and inspection. Prior intimation shall be given to enable the Company to depute its representative to witness the test.

7. CERTIFICATION

(1) Photometric type test certificates by UL or NVLAP (National Voluntary Laboratory Accreditation Program, USA) accredited laboratory for the following shall be provided:

(a) LM-79 (IESNA) for
- Total Luminous Flux
- Luminous Intensity Distribution
- Electrical Power Characteristics
- Luminous Efficacy (calculated)
- Color Characteristics

(b) LM-80 (IESNA).

(2) IP/NEMA rating certificates issued by UL or FM or IECEx or ATEX.

(3) All the light fixtures and junction boxes offered or those with similar design & construction features, manufactured by same vendor:
- Shall have been type tested by an accredited authorized agency/lab.
- Shall be having current certification/approval/listing UL/FM/BASEEFA/ CIMFR/ ATEX/IECEx.

8. PACKING AND TRANSPORT

The offered equipment along-with their accessories shall be packed suitable for shipping to yard site, marine transport to offshore site. The packing shall ensure that there is no damage to the surface finish & product. The packaging shall have skid bottom for handling.

FUNCTIONAL SPECIFICATION

FOR

ELECTRICAL CABLE GLANDS

PREPARED / REVISED BY	REVIEWED BY	APPROVED BY	TOTAL NO. OF PAGES INCLUDING THIS PAGE	DATE	REV.
KL	AK	PK	4	24.01.2022	1
BR	KR	AS	4	19-09-2016	0

1.0 SCOPE

This Specification defines the minimum requirements for design, manufacture, supply and testing of electrical cable glands and accessories for high voltage and low voltage systems for armoured and non armoured cables on the offshore process / well platforms.

2.0 CODES AND STANDARDS

2.1 Codes Standards and Regulations:

All cables, accessories and other components supplied, shall comply with the latest revisions of the applicable codes and standards.

IEC	IEC 60079.
British Standard	BS EN 50262
B I S	Bureau of Indian Standards

Further the Contractor shall ensure all mandatory Indian statutory requirements are complied with.

3.0 SITE CODITION

All the cables and accessories shall be designed to withstand the conditions specified in the General Design Criteria enclosed elsewhere in the bid package.

4.0 SCOPE OF SUPPLY

The Scope of this specification includes the following items:

4.1 The complete cable gland system including body, washer, nut, seal etc.

5.0 TECHNICAL REQUIREMENT

GENERAL:

5.1 All cable glands shall be made of brass and finishing shall be nickel plated.

5.2 Cable Glands shall be double compression type except where soft bedded cables i.e. typically fire resistant are used. For such cables the sealing method should be chosen to avoid cold flow. Examples of such types are displacement and diaphragm types.

5.3 To meet the ingress protection requirement the space between cable glands, adapters and blanking elements and the enclosure shall be sealed by means of a sealing washer or thread sealant.

5.4 The cable gland shall be selected to match the cable diameter. The use of sealing tape, heat shrink tube or other materials is not permitted to make the cable fit to the cable gland.

FOR CLASSIFIED AREA:

5.5 Cable glands, when installed shall not invalidate the specific characteristics of the type of protection of the electrical equipment on which they are mounted.

5.6 Cable glands shall be in accordance with IEC 60079-0 and shall be selected to maintain the requirements of the protection technique according to Table 10 of IEC 60079-14.

5.7 Where glands are used for unarmored cables in a hazardous area, care should be taken to ensure that they are able to fulfil the requirements of IEC60079-0. Where the product has limitations denoted by an X on the certificate, additional measures should be taken to secure the cable.

5.8 The cable entry system shall comply with one of the following for Ex ‘d’ equipment:

a) Cable glands sealed with setting compound (barrier cable glands) in compliance with IEC 60079-1 and certified as equipment;

b) Cables glands with IEC 60079-1 and are certified as equipment

5.9 Cables glands and accessories shall be suitable for operations without periodic maintenance applied throughout the life span of the equipment.

5.10 **Units and information.**

All quantities and dimensions shall be expressed in metric units. All information, manual, certificates, data and inscriptions shall be in the English language.

6. INSPECTION AND TESTING

(a) Test certificates for all cable glands shall be provided to the Company.

(b) Random inspection of cable glands installed at yard or site shall be carried by company or authorized representative.

7. CERTIFICATION

All cables glands and gland accessories of similar design and construction features manufactured by the same Vendor:

a) Shall have been type tested by an authorized testing agency/lab viz. CPRI, ERDA etc.

b) Shall be having current certification/approval/listing by UL/FM/ATEX/IECEx etc.

आफशोर के लिए इंस्ट्रूमेंट्स एवं कंट्रोल्स के डिजाइन हेतु मानक विनिर्देश

STANDARD SPECIFICATION FOR DESIGN OF INSTRUMENTS AND CONTROLS FOR OFFSHORE

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
2	06.04.21	Revised and Reissued	SD	PKS	MN	SM
1	15.06.15	Revised and Reissued	SD	PKS	RG	SC
0	20.09.10	Issued as Standard Specification	SD	RG	TGM/JMS	ND

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Abbreviations:

AC	:	Alternating Current
AMS	:	Asset Management System
AWG	:	American Wire Gauge
CPU	:	Central Processing Unit
CRCA	:	Cold Rolled Cold Annealed
DC	:	Direct Current
DCS	:	Distributed Control System
DPDT	:	Double Pole Double Throw
DPST	:	Double Pole Single Throw
ESD	:	Emergency Shutdown
F&G	:	Fire & Gas
FRP	:	Fibre Glass Reinforced Plastic
FSD	:	Fire Shutdown
HART	:	Highway Addressable Remote Transducer
HMI	:	Human Machine Interface
HMS	:	HART Maintenance System
I/O	:	Input /Output
IP	:	Ingress Protection
LCD	:	Liquid Crystal Display
LED	:	Light Emitting Diode
MAWP	:	Maximum Allowable Working Pressure
MCB	:	Miniature Circuit Breaker
MCC	:	Motor Control Center
NB	:	Nominal Bore
NC	:	Normally Close
NO	:	Normally Open
NPT	:	National Pipe Threads
P&ID	:	Piping and Instrumentation Diagram,
PID	:	Proportional, Integral and Derivative
PB	:	Push Button
PC	:	Personal Computer
PLC	:	Programmable Logic Controller
PTC	:	Performance Test Code
PTFE	:	Poly Tetra Fluoro Ethylene
PVC	:	Poly Vinyl Chloride
RAM	:	Random Access Memory
RB	:	Rockwell Hardness in B Scale
RF	:	Radio Frequency
RP	:	Recommended Practices
RTD	:	Resistance Temperature Detector
RTU	:	Remote Terminal Unit
SOV	:	Solenoid Valve
SPDT	:	Single Pole Double Throw
SS	:	Stainless Steel

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TFT	:	Thin Film Transistor
TIC	:	Telemetry Interface Cabinet
TMR	:	Triple Modular Redundant
UCP	:	Unit Control Panel
UPS	:	Uninterrupted Power Supply
URV	:	Upper Range Value
VDU	:	Visual Display Unit
WC	:	Water Column

Instrumentation Standards Committee

Convener : Mr. Mainak Nandi

Members : Mr. Arupjyoti Saikia
Mr. Sachin Garg
Mr. Jerry Johnson
Mr. Mayank Kaushik
Mr. Sourav Mukherjee
Ms. K.Pratheepa
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1.0 GENERAL

1.1 Scope

1.1.1 This specification covers the design and engineering of instruments, control systems, accessories and materials, along with special test equipments, where the supply is part of an offshore project. This specification is general and all instruments/systems included may not be required for the proposed project. The scope shall be as defined in Scope of Supply of Instrumentation.

1.1.2 The detailed Scope of work, Specific Job requirements, exclusions, deviations, additions etc. if any, will be indicated in Scope of Supply of Instrumentation and Job Specification of Instrumentation included in Requisition/Tender.

1.1.3 This specification provides only the qualitative specifications of commonly used instruments. Specifications of Instruments not covered in this specification shall be submitted by bidder for review by Purchaser.

1.1.4 Bidder shall be fully responsible for design, material selection, sizing and selection of the proper instruments for their system. The compliance to this specification does not absolve the bidder of the responsibility towards contractual obligations with regards to completeness, proper selection, satisfactory operation and easy maintenance of the platform/package.

1.1.5 All instruments and control systems, Distributed Control System (DCS), Programmable Logic Controller (PLC) and other system oriented items like Multiphase Flow Meter , Water Cut Meter, machine monitoring system, speed governor control, antisurge / performance control system, analyser system etc. offered shall be field proven and should have completed trouble free satisfactory operation for a period of minimum 6 months on the bid due date in a similar application in offshore installation with the process conditions similar to those specified in Purchaser's datasheets. Items with proto-type design or items not meeting provenness criteria specified above shall not be offered.

1.1.6 In the event of any conflict between this specifications, data sheets, related standards, codes etc., bidder shall refer the matter to the purchaser for clarification and only after obtaining the same should proceed with the manufacturing/engineering of the item in question. DCS/PLC specifications is not included in this document. Bidder shall refer Standard specifications for DCS and PLC attached separately in case DCS and PLC are included in Scope of supply.

1.2 Applicable National / International Standards

1.2.1 Design, terminology and selection shall comply, as a minimum, with the latest edition of following codes, standards and publications.

AGA American Gas Association, Gas Measurement Committee

Report No.3	Orifice Metering of Natural Gas and Other Related Hydrocarbon Fluids – Concentric, Square-Edged Orifice Meters
Report No.7	Measurement of Natural Gas by Turbine Meter
Report No.8	Compressibility factors of Natural Gas and other related Hydrocarbons gases
Report No.9	Measurement of Gas by Multi path Ultrasonic Meters
Report No.11	Measurement of Natural Gas using Coriolis Mass Flowmeters

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AG181 Foundation Fieldbus System Engineering Guidelines

ASME American Society of Mechanical Engineers.

B 1.20.1	Pipe Threads, General Purpose
B 16.34	Valves – Flanged, Threaded and Welding End.
B 16.47-B	Large Diameter Steel Flanges : NPS 26 through NPS 60 Metric / Inch standard
B16.5	Pipe Flanges and Flanged Fittings : NPS ½ through NPS 24 Metric / Inch Standard
B 16.20	Ring joint and Grooves for Steel Pipe Flanges
B40.3	Bimetallic Actuated Thermometers

ANSI/ FCI American National Standards Institute/ Fluid Controls Institute

70.2 Control valve seat leakage classification.

API American Petroleum Institute

6A	Specification for Wellhead and Tree Equipment
6D	Specification for pipeline valves (Gate valve, Plug, Ball and check Valves)
6FA	Specification for Fire Test for Valves
607	Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats
RP 14C	Recommended Practice for Analysis, Design, Installation and Testing of Basic Surface Safety Systems for Offshore Production Platforms
RP 14F	Design, Installation and maintenance of Electrical Systems for Fixed and Floating Offshore Petroleum facilities for unclassified and Class 1, Division 1 and Division 2 locations
RP 14FZ	Recommended Practice for Design, Installation and maintenance of Electrical Systems for Fixed & Floating Offshore Petroleum Facilities for unclassified and Class 1 Zone 0, Zone 1 & Zone 2 locations.
RP 14G	Recommended Practice for Fire Prevention and Control on Open Type Offshore Production Platforms
RP 14J	Recommended Practice for Design and Hazard analysis for offshore production facilities.
RP 520	Sizing, selection and installation of Pressure Relieving system in refineries. Part-I - Sizing and selection Part-II - Installation
RP 521	Guide for Pressure Relieving and Depressurising systems.
RP 526	Flanged Steel Safety Relief Valves.
RP 527	Seat Tightness of Pressure Relief Valves
MPMS	Manual of Petroleum Measurement Standards Chapter 1 Vocabulary Chapter 5 Metering 11.2.1 Compressibility factors for HC
RP 551	Process Measurement Instrumentation. Part 1 - Process Control and Instrumentation
RP 552	Transmission Systems
RP 554	Process Instruments and Control
RP 555	Process Analyzers
598	Valve Inspection and Testing
S 670	Machinery Protection System

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S 2000	Venting Atmospheric and low pressure storage tank
1101	Measurement of Petroleum liquid hydrocarbons by Positive Displacement meter

ASME American Society of Mechanical Engineers

-	ASME Boiler and Pressure Vessel Code (BPVC), Section VIII, Division 1: Rules for Construction of Pressure Vessels.
PTC 19.3 TW: 2010 & ERATA 2012-THERMOWELLS Performance Test	
MFC-3M	Measurement of Fluid Flow in Pipes using Orifice, Nozzle and Venturi
MFC 5M	Measurement of Liquid Flow in Closed Conduits Using Transit-time Ultrasonic Flowmeters
MFC 6M	Measurement of Fluid Flow in Pipes using Vortex Flowmeters
MFC 11M	Measurement of Fluid Flow by Means of Coriolis Mass Flowmeter
MFC 14M	Measurement of Fluid Flow in Pipes using Small Bore Precision Orifice Meters
MFC 16M	Measurement of Fluid Flow in Closed Conduits by Means of Electromagnetic Flowmeters
MFC SC17	Measurement of Fluid Flow in Closed Conduits using Multipoint Averaging Pitot Primary Meters

ASTM American Society for Tests and Materials

E-84	Standard test method for Surface Burning Characteristics of Building materials.
A269	Standard Specification for Welded Austentic Stainless Steel tubing for general service.

ATEX European Union directive for protection against explosive atmospheres

BS British Standard

6364	Specifications for Valves for cryogenic service

EN European Standards

10204	Inspection Document for Metallic Products
60079-29	Gas Detectors

EPA Environment protection Agency

IEC International Electro-Technical Commission

60079	Electrical Apparatus for Explosive Gas atmosphere.
60529	Classification of Degree of Protection Provided by Enclosure
60332	Tests on electric and optical fibre cables under fire conditions - Part 3 (Test for vertical flame spread of vertically-mounted bunched wires or cables- Category A)
60331	Tests for Electric cables under Fire conditions60534-2 Industrial Process Control Valves-Flow capacity
60584-2	Thermocouples - Tolerances

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	60584-3	Thermocouples - Extension and compensating cables - Tolerances and identification system.
	60751	Industrial Platinum Resistance Thermometer and Platinum Resistance Sensors
	61000-4	Electromagnetic compatibility for Industrial Process Measurement and Control equipment.
	61508	Functional Safety of Electrical/ Electronic/ Programmable Electronic Safety Related Systems
	61511	Functional Safety- Safety instrumented systems for the process industry sector
	61158	Foundation Fieldbus Specifications
	61131	Programmable Controller Language
	61804-3	Functional blocks for process control — part 3 Electronic Device Description Language (EDDL)
	62591	Industrial networks-Wireless communication network and communication profiles-Wireless HART
IS	Indian Standards	
	5	Colours for ready mixed paints.
	3624	Specification for Pressure and Vacuum gauges
	8784	Thermocouple compensating cables.
	6746	Unsaturated polyester resin system
ISA	International Society of Automation	
	S-5.1	Instrument Symbols and Identification
	S-5.2	Binary logic diagrams for Process Operations.
	S-5.4	Instrument Loop Diagrams
	S-7.3	Quality standard for Instrument Air.
	18.1	Annunciator Sequences & Specifications
	S-20	Instrument Specification form
	S-75.01	Control Valves sizing equations
	S 75.03	Uniform Face to Face Dimensions for Flanged Globe Style Control valve bodies
	S12.13	Part I: Performance Requirements, Combustible Gas Detectors Part II: Installation, Operation and Maintenance of Combustible Gas Detectors
	S12.15	Part I: Performance Requirements, H2S Gas Detectors Part II: Installation, Operation and Maintenance of H2S Gas Detectors
	100.11a	Wireless Standards for Industrial Automation
ISO	International Organization for Standardization	
	5167-1	Measurement of Fluid Flow By Means Of Pressure Differential Devices - Part 1: Orifice Plates, Nozzles and Venturi Tubes Inserted in Circular Cross-Section Conduits running full
	5167-2	Measurement of Fluid Flow in Circular Cross-Section Conduits running full using Pressure Differential Devices - Part 2: Orifice plates
	5208	Industrial Valves- Pressure Testing of Metallic Valves

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10497	Testing of valves — Fire type-testing requirements
16852	Flame arresters — Performance requirements, test methods and limits for use
10790	Measurement of fluid flow in closed conduits- Coriolis Mass Flow Meter
17025	Testing and calibration laboratories

NACE National Association of Corrosion Engineers

MR0175	Material for use in H2S containing environments in Oil and Gas Production.

NEC National Electric Code

NEMA National Electrical Manufacturers Association

FG I	Fibre Glass cable tray system

NFPA National Fire Protection Association.

NFC-72	National Fire Alarm and Signalling Code
496	Purged and pressurised enclosures for electrical equipment.

OSHA Occupational Safety and Health Authority.

RAL Reichsausschuß für Lieferbedingungen und Gütesicherung

UL 1709	Standard for Rapid Rise Fire Tests of Protection Materials for Structural Steel
VDI/VDE 3513	Variable Area Flowmeters

2.0 DESIGN PHILOSOPHY

2.1 All field as well as control room instruments shall be suitable for use in a humid and tropical offshore climate. These shall be furnished with all necessary weather and anticorrosion protection to prevent damage from saline and corrosive process atmosphere. PLCs and Control Systems located in local panels shall be suitable for continuous operation in specified ambient temperature and humidity conditions without any deterioration in performance.

2.2 Statutory Approvals

a) Bidder shall be responsible for obtaining all statutory approvals, as applicable, for all instruments and control systems.

i) For all intrinsically safe / explosion proof / flameproof equipments / instruments/ systems or equipments with any other type of protection allowable which are manufactured abroad and certified by any statutory authority like Laboratorie Central Des Industries Electriques (LCIE), British Approval Service for Electrical Equipment in Flammable Atmospheres (Baseefa), Factory Mutual (FM), Underwriters Laboratories (UL). , ATEX, CIMFR etc.

ii) All instruments Panels and Junction Boxes located in open deck areas shall be certified for use in Class I, Division 1/ Zone 1 area irrespective of the actual hazardous area classification. The methods of protection shall generally be as specified below. Any alternative method, if required due to any special technical reasons, shall be used only with a categorical approval from Purchaser.

1) All field transmitters and valve positioners shall be intrinsically safe type based on entity concept. Intrinsic safety shall be achieved by means of external Intrinsic

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Active Safety barriers having three port isolation for the Process platforms and Zener Barriers for the Unmanned Wellhead platforms.

2) All process switches shall be flameproof certified to NEMA 7/ Ex d. Solenoid valves protection shall be as indicated in Job specification/ data sheets.

3) Junction boxes shall be flameproof to NEMA 7 / Ex d for explosion-proof instruments and increased safety type for intrinsically safe instruments and safe area instruments. Material of Construction for JBs shall be epoxy coated Copper Free Aluminium (Cu content < 0.4%) for Ex-proof JBs and SS 316 for Increased Safety Junction Boxes.
The above Instruments and JBs shall be Weatherproof (NEMA 4X/ IP 65 Equivalent) in addition to being increased safety or Explosion proof.

4) A copy of hazardous area certification from a statutory body shall be submitted by Bidder for each instrument and control systems.

5) All panels and cabinets installed in classified area shall be flameproof as per NEMA 7/ Exd, as a minimum.

2.3 All signals to and from central control room shall be electronic. The standard analog signal shall be 2 wire, 4-20 mA DC Smart HART Type having superimposed digital signal for both transmitters and control valve positioners.

2.4 Electrical control and alarm switches shall have SPDT snap acting micro switches rated for 110V/240V AC, 5 AMPS and 24 VDC,1 amps and shall be suitable for inductive load. Limit switches shall be proximity type only.

2.5 Instrument ranges shall be selected such that the normal operating point is middle third of the instrument total range. Standardisation of such items as scale ranges, length of level gauges/level controllers shall be considered.

2.6 All level instruments on vessels/tanks shall have flanged connections. All inline instruments including control valves, flow instruments etc shall have flanged end connections. Standpipes wherever applicable shall be provided with isolation valves.

2.7 Nozzle requirement for Temperature Instruments

a) The nozzle projection on vessels for all instruments and nozzle length for temperature instruments on piping will be minimum 8" and to be verified during detailed engineering stage.

b) For Thermowell installation, minimum pipe size shall be 4". Less than 4" pipe size shall be blown upto 4".

2.8 Tube fittings shall be of the double ferrule, pressure seat, no torque type for use on heavy or thin wall tubing.

Instrument materials shall be resistant to highly corrosive marine saline atmosphere. The case material of all instruments shall be SS 316. Wherever SS 316 is not available, epoxy painted Copper Free Aluminium (Cu content < 0.4%) cases suitable for offshore corrosive environment shall be provided. Mounting accessories for all instruments shall also be SS 3316 or superior material. Canopy shall be provided for all electronic type instruments (transmitters & positioners), Fire & Gas devices and junction boxes, panels located in open or under grating area to protect from rain and sun without impairing access or visibility.

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2.10 Use of asbestos, mercury, lead and cadmium in instruments shall be prohibited due to hazards associated with them.

2.11 Connection for air or gas supply, transmission and control signals shall normally be internally threaded 1/4" NPT.

2.12 Instruments process measurement connections shall be threaded 1/2" NPT minimum.

2.13 Shutdown shall be actuated by independent tripping devices with independent tapping points & independent instrument root valves.

2.14 Terminals for electrical connections including thermocouple extension wires, shall be numbered and identified to indicate polarity, ground connections, test connections and any other pertinent information.

2.15 Terminal blocks shall be non-hygroscopic. Vibration proof and shall use captive screws for terminals. Terminals shall be tinned or Silver-plated. All terminations shall be through use of wire lugs.

2.16 All venting instrument including solenoid valves and pilot valves shall have bug screens fitted to atmospheric vents.

2.17 All moving parts exposed to atmosphere such as linkages, bellows, springs and coils shall be resistant to the atmospheric conditions Cadmium plating is not acceptable for this service.

2.18 Instruments shall be supplied in sealed heavy plastic bags containing desiccant material during shipment. All parts, subject to moisture, shall be suitable for tropical conditions.

2.19 All electronic instrumentation shall be immune to RF interference due to walkie-talkie, paging system, communication system etc. Also all electronic modules shall be designed for protection against short circuit.

2.20 A means for calibration of transmitters, indicators and transducers shall be provided to permit adjustment of the zero output and span output. Calibration and alignment of all instruments shall be performed by the manufacturer before shipment.

2.21 Instruments Operating Conditions

2.21.1 Pneumatic instrumentation operating conditions

Signal Range	0.2 - 1 kg/cm²g for control, transmission
Alarm Logic	3.5 kg/cm²g (Normal)
Alarm Logic	0 kg/cm²g (Abnormal/Shutdown)

2.21.2 Electronic Instrumentation operating conditions

Signal Range	
Transmission	4-20 mA DC. 2 wire system superimposed with HART signal
Switch	SPDT dry
Normal	Close
Alarm/Shutdown	Open

2.21.3 **Power Supply**

Unmanned well platforms

24 to 26V DC Negative earthed. Solar/ wind turbine panel for all Instruments and Controls.

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Process Platforms and bridge connected well platforms : (as specified in datasheet)
24V ±5% with batteries back up for instrumentation including ESD PLC and F&G PLC.
110V AC ± 10%, 50 Hz ± 3% UPS.
240V AC ± 10%, 50 Hz ± 3% UPS.
240V AC ± 10%, 50 Hz ± 3% Non UPS for lighting of panels.

2.21.4 Site conditions and Climate Data.
For site conditions and climate data refer to –"Site Conditions and Climate" attached elsewhere in the bid.

2.22 Field Transmitter

2.22.1 Pneumatic Field Transmitter

2.22.1.1 Pneumatic Transmitters, indicators, controllers and transducers shall meet the following requirements

i) Reference accuracy shall equal or better than ±0.5% of span. Controller output indications shall be better than ±2% of span.

ii) Hysteresis shall not exceed ±0.5 % of span.

iii) Repeatability shall not exceed ±0.5 % of span.

iv) Output change caused by 100°F temperature change shall not exceed ±1% of span.

2.22.1.2 Pneumatic Transmitters shall be generally force balance type. The transmitters shall have an integral indicator.

2.22.2 SMART Field Transmitter

2.22.2.1 Electronic transmitters shall be SMART type with 4 - 20 mA DC output with superimposed digital signal having simultaneous analog and digital communication. Communication protocol shall be HART only. The following specifications shall also apply:

i) Transmitters shall be provided with external zero & span adjustment and shall have integral vent/drain.

ii) All transmitters shall be provided with integral intrinsically safe output meter having LCD display. The transmitter shall be microprocessor based and it shall incorporate a non-volatile memory which shall store complete configuration data of transmitter. All necessary signal conversions, including conversion to produce output with the required protocol shall be carried out in the transmitter electronics.

iii) Transmitter shall also run complete Sub-diagnostic routine. In the event of detection of failure, the output shall be driven to predefined value, which shall be field configurable.

iv) Temperature sensors shall be provided to compensate for process temperature and ambient temperature variations.

v) The transmitter shall be able to assure an accuracy of ±0.065% of span for a rangeability of 1:10. The accuracy shall include the combined effect of linearity, hysteresis and repeatability. However, for diaphragm seal instruments, the accuracy of ±0.25% is acceptable.

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vi) HART protocol of latest version shall be used and shall be suitable for HART maintenance system with the following features as minimum:

a) It shall allow multi masters (two for example, primary and secondary) for configuration, calibration, diagnosis and maintenance. The primary could be control system or host computer and the secondary could be the hand held communicator or a maintenance computer.

b) It shall be capable of implementing universal commands. It must be possible to communicate all the commands of HART maintenance system to field transmitters and SMART transmitters shall be selected accordingly.

vii) Unless recommended otherwise by compressor manufacturer, field transmitters used in anti-surge control loop shall be non-SMART and flameproof type. The flameproof transmitters shall have their calibration adjustment from outside without any need to remove the cover.

2.22.3 Hand Held Field Communicator

Bidder shall supply Universal Hand Held Field Communicators for SMART transmitters and smart positioners for platforms to perform routine configuration, calibration, display process variable, diagnostics etc. Universal Hand Held Field Communicators suitable for all makes of transmitters and positioners shall be supplied. It shall be possible to perform all the above functions on line. The loop function shall remain unaffected while communication is going on between transmitter and the field communicator; No output interruption should occur.

Field communicator shall meet the following requirements.

i) They shall be battery powered with replaceable and rechargeable batteries. Re-charger power supply shall be 240 V AC ± 10 %, 50 Hz ± 3%.

ii) Plug in type connections shall be provided with field communicator. Necessary interconnection cable and any other accessories required for connection shall be supplied by bidder.

iii) Offered communicator shall be, certified intrinsically safe and weatherproof suitable for outdoor location. Carrying case with charger shall be supplied with each communicator.

2.22.4 Diaphragm Seals

Seal systems shall be used to isolate instrument from the process fluid encountered in the following services:

a) Process fluids that vaporize, condense or solidify under operating pressure and ambient temperature.

b) Process fluids that will subject the element to high temperature.

c) Corrosive process conditions.

d) Viscous & congealing liquids.

2.23 Material Selection

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2.23.1 Material selection shall be done based on Piping Material Specification (PMS) / Valve Material Specification (VMS) for the project. However minimum trim/ wetted parts material for instruments including inline instruments shall be SS316. Testing requirements including NACE MR 0175 compliance shall be as per Piping Material Specification (PMS) / Valve Material Specification (VMS). For vessel/ tank mounted instruments Vessel/ tank metallurgy shall be adhered for the Cage/ body of instruments however wetted part materials shall be minimum SS316. Any material specified in Job specification/ data sheets part of Bid package shall override materials requirements specified in this document.

2.23.2 Wherever sea water containing chlorides is handled, the instruments wetted part materials shall be Monel.

2.23.3 Materials shall be selected to avoid galvanic corrosion due to dissimilar materials.

2.23.4 Where ever welding is involved, SS316L material as a minimum to be considered for wetted parts.

2.23.5 For high velocity and erosion services hardened trim material shall be considered.

2.23.6 Aluminium instrument housing if used (in case SS316 housing is not available), the same shall be epoxy coated (Copper content shall be less than 0.4%).

3.0 GENERAL SPECIFICATION OF FIELD INSTRUMENTS

3.1 Flow Instruments

3.1.1 General

3.1.1.1 Generally a Piezoelectric/ Capacitance type Electronic Smart transmitter coupled with orifice plate flow element shall be used for flow measurement. Variable area meters may be used for conditions described in 3.1.2.2.

3.1.1.2 The static pressure element shall be connected to the downstream pressure tap on gas flow measurement. The static sensing point shall be independent of the differential sensing points.

3.1.1.3 Differential pressure instruments shall withstand maximum pressure equal to full line pressure without zero or calibration change.

3.1.2 Primary Element

3.1.2.1 The standard primary measuring element shall be a concentric orifice plate designed for flange taps except for particular process requirements.

Quadrant edge or quarter circle orifice plates shall be used for highly viscous liquids and for pipe Reynold number below 10,000.

3.1.2.2 **Variable area flow meters**

3.1.2.2.1 Variable area flow meters may be used when:

a) Only local flow indications required.

b) Linear flow rate signal characteristics are required.

c) Extremely low flow rate measurement.

d) Rangeability greater than 3 to 1 and as identified in P&ID.

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3.1.2.2.2 Linear-scale rotameters shall be marked for direct reading of flow and manufacturer shall furnish a guaranteed calibration curve.

3.1.2.2.3 It shall be metal tube type except for non hazardous, non toxic and non viscous service such as Instrument air / Nitrogen service where glass tube can be used.

3.1.3 Measuring Instruments

Where orifice, flow nozzle, venturi, averaging pitot tube is used for flow measurement, Differential Pressure transmitters shall be generally used. A five (5) valve manifold with proper drain connections shall be utilised with differential pressure instruments.

3.1.4 Range Selection

3.1.4.1 The orifice meter differential range shall be 500, 1250, 2500 or 5000mm of water (20, 50, 100 or 200 inches of water). The range 2500 mm (100 inches) is desirable wherever possible.

3.1.4.2 For orifice meters the normal flow should be between 60% and 80% of the meter maximum.

3.1.4.3 For rotameters / variable - area flow meters the normal flow rate shall be between 50% and 80% of the instrument range.

3.1.4.4 Multistage Restriction Orifice shall be considered for high pressure drops. Thickness of Restriction Orifice shall be selected based on the pressure drop for each stage.

3.1.5 **Orifice Flanges**

3.1.5.1 Orifice flanges shall be in accordance with piping classification with a minimum flange rating of ASME 300 class and for 1½" and below line size integral orifice with meter run assembly shall be considered. Two pairs of flange taps located diametrically opposite shall be provided for each orifice assembly.

3.1.5.2 D – D/2 taps should be considered for line size greater than 14".

3.1.6 **Orifice Meter Runs**

3.1.6.1 All meter runs shall be designed for flange tap differential.

3.1.6.2 Orifice fittings which permit changing of the orifice plate without interrupting the flow through the pipeline and without the use of a bypass shall be used, wherever necessary.

3.1.6.3 The piping layout, wherever possible, should be arranged so that straightening vanes are not required.

3.1.6.4 It is preferred that metering orifices be located in horizontal lines. Sizing of Orifice shall be as per AGA 3 Standard for Natural Gas service and ISO 5167 standard for liquid service. Upstream and downstream straight lengths shall be as per applicable codes and standards.

3.1.6.5 Orifice meter runs shall not be less than 1½ inch nominal pipe size.

3.1.6.6 Mating flanges shall be so aligned that jack screws will be diametrically opposite.

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3.1.6.7 **Senior Orifice Plate Assembly**

Senior orifice assembly shall include as a minimum supply of orifice plates, plate carrier, flanges, senior orifice fittings, studs, bolts, nuts, hand wrench and plugs. Senior orifice fittings shall include sliding valve, equaliser, stuffing box and sealing unit and all other system to make the system complete. Senior orifice assembly shall be flanged at both ends.

The orifice plate rangeabilty is to be taken as 1:3 while deciding the number of plates required to cover the required range. In addition, adequate overlap should be provided between two consecutive flow ranges.

Straightening vanes with meter runs required shall be provided by vendor. Straightening vanes shall be flanged type. Upstream and downstream straight lengths for orifice plates shall be as per applicable sizing codes and standards (AGA 3/ ISO 5167).

3.1.7 **Positive Displacement Flow Meters**

3.1.7.1 Flow Meter / Transmitter

3.1.7.1.1 The following performance characteristic shall be satisfied as a minimum by the flow transmitter:

- Repeatability of ± 0.05 % of reading in the rated linear flow range.
- Accuracy of ±0. 2 % over the rated flow range
- For high accuracy applications Positive Displacement Flow meter having accuracy of ±0.1% shall be provided

3.1.7.2 The flow direction shall be clearly stamped or cast on the body.

3.1.7.3 The flow meter shall have an over-range protection of at least 130%.

3.1.7.4 Pick up coil shall be hermetically sealed.

3.1.7.5 PD Meter shall be used for high viscosity and congealing services. PD meter shall be provided with strainer and air eliminator in crude oil service and wherever necessary. The air eliminator shall be supplied complete with pressure gauge, safety valve etc. and shall be located upstream of strainer.

3.1.7.6 Strainer shall be provided with Differential Pressure Gauge for choked flow indication.

3.1.7.7 Each Positive Displacement meter shall be calibrated using a suitable method as per API-MPMS.

3.1.8 **Turbine Meters**

3.1.8.1 Turbine meter output shall be 4-20 mA dc / pulse with power supply 24V DC. Necessary preamplifier shall be provided integral with meter. Local electronic totalizer, transmitters (pulse to current converter) with 4-20mA dc output shall be provided. Accuracy for the turbine meter shall be ±0.2% of FSD. For high accuracy applications, Turbine meters having accuracy of ±0.15% shall be provided. Turbine meters shall have flanged end connections.

3.1.8.2 Output for turbine meter shall be compatible with the control room instrumentation.

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3.1.8.3 Bidder/Vendor shall be responsible for following:

a) Proper selection of material compatible with fluid and operating conditions for Turbine meter body and trim and provide accessories like flow straighteners, strainers, air eliminators etc.

b) Proper sizing of turbine meter, strainer and other accessories. Cavitation checks shall be performed for Turbine meters in liquid service. Turbine meter for Gas services shall be specifically designed for gases.

3.1.8.4 Maximum pressure loss through the turbine meter and its accessories like strainer shall be within the limit indicated in Instrument process data sheets.

3.1.8.5 Flow meter shall have over range protection of at least 30% of required range. Suitable protection shall be provided against over speeding.

3.1.8.6 Viscosity compensated rotors shall be provided wherever wide variation in fluid viscosity is present.

3.1.8.7 Accessories

a) Air eliminator as per ASME Section VIII wherever required shall be complete with all accessories such as pressure gauge, differential pressure gauge, pressure relief valves, sight flow glass, mounting brackets etc. as applicable.

b) Strainer shall be basket type. Strainer shall be selected by vendor suitable for service conditions and turbine meter. Strainer shall meet ASME section VIII as applicable. Accessories mentioned in air eliminator such as differential pressure gauges are also applicable as required.

c) All accessories shall be of the same size and rating as turbine meters.

d) All accessories shall have flanged end connections.

e) Turbine meter bearing material shall be tungsten carbide or suitable hard material.

3.1.8.8 Turbine meters shall be calibrated using mechanical displacement meter provers as per API Manual of Petroleum Measurement Chapter-4.0.

3.1.9 **Flow Straightener**

The flow straightener may be of either tube type or vane type. Material of Construction shall be 316 SS or superior material and supplied from meter vendor.

3.1.10 **Averaging Pitot Tube**

Averaging Pitot tube shall have on-line insert-retract/ removal mechanism.

3.1.11 Cavitation checks shall be performed for all Coriolis Mass Flow Meter in liquid service.

3.1.12 Water Cut Flow meters shall be microwave type unless otherwise specified in Job specification/ data sheets. Water cut Flow meter shall include meter sensors, Flow meters electronics with local indicator and integrator, special cables between sensor and electronics. Flow meter sensor and electronics shall be certified suitable for hazardous area. Sensor shall be powered from Water Cut meter electronics. Output of water cut meter shall be 4-20 mA corresponding to water cut. Necessary proprietary software for water cut shall be supplied by bidder. Diagnostics for Water cut meter shall be provided in control room.

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3.2 Pressure Instruments

3.2.1 General

3.2.1.1 The following considerations should be applied in the selection of pressure instruments:

a) Pressure gauge measuring element shall generally be the Bourdon tube type. For measurement of slurries, viscous and corrosive fluids, diaphragm seals shall be used.

b) Case material for all Pressure gauges, transmitters, switches shall be SS 316.

c) Gauges and instruments shall be capable of withstanding 130 % of the maximum range.

d) Gauges and instruments subject to damage from pulsating pressures shall be provided with snubber type pulsation dampeners. Gauge savers (Pressure Limiting Valve) shall be provided wherever necessary for protection of the pressure gauge. Material shall be 316 SS or better as required for the service.

e) Indicating scales shall be direct reading in engineering units.

f) Ranges shall be selected so that the normal operating pressure will be in the middle third of calibrated range.

3.2.1.2 The process sensing point shall be located in an undisturbed pressure zone with the pressure instrument located as close as possible.

3.2.1.3 Pressure Switches shall be avoided as far as possible except for packages where there are space constraints. Pressure switches shall have field adjustable set points. Adjustments shall be internal and tamper proof. Electrical pressure switches shall have set point indicator.

3.2.1.4 For pressure differential instrument a (5) Five valves manifold or a (3) valve manifold with proper drain connections shall be provided.

3.2.2 Controller

3.2.2.1 Local controllers shall be pneumatic if identified in the P&ID for well platforms and shall normally include adjustable proportional band plus reset action and auto-manual bypass assembly.

3.2.2.2 When blind pressure pilots are used, indicating pressure gauges shall be mounted adjacent to the pilot for indication of pilot output.

3.2.3 Pressure Gauges

3.2.3.1 Pressure gauge case shall be liquid filled type with external zero adjustment.

3.2.3.2 Case diameters shall be 6" (150 mm).

3.2.3.3 Instrument air/gas supply gauges used as output gauge may have 50 mm (2") dials.

3.2.3.4 Movement shall be 316 SS and 1/2" NPT bottom connection.

3.2.3.5 Accuracy shall be ±1 % of full scale.

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3.2.3.6 Gauges shall have micro-meter pointers and white dial with black numerals and markings. Gauge pointers shall be adjustable without removal from shaft.

3.2.3.7 Pressure instruments in fields shall have 1/2" NPT (M) connections. Diaphragm seals with capillary shall generally be 1 1/2" flanged.

3.2.3.8 All pressure gauges shall be with shatterproof glass.

3.2.3.9 All pressure gauges with maximum operating pressure exceeding 60 kg/cm^{2}g and gauges in toxic service shall be solid front type.

3.2.3.10 Surface mounted type pressure gauges shall be used for high temperatures exceeding 200^{0}C.

3.3 Level Instruments

3.3.1 Displacer type Level instrument to be avoided. Guided wave Radar type Level transmitters to be used as fas as possible. Non Contact Radar type Level transmitters to be used where service is viscous, congealing or cryogenic. Radar Antenna shall be selected by vendor based on process conductivity suitability and the dielectric constant. Echo rejection circuit shall be provided in Non contact radar electronics. All Radar level instruments shall be provided with local digital level indication. All Radar level instruments shall be provided with diagnostics in control room. Guided wave radar shall be external type except where vessel conditions/ dictates otherwise.

3.3.2 External displacer cages shall be supplied with side/side connections. Where this is not possible top/ bottom or side/ bottom connections may be provided. Displacer instruments shall be Torque Tube type construction.

3.3.3 Differential pressure level measurement is acceptable for remote level indication for level range of 1524 mm (60") and above. Seals and wet leg compensation shall be provided as required. These instruments shall have zero elevation/ suppression adjustment wherever required.

3.3.4 External displacer connections on transmitters/ switches shall not be common with other service connections, such as water drain or pump-out lines.

3.3.5 All level instruments in toxic and hydrocarbon service shall have blow down connections to a closed drain.

3.3.6 Rotatable heads shall be provided for displacer type instruments.

3.3.7 For external cages, block valves are to be installed in horizontal position of both lines at the vessels. If a vertical nozzle is used the valve shall be arranged so that no sediment pockets are present. A valve shall be installed to permit draining each displacer cage and piped to drip pans or deck drains.

3.3.8 Units for level indicating scales shall be- 0-100 percent.

3.3.9 **Liquid Level Switches**

3.3.9.1 Field mounted level switches shall be avoided as far as possible. Field level transmitters shall be used for alarm and shutdown.

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3.3.10 **Tank Gauges**

3.3.10.1 Level instruments for pressure and non-pressure storage tanks where measurement is not for accounting purposes, may be by differential head measurement or by a float and tape type level gauge. Float and tape type level gauge measurement accuracy shall be ± 2 mm. For remote transmission or difficult congealing, viscous, corrosive, high temperature services, cryogenic services etc., non-contact Radar type Level transmitters to be used.

3.3.11 **Magnetic Level Gauges**

3.3.11.1 Type of level gauges shall be magnetic only. Body (Float chamber, Float, Float Guide tube & spring assembly) shall be of minimum 316 SS with SS or Aluminium indicating tube with flapper colour change design.

3.3.11.2 Float design i.e. its type, size & material requirements shall consider worst case process conditions also like specific gravity, temperature & pressure specified in process datasheets.

3.3.11.3 Where magnetic level instrument is specified as external mounting type, the instrument shall be provided complete with external cage (float chamber).

3.3.11.4 Float chamber or cage shall be provided with ¾: flanged end vent & drain connections.

3.3.11.5 The design shall also facilitate float removal.

3.3.11.6 Mechanism to prevent float rotation & float damage due to rubbing against the chamber wall shall be provided.

3.3.11.7 Flanges of Magnetic level gauges shall be minimum SS316, Carbon steel flanges shall not be used. End connections shall be threaded to NPT as per ASME B1.20.1, Flanged as per ASME B 16.5 & Grooves of ring type joint flanges shall be octagonal as per ASME B 16.20.

3.3.11.8 Orientation of measurement of scale for the offered magnetic level instruments shall be site adjustable to any angle without hot working/ welding.

3.3.11.9 The design shall ensure that the magnetic properties of the applicable parts of the magnetic level gauge shall not degrade under the influence of the design temperature given on datasheets.

3.4 Temperature Instruments

3.4.1 **General**

3.4.1.1 Filled thermal systems with armoured capillary shall be used for local mounted pneumatic transmitters, temperature switches (only as per package vendor recommendation) and controllers. Capillary tubing lengths over 8 meters (25 feet) shall not be used. Each thermal system shall be selected to withstand the maximum temperature to be encountered. Mercury filled system is not permitted.

3.4.1.2 RTD type temperature sensing elements are to be used for all electronic temperature transmitters and measuring systems. The sensing element shall be Platinum 100 ohms at 0°C to IEC 60751, 3 wire system.

3.4.1.3 Electronic Temperature transmitters are to be head mounted at the sensing source with two wire transmission to the central control room. The output signal shall be 4-20mA DC with superimposed digital HART signal.

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3.4.1.4 Indicating scales shall read out in degrees Celsius.

3.4.2 **Temperature Indicators**

3.4.2.1 Local temperature indicators, in general, shall be heavy duty, weatherproof to NEMA 4X/ IP65, dial type bi-metal thermometers with 150 mm (6 inch) dials and rigid stems.

3.4.2.2 Case material for all temperature gauges shall be SS 316.

3.4.2.3 The range shall be selected so that the normal operating temperature shall fall in the middle third of the span.

3.4.2.4 The maximum error shall not exceed ±1% of the span.

3.4.2.5 Gas Filled system dial thermometers may be used when the operating temperature is between 0°C and -45°C. However mercury filled systems are not permitted.

3.4.2.6 Separable stainless steel thermowells shall be provided for dial thermometers.

3.4.3 **Thermocouple and Thermowells**

3.4.3.1 Thermowells shall be installed for all temperature sensing elements to permit their removal during operation.

3.4.3.2 All thermowells shall be machined from type 316 SS (Or superior material) bar stock.

3.4.3.3 Thermowell shall have 1/2" NPT female thread for element and 1½" flanged connection for vessel or piping connection. All flanges for thermowell shall be forged type.

3.4.3.4 Immersion length of thermowells shall be as follows:

Line Size	Immersion length
From 4" to 6"	280 mm
From 8" onwards	320 mm
Vessels / columns	400 mm

In special applications, not covered above, bidder shall decide the immersion length based on actual requirements. Immersion length is based on 200 mm length between flange face and outer wall of pipe.

3.4.3.5 Test thermowells shall have screwed plug and chain.

3.4.3.6 Thermowells for insulated vessels and lines shall have extension necks.

3.4.3.7 The thermowell tip shall be at centerline of pipe but immersion length shall not exceed 400 mm for higher size pipes.

3.4.3.8 Thermocouples, wherever specified, shall be as per IEC-60584-2.

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3.5 Pressure Relief Valves

3.5.1 General

3.5.1.1 The basis of the selection of pressure relieving devices shall be in accordance with the requirements set forth in the API Standard, RP 520 and the latest revisions of the ASME-Power Boiler Code", and the ASME' "Code for Unfired Pressure Vessels, as well as local codes.

3.5.1.2 Venting and breathing equipment for low pressure above ground storage tanks shall be in accordance with API Std. 2000.

3.5.1.3 All pressure relief valves shall be marked and Certified in accordance with the ASME Boiler and Pressure Vessels Code.

3.5.1.4 Lifting levers and bellows shall be specified as service requirements where codes dictate. Lifting lever shall be provided for all valves in air services and hot water services.

3.5.1.5 Testing for tightness shall be in accordance with API RP 527.

3.5.1.6 Pressure Relief valves sizing calculations shall be furnished to purchaser for review and approval.

3.5.1.7 All safety relief valves shall be provided with test gag.

3.5.2 Types and Materials

3.5.2.1 All relief valves of size 1" and above shall be full nozzle full lift type. Relief valves with size less than 1" shall be modified nozzle type.

3.5.2.2 Relief valves 1" and larger shall be of the type having a nozzle bushing which extends through and beyond the base of the inlet bolting flange, and which forms the gasket bearing surface for the inlet flange.

3.5.2.3 Flanged relief valves for process piping, excluding steam and air pressure piping, shall be either the enclosed spring type or pilot-operated type.

3.5.2.4 Conventional valves shall be specified for constant back pressure while bellows seal type valves shall be specified for variable back pressure more than 10% of set pressure or where the service is corrosive. Pilot operated pressure relief valves shall be used for special services and where set pressure is closer than 10% of the operating pressure, in general. Balanced bellow materials shall be SS 316L or superior material.

3.5.3 Connections

3/4" X 1" thermal relief valves may have screwed ends. For variable back pressure and viscous services, size of thermal relief valves shall be 1" X 2" with D orifice. All process relief valves 1" size and larger shall have flanged inlet and outlet connections The outlet flange of relief valves discharging into closed systems shall be at least equal to the pressure rating of the outlet system at operating temperature.

3.6 Control Valves and Shutdown Valves

3.6.1 Control Valve Selection

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3.6.1.1 In general, control valves shall be sized for maximum flow at not more than 90% opening of the valve and normal flow around 70% opening of the valve.

3.6.1.2 For Two-phase flow the control valve CV shall be the summation of the CV of the Liquid phase and CV of the vapour phase. The quantities and physical properties of liquid and vapour should be established on the outlet conditions of pressure and temperature.

3.6.1.3 Valve bodies should be the maximum size required for any anticipated operating condition. If a substantial increase in future flow rate is indicated, reduced valve trim shall be used.

3.6.1.4 Control valve sizing calculations shall be furnished for review and approval.

3.6.1.5 Standard control valve body sizes are 1, 1½, 2, 3, 4, 6, 8, 10, 12, 14 and 16 inches. Valves with 2½ inch and 5 inch body size shall not be used.

3.6.1.6 Globe valves shall be used where throttling is required.

3.6.1.7 Butterfly valves may be used when low pressure loss and high capacity flow is desirable.

3.6.1.8 All control valves must have sound levels less than 85 db at 1 meter from valve. Source treatment shall be performed for noise control as far as possible. For very high noise levels Source treatment in conjunction with Path treatment shall be performed.

3.6.1.9 Anti-cavitation trims shall be provided for valves with cavitation in liquid service.

3.6.2 **Control Valve Trim Characteristic**

3.6.2.1 Equal percentage characteristic shall be used in most applications where throttling is required. Other characteristics such as linear or which fall between the two shall be used when special types of valves, such as butterfly or ball valves, are specified.

3.6.2.2 Quick-opening/ closing trim shall generally be used for on-off service.

3.6.2.3 Control valves used in safety shut down service shall be single seated tight shutoff.

3.6.3 **Guiding**

Generally valves shall be single seated for viscous, congealing services or where clogging of cage can occur, Small single-seated valves and angle body valves may be stem-guided. Cage guiding shall be used in clean gas service.

3.6.4 **Shutdown Valves**

3.6.4.1 Shut down valves shall be of Flanged Full Bore Ball Valve with actuator.

3.6.4.2 All Shut down valves shall be certified Fire safe as per API 607/ ISO 10497 by a recognized Third Party Testing agency. For Christmas tree valves (API6A), API 6D valves and metal seated valves, API 6FA certification is also acceptable.

3.6.4.3 Shut down valves shall be Soft seated type with Class VI tight shut-off as per FCI 70.2. Rate A as per ISO 5208 shall be provided where specified in data sheets. For triple offset butterfly valves, Zero leakage valves shall be considered.

3.6.4.4 Actuator type shall be spring return piston type with Quarter turn operation.

3.6.4.5 Hand wheels shall not be provided on Shutdown Valves unless specified in

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P&IDs/ Process data sheets. In case hand wheels considered, the same shall be detachable type so that same can be removed at site after commissioning as required.

3.6.5 **Valve Bonnets**

Radiating fins or extended bonnet shall be provided when the temperature exceeds 200°C (392° F) or below 0° C (32° F).

3.6.6 **Material of Construction**

3.6.6.1 Flanged end body connection shall be used for all valves.

3.6.6.2 Valve plugs, seats and trim shall be AISI type 316 SS or superior material except if the normal pressure drop exceeds 7 kg/cm²(100 psi) or the fluid is flashing, hardened trim such as stellite shall be used. SS 440C and Tungsten carbide shall be used for erosive services and very high pressure drops.

3.6.6.3 Generally, valves shall be furnished with bolted packing glands and Teflon packing up to 200 deg C and Graphoil packing above 200 deg C.

3.6.6.4 Live loaded packing for valves shall be considered for toxic service against fugitive emission as per applicable standards.

3.6.7 **Valve Positioners**

3.6.7.1 Smart Electro pneumatic Valve Positioners shall be provided for all valves on regulatory control applications in Process Platforms.

3.6.7.2 The smart positioner shall be a two wire device having 4-20 mA output signal from the controller superimposed with two way digital communication in HART protocol format. All engineering, configuration, diagnostic and maintenance related data shall be provided by the positioner. Positioner shall be provided with position feedback

3.6.7.3 The smart positioner shall provide HART protocol of latest version and shall be suitable to operate with HART maintenance system (HMS).

3.6.7.4 In addition to operational and maintenance data provided by the positioner like actual stem travel, input signal, pneumatic pressure to actuator etc the positioner shall also provide cycle counting and store accumulation data to help keep the control valve stem leakage in control.

3.6.7.5 The positioner shall also have built-in manual stroke button for stroke open or close the valve for quick verification.

3.6.7.6 Valve's operating signatures in the form of hard copy and soft copy shall also be provided for each control valve.

3.6.7.7 Valve Positioners shall have minimum two nos. Air/Gas gauges.

3.6.7.8 Air/gas-filter/regulator shall be factory mounted on the control valve yoke.

3.6.8 **Actuators and Accessories**

3.6.8.1 Actuators shall be sized for control valves and shutdown valves at shut-off pressure indicated in Process Data Sheets. Actuators shall be painted green for fail open service and red for fail close valves.

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3.6.8.2 Two no. of Limit switches shall be provided for each Shut Down valve and for Control valve (wherever applicable) for open/ close status indication. All limit switches shall be proximity type.

3.6.8.3 In general the actuators shall be sized for standard spring range of 0.2 to 1.0 kg/cm²g.

3.6.8.4 Solenoid valve shall be 3 way Universal type having ½"NPT cable entry, ¼" pneumatic connection, SS 316 material of construction, and Weatherproof to NEMA 4X/ IP65, certified by a statutory body suitable for continuous service. All solenoid valves shall be with manual reset facility.

3.6.8.5 Limit Switches contact output shall be rated for 110V/240V AC, 5 AMPS and 24 V DC, 2 AMPS, suitable for inductive load, and shall have ½" NPT cable entry. Proximity Limit Switches shall be certified Intrinsically Safe and weatherproof to NEMA 4X/ IP65 certified by a statutory body. Necessary converters shall be supplied by bidder. Proximity type Limit Switches shall be as per NAMUR standard. Amplifiers shall be provided for proximity switches and located in control room.

3.6.9 **Self-Actuated Control Valves**

Self-Actuated Control Valve droop shall be less than 10% of set point. For regulated pressure under 10 psig, a packless valve shall be used if available. Materials of diaphragm shall be SS316 minimum for hydrocarbon/ high pressure services.
Material used for seats/ packing of valves and diaphragm of Self actuated control valves shall be durable and shall have minimum operating life of 5 years.

3.6.10 **Stem Indicators**

All control valves shall be equipped with stem indicators where valve construction permits.

3.6.11 **Control Valve Manifold**

3.6.11.1 Block and bypass valves shall be provided for all control valves as per process requirements. Valve sizes and manifold arrangement shall be as per API recommendations.

3.6.11.2 The bypass valve shall be correctly sized for the process requirement and when fully open shall not permit an excessive flow beyond the capability of the relief system.

3.7 Flame Arrestors

Body shall be with retractable element bank. End connections shall be flanged. Weather hood shall be provided, wherever necessary. Flame arrestor shall be designed as per ISO 16852 and certified by a recognized Third party testing agency.

3.8 Analysers

3.8.1 The analysers shall be continuous on line measurement type.

3.8.2 Vendor shall not offer any prototype analyser. All analysers offered should have been proven in offshore application. In general analyser and sampling systems shall be designed and constructed in accordance with API standards, as applicable.

3.8.3 Material of all components wetted by sample shall be 316 SS or superior material for general applications and Monel for Sea Water applications.

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3.8.4 Analyser enclosures and accessories shall be suitable for the applicable electrical area classification.

3.8.5 The sampling system, analyser etc. shall be supplied pre-mounted on a self-standing panel, in general.

3.8.6 The analyser shall provide isolated analog current output as specified in data sheet.

3.8.7 Sampling system design shall be such that the sample drawn for analysis is truly representative of the process stream. Extra care shall be exercised in the selection of the various components forming the sampling system.

3.8.8 The sampling system shall be designed to move the sample from the process to the analyser proper in the shortest possible time. Due consideration shall be given to the size of the sample tubing to avoid possible plugging.

3.8.9 Selection of materials for all the parts of the analyser system so as to be compatible with the process stream and surrounding atmosphere.

3.8.10 Bidder shall furnish detailed specification sheet for each analyser system alongwith the Purchase Specification, which shall contain the following information: -

- All the details regarding type, construction, materials, accessories etc. of the analyser.
- Detailed sketch showing various components of sampling and sample conditioning system
- Sample flow rates required to achieve response time and the recommended length and size of the sampling tube between process tap and sampling system.
- An interconnecting wiring diagram for the complete analyser system showing locations (e.g. field, control room etc.) of various units.

3.8.11 Continuous Emission Monitoring System (CEMS) shall be provided as per statutory requirements where applicable.

3.9 **CCTV Cameras**

3.9.1 CCTV cameras shall be provided with remote pan, tilt, zoom operation.

3.9.2 CCTV cameras shall be provided with internet protocol (IP) for communication.

3.9.3 Necessary mounting accessories for mounting of CCTV cameras including Pole etc shall be supplied by bidder.

4.0 FIRE AND GAS DETECTION

4.1 All Fire and Gas detectors shall be FM/UL listed (or equivalent approved agencies) in addition to being certified suitable for hazardous area.

4.2 Voting logic for the Fire and Gas detectors shall be provided for each zone/location. Voting Logic shall be 2 out of 3 for Gas Detection and 2 out of 2 for Fire detection.

At least three flame detectors shall be provided for coverage of any area, for coincident operation. Any two detectors need not be next to each other physically, but may be separated from each other by a minimum distance, as recommended by the Vendor but not exceeding three meters.

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4.3 Quantities of Smoke detectors in a room shall be as per NFPA NFC-72.

4.4 Manual ESD and FSD stations shall be provided for manual Emergency Shut Down and Fire Shut Down and ESD/ FSD alarms.

4.5 Fusible Plugs shall be provided as per the recommendation of API RP 14C.Where high ambient/ surrounding temperature are encountered, melting point of fusible plug shall be considered as 135 Deg C (higher than 85 Deg C).

4.6 Minimum 2 nos. Hooters per deck shall be provided at opposite ends.

4.7 Beacons shall be provided in high noise areas where high noise equipment and machinery is located and on helidecks and boat landing.

4.8 Manual call-points in appropriate quantity shall be supplied by bidder suitable for hazardous area and installed at every exit of the enclosed area.

4.9 Abandon platform switches shall be provided for manned platforms. Platform abandon switches shall be supplied by bidder suitable for hazardous area and mounted adjacent to each survival craft.

4.10 IR type HC detectors and Catalytic diffusion type H2 detectors shall be provided. Catalytic diffusion Sensors shall be resistant to catalyst poisoning and shall be " Plug in" type. H2 Detectors shall be provided in battery rooms and where H2 is handled.

4.11 Diffusion/Adsorption or electrochemical type "plug-in" H2S Gas Detectors shall be provided.

4.12 Construction of Gas Detectors shall be 316 SS Body with SS316 flame arrestor/splash guard/ dust shield with weather protection equivalent to IP 65 for outdoor location. Flying leads shall not be provided.

4.13 Where monitors are used, Type of the Gas Monitors shall be Flush panel mounted in modular construction with single channel monitors. Set point for alarm points shall be adjustable over the entire range.

4.14 Flame Detector shall be Triple IR type. Flame detector shall not be sensitive to direct or reflected sunlight or normal artificial lighting and shall be provided with automatic optical integrity.

4.15 Smoke Detector shall be Ionisation type, Plug-in, dual chamber type with time delay actuation.

4.16 Rate Compensated/Fixed Temperature Thermal Detectors shall be provided with a 57°C (135°F) rating.

4.17 Complete calibration equipment for the F&G detectors including calibrators, cylinders and accessories as applicable shall be provided.

5.0 CONTROL PANELS (LOCAL AND CONTROL ROOM)

5.1 Instruments shall be located on panels for convenient observation and operation. As far as possible, the horizontal instrument arrangement shall follow the piping and instrumentation diagram from left to right. No panel front item shall be located at an elevation less than 700 mm.

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5.2 Instrument density on central control panel shall allow for at least 10% space for future instruments.

5.3 Engraved identification plates shall be provided for all instruments on rear of the panel face instruments shall have integral nameplate on the panel front.

5.4 Minimum spacing shall be used between the instruments mounted on the panels, within the limitations of maintenance requirements.

5.5 Operating temperature of electronic instruments shall not be affected because of high-density layout.

5.6 Panel wiring shall be 1 mm^2 copper standard conductors with flame retardant PVC insulation Rear of the panel wiring shall not interfere with future instrument installation on the panels, or accessibility for maintenance. Interconnection between the panels shall be through cables.

5.7 All panels shall be of self-standing, totally enclosed construction Enclosures shall be fabricated from minimum of 3-4 mm thick CRCA steel sheet. Entire steel structure shall be sand blasted/chemically treated and phosphatized and painted inside/outside with 2 coats of primer and surfacer. The final exterior finish shall be a minimum of three coats of non-glossy epoxy paint of colour same as other panels in Control/ TIC Room. Panel interior finish shall be pale cream. Local control panels in open areas shall be of 316 SS material of construction.

5.8 All doors shall have continuous stainless steel piano hinges Panels over 1200 mm (48") wide and above shall have double rear doors with heavy duty T-bar latches. Panels smaller than 1200 mm (48") wide shall have single rear door with heavy duty 1-bar latch. Central control room panels shall be of the cubicle type with walkway and side doors.

5.9 Panels shall be fitted with four (4) lifting lugs at the top, capable of easy removal after installation.

5.10 Bulk head fittings for signals shall be 1/4", 316 stainless steel. Bulkhead fittings for supply and shutdown loops shall be 3/8", 316 stainless steel.

5.11 It shall not be necessary to remove any component or tubing from a control panel in order to gain access to or remove any other components.

5.12 A system of plastic wire ducts shall be provided for routing wiring inside panels Special precautions shall be observed to maintain the integrity of intrinsically safe wiring. Plastic wire ducts shall be flame retardant type.

5.13 Central control-room alarm annunciators shall be of electronic type. Central control-room alarm annunciator components, except the alarm horn shall be mounted integrally in a protective enclosure. Mounting clamps which present a flat surface to the front panel face shall be provided for panel mounting the assembly. All field connections shall be made on screw type terminal strips, located at the rear of the annunciator assembly sufficient spare terminals shall be provided to accommodate all future alarm points in the display.
The annunciators display shall consist of translucent hack-lighted windows. Colour of the windows shall be as follows

White	-	Alarm
Amber	-	Pre Trip
Red	-	Trip

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At least two lamps shall be provided in each window. The legend shall be engraved in black lettering.
Sequence shall be as per ISA 18.1.

5.14 Local shutdown systems shall be of pneumatic type. Shutdown systems located in the central control room shall be PLC based.

5.15 Special precautions shall be observed to ensure proper grounding of all electronic and power supply circuits and shield on electronic signal cables as per applicable codes and practices. All intrinsically safe barriers must be grounded.

Panel ground and signal ground shall be separate. Signal ground shall be provided for circuit grounds of electronic instruments, drain wires of cable shields, Intrinsically Safe barrier earth etc. For this purpose an insulated grounding bus bar 25 mm wide X 6 mm thick of copper material shall be provided which shall run along the entire panel length. Ground wiring shall be directly connected to the ground bus bar by means of suitable wire lugs. Intermediate terminals are not allowed. This ground bus bar shall be provided with grounding terminals at both ends. Terminal shall be compression type suitable for AWG No 4 copper ground cable. The same shall be connected to platform ground.

Earthing lugs shall be provided in each panel structure for panel grounding. Creation of multiple grounds in a loop shall be avoided and if it is unavoidable, signal isolators shall be used. Shields of thermocouple extension cables for grounded thermocouples shall not be grounded in control room.

Special precautions shall be observed to ensure proper grounding of Intrinsically Safe barriers. Two independent and distinctly identified wires shall be run from barrier earth bus bar to ground. The resistance between the barrier earth and the ground shall be less than one (1) ohm.

5.16 Minimum of 20% spare terminals shall be provided with uniform spare capacity distribution in panels.

5.17 ESD and FSD levels of shut down shall be annunciated by means of public address system tone generator wherever central control panel is located.

5.18 Upon the loss of a pneumatic signal from a shutdown sensor, the pneumatic shutdown panel shall do the following as applicable:

- Indicate which sensor caused the shutdown by a pneumatic first out indicator and an audible horn.
- Depressurise the outputs (if any) to various shutdown valves as per logic.
- Depressurise the output (if any) to the master pneumatic panel or other local panels.
- Shutdown a motor (if any) by tripping a low pressure switch which is connected to the motor control center via the electronic control panel The pressure switch shall be hermetically sealed and shall have DPDT contacts which open on shutdown.
- Trip a single low pressure switch that is wired to an electronic annunciator on the central control panel to signify that a shutdown has occurred.

These local panels shall be equipped with a master bypass for testing and maintenance and individual bypass for testing individual sensors. Emergency shutdown signals shall

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not be bypassed by either master or individual bypass. Each local panel must remain locked out after detection of a shutdown until manually placed back into service. Panel lights shall be provided on the central control panel to alarm while an individual panel is bypassed.

5.19 Power Supply System

5.19.1 Components of power supply system shall be of highest available quality for reliability and long service life.

5.19.2 Power supplies for all transmitters, controllers, signal converters, electric system and components in Distributed Control System and Emergency shutdown system shall be supplied from uninterrupted power supplies (UPS). For details of type and levels refer Job Specification.

5.19.3 Power distribution to each consumer shall be through proper MCBs and MCCBs.

6.0 CONTROL ROOM INSTRUMENTATION

6.1 General

6.1.1 Specifications for Distributed Control System, Programmable Logic Controller (PLC) based Emergency Shutdown (ESD) system and Fire and Gas System shall be as per the specifications attached elsewhere.

6.1.2 Hardwired Instruments shall be high-density electronic microprocessor based programmable type and suitable for mounting in multibins.

6.1.3 For all Intrinsically safe installations, all equipments, material and installations procedures shall comply with requirements of intrinsic safety loop as per applicable codes.

6.1.4 Controllers, indicators, etc., shall operate on voltage signal measured across precision calibrated resistors. Resistors shall remain in the loop when any of these devices are disconnected for servicing.

6.1.5 Intrinsic Safety Barriers shall be active isolating type for Process Platforms and bridge connected well platforms. Barriers shall be Passive type for unmanned Wellhead Platforms.

6.1.6 One 24 hour digital clock shall be furnished for each control room and TIC Room.

6.1.7 Vibration monitoring system for machines shall be designed, manufactured, installed and tested as per API 670 (both probes and vibration monitors shall be as per API 670 to avoid any compatibility issue)

6.2 Electronic Control Components

6.2.1 Control system constants such as gain, reset and rate shall be adjustable without removing modules from automatic control and shall have calibrated scales.

6.2.2 Anil-reset wind-up feature shall be included in all reset controllers.

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6.2.3 Electronic circuit modules shall be designed for short circuit modules shall be designed for short circuit protection.

6.2.4 Auto/manual changeover shall be procedure less/bump less Removal of control station for maintenance without system upset or disturbance should be possible.

7.0 TELEMETRY INTERFACE CABINET

Telemetry interface cabinet (TIC) shall be provided at Well and Process platforms for signals going to Remote telemetry unit (RTU). Telemetry interface cabinet shall be located in telemetry room or in process control room for interfacing all signals between the wellhead platform equipment and company supplied RTU/telemetry/telecontrol and data gathering equipments. All cabling between Instruments and TIC, TIC and RTU shall be in bidder's scope.

8.0 PNEUMATIC SHUTDOWN PANEL

8.1 The pneumatic shutdown panels, located outdoors on the Process platform, in addition to pressurising/controlling the fire shutdown (FSD), Emergency shutdown (ESD) and Fusible plug loops, shall coordinate all the pneumatic shutdowns logics as specified The FSD/ESD system is a pressurised system (3.5 Kg/ cm^2). Refer Specification for Shutdown Panel attached elsewhere in Bid Package.

9.0 PORTABLE CALIBRATORS

9.1 For testing, calibration, auto ranging of instruments, bidder shall provide portable user friendly advance calibrators (these shall be in addition to hand held configurators being provided for smart transmitters) for calibration of pressure, temperature, level instruments etc. at well platforms. These portable calibrators shall be supplied loose and calibrator shall be digital type with current source and shall be light weight. Detailed specifications for the same shall be generated by bidder.

10.0 INSTALLATION MATERIALS

10.1 Bidder shall ensure and supply all erection hardware required for the installation of complete instrumentation which form a part of the package unit. This includes items like cables, cable glands, junction boxes, instrument valves and manifolds, mounting accessories, impulse piping/tubing, pipe/tube fittings, pneumatic signal tubes, air line pipes and fittings, filter regulators, steam/electrical tracing, insulation materials, cable trays, conduits, prefabricated canopies for instruments, identification tags, structural material required for instrument supports and trays etc.

10.2 Clause 10.1 above broadly covers the items required for any typical plant, however bidder shall supply all necessary items to make the installation and commissioning work complete in all respect, irrespective of whether these have been explicitly included in their scope or not.

10.3 Salient feature and minimum requirement for some of the main installation materials is being described in the following paragraphs. For items for which no specification has been provided, bidder may follow their own specifications and prevailing international standards.

10.4 Instrument Cabling

10.4.1 **Construction**

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10.4.1.1 Signal and Alarm cables including RTD cables:

10.4.1.1.1 These Cables shall be Twisted single pair/triad cables rated for 600 V and made of 1.5 Sq. mm. XLPE covered, 7 stranded Copper conductor.

10.4.1.1.2 A pair/triad shall be twisted cores and number of twists shall be not less than 10 per meter. Colour of core insulation shall be black.

10.4.1.1.3 Individual pair and triad shall be shielded and shield shall be Aluminium backed by mylar/polyester tape with the metallic side down helically applied with either side 25% overlap of 100% coverage. Minimum shield thickness shall be 0.05 mm2.Drain wire shall be 0.5 sq. mm single or multistrand bare tinned annealed copper conductor. The drain wire shall be in continuous contact with aluminium side of the shield.

10.4.1.1.4 These shall have XLPE inner and outer jacket.

10.4.1.1.5 The cable shall have galvanised steel armouring.

10.4.1.1.6 Oxygen Index shall be minimum 30 and Temperature Index 250°C.

10.4.1.1.7 Mutual capacitance of the pair or adjacent cores shall not exceed 250 pF/meter at a frequency of 1 kHz. The capacitance between any core and screen shall not exceed 400 pF/meter at a frequency of 1 kHz.

10.4.1.1.8 A pair or triad identification shall be with numbers at interval of not more than 250 mm as per vendor's standard.

10.4.1.1.9 Ripcord shall also be provided.

10.4.1.1.10 The Cable shall be fire retardant as per standard IEC 60332-3 Part 3 Cat. A.

10.4.1.1.11 Jacket colour shall be Light Blue for Intrinsically Safe applications and Black for others.

10.4.1.1.12 Multipair cables shall have following additional features:

- Pair identification by colour coding/numbering.
- Individual pair shielding apart from overall shielding.
- One pair of 22AWG, 7 stranded copper wire with XLPE insulation, for communication.

10.4.2 **Control Cables**

These shall be same as signal cables except that individual pair shielding may not be required in case of multipair cables and primary insulation is required to meet the fire resistance. In addition these shall be listed by UL/FM or equal in fire service and shall be fire resistant as per IEC 60331 in addition to fire retardant as per standard IEC 60332-3 Part 3 Cat. A.

10.4.3 **Thermocouple Compensating Cables**

These cables shall be generally same as signal cables except conductor shall be solid 16 AWG in single pair and solid 20 AWG in case of multipair cables. Colour coding, thermoelectric characteristics and limits of errors shall be as per IEC-60584/ IS-8784.

10.4.4 **Special Cables**

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Cables for special instruments like analyser, turbine meter, Coriolis flow meter, F&G Detectors, serial interface etc. shall be as per individual manufacture's recommendation and specifications.

10.4.5 **Fire and Gas detection/protection Cable with Fire-rated Insulation**

This shall be generally same as signal cables excepting primary insulation. The primary insulation is to meet the fire resistance. But in addition these shall be listed by UL/FM or equal in fire service and shall be fire resistant as per IEC 60331 in addition to fire retardant as per standard IEC 60332-3 Part 3 Cat. A.

10.4.6 Power cables shall be 3 core, minimum 2.5 mm^2 copper conductors with PVC insulation, galvanised steel armouring and overall PVC sheathing. Power cables used in Fire and Gas system shall be Fire Resistant as per IEC-60331 in addition to fire retardant as per standard IEC 60332-3 Part 3 Cat. A.

10.4.7 All cables shall be terminated using double compression type glands on explosion proof equipment. These glands shall be explosion proof, certified by a statutory body. All cable glands in outdoor service and on equipment subjected to salt water spray shall be weather proof also.

10.4.8 Separate junction boxes shall be used for intrinsically safe wiring. Junction boxes in any cable path shall he limited to one only. Instrument junction boxes shall not be used for electrical cables.

10.4.9 Terminals for intrinsically safe cables shall be distinctly coloured (Light Blue).

10.5 Junction Boxes and Cable Glands

10.5.1 **Junction Boxes**

10.5.1.1 For Explosion proof instruments Junction boxes shall be certified weather proof as per NEMA-4X and explosion proof as per NEMA-7 for the area classification specified.

10.5.1.2 For intrinsic safe instruments Junction boxes shall be certified weather proof as per NEMA-4X and certified increased safe type for the specified area class.

10.5.1.3 Material of Construction for JBs shall be Copper Free Aluminium (Cu content < 0.4%) epoxy painted for explosion proof junction boxes and SS 316 for weatherproof, increased safe type junction boxes.

10.5.1.4 All terminals inside junction boxes shall be spring loaded, vibration proof, clip-on type mounted on nickel plated steel rail complete with end cover and clamps for each rows.

10.5.1.5 Junction boxes shall be provided with external earthing lugs.

10.5.1.6 All junction boxes shall be provided with telephone sockets and plugs for connection of hand powered telephone sets.

10.5.1.7 Spare entries of the JBs shall be provided with certified explosion proof plugs.

10.5.2 **Cable Glands**

10.5.2.1 Cable Glands shall be Double Compression Type.

10.5.2.2 Material of construction of cable glands shall be as follows:

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a) Nickel Plated brass for copper free Aluminium junction boxes.

b) 316 SS for 316 SS junction boxes

10.5.2.3 Type of protection for Cable Glands shall be Explosion proof and Weatherproof for explosion proof JBs and increased safety and weatherproof for Increased safe type JBs.

10.5.2.4 All cable glands shall be protected by heat shrink shrouds.

10.6 Tubes and Tube Fittings

10.6.1 All tubing and fittings shall be imperial sizes, expressed in nominal outside diameter (OD) and all threads shall be NPT.

10.6.2 The tubes shall be 316L Stainless Steel fully annealed, seamless and cold drawn as per ASTM A269.Tube fittings shall be of 316 SS to ASTM A269. For applications with high chloride contents in environment, composition of chromium/ Nickel/ Molybdenum in SS316L tube/ SS316 fittings shall be suitably increased to resist the environmental corrosion.

10.6.3 The minimum size for Tube and Tube fittings shall be ¼" OD.

10.6.4 Tubing runs shall be supported and protected.

10.6.5 Tube fittings shall be of double ferrule, pressure seat, no torque type for use on heavy or thin wall tubing, and shall be of reputable makes (such as Swagelok or Parker).

10.6.6 Ferrule and nut shall be of the same material as the fittings. Flare type fitting shall not be used.

10.6.7 Sizes shall be as follows:

S. No.	Application	Size
1	High-pressure hydraulic lines	¼ "O.D. X 0.035" W.T. 3/8" O.D. X 0.035" W.T. and 3/8" O.D. X 0.049" W.T.
2	ESD/FSD loop	3/8" O.D. X 0.035" W.T.
3	Pneumatic supply to shutdown valves	3/8" O.D. X 0.035" W.T. 316 SS tube
4	Process impulse lines	1/2" O.D X 0.065" W.T. 316 SS tube

10.6.8 Threaded connections shall be NPT for all components and piping and tubing systems for process and utilities connections. TFE threads sealant shall be used on all threaded connections. Tape shall not be used.

10.6.9 Ferrule and nut shall be of the same material as the fittings.

10.6.10 Hardness for tubes shall not exceed HRB 70 – 79 and bidder shall ensure hardness for fittings (ferrules) shall be such that, there is a minimum hardness difference of 5 to 10 between tube and fittings for better sealing.

10.6.11 **Tubing Testing Requirements**

Testing requirement for tube and tube fittings (Both Hydraulic and Pneumatic):

- The tests shall be according to ASTM A450.

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- The tests specified below shall be carried out on samples drawn at random.
- The test fluid for hydraulic testing shall be mineral oil having viscosity grade of ISO VG46 and ISO VG32.
- Tests to be carried out are:

10.6.11.1 **Proof Pressure Test**

Test assemblies (tubes and fittings) selected at random shall be subjected to a pressure of 1.5 times the maximum working pressure applied at the rate of 200 kg/cm^2 per minute and maintained at the final pressure for five minutes without leak.

10.6.11.2 **Dismantling and Re-assembly Test**

A minimum of three test assemblies that have successfully completed 'Proof Pressure Test' as above shall be disassembled and assembled ten times after which they must pass the 'Proof Pressure Test' again.

10.6.11.3 **Hydraulic Impulse Cum Vibration Test**

A minimum of three test assemblies that have successfully completed the 'Proof Pressure Test' shall be mounted on a hydraulic impulse cum vibration test bench. They shall be subjected simultaneously to pressure impulse of 1.5 times the maximum working pressure at 30 to 100 cycle per minute and vibration in two mutually perpendicular planes at 1300 to 2820 cycles per minutes for a minimum of 5 X 10 pressure impulses and 20 X 10 vibration cycles. The only retightening is after the first 1000 pressure impulses to allow for bending. During the testing the leakage shall not exceed $1cm^3/Hr/Assembly$.

10.6.11.4 **Vacuum Test**

A minimum of three test assemblies that have successfully completed the 'Proof Pressure Test' shall be subjected a negative pressure of 700m Bar and then disconnected from Vacuum Pump. The assembly must maintain the vacuum pressure for fifteen minutes.

10.7 Instrument Cable Trays

10.7.1 All cable trays and accessories shall be made of the fibre glass and shall be corrosion resistant and fire resistant (Class 1, Fire rating). The cable trays shall be designed and manufactured meeting the requirements of NEMA FG I, IS-6746 and ASTM E-84 standards. An additive material shall be mixed with the fibre glass to make them resistant to ultraviolet light.

10.7.2 The minimum glass content in FRP material shall be 45%.

10.7.3 The ladder type trays shall have side rails and horizontal rungs. Both of these shall be of same material i.e. the Fibre glass with additive. The rungs shall be placed 300 mm apart from each other.

10.7.4 The cable trays shall be supplied in standard lengths of 3000 mm and clear width of trays shall be 150,300,600 and 750 mm.

10.7.5 The inside depth of the cable trays shall not exceed 150 mm.

10.7.6 The tray construction shall have high degree of finish and shall be such as to facilitate easy handling and also to ensure easy laying of cables without causing damage to cables.

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The inside surface shall be perfectly smooth and free from sharp edges, burrs or projections, etc.

10.7.7 Each section of tray shall be complete with necessary connector plate and hardware. The hardware shall conform to NEMA FGI-198X and shall be able to withstand the loading as specified below. The bends, tees, reducers, crosses and droppers shall have required bending radii but not less than 300 mm.

10.7.8 For tray system design, in addition to self load, following guideline for design shall be considered:

a) Support Span 2000 mm.

b) Cable load for

i) 150 mm width 30 kg/Linear Metre.

ii) 300 mm width 60 kg/Linear Metre.

iii) 600 mm width 75 kg/Linear Metre.

iv) 750 mm width 75 kg/Linear Metre.

In addition to this, 70 kg concentrated load at centre span with a safety factor of 2 shall be considered.

10.7.9 The side runners and all accessories shall have the oblong holes on each end for fixing the connector plates. The connection between two sections of the cable trays or a section of cable tray with associated accessories shall be done by bolting only. Nuts, bolts and washers shall be of stainless steel SS 316.

10.7.10 Fibre glass type of trays shall be tested as per NEMA-FGI-198X. Following tests, as a minimum, should have been performed on each type of tray system:

a) Destruction Load Test
b) Deflection Test
c) Drop Test
d) Vertical burning Test

Type test certificates for the same shall be furnished.

10.7.11 All offered cable trays or those of similar designs manufactured by the same vendor:

- Shall have been type tested by an authority approved by the Purchaser.
- Shall have been in continuous satisfactory service in off-shore for a minimum period of two years.
- Shall be having current certification/approval/listing by an approved agency or UL or FM.

10.7.12 Bidder shall supply 20% hardware such as nuts, bolts, splice/connector plate for each type and size of cable trays and accessories. These shall be properly wrapped and packed separately, clearly marked as "spare hardware" for cable trays.

10.7.13 HDPE conduit shall be considered for Fibre Optic cables. These conduits shall be dedicated for Fibre Optic Cable.

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11.0 INSPECTION AND TESTING REQUIREMENT

11.1 Requirements of non-destructive testing like, radiography, magnetic particle test, hardness test, hydrotest for items such as control valves, safety valves, orifice flange assembly, level gauges etc., shall be carried out strictly as per respective piping class. For all instruments, welding and testing shall be carried out as per piping specifications. Post weld heat treatment shall be carried out for all weldings of instrument as required in accordance with the requirements of piping material specifications.

11.2 For items like control valves, shutdown valves, pressure relief valves, desuperheaters etc. radiography procedure and area of casting to be radiographed shall be as per ANSI B16.34 and acceptance criteria shall be as per ANSI B16.34 Annexure B, in general. However areas of casting to be radiographed for type of valves/desuperheaters not covered in ANSI B16.34 shall be in line with ANSI B16.34.

100 % radiography shall be done on the following areas of each item requiring radiography:-

- Inlet flange neck
- Outlet flange neck
- cast nozzle (in case of safety valves)
- All weld joints

Two shots shall be taken for each area to be radio graphed, as a minimum.

11.3 Following test shall be carried out by bidder at vendor's works and test certificate shall be furnished.

11.3.1 Material test certificate including NACE requirements for all instruments and items.

11.3.2 *Calibration/ test certificates for all instruments including control valves, safety valves.

11.3.3 *Seat leakage test for control valves and shutdown valves.

11.3.4 *Test certificate for safety valve set pressure and reseat pressure.

11.3.5 *Hydrotest certificate for all inline instruments, pipes, tubes, fittings etc.

11.3.6 *Radiography of control valve and pressure relief valves of rating ASME 600 Class or above.

11.3.7 Conformity certificate from sub-vendor.

11.3.8 Capacity certificate for control valves duly approved by third party.

11.3.9 NACE compliance certificate.

11.3.10 Fire Safe certificates for Shutdown valves

All tests prefixed by (*) shall be witnessed by owner/owner representative /third party. The above is minimum requirement. Inspection agency engaged for the job shall detail out inspection and detailed testing procedure for individual items after contract is awarded.

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11.4 Functional and simulation tests for all instruments supplied by the bidder shall be carried out at sub-vendor's works and witnessed by Purchaser/purchaser's representative's representative.

11.5 Analyser System

Each components and the system shall be tested separately at the vendor's works and the test certificates like material test certificate for all wetted parts shall be provided. Certificates from statutory bodies for use in area classification indicated in data sheet also to be provided to purchaser.

11.6 Specialty Service

Bidder shall engage specialty services from the sub-vendors for special instruments and systems. This shall include engagement of qualified engineers and technicians from vendors for following items for engineering, installation, field testing and commissioning.

- Compressor/Turbine/Pump/ filters/ chlorinators instrumentation
- DCS/ESD PLC
- F & G System
- Machine Monitoring system
- Pneumatic Shutdown Panel
- Multiphase Flow meters
- HIPPS (High Integrity Pressure Protection System)
- Non Contact Radar Level instruments
- Analysers
- Continuous Emission Monitoring System
- CCTV System

12.0 NAME PLATES

12.1 Name plates and identification tags shall be provided to properly identify each piece of instrumentation equipment.

12.2 All instruments shall have SS tag or nameplate of a permanent type with identification number and service description.

12.3 All panel instrument nameplates shall be constructed of white laminated plastic plates with black engraved lettering and securely fastened with 316 SS screws.

12.4 All front panel mounted equipment shall also be identified with a metal or plastic nameplate attached to the rear of the device, and easily visible via the rear access doors.

12.5 All wiring, cables, tubes, multitude bundles, junction boxes and auxiliary equipment shall be suitably identified as per applicable codes and practices Plastic adhesive tapes shall not be used for identification. Identification and details of all such equipment and wiring/tubing shall also be clearly marked on appropriate drawings. All wiring shall be tagged with slip on or clip-on wire marker at both ends, with the wire number specified on the drawings.

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ऑफशोर के लिए इन्स्ट्रूमेंट्स एवं कंट्रोल्स की स्थापना हेतु मानक विनिर्देश

STANDARD SPECIFICATION FOR INSTALLATION OF INSTRUMENTS AND CONTROLS FOR OFFSHORE

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
2	04.09.20	Revised and Re-issued	SD	PKS	RG	SM
1	26.06.15	Revised and Re-issued	SD	PKS	RG	SC
0	20.09.10	Issued as Standard Specification	SD	RG	TGM/JMS	ND

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Abbreviations:

DC	:	Direct Current
F&G	:	Fire and Gas
IP	:	Ingress Protection
IS	:	Intrinsically Safe
OD	:	Outer Diameter
RTD	:	Resistance Temperature Detector
SS	:	Stainless Steel
TIC	:	Telemetry Interface Cabinet
WT	:	Wall Thickness

Instrumentation Standards Committee

Convenor : Mr. Rajiv Gupta

Members : Mr. Arupjyoti Saikia
Mr. Sachin Garg
Mr. Jerry Johnson
Mr. Mayank Kaushik
Mr. Sourav Mukherjee
Ms. K. Pratheepa
Mr. R.K. Gupta (Proj.)

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1.0 GENERAL

1.1 This specification defines the minimum requirements for installation and testing of Instruments and Controls for Offshore.

1.2 The bidder shall carry out the work in accordance with the codes, standards & recommended practices listed in Instrumentation Design Basis/ Criteria, Instrumentation Job Specifications, Standard Specifications, this Specification, and any "Statutory Regulations".

2.0 APPLICABLE NATIONAL/ INTERNATIONAL STANDARDS

The installation of instruments and controls shall comply, as a minimum, with the latest editions of following codes and standards:

API American Petroleum Institute
- RP 14C Recommended Practice for Analysis, Design, Installation and Testing of Basic Surface Safety Systems for Offshore Production Platforms
- RP 14F Design, Installation and Maintenance of Electrical Systems for Fixed and Floating Offshore Petroleum facilities for unclassified and Class 1, Division 1 and Division 2 locations
- RP 14G Recommended Practice for Fire Prevention and control on Fixed open type Offshore Production Platforms
- RP 14FZ Recommended Practice for Design, Installation and Maintenance of Electrical Systems for Fixed & Floating Offshore Petroleum Facilities for unclassified and Class 1 Zone 0, Zone 1 & Zone 2 locations
- RP 552 Transmission Systems

IEC International Electro-Technical Commission
- 60079 Electrical Apparatus for Explosive Gas Atmosphere

ISA International Society of Automation
- S 7.0.01 Quality Standard for Instrument Air
- S12.13 Part I: Performance Requirements, Combustible Gas Detectors
  Part II: Installation, Operation and Maintenance of Combustible Gas Detectors
- S12.15 Part I: Performance Requirements, H_2S Gas Detectors
  Part II: Installation, Operation and Maintenance of H_2S Gas Detectors

NEMA National Electrical Manufacturer's Association

3.0 INSTRUMENT INSTALLATION REQUIREMENT

Bidder is responsible for installation of all items in their scope of supply including supply/ fabrication of all installation/ erection material. The scope of installation shall include instruments on platforms, Package item instrumentation, Pneumatic shutdown panels, control system instruments like Hardwired Relay Based Control Panels, Local Control Panels etc. as required to complete the project in all respects.

The bidder shall prepare hook-up and installation detail drawings based on indicative hook-up drawings regarding each type of instrument for Purchaser's approval and shall carry out the installation in accordance with these approved drawings.

The bidder shall install instruments and equipment with due consideration of the following:

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3.1 No instrument with the exception of pressure gauges and temperature indicators shall be installed in such a way that it depends for support on the impulse piping or electrical connections to it.

3.2 Positioning of equipment shall not constitute a safety hazard. Where possible, instruments shall be mounted so that they are protected from the effects of rain and sun, while maintaining the requirements for access and visibility. Bidder shall provide canopyfor all instruments (electronic type) and junction boxes located in open or under grating areato protect from rain and sun without impairing access or visibility.

3.3 Ease of access shall be provided for lifting heavy items or equipment such as valves, panels etc.

3.4 Instruments shall be mounted/ connected so as not to stress vessel nozzles or pipe tappings. Isolation valves for maintenance purpose shall be provided for standpipes.

3.5 Instruments shall be properly supported on brackets or mounted on sub-plates or placed on a suitable pedestal, pipe stand or structural support. Pipe or structural stands may be welded directly onto platform plate, with a suitable penetration in the grating, where applicable.

3.6 Instruments, tubing, cables and cable ladder shall not be fixed to gratings or handrails.

3.7 Fittings such as instrument isolating valves and instrument air or gas regulators shall be supported either on the instrument stand or close-coupled to the instrument in a manner that no undue stress is imposed on the tubing or instrument.

3.8 Separate local panels for Instrumentation and Electrical shall be considered as far as possible. Instrument stands or panels shall be in accordance with the approved drawings, with consideration for:

a) The direct routes for minimum tubing and piping to and from the stand, using common tubing runs and avoiding crossovers.

b) Ease of inter-connections between instruments.

c) Ease of access for on-site calibration and/ or removal of instruments.

d) Minimum interference between tubing, piping and cabling to instruments.

3.9 A filter regulator with gauge shall be provided for each instrument requiring regulated gas or air supply.

3.10 All open ports (vents/ drains) on field instruments including solenoid valves shall be made to face downwards and fitted with bug screens.

3.11 Local mounted instruments shall be mounted on brackets, sub panels or placed on a suitable pedestal. They shall be easily accessible from grade, ladder or platform. Pressure gauges and other local indicators shall be readable from grade or operating level and if used for manual control, shall be visible from the related valve.

3.12 Wherever possible, local mounted instruments shall be mounted at approximately 1.3 meters above the platform or floor in an accessible manner. Instruments shall not impede process operation, e.g., masking valves, etc.

3.13 Instruments shall be mounted as close to the sensing points as possible.

3.14 Orifice meters shall not be installed on the top of orifice fitting, where it is necessary for a person to stand above it to change an orifice plate. On horizontal lines tap holes shall be as follows:

a) On top for air and gas service.

b) Bottom for liquid

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c) Horizontal for condensable vapour service.

3.15 Instruments and impulse lines shall be protected against mechanical damage.

3.16 Fire and Gas Detectors elevation shall be decided based on whether hazardous gas/ vapour is lighter or heavier than air.

3.17 Manual Call Point shall be provided at all entrance and exit location of the Offshore platform including staircase.

3.18 Suitable access / platform shall be provided for all instruments/ field mounted panel as required to maintain or to remove instruments/ panel.

4.0 TUBING/PIPING INSTALLATION

4.1 The primary instrument block valves for all instruments shall be as per piping specifications. All the isolation and drain valves material shall be as per hook-up and installation detail drawings.

4.2 Unless otherwise specified in the drawings, impulse lines shall have a slope of not less than 1 in 12 on the horizontal runs. Direction of slope is to be downward from the process for liquid service and upward from the process for gas service. For High pressure applications tubing hookups shall be used.

4.3 Tubing/ Piping service water connection for CCTV camera wipers shall be provided unless specified that it is not required.

4.4 Tubing shall be joined by compression fittings. Piping shall be joined by pipe fitting and flanges as per piping specification.

4.5 Instrument lines shall run in the vertical plane as far as possible and shall run with the minimum number of changes of directions consistent with good practice and neat appearance.

4.6 Tubing shall be bent with correct size tubing bender when possible to avoid the use of fittings. Tubing cutter shall always be used to cut tubing. The use of short lengths of tubing in long runs shall be avoided, to avoid the use of fittings.

4.7 All tubing shall be run in such a manner as to give the maximum protection against mechanical damage. Tubing runs shall be grouped together and clamped where possible. Tubing shall be arranged so that couplings can be tightened without distorting lines. Instrument tubing shall not run on trays intended for cables and shall not share the same transit.

4.8 Tubing run shall not have joints.

4.9 Where tubing is run in permanent enclosures, the bidder shall ensure that entry and exit of such enclosures is clean and smooth.

4.10 Pipes and tubes installed but not connected, shall have the ends closed to prevent the entry of foreign material.

4.11 All reasonable precautions shall be taken to prevent foreign material entering piping before and during erection.

4.12 No pipe or tube shall be left with mechanical strain on it.

4.13 Where permanent enclosures are left with space for instrument tubing to be pulled in at a future date, a galvanized pull wire of adequate size shall be left in the tray.

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4.14 Piping and Tubing Supports

Piping and tubing shall be adequately supported and fixed at distances not exceeding those in the following table:

Single Tubing	Max. Distance between Supports
⅜" O.D. and less	Continuous
½" to <¾" Nominal Size	2 meters (6 feet)
≥¾"to 1" Nominal Size	3 meters (9 feet)
Multi-tube bundles	3 meters (9 feet)

4.15 All field mounted instrument air tubing shall be supported with galvanized steel angles or channels of minimum ⅛" thickness fabricated to present a neat appearance.

4.16 All instrument tubing supports shall be galvanized prior to installation.

4.17 All mechanical ferrule seater shall be used on tubing for 140 kg/cm2 (2000 psi) and over.

4.18 Pipe bushing shall not be used.

4.19 Plugs shall be of bar-stock with hex heads.

4.20 Signal transmission shall be through ¼" O.D X 0.035" WT tubing run singly or in bundles.

If extended lengths of multiple tubing are to be run, multiple bundles and bulk heads with weatherproof entries may be used to the best advantage.

4.21 Manufacturer or fabricator supplying skid-mounted equipment or vessels with instrumentation, which provides or utilizes outgoing skid alarms, shutdowns, or control functions shall tube signals to a central bulkhead. Central bulkhead shall be located near skid boundary, available for hook up by the bidder for connection to equipment outside skid. Location of the bulkhead shall be noted on bidder drawings.

4.22 Differential or static pressure sensing lines shall not exceed 6meters (20 feet) for direct connected or locally mounted instruments.

4.23 Tube unions, as required, shall be provided by bidder (though not indicated in hook-up standards).

5.0 INSTRUMENT AIR/GAS SUPPLY

5.1 Instrument air/gas main lines shall be in accordance with the piping specification of appropriate service and operation rating. Instruments, piping, tubing, fitting, valves etc. pertaining to Gas (containing H2S composition) application shall be NACE MR0175 compliant.

5.2 All take-off for branch lines are to be from the top of the main header with block valves equal in size to the branch line. All low points shall have a ½" valve installed as a drain and blowdown point. All take off lines including block valves from the header shall be 316 SS.

5.3 The branches to individual instruments shall be ⅜" O.D. X 0.035" WT tube terminating not more than 900 mm from the instrument. Tubing from the isolation valve to the instrument shall be ¼" or ⅜" O.D X 0.035"WT as required. .

5.4 Filter regulator shall be provided for individual air/gas supplies to field mounted instruments and shall be complete with an output gauge. Pneumatic pressure switches, temperature switches and level switches shall be provided with filter regulators to provide control signal pressure to panels.

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5.5 Manufacturer or fabricator supplying "skidded" equipment or vessels which incorporate instrumentation requiring pneumatic supply shall install a supply header, of the proper size, on the skid. The non-skid supply piping shall terminate at skid boundary and the location and size of the supply connection shall be noted on bidder drawings.

6.0 INSTRUMENT CABLE INSTALLATION

6.1 All instrument signal cables shall be continuously screened, with screens grounded at the equipment room end only and securely insulated at the field end. All cables exiting equipment rooms shall be armoured, including Optical Fibre, serial link and data highway cables. Optical Fibre Cable for subsea application shall be triple armoured type.

6.2 A neat loop of cable 250 mm diameter shall be left immediately adjacent to all instrument devices in the field.

6.3 Cabling to locally mounted instruments shall not be routed across normal or emergency walkways nor interfere with access to equipment.

6.4 Multicable transits shall be used to provide fire resistant, water tight, gas-tight cable penetrations through firewalls, control rooms, deck penetrations, TIC room and other rooms in Offshore platforms.

6.5 All instrumentation cables shall be supported in trays. Power cables shall not be run in trays meant for signal cables. The physical separation of power and signal cables shall be as per API RP 552. Cables in intrinsically safe circuits shall not run in the tray for cables in non-intrinsically safe circuits. If run in the same tray, a metallic earthed separator shall be provided. Conduits carrying intrinsically safe cables shall be painted with the blue color bands at each end. Armoured Optical fibre cable shall be laid in dedicated solid tray with cover.

6.6 Splices shall be made only at terminals, in instruments or approved equipment/junction boxes in any cable path and shall be limited to one only.

6.7 All wiring, cables, tubes, multitube bundles, junction boxes and auxiliary equipment shall be suitably identified as per applicable codes and practices. Plastic adhesive tapes shall not be used for identification. Identification and details of all such equipment and wiring/tubing shall also be clearly marked on appropriate drawings. All wiring shall be tagged with slip on or clip-on wire marker at both ends with the wire number specified on the drawings.

6.8 Separate junction boxes shall be used for the following type of signals:

a. 4 - 20 mA DC Intrinsically safe
b. 4 - 20 mA DC Non-Intrinsically safe
c. Contacts
d. RTDs
e. Solenoid valves
f. F&G Detectors
g. Special instruments such as analysers, ultrasonic flow meters as applicable
h. Vibration probes
i. Proximity limit switches

Instrument junction boxes shall not be used for electrical cables.

6.9 Only one Junction box shall be used in the complete cable run from field upto control room.

6.10 For multipair/ multi triad/ multicore cables, maximum of upto 12 pair/8 triad/ 24 core cables shall be used.

6.11 Terminals for intrinsically safe cables shall be distinctly coloured (light blue). F&G cables shall have orange outer sheath colour.

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6.12 Cable Termination

6.12.1 All cables shall be terminated using double compression type cable glands. All cable glands shall be weather proof to IP 65 and flameproof to Ex d for flameproof instruments and increased safety for intrinsically safe instruments in hazardous area. All terminations shall be screw clamp type terminals for 2.5 mm^2 conductors minimum (with no flying leads) and properly certified where necessary. Insulated crimp lugs shall be used for all cable core connections, with only one conductor per terminal side.

6.12.2 No wire shall be terminated/left with mechanical strain on it in any conduit or trunking.

6.12.3 Cables shall be terminated in 316 SS glands for intrinsically safe instruments and Nickel Plated brass for flame proof instruments suitably certified. The inner sheath shall be left on the cable after the gland and removed at the point of entry into wiring duct.

6.12.4 Cable screens and drain wires shall be securely insulated at the final field termination.

6.12.5 The communications conductor of a cable shall be terminated in the bottom terminal(s) of arrow of terminals.

6.12.6 Printed sleeve-type ferrules shall be fitted to all cables on both sides. Cross ferruling philosophy shall be followed for termination of cables.

6.12.7 Only one conductor shall be used per terminal side.

6.12.8 Terminal blocks shall be non-hygroscopic, vibration proof, stack on type and shall use captive screws for terminals. Terminals shall be tinned or silver plated. All terminations shall be through use of wire lugs. Terminal blocks shall be flame retardant type.

6.12.9 **Cable Termination for Skid Items**

Manufacturer or fabricator supplying skid mounted equipment or vessels with instrumentation which provides or utilizes off-skid alarms/ shutdowns/ control functions shall terminate signals on junction boxes, near skid boundary, available for hook-up by bidder. The location of these junction boxes shall be noted on bidder drawings.

7.0 MARINE ENVIRONMENT SUITABLITY AND PAINTING

SS 316 enclosure shall be used as far as possible. In case Stainless Steel enclosure is not available for any instrument items, Copper free (less than 0.4 % Cu) Aluminum enclosure with epoxy coating shall be provided.

All painting shall be in accordance with Protective coating Specifications attached in the Bid package.

8.0 PROTECTION

8.1 The bidder shall cover all instruments with polythene bags to ensure that the instruments are adequately protected from the weather. The bags shall be kept in place at all times except during work on the device they protect, until they are removed for start-up. Drying agent with humidity indicator shall be put inside the polythene bag and it shall be replaced when color changes from blue to pink.

8.2 The bidder shall protect all instruments, tubing and piping, during construction or while in storage from physical damage, contamination by dirt, sand or any other foreign material.

9.0 IDENTIFICATION OF LINES / INSTRUMENTS

9.1 All site-mounted instruments, junction boxes, air/gas headers, tubing, and wiring terminations shall be labeled or tagged in a manner approved by the Purchaser. Instruments shall be furnished with stainless steel name tags containing tag number, manufacturer's name, model number and serial number. This tag shall be approximately 3" x 1" and shall be attached to the instrument with 16 gauge stainless steel wire.

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10.0 TESTING AND CALIBRATION

10.1 The bidder's quality plan shall include a comprehensive fully documented inspection and testing plan specific to the project.

10.2 The procedures shall include inspection specifically for compliance with hazardous area requirements, including current certificate, without which no circuit or loop shall be energized.

10.3 After calibration at factory/ vendor's shop, recalibration of all instruments shall be carried out at platform / yard/ barge to prevent drift in accuracy over a period of time. All testing, calibration and pre-commissioning shall be done by the bidder. The bidder shall also provide assistance as required during commissioning activities.

10.4 Purchaser reserves the right to reject any or all test and calibration work if found not complying with the Specification requirement. The bidder shall complete and submit documentation for all calibration, testing and pre-commissioning. Purchaser/ Representative prior to shipment shall check out panels, consoles and packaged instrument assemblies for their compliance with specification requirements.

10.5 Instrument air/gas piping and tubing shall be disconnected upstream of all filter/regulators and blown down to remove water, slag and mill scale from lines.

10.6 All process impulse lines shall be disconnected and flushed with potable water. Air lines shall be blown down with filtered air. Hydraulic lines shall be flushed with hydraulic oil.

10.7 After flushing, process impulse lines shall be isolated from the instrument and pressurized hydraulically to 1½ times maximum working pressure corrected for ambient temperature. They shall then be isolated from the pressure source and the pressure reading on a test gauge shall not fall at a rate exceeding one psig/hour.

For direct mounted instruments such as level gauges, level transmitters (displacer type), level switches etc., the installations shall be pressurized to maximum operating pressure slowly and steadily with the instruments. The installations shall then be isolated from main pressure source. The pressure shall not fall at a rate exceeding one psig/hr.

10.8 Instrument air/gas tubing and piping shall be hydrostatically tested.

10.9 Instrument air supply lines shall be blown with instrument air prior to connecting to instruments. Instrument air/gas mains shall be isolated from the instrument and pressurized to 1½ times maximum working pressure with instrument air, they shall be isolated from the pressure source and the pressure reading on a test gauge shall not fall by more than one psig in ten minutes.

10.10 Instrument signal lines shall be blown with instrument air prior to connecting to instruments. All air/gas tubing shall be tested and inspected by one of the methods given in ANSI/ISAS 7.0.01. Clean, oil free instrument air shall be used for the test.

10.11 Wiring shall be checked to ensure that it is correctly connected and properly grounded. Insulation test shall be carried out on all wiring taking necessary precautions.

10.12 Correct connections of all electric or pneumatic switches shall be checked.

10.13 The bidder shall provide written results of all above tests and if required by the purchaser, provide reasonable evidence of the satisfactory condition of test equipment.

10.14 All errors of faulty workmanship discovered during this testing shall be corrected to the satisfaction of purchaser.

10.15 Equipment shall be calibrated by the bidder's instrument personnel. This calibration shall, when possible, be done with the instrument or system in place, otherwise calibration prior to installation or removal for calibration shall be done. The bidder shall provide written

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results of all instrument calibration in prescribed format. The status/progress of testing calibration for each loop shall be reported to the purchaser.

In general, all tests shall simulate, as closely as possible, design process conditions by use of manometers, potentiometers, deadweight testers, test pressure gauges, etc. utilizing hydraulic and pneumatic supplies. Three (3) point calibration shall refer to the input signal to an instrument equivalent to 0, 25, 75 and 100 per cent of the instrument range upscale (rising), and 75,25 and 0 per cent of the instrument range downscale (falling). All instruments, unless otherwise noted, shall be calibrated in upscale and downscale directions and, if necessary, adjusted until their accuracies conform to those limits stated by the manufacturer. Upon completion of these tests, the instruments shall be drained, the components removed and the shipping stops replaced.

10.16 Requirements of Calibration/ Testing for Individual Instruments

10.16.1 Temperature Instruments

a) Transmitters and controllers shall be three (3) point bench-calibrated using a temperature bath prior to installation.

b) Bi-metallic dial thermometers shall be three (3) point bench-checked for proper operation and calibration using a temperature bath prior to installation

10.16.2 Pressure Instruments

a) Pressure and differential pressures transmitters, switches, and transducers shall be three (3) point calibrated using a hydraulic or dead-weight tester or precision pneumatic calibrator prior to installation. A precision output meter, gauge or manometer shall be used to monitor the output.

b) Direct connected bourdon-type pressure gauges shall be dead-weight tested or tested against a test gauge prior to installation.

c) Receiver type pressure gauges shall be three (3) point calibrated using a manometer or precision pressure gauge and precision air regulator.

10.16.3 Flow Instruments

a) Differential pressure type flow instruments shall be calibrated against a manometer or precision pneumatic calibrator at three (3) points, prior to installation. A precision output meter, gauge or manometer shall be used to monitor the output of transmitter.

b) Rotameters shall be installed as received. A check shall be made to confirm that shipping stops have been removed. Rotameters have transmitting mechanism, the float shall be raised and lowered mechanically and the output shall be checked. A check shall be conducted with a level for a plumb installation. Shipping stops shall be installed prior to shipping of module to the installation site.

c) Turbine meters and/or positive displacement meters shall be installed as received.

d) Orifice plates shall be checked visually for nameplate data and for an upstream sharp edge. Bore shall be calipered to check for compliance with specifications.

10.16.4. Level Instruments

a) Displacer type instruments shall be checked by raising and lowering mechanically the displacer and checking the pilot or transmitter action. Check transmitters with output gauges or meters for smooth and full output range. Check operation of controllers and set actions and responses. Reinstall shipping stops prior to shipping of module to the installation site.

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b) Differential pressure type level instruments shall be calibrated against a manometer or precision pneumatic calibrator at three (3) points prior to installation. A precision output meter, gauge or manometer shall be used to monitor the output of transmitters.

10.16.5 Control Valves, Shutdown Valves and Regulators

a) All diaphragm and piston-operated control valves shall be stroked pneumatically using a pressure regulator and pressure gauge against the spring range shown on the nameplate. Mechanical seating and travel shall be checked against the name plate. This check shall be made in the shop prior to installation.

b) Valve positions shall be calibrated on the control valve in accordance with the nameplate data and instrument specifications. Split-range or reverse acting positioners shall not have bypasses and shall be checked carefully. Zero position shall be a "live" zero (just off the seat at minimum setting, on the seat with no air).

c) Volume bottles, where used, shall be checked for proper filling. The signal line shall be bled to zero and failure action confirmed.

d) Control valve accessories such a hand wheels, boosters, relays, etc., shall be checked operationally. Declutchable handwheels shall be operable both with and without an air signal to the diaphragm.

e) All control valves shall be checked carefully against nameplate data and specification.

f) Self-contained pressure regulators shall be installed as received, checking inlet and outlet ports and nameplate data. Regulators with external pressure connections shall be inspected for proper installation.

g) Butterfly valves shall be checked carefully to see that the vane moves freely into the upstream and downstream piping. Proper vane movement to stroke shall be confirmed. Multi-position actuators shall be checked for proper cross tubing and/or piping.

h) All control valves and regulators shall be removed from the line prior to flushing during hydrotest.

i) Partial Stroke testing facility shall be provided for all Shutdown Valves at Offshore and same shall be tested for PST.

10.16.6 Pressure Relief Valves

a) Pressure relief valves shall be installed after carefully checking nameplate details against specifications and then testing/setting for set pressure, blow down and seat tightness.

b) Compressed air or nitrogen shall be used for testing pressure relief valves and shall be furnished by the bidder.

10.16.7 Switches and Electrical Devices

a) Level switches, float or displacer-operated, shall be actuated mechanically for switch operation.

b) Pressure switches shall be calibrated using a hydraulic or dead weight tester or precision air regulator and output gauge and set to required alarm and/or trip point.

c) Temperature switches shall be calibrated using a temperature bath prior to installation and set to the required alarm and/or trip point.

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d) Pneumatic relays, timers, push-buttons, switches and other miscellaneous pneumatic devices shall be checked for proper operation.

e) Annunciators shall be checked for lamp operation using dummy signals and for proper engraving.

10.16.8 Analytical Instruments and Miscellaneous

a) Calibrate and check completely all analyzers and sampling system, using gas or liquid samples when furnished.

b) Package instrument systems such as compressors, heaters, special pumps, etc. shall be checked completely with special care given to calibration and set pressure of switches, etc. to Package Vendor's specifications.

10.16.9 Other miscellaneous devices shall be checked against specifications and for proper operation.

10.16.10 Commissioning assistance shall be provided by OEM for all sophisticated and special field items such as Multiphase Flow Meter, Water Cut Meter, Radar Level Instruments, CCTV systems, Ultrasonic Flow Meter, Shutdown panel, Special Analyzers etc.

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TECHNICAL COMPLIANCE (INSTRUMENTATION & CONTROL)

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
0	29.09.2025	ISSUED FOR TENDER	RN	SKA	RK
A	03.09.2025	ISSUED FOR COMMENTS	RN	SKA	RK

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Sl. No.	COMPLIANCE STATEMENT/QUERY	BIDDER'S CONFIRMATION/ ANSWER
	Please note that it is mandatory to furnish replies to the technical queries listed below, without which your offer will be considered incomplete.	
1.	Bidder to submit this Technical Questionnaire with categorical confirmation against each of the clauses. No deviation of any type is acceptable. Bidder to confirm compliance with all provisions of bid package which shall be deemed to be included in the scope of bidder's.	
2.	Confirm categorical compliance to the Scope of Work & Supply for Instrumentation & control (B774-000-16-51-SOW-1001) specified in the bid without any deviations of any kind.	
3.	Confirm categorical compliance to various other Instrumentation & Control documents attached in the bid package, and all other equipment, materials and work not explicitly mentioned but nevertheless required to fulfill the functional requirements with no additional cost and time implication to the owner.	

NOTES:

1. Bidder shall indicate his reply in the space provided in the Technical Questionnaire. In case space provided is not adequate, the reply may be furnished separately under suitably numbered annexes/ attachments duly referred against the comment/ query.

2. The Compliance Statement is required to be categorically confirmed / answered by the bidder and the completely filled in compliance shall be submitted together with the Bid.

Bidder's Signature & Stamp	

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LOGISTICS SUPPORT FORMAT

(INSTRUMENTATION & CONTROL)

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
0	29.09.2025	ISSUED FOR TENDER	RN	SKA	RK
A	25.08.2025	ISSUED FOR COMMENTS	RN	SKA	RK

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Logistic Support Certificate

1. Vendor must furnish Certificate for providing necessary support services for all Instrumentation and Control system listed as part of Instrumentation scope of work (Doc no. B774-000-16-51-SOW-1001) in favour of the Owner (M/s ONGC) committing themselves to provide logistic support and performance guarantee to Purchaser in the format furnished.

2. The Certificate must be signed with seal by the official signatory on the company's letter head.

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Certificate for logistics support for Various Instrumentation and control system

A) (To be signed by Principal's corporate level signatory on company's letterhead)

I, on behalf of M/s_____confirm that the____*_____Model No. _____ quoted by M/s _____ for Bridge replacement project of ONGC shall continue to be supported by us. The quoted item shall not be withdrawn from Indian market as a matter of our corporate policy.
I further confirm that in case of placement of order by ONGC on M/s _____, we shall continue to support M/s_____in providing back-up engineering, maintenance support and spare part to ONGC for a period of 10 years from the date of placement of order.

SIGNATURE WITH SEAL
AUTHORIZED, SENIOR MANAGEMENT LEVEL

B) (To be signed by Vendor's corporate level signatory on company's letterhead)

I, on behalf of M/s_____ confirm that the____*_____Model No. _____ quoted by M/s _____for Bridge replacement project of ONGC shall continue to be supported by us and our__*____Principal's.
The quoted item shall not be withdrawn from Indian market as a matter of our corporate policy as supported by attached certificate from our Principal's M/s._____.

I further confirm that in case of placement of order by ONGC on M/s _____, we shall continue to support M/s_____ in providing back-up engineering, maintenance support and spare part to ONGC for a period of 10 years from the date of placement of order.

SIGNATURE WITH SEAL
AUTHORIZED, SENIOR MANAGEMENT LEVEL

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MANDATORY SPARE PARTS
(INSTRUMENTATION & CONTROL)

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
0	29.09.2025	ISSUED FOR TENDER	RN	SKA	RK
A	03.09.2025	ISSUED FOR COMMENTS	RN	SKA	RK

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Mandatory Spares List
(To be supplied loose)

General

1. This section defines the Philosophy for mandatory spares to be supplied by the bidder as part of the contract.

2. The list below indicates the Minimum requirement for mandatory spares for Instruments / Accessories and systems. However additional mandatory spares if and as recommended by Instrument / System Manufacturer shall be supplied by bidder without any cost and time implication.

3. All spares shall be marked with the manufacturer's part number, name and / or reference number.

4. All spares shall comply with the same standards and specifications as the original equipment and shall be fully interchangeable with original parts.

5. All spares shall be packed separately from the main instrument, and shall be properly protected to prevent deterioration and damage during shipment and storage.

6. Bidder shall also refer notes given at the end of this section for working out mandatory spares requirement.

7. Mandatory spares shall be as per the following table for the applicable items.

8. The spares mentioned below is with respect to each particular platform.

S. No.	PART DESCRIPTION	QUANTITY REQUIRED
1	**Field Instruments**	
1.1	Fire & Gas devices including detectors, hooter, beacon, ESD/ FSD Station, Manual Call points etc.	10% subject to minimum 1 No. of each type
1.2	Pressure Gauges, Temperature Gauges, Differential Pressure Gauges, Draft Gauges, Gauge Field Indicators, RTD / Thermocouple (with Thermowell) monitors and local indicators (As applicable).	10% subject to minimum 1 No. of each type
1.3	Smart Transmitters for Pressure, Level, Flow, Temperature & Differential Pressure(As applicable).	10% subject to minimum 1 No. of each type
2	**Manual Control Stations / Local Panels**	
2.1	All items like push buttons, hand switches, lamps, holders etc. mounted in the local control station/ local panels.	10% subject to minimum 1 No. of each type
4	**Installation Material**	
4.1	Junction Boxes	10% subject to minimum one number, whichever is higher, of each type

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S. No.	PART DESCRIPTION	QUANTITY REQUIRED
4.2	Instrument Valves and Manifolds	10% subject to minimum one number, whichever is higher, of each type
4.3	Pipe/ Tube fittings	10% subject to minimum one number, whichever is higher, of each type
4.4	Pipe/ Tubes	10% of length of each type
4.5	Cables	10% of length of each type
4.6	Cable Glands	10% subject to minimum one number, whichever is higher, of each type
4.7	Instrument airline fittings, valves	10% subject to minimum one number, whichever is higher, of each type
4.8	Cable Trays/ Duct	10% of length of each type
4.9	MCT Blocks for Buildings/ Cabinets/Panels (As applicable).	10% subject to minimum one number, whichever is higher, of each type
5	**Line Mounted Instruments**	
5.1	Air filter Regulator, Solenoid valve, Smart Positioner, Volume booster, Lock Relay (As applicable).	10% subject to minimum one number, whichever is higher, of each type
5.2	Control Valves / Shutdown Valves / Safety valve / Selg actuated Regulator / Rotameter internals including Gland packing, Bonnet Gasket, Plug, Seat, Piston O ring, Seal Ring, Bearing & Liner, Disc, Bellow, Spring, Gasket set, diaphragm, Float, Glass tube (As applicable).	10% subject to minimum one number, whichever is higher, of each type
6	CCTV Camera	10% subject to minimum one number, whichever is higher, of Cameras
7	Startup and Commissioning spares	As required and as recommended by bidder / vendor shall be supplied by the Bidder
8	Consumables	For Instrumentation and controls for at least one year operation.

NOTES:

1. The word ‘TYPE’ means the Make, Model no., Type, Range, Size/ Length, Rating, Material as applicable.

2. Wherever % age is identified, Contractor shall supply next rounded figure.

3. The terminology used under ‘Part Description’ is the commonly used name of the part and may vary from manufacturer to manufacturer.

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4. Commissioning spares are part of contractor scope of supply. Mandatory spares as indicated above do not cover commissioning spares.

5. Mandatory spares shall be applicable for instrumentation items of sub-packages (as applicable) as per mandatory spares philosophy.

6. Spares for instruments shall be supplied along with applicable accessories as mandatory spares. No accessories shall be supplied separately as mandatory spares.

7. Any special tools or test equipment needed shall also be supplied by the Bidder.

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SCOPE OF WORK & SUPPLY (INSTRUMENTATION & CONTROL)

PROJECT : ENGINEERING / TECHNICAL CONSULTANCY SERVICES FOR PREPARATION OF EXECUTION METHODOLOGY, SOW AND COST ESTIMATES FOR REPLACEMENT / REFURBISHMENT OF BRIDGES IN MH ASSET

LOCATION: MH ASSET (NORTH AND SOUTH FIELD)

OWNER : ONGC, MUMBAI

JOB NO : B774

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
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DEFINITIONS AND ABBREVIATIONS

BIS	Bureau of Indian Standards
CCTV	Closed Circuit Television
EIL	Engineers India Limited
EU	European Union
ESD	Emergency Shutdown
FAT	Factory Acceptance Test
F&G	Fire & Gas
FRP	Fiber-Reinforced Polymer
IEC / IECEx	International Electrotechnical Commission / IEC Certification to Standards Relating to Equipment for Use in Explosive Atmospheres
ISA	International Society of Automation
LCP	Local Control Panel
MCP	Manual Call Point
MCT	Multi Cable Transit
OEM	Original Equipment Manufacturer
ONGC	Oil and Natural Gas Corporation
Owner	ONGC
PLC	Programmable Logic Controller
SAT	Site Acceptance Test
TPI	Third Party Inspection

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1.0 INTRODUCTION

This document describes the requirements of Instrumentation & controls, necessary for replacement / refurbishment of bridges envisaged in this project including all hook-ups and tie-ins for the associated work. This document covers the minimum requirements of design of Instrumentation and controls , engineering, procurement, supply, calibration, installation, hook-up, testing, loop checking, logic implementation & verification, integration and commissioning of instrumentation and controls envisaged for this project as per specifications / documents in the bid package.

2.0 CODES & STANDARDS

Design, terminology and selection shall comply, as a minimum, with the latest editions of codes and standards, recommended practices and publications as specified in this document & respective requisition / tender.

All instruments and equipment required shall be suitable for use in specified site climatic conditions and industrial environment in which corrosive gases and / or chemicals may be present.

The design of electronic instruments shall be in compliance with the electromagnetic compatibility requirements as per codes & standards.

Equipment /instrument/systems located in the hazardous area shall be suitable for the area classification and shall be certified as per following clauses. Instruments used in hazardous area shall generally be intrinsically safe. In case of non-availability of Intrinsically safe instruments, flameproof enclosure can also be considered. A copy of approval for Intrinsically safe / Flameproof enclosure as referred below shall be provided:

i) For all electrical items manufactured in India, the testing shall be carried out & test report issued by Indian test laboratory recognized by the Chief Controller of Explosives and valid copy of BIS license (only for Ex Electric Apparatus having flameproof type protection) or IECEx Certificate of Conformity, valid copy of IECEx Quality Assessment Report (summary) and IECEx Test Report.

ii) For all electrical items manufactured outside India, Test report issued by Indian test laboratory recognized by the Chief Controller of Explosives and valid copy of BIS license or IECEx Certificate of Conformity, valid copy of IECEx Quality Assessment Report (summary) and IECEx Test Report or EU Type Examination Certificate (Applicable for manufacturers having manufacturing locations in countries covered under the European Union), valid copy of Production Quality Assessment / Assurance Notification

3.0 GENERAL

a. This document covers details of Scope of Work, installation, Inspection, Commissioning, FAT, SAT, Trial run etc. pertaining to Instrumentation & Control system components necessary for replacement / refurbishment of bridges envisaged in this project including all hook-ups and tie-ins for the associated work.

b. This scope of work shall be read in conjunction with Standard specification and other Documents / Drawings which are part of this Tender.

c. The description and requirements contained in this document are concise by necessity and cannot include all details. However, it is the responsibility of the bidder to execute the job on totality in accordance with the tender requirement and internationally recognized

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good engineering practices for smooth and successful operation of various Instrumentation & control system items.

d. Any activity specifically not listed in this document, does not absolve the bidder of their responsibility to include such activities in their scope of work and supply, which otherwise is necessary, to complete Instrumentation & Control system item as per tender.

e. All Instrumentation & control system items shall be supplied from Owner approved vendors. Items for which vendor list is not specifically identified / provided, Bidder may propose vendor along with suitable Proven Track Record (PTR) for Owner review and shall proceed only after getting necessary approval from Owner.

f. The fieldwork shall be carried out under strict safety regulations and work permits as laid down by Owner.

g. The ferruling for cables shall be done with tag number of instruments for individual pair. Cross ferruling shall be done for cable ferruling in field and in the control room as per the instruction of the Owner.

4.0 SCOPE OF WORK / SUPPLY

This Scope of Work shall be read in conjunction with the various specifications, standards, documents, drawings and other requirements enclosed in the bid package.

Bidder's scope of work for Instrumentation and Control System include replacement / reuse / relocate / reroute / dismantle/ re-installation of the existing instrumentation and control system items including cables / trays / junction boxes etc. due to the Replacement of bridges / walkways, new bridges / walkways, refurbishment / strengthening of bridges / walkways & other equipment / items (as defined elsewhere in the bid) as a minimum but not limited to the following:

The bridge / walkway details with respect to new / replacement / strengthening shall be defined elsewhere in the tender.

Sr. No.	Description	By Bidder	By Owner
1.0	**Overall Scope of work**		
1.1	Pre-bid site survey, Design, engineering, procurement, Inspection / testing / calibration, Integration, Training, Factory Acceptance Testing, Supply & Transportation.	YES	
1.2	Site Calibration / testing	YES	
1.3	Site Installation	YES	
1.4	Site Acceptance Testing	YES	
1.5	Commissioning	YES (Note-22)	
1.6	Trial Run	YES	

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1.7	Obtaining all statutory clearances, from respective statutory Govt. authorities / organizations and maintaining liaison as required for implementation of the project.	YES	
2.0	**Detailed Scope of Work**		
2.1	Basic Engineering Activities	YES	
2.1.1	Pre-bid site survey to assess the actual site conditions / requirements, collecting existing data, ascertain the exact quantum of work etc.	YES (Note-1)	
2.1.2	Basic design, sizing, selection of Instrumentation components (Type, material, range etc.), preparation of specifications / data sheets, preparation of engineering drawings and documents (as per contractor data requirement).	YES	
2.2	Instrumentation & Control system requirement pertaining to replaced BHS-SLQ bridge / walkway, New ICW-ICG bridge / walkway & New ICP-ICG bridge / walkway.	YES	
2.2.1	Collecting details of existing cables / tubes / trays / junction boxes / CCTV / F&G Detectors & other F&G devices (Including ESD / Beacon / Hooter / MCP) etc.	YES (Note-2)	
2.2.2	Removal / dismantling of existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / & other F&G devices (Including ESD / Beacon / Hooter / MCP) etc. & handing over the same to owner.	YES (Note-3)	
2.2.3	Supply of all new & replaced existing cables / tubes / trays / junction boxes / CCTV / F&G Detector /& other F&G devices (Including ESD / Beacon / Hooter / MCP) etc. with one to one new.	YES (Note-4,5,6,7,23,24)	
2.2.4	Relocating & rerouting of cables / tubes / trays / junction boxes / CCTV / F&G Detector / /& other F&G devices (Including ESD / Beacon / Hooter / MCP) etc.	YES (Note-5,23)	

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2.3	Instrumentation & Control system requirement pertaining to refurbishment / strengthening of Bridges / walkways / other equipment & items.	YES	
2.3.1	Collecting details of existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc.	YES (Note-2)	
2.3.2	Removal / dismantling of existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc.& handing over the same to owner.	YES (Note-3)	
2.3.3	Supply of all replaced existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc. with one to one new.	YES (Note-7,8,24)	
2.3.4	Relocating & rerouting of existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc.	YES	
2.4	Testing and Inspection of Bidder supplied Items.	YES	
2.4.1	Testing and acceptance as per requisition as applicable for the bidder supplied items; a) Radiography. b) Hydraulic testing. c) Dye penetration / magnetic particle test. d) Megger testing. e) Fire safe testing. f) Fugitive emission testing.	YES	
2.4.2	Inspection for bidder supplied items.	YES	
2.5	Installation of all bidder supplied cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD /	YES	

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	Beacon / Hooter / MCP) etc. and making it operational.		
2.6	Installation of Relocated & rerouted existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc. and making it operational.	YES	
2.7	Power supply distribution, as required for bidder supplied items & other modified system / items.	YES	
2.8	Earthing for bidder supplied control room mounted items & field mounted items and other modified system / items.	YES	
2.9	Field testing, calibration, loop checking for bidder supplied items & other modified system / items.	YES	
3.0	Supervision of installation, erection, field testing, calibration, & loop checking activities for bidder supplied items & other modified system / items.	YES	
3.1	Site acceptance testing for bidder supplied items and other modified system / items.	YES	
3.2	Co-ordination activities; a. Coordination with owner for implementation of interface with existing system including serial interface configuration etc. b. Coordination with owner / owner representative during engineering, procurement, factory testing, site activities and site acceptance testing. c. Coordination with various sub-vendors / package vendor / manufacturers/ Existing Instrumentation & control system OEM etc. associated with bidder supplies and services. d. Coordination with suppliers of special instruments / instrumentation items, control system, telemetry system etc. for availability of their specialist during start up and commissioning.	YES YES YES YES	
3.3	Documentation as per Vendor data requirement attached in the requisition.	YES	

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3.3	Spare parts as per scope of work.	YES	
3.4	If any other items supply not included but required to complete the scope and other documents attached in the tender.	YES	
3.5	Warranty for bidder supplied Instrumentation items.	YES	

Note-1: Bidder shall carry out pre-bid site survey on the existing bridges / walkways, fallen bridges / walkways & other equipment / Items (as defined elsewhere in the bid) to ascertain the exact quantum of work to ascertain the extent of modifications envisaged. Bidder shall collect required details from platforms / Bridges / Walkways for proceeding with detailed engineering and also prepare a survey report and obtain owner's approval before commencement of engineering works. Any additional scope of work & supply as identified during site survey / engineering shall be carried out by Bidder without any implications for safe implementation of the project.

Note-2: Contractor thoroughly must check information collected from platform for correctness and update it to as-built status (if not updated yet) before using it to carry out the replacement / relocation / reuse / rerouting / dismantling / Integration of Existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc. This activity requires close interaction with Owner including plant operation and maintenance personnel so that all "known" changes are captured efficiently. The above details shall also be included in the site survey report including proposed tagwise existing replaced / relocated / rerouted / reused / dismantled cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc. by meeting all other scope of work & supply requirement.

Note-3: Dismantling and removal of existing cables / tubes / trays / junction boxes / CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc., if any, associated cables, trays, junction boxes, tubing, their supports etc. and accessories, creation of space for placing the new instruments / equipment / systems as applicable shall be under the scope of bidder. Contractor to get prior approval from owner for removal / dismantling of existing Instrumentation & control system items with proper document having details like tag no. etc. along with proper justification for removal / dismantling. Packing, loading, shifting & hand overing of all the dismantled Instrumentation & control system items as per philosophy defined elsewhere in Bid package is in bidder's scope.

Note-4: All the existing Instrumentation & control system items on the replaced BHS-SLQ Bridge and other replaced bridge / walkways as applicable shall be replaced one to one with new components. All the existing ESD / FSD stations, Fire and Gas detectors including Hooter/ beacon, Manual call points, CCTV & other items on these bridge as applicable shall be replaced one to one with new components along with required main cable, branch cables, pneumatic tubing, tray, duct, Junction box, canopy, hook-up material, mounting / support material, painting, insulation, glanding / termination, ferruling and shall be made operational after required connectivity to control system / power supply is established.

Bidder to ensure the all the Instrumentation & control items on the fallen bridges / walkways based on available data, Instrumentation & control item including the CCTV / F&G Detector / F&G devices (Including ESD / Beacon / Hooter / MCP) etc. as required as per latest codes / standards shall be supplied as new for the new bridges / walkways along with required main cable, branch cables, pneumatic tubing, tray, duct, Junction box, canopy, hook-up material,

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mounting / support material, painting, insulation, glanding / termination, ferruling and shall be made operational after required connectivity to control system / power supply is established.

Note-5: In case existing Instrumentation and Control System are in healthy conditions, the same may be reused / relocated / rerouted by bidder with the consent of Owner provided the existing instrument is not obsolete & meeting latest statutory / codes & standards requirements. Bidder's scope includes ensuring the item is made operational after required connectivity to control system / power supply is established.

Note-6: The new location of replaced bridges / walkways / other equipment shall be taken in to account while designing / engineering / procurement of Instrumentation & control system components replaced with new / reused.

Note-7: All the new Instrumentation & control system items supplied by contractor shall be compatible with the existing system (Including DCS / ESD / F&G PLC / CCTV System). Contractor to avoid introduction of any joint in the cable along the cable routing that requires replacement or rerouting. Contractor to supply and install complete end to end cable wherever complete cable replacement is envisaged. Introduction of any new junction boxes in between, instead of complete cable replacement is also acceptable.

Note-8: In general replacement of existing instrument with new is not envisaged for the items on the refurbished / strengthened Bridges / walkways / other equipment & items. However, if any existing item is recommended to replace with new (due to constructability / relocation / reroute) by contractor same is in bidder's scope. However, the replaced items shall meet required specification along with latest statutory / codes & standards requirement. Bidder's scope includes ensuring the item is made operational after required connectivity to control system / power supply is established. The new location of Instrumentation & control system items shall be taken into account while design / engineering / procurement.

Note-9: Bidder shall verify the space availability on / below / top / side of the Replaced / refurbished / strengthened bridges / walkways & other equipment before replacement / relocate / reroute of Instrumentation & control system items. In case of any space constraint, the same needs to be addressed and resolved during design & engineering stage. Any modification, including structural, so as to house instrument items, in case space constraints are encountered, shall be carried out by the bidder without any implication.

Note-10: If any other existing facilities needs upgradation for Instrumentation & control system item replacement / relocation / rerouting requirement by the contractor, which is identified during the execution of contract for completing the project, the same is in contractor's scope. No time or cost implication will be applicable for the same.

Note-11: Any damage caused to the existing facilities during execution of the job shall be to bidder's account. Any damage caused by contractor / sub-contractors / vendors during execution of the job shall also be to bidder's account. Bidder shall repair and /or replace the damaged equipment / items with new equipment / items to the satisfaction of Owner, at no additional cost. All such works shall be carried out without any loss of time or any cost implication.

Note-12: Bidder shall carry out complete analysis for power adequacy for the required replaced / rerouted / relocated Instrumentation & Control system items. Any new Power supply arrangement or modification in existing power supply system required for instrumentation and control system items shall be done by bidder without any implication. Additional power, if required, to cater to instrumentation loads shall be provided by the bidder as part of this project scope.

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Note-13: Developing drawings / documents as per list as attached elsewhere in the bid separately shall be part of bidder's scope.

Note-14: Area classification for platforms for selection of Instrumentation items shall be as per approved Area Classification drawings.

Note-15: Testing, Loop checking, Logic Implementation and Verification, Integration, Site acceptance testing and Commissioning of replaced with new / reused / relocated / rerouted Instrumentation and control system item is in bidder's scope.

Note-16: The relocation / rerouting of existing Instrumentation & control system Items shall be decided with the consent of Owner.

Note-17: Procurement, Factory Calibration, Shop testing, Inspection and Supply of all Instrumentation items for safe and smooth implementation of the project shall be in the scope of Bidder's.

Note-18: All required logics / graphics for the changes & requirement envisaged for this project scope shall be implemented with full assistance of OEM and shall be made its operational. In case of any fault in the logic / graphics detected during the erection & commissioning phase due to changes & requirement envisaged for this project scope, necessary logic modifications shall be carried out by the Bidder in coordination with OEM without any time or cost implication. Any modifications in logic / graphics required during erection / commissioning as per platform personnel recommendations for the project scope shall be carried out by Bidder without any implication**.**

Note-20 Bidder shall take cold / hot work permits from Owner personnel before commencing any cold / hot work.

Note-22: Replacement / Relocation / Reroute / Reuse / Integration philosophy & Execution Methodology shall be submitted by bidder for ONGC review and approval. Bidder to avoid any shutdown requirements for execution of projects, however in case any shutdown is required, bidder to execute the site activities in any planned shutdown or inform M/s ONGC for their review and approval.

Note-23: Other instrumentation items including valves etc. coming on the BHS-SLQ bridge shall be replaced / reused / reinstalled / rerouted as applicable along with required main cable, branch cables, pneumatic tubing, tray, duct, Junction box, canopy, hook-up material, mounting / support material, painting, insulation, glanding / termination, ferruling and shall be made operational after required connectivity to control system / power supply is established.

Note-24: All the bidder supplied Instrumentation & control items shall be designed as per the Instrumentation design criteria & functional specification attached in the tender by incorporating the existing design & functional requirement in concurrence of Owner.

5.0 INSTALLATION

Bidder's scope of installation for Instrumentation and control system shall include but not to be limited to supplied items under the bidder's scope and defined below:

5.1 Installation of ESD / FSD stations, Fire and Gas detectors including Hooter / beacon, Manual call points, CCTV, MCT & other items along with required main cable, branch cables, pneumatic tubing, tray, duct, Junction box, canopy, hook-up material, mounting / support material, welding, painting, insulation, glanding / termination, ferruling and shall be made operational after required connectivity to control system / power supply is established.

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5.2 Relocation / rerouting of existing Instrumentation & control system Items with the consent of Owner.

5.3 Laying & termination of all single pair cables, cable glands and plugs for interconnection between all field instruments and junction boxes / local panels. Multipair cables from field junction boxes shall be supplied, laid and terminated at Control room by bidder as required.

5.4 Installation of FRP cable tray / tube trays along with all accessories for complete cabling Minimum 20% free space shall be left in the cable trays for future use.

5.5 Installation of LCPs (as applicable), Shifting / reinstallation of local control panels and laying & termination of cabling between LCPs / Control panels (as applicable) and Control room.

5.6 Installation of Pipe, pipe fitting, tube & tube fittings and valves for Instrument air / gas supply distribution to pneumatic instruments etc.

5.7 Installation of MCTs as applicable with cables for deck penetration and Control room cable entries for the new cabling system complete with steel frames, insert blocks, solid blocks, compression plates etc. as required for offshore platforms. MCT Blocks for instrumentation cables shall be separate from the Electrical Cables.

5.8 Installation of complete earthing including supply of materials i.e. earthing conductor / cable, nuts & bolts, lugs, washers etc. as required for intrinsically safe, signal & personnel safety earthing.

5.9 Interconnection, integration of different system / items as per the requirement to complete the job.

5.10 Any scope of installation not specifically mentioned here but required as per Bid documents for safe and proper operation of the facility.

5.11 All works shall be of the highest quality craftsmanship and shall conform to the best applicable engineering practices and relevant codes referred in the bid document. Bidder shall carryout installation of all instruments as per details furnished in Standard Specification for Installation of Instruments and Controls for Offshore doc. no. 6-52-0086.

5.12 Bidder shall prepare hook-up and installation detail drawings regarding each type of instrumentation and control system items as per existing philosophy for the owner approval and shall carry out installation in accordance with these approved drawings.

6.0 TESTING AND CALIBRATION

6.1 Calibration of all instruments, testing of all instrumentation & control items shall be done by the Bidder. All instruments shall be factory calibrated and shall be recalibrated at platforms during installation / commissioning.

6.2 The Bidder quality plan shall include a comprehensive fully documented inspection and testing plan specific to the project.

6.3 The procedures shall include inspection specifically for compliance with hazardous areas requirements, including current certificate, without which no circuit or loop shall be energized.

6.4 The Bidder shall provide suitable workshop facilities and shall provide all necessary test and calibration instruments and equipment.

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6.5 The owner reserves the right to reject any or all test and calibration work if found not complying with the specification requirement. The Bidder shall complete and submit documentation for all calibration, testing and pre-commissioning. Owner Representative / TPI prior to shipment shall check out major instrumentation items for their compliance with specification requirements. The Bidder shall verify calibration and loop testing for all instruments in field including local and Control room panels are complete and operable as per specification requirement in the presence of the Owners Representative. All testing and calibration shall be subject to approval of the Owner.

6.6 The Bidder shall provide written results of all above tests and if required by the Owner, provide reasonable evidence of the satisfactory condition of test equipment. All errors of faulty workmanship discovered during this testing shall be corrected to the satisfaction of the Owner.

6.7 Testing of impulse lines, instruments, instrument air lines, and instrument cables including special instruments / items if any, and calibration of all instruments shall be carried out as per the requirements in various specifications in the bid document & vendor recommendation.

6.8 In case of any delay between installation and commissioning of Bridges / walkways / other equipment & items, necessary recalibration of all field instruments shall be done by the Bidder without any cost / time implication.

7.0 COMMISSIONING

7.1 Bidder shall be responsible for complete loop checking, logic implementation & verification, testing, pre commissioning, & commissioning including but not limited to following as minimum in order to check the functional performance of all elements comprising the loop and ensure proper interconnection configuration, functioning and implementation.

7.2 The co-ordination and start-up, pre-commissioning, commissioning and stabilization of system / item with Owner / Various vendors / contractors shall be the total responsibility of Bidder.

7.3 Bidder shall be responsible for Testing, Loop checking, Logic Verification, Integration, Site Testing and Commissioning of Instruments and control systems.

7.4 Engaging OEM personnel for configuration, modification, troubleshooting & commissioning of Instrumentation & control system item is in bidder’s scope.

8.0 DETAIL ENGINEERING

Bidder shall carry out design and engineering of instrumentation and controls based on bid package and other technical requirements described in various volumes of this bid package specification. In carrying out design and engineering, Bidder shall do detailed calculation, residual engineering, prepare data sheets, etc. for all types of supply items.

9.0 PROCUREMENT AND SHOP TESTING

9.1 Engineering for procurement shall include selection and preparation of technical specification for instruments based on bid package, other specifications, Standards etc. evaluation of offers received from different vendors, preparation of purchase specification and review / approval of vendor drawings.

9.2 The Purchase Specification (PS) for each item shall be prepared by using ISA data sheet formats which shall contain all information including the make and model no. of each instrument, in addition to the material specification and other details. In case, the bidder

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intends to follow their own format, it shall necessarily contain all information as in ISA format, as a minimum. Selected vendor's offer shall be attached in Purchase Specifications.

9.3 The bidder shall carry out shop testing and calibration of all instruments and control systems. Owner and/or third party, as the case may be, may witness any or all items testing in stages during manufacture and / or at final stage before shipment. Site calibration / testing shall be carried out as required and as per the recommendation of Platform Personnel.

10.0 LIST OF ATTACHMENTS

S. No.	Document	Document No.
1.	Scope of Works & Supply (Instrumentation & Control)	B774-000-16-51-SOW-1001
2.	Vendor Data Requirement	B774-000-16-51-VDR-1001
3.	Mandatory Spare Parts	B774-000-16-51-SL-1001
4.	Compliance Statement	B774-000-16-51-CF-1001
5.	Logistic support certificate	B774-000-16-51-LSF-1001
6.	Standard Specification for Design of Instruments and Controls for Offshore	6-52-0085
7.	Standard Specification for Installation of Instruments and Controls for Offshore	6-52-0086
8.	Standard DCI (Document Control Index) /MCI (Material Control Index)	Appendix C-8
9	Instrumentation Design Criteria	Volume-II
10	Functional Specification	Annexure-1

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DOCUMENTS / DRAWINGS FOR OWNER'S REVIEW (INSTRUMENTATION & CONTROL)

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
0	29.09.2025	ISSUED FOR TENDER	RN	SKA	RK
A	04.09.2025	ISSUED FOR COMMENTS	RN	SKA	RK

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CONTRACTOR INSTRUMENTATION DOCUMENTS AND DRAWINGS REQUIRED FOR OWNER's REVIEW / RECORDS

Sl. No.	Description	Approval (A)	Review (R)	Information (I)	Remarks
1.	Drawing and Document Schedule/Index (DCI/MCI)		✓		
2.	Survey Report		✓		
3.	Execution work philosophy		✓		
4.	Power supply distribution scheme for Instrumentation & Control system items (as applicable for the modification)			✓	
5.	Purchase Specification - F&G Detectors, Beacons, Hooters	✓			
6.	Purchase Specification -CCTV	✓			
7.	Purchase Specification - Manual ESD / FSD	✓			
8.	Material Take off i) Cables, Tray, JB, glands etc. ii) Tube, tube fittings, pipe etc. iii) F&G Devices, ESD/FSD Station iv) CCTV etc			✓	
9.	Purchase Specification – Instrumentation Tube	✓			
10.	Purchase Specification – Instrumentation Tube Fittings	✓			
11.	Purchase Specification – Instrumentation Cable	✓			
12.	Purchase Specification – Instrumentation Tray	✓			
13.	Purchase Specification – Instrumentation Junction box with cable gland	✓			
14.	MCT Layouts & Schedule (as applicable for the modification)			✓	
15.	CCTV System Block Diagrams	✓			
16.	CCTV Layouts	✓			
17.	ESD, FSD loop layouts	✓			
18.	Fire and Gas detectors layout	✓			
19.	Interconnection Diagram for CCTV System			✓	
20.	Instrument And F&G Cable Block Diagram			✓	

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21.	Instrument And F&G Cable Schedule			✓	
22.	Inst. Loc. & JB Location Layout			✓	
23.	Instrument Cable Tray Layout			✓	
24.	Instrument & F&G Index			✓	
25.	Instrument & F&G I/O List			✓	
26.	Instrument Installation Details			✓	
27.	Tubing Routing, Tubing Tray Layout			✓	
28.	Instrument Hook Up Drawings			✓	
29.	Logic Diagrams			✓	
30.	Inst. Loop Schematics			✓	

NOTES:

1. Bidder shall include other applicable drawing / documents as per Appendix C-8 attached in the tender.

2. The submission of document / drawing shall be sequenced as per Appendix C-8 attached in the tender .

3. Document numbering system & review cycle shall be as per Appendix C-8 attached in the tender.

4. All the drawings / documents listed shall be converted into "As Built" after commissioning of Project. "As Built" shall include documents generated by Contractor & documents generated / submitted by various Sub-vendors.

5. All the above documents, as applicable, are also required for the package items / instruments supplied by the Contractor.

6. Document category shall be strictly followed as per Appendix C-8 attached in the tender.

7. For system-oriented items supplied by the contractor all drawings / documents required from respective system vendor as vendor data shall also be applicable.

8. Any other document / drawing not listed above but required to carry out the engineering and for integration of other equipment packages sub-ordered by contractor shall also be prepared and furnished to owner.

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Instrumentation Design Criteria

OIL AND NATURAL GAS CORPORATION LTD.

MUMBAI

PREPARED	CHECKED	APPROVED	PAGES	DATE	REV. NO.
Rakhi Saxena 8/6/23	8/6/23	08/06/23	**59**	**08.06.2023**	**09**
RS	YD	SA	59	08.06.2023	09
SSK/VKS	ARD	KM	43	06.09.2016	08
SRS	ARD	AKR	41	31.08.2012	07
SRS	ARD	AKR	41	28.11.2011	06
SRS	ARD	AKR	40	22.01.2010	05a
SRS	ARD	AKR	40	08.12.2009	05
RS/ BK	S.S.K	G.R.P	36	06.02.2008	04
VB	BK	G.R.P	70	03.12.2007	03
ET	MC	RK	63	21.06.2005	02
E.T.	M.C.	R.K.	63	10.01.2005	01
E.T.	M.C.	A.C.	63	13.08.2003	0

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3.6.1 SCOPE OF THIS DOCUMENT

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This document defines the Design Criteria for the Instrumentation and Control System envisaged for the Offshore Platforms/ Mobile Offshore Production Unit (MOPU)/ Packaged Equipment/ Free Issue Materials (FIM) etc. for design, selection, testing, installation and commissioning of the instrumentation systems. This Instrumentation Design Criteria (IDC) is a guideline and general in nature.

This document shall be read in conjunction with Basic Bid Work, Process and Equipment requirements, P&IDs, Functional Specifications (FS), System Architecture etc. to complete the Instrumentation Scope of Work (SOW), as applicable in the project.

3.6.2 PURPOSE OF INSTRUMENTATION

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The purpose of Instrumentation is to provide a complete and safe monitoring, control and shutdown system for process and associated systems such that all the required information/ data/ signals are available in the desired form and place at Control Room and Monitoring Centers as envisaged in the scope of work in required form following the Basic Bid Work, P&IDs and the Functional Specifications in the tender document.

The intent of instrument specifications, data sheets and general guidelines is to provide the basic system requirements and establish a level of quality and service conditions.

3.6.3 STANDARDS AND CODES

09

Latest editions of the Codes/ Standards/ Specifications/ Recommended Practices (RP) enlisted below shall be followed as applicable:

American Gas Association (AGA)	
AGA Report 3	Orifice Metering of Natural Gas
AGA Report 7	Measurement of Gas by Turbine Meters
AGA Report 8	Compressibility Factors of Natural Gas and Other Related Hydrocarbon Gases
AGA Report 9	Measurement of Gas by Multi-path Ultrasonic Meters
AGA Report 10	Speed of Sound in Natural Gas and Other Related Hydrocarbon Gases
AGA Report 11	Measurement of Natural Gas by Coriolis Meter
American National Standards Institute (ANSI)/ American Society of Mechanical Engineers (ASME)	

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ANSI B 2.1	Pipe Threads
B 16.5 (09)	Pipe Flanges and Flanged Fittings
B 16.10	Face to Face and End to End Dimensions of Ferrous Valves
B 16.34 (09)	Valves – Flanged, Threaded and Welding End
B 16.36	Orifice Flanges
ANSI FCI 70.2 (09)	Control Valve Seat Leakage
ANSI MC 96.1	Temperature Measurement Thermocouples
ANSI B 1.20.1	Pipe Threads, General Purpose
ASME PTC 19.3 TW (09)	Performance Test Code on Thermowells
ASME Section VIII	Sizing and Selection of Pressure Relief Valves
American Petroleum Institute (API)	
API SPEC 6D	Specification for Valves
API STD 6FA	Standard for Fire Test for Valves
API RP 14C	Analysis, Design, Installation and Testing of Basic Surface Systems for Offshore Production Platforms.
API RP 14F (09)	Design, Installation, and Maintenance of Electrical Systems for Fixed and Floating Offshore Petroleum Facilities for Unclassified and Class I, Division 1, and Division 2 Locations
API RP 14G	Fire Prevention and Control on Fixed Open-type Offshore Production Platforms
API RP 17A	Design and Operation of Subsea Production Systems - General Requirements and Recommendations
API SPEC 17E	Specification for Subsea Umbilicals
API RP 505	Classification of Locations for Electrical Installations at Petroleum Facilities Classified as Class I, Zone 0, Zone 1 and Zone 2
API RP 520 (09)	Design and Installation of Pressure-Relieving Systems in Refineries; Part I—Design Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries; Part II—Installation
API STD 521	Pressure-relieving and Depressuring Systems
API STD 526	Flanged Steel Pressure-Relief Valves

API STD 527	Seat Tightness of Pressure Relief Valves
API RP 551	Process Measurement
API RP 552	Transmission Systems
API RP 554-1	Process Instrumentation and Control - Functions and Functional Specification Development
API RP 555	Process Analyzers
API STD 598	Valve Inspection and Testing
API STD 2000	Venting Atmospheric and Low-Pressure Storage Tanks
API MPMS	API Manual of Petroleum Measurement Standards **Chapter 1 -** Vocabulary **Chapter 4** – Proving - Guide for the design, installation, calibration, and operation of meter proving systems. **Chapter 5** – Metering- Dynamic measurement of liquid hydrocarbons **Chapter 6** – Metering Assemblies - Design, installation, and operation of metering systems for coping with special situations in hydrocarbon measurement **Chapter 12** - Calculation of Petroleum Quantities **Chapter 14** - Natural Gas Fluids Measurement - Standardizes practices for measuring, sampling, and testing natural gas fluids **Chapter 20** - Allocation Measurement of Oil and Natural Gas **Chapter 21** - Flow Measurement using Electronic Metering Systems **Chapter 22** – Testing Protocols
American Society for Testing and Materials (ASTM)	
ASTM A269	Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A276-316L	Stainless Steel Fittings
ASTM 370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM 450	General Requirements for Carbon and Low Alloy Steel Tubes
ASTM F 1387	Performance of Piping and Tubing Mechanically Attached Fittings
ASTM G48	Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steel

British Standards	
BS 1904	Industrial Platinum Resistance Thermometer Sensors
BS 4937	International Thermocouple Reference Tables
BS 5501	Electrical Apparatus for Potentially Explosive Atmospheres
BS/ IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)
BS EN 12874	Flame Arrestors
International Electrotechnical Commission (IEC)	
IEC 61000-4	Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques – Electrostatic Discharge Immunity Test Part 4-3: Testing and measurement techniques – Radiated, radio-frequency electromagnetic field immunity test
IEC 60079-11	Explosive atmospheres – Part 11: Equipment protection by intrinsic safety "I"
IEC 60092-373	Shipboard flexible coaxial cables
IEC 60092-359	Sheathing Materials For Shipboard Power And Telecommunication Cables
IEC 60227	Polyvinyl chloride insulated cables of rated voltages up to and including 440/ 750 V
IEC 60331	Tests for Electric Cables Under Fire Conditions – Circuit Integrity
IEC 60332-1	Tests on electric cables under fire conditions Part I: Tests on single vertical insulated wire or cable
IEC 60332-3	Tests on electric cables under fire conditions Part II: Tests on single small vertical insulated copper wire or cable
IEC 610 86-1	Coatings for loaded printed wire boards (conformal coatings) - Part 1: Definitions, classification and general requirements
IEC 61131-3	Programmable Controllers – Part 3: Programming languages
IEC 61508-1-7	Functional Safety on Electrical/ Electronic/ Programmable Electronic Safety-Related Systems
IEC 61511 Part-1 to 4	Functional Safety – Safety Instrumented Systems for the Process Industry Sector
IEC 60654-4	Operating Conditions for Industrial – Process Measurement and Control Equipment Part 4 - Corrosive and Erosive Influences
Institute of Electrical and Electronics Engineers (IEEE)	

IEEE 472	Guide for Surge Withstand Capabilities (SWC) Tests
IEEE C37.90.1	Standard Surge Withstand Capability (SWC) Tests for Protective Relays and Relay Systems
IEEE 730.1	Guide for Software Quality Assurance Planning
IEEE 828	Configuration Management in Systems and Software Engineering
IEEE 1042	Guide to Software Configuration Management
Instrumentation Systems and Automation Society (ISA) **Note:** A lot of ISA Standards are adopted by ANSI and known as ANSI/ ISA Standards.	
ISA 5.1	Instrumentation Symbols and Identification
ISA 5.4	Instrument Loop Diagrams
ISA 7.0.01	Quality Standard for Instrument Air
ISA RP 12.4	Pressurized Enclosures
ISA RP 12.6	Wiring Practices for Hazardous (Classified) Locations Instrumentation Part I: Intrinsic Safety
ISA RP 42.00.01	Nomenclature for Instrument Tube Fittings
ISA 50.02-2	Fieldbus Standard For Use In Industrial Control Systems Part 2: Physical Layer Specification And Service Definition
ISA RP 60.1	Control Center Facilities
ISA RP 60.2	Control Center Design, Guide and Terminology
ISA RP 60.4	Documentation for Control Centers
ISA RP 60.6	Nameplates, Labels & Tags for Control Centers
ISA 51.1	Process Instrumentation Terminology
ISA 12.13.01	Performance Requirements for Combustible Gas Detectors
ISA RP 12.13.02	Installation, Operation and Maintenance of Combustible Gas Detection Instruments
ISA 92.00.01	Performance Requirements for Toxic Gas Detectors
ISA 92.00.02	Installation, Operation and Maintenance of Toxic Gas Detection Instruments
ISA 71.01	Environmental Conditions for Process Measurement and Control Systems: Temperature and Humidity
ISA 71.04	Environmental Conditions for Process Measurement and

	Control Systems: Airborne contaminants
ISA 75.01.01	Flow equations for sizing control valves
ISA 75.02.01	Control Valve Capacity Test Procedures
ISA 75.08.01	Face to Face Dimensions for Integral Flanged Globe Style Control Valve Bodies
ISA 75. 19.01	Hydrostatic Testing of Control Valves
ISA 100.11a	Wireless Systems for Industrial Automation: Process Control and Related Applications
ISA 18.1	Annunciator Sequences and Specifications
ISA 18.2	Management of Alarm Systems for the Process Industry
ISA TR84.00.02 Part 1 to 4	Safety Instrumented System – Safety Integrity Level Evaluation Techniques
ISA TR84.00.07 Part 1 to 4	Guidelines for Evaluating the Effectiveness of Fire, Combustible Gas and Toxic Gas Systems.
International Organization for Standardization (ISO)	
ISO 5167-1	Measurement of fluid flow by means of pressure differential devices inserted in circular cross-section conduits running full - Part 1: General principles and requirements
ISO 5167-2	Measurement of fluid flow by means of pressure differential devices inserted in circular cross-section conduits running full - Part 2: Orifice plates
ISO 5167-3	Measurement Of Fluid Flow By Means Of Pressure Differential Devices Inserted In Circular Cross Section Conduits Running Full - Part 3- Nozzles And Venturi Nozzles
ISO 5167-4	Measurement Of Fluid Flow By Means Of Pressure Differential Devices Inserted In Circular Cross Section Conduits Running Full - Part 4-Venturi Tubes
ISO 5167-5	Measurement of fluid flow by means of pressure differential devices inserted in circular cross-section conduits running full - Part 5: Cone meters
ISO TR 12748	Wet gas flow measurement in natural gas operations
ISO TR 11583	Measurement of wet gas flow by means of pressure differential devices inserted in circular cross-section conduits
ISO 13628-5	Petroleum and natural gas industries - Design and operation of subsea production systems - Part 5: Subsea umbilicals

ISO/ IEC 17025	General requirements for the competence of testing and calibration laboratories
ISO 17089-1	Measurement of fluid flow in closed conduits- Ultrasonic meters for gas - Part 1: Meters for custody transfer and allocation measurement
ISO 17089-2	Measurement of fluid flow in closed conduits- Ultrasonic meters for gas - Part 2: Meters for industrial applications
National Association of Corrosion Engineers (NACE)	
ANSI/NACE MR 0175/ ISO 15156	Petroleum, Petrochemical and Natural Gas Industries- Materials for Use in H2S-Containing Environments in Oil and Gas Production
National Electrical Manufacturers Association (NEMA)	
ANSI/NEMA 250	Enclosures for Electrical Equipment (1000 Volts maximum)
National Fire Protection Association (NFPA)	
NFPA 70	National Electrical Code (NEC)
NFPA 1	Fire Code
NFPA 72	National Fire Alarm and Signaling Code
NFPA 86	Ovens and Furnaces
NFPA 87	Fluid Heaters
NFPA 496	Purged and Pressurized Enclosures for Electrical Equipment
International Telecommunication Union (ITU)	
ITU-T Reco. G.650	Definition and test methods for the relevant parameters of single-mode fibres.
ITU-T Reco. G.652	Characteristics of a Single-Mode Optical Fibre and Cable
ITU-T Reco. G.655	Characteristics of a non-zero dispersion-shifted single-mode optical fibre and cable.
ITU-T Reco. G.656	Characteristics of a fibre and cable with non-zero dispersion for wideband optical transport.
Society of Automotive Engineers (SAE)	
SAE J343	Test and Test Procedures for SAE 100R Series Hydraulic Hose and Hose Assemblies
<u>Other Bodies</u>	
Norsok Standards: U-CR-006 Sub-sea Production Control Umbilical	

DIN Standards: DIN 3381: Safety Devices for gas supply installations, pressure relief governers and safety shut off devices.
Engineering Equipment Materials Users Association (EEMUA) publication No. 191, Alarm Systems – a Guide to Design Management and Procurement
Normenarbeitsgemeinschaft für Mess-und Regeltechnik in der chemischen Industrie (User Association of Automation Technology in Process Industries)- **NAMUR NE43:** Standardization of the Signal Level for the Failure Information of Digital Transmitters
OISD: Oil Industry Safety Directorate relevant standards
All goods and services supplied shall meet all applicable local and international regulations on health, safety and environmental issues.

Note: In case of a conflict among various standards, the most stringent standard shall be followed**.**

3.6.4 INSTRUMENTATION SYSTEM PHILOSOPHY

3.6.4.1 INSTRUMENT PHILOSOPHY

3.6.4.1.1 Units of Measurement

Following units of measurement shall be used, unless otherwise specified:

Flow (Liquid)	Kg/hr, m^3/hr
Flow (Gas)	Sm^3/hr, MMSCMD
Temperature	Deg. Centigrade
Pressure	Kg/cm^2g
Differential Pressure	inch/ mm of Water Column

3.6.4.1.2 All electronic and pneumatic instrument systems shall be designed for "Fail Safe" operation.

3.6.4.1.3 All the offered instruments/ equipment or equipment of similar design manufactured shall have been:

a) Type tested by the accredited authority.

b) Of proven design.

c) Instruments, bulk materials, accessories etc., shall conform to the various relevant Functional Specifications.

d) The criteria, guidelines and specifications do not absolve the Vendor/ Contractor/ Packager of his responsibilities of providing the instruments with all necessary hardware, cards and accessories for proper measurement and display at field/ control room, computation, power supplies, transmission of data/ signals for proper operation and for the intended purpose.

e) OIML (Organisation of Legal Metrology) Certification shall be furnished where the measurements are for custody transfer applications.

3.6.4.1.4 Field Instruments

a) *Wellhead Platforms with Solar Power:* Field Instruments connected to Well-Head Control Panel shall be Pneumatic, unless otherwise specified in the scope of work for the project. The field instruments connected to RTU for monitoring, control, telemetry and remote control shall be electronic SMART type with HART communication protocol.

b) *Smart Wellhead Platforms with External Power:* All field instruments for well monitoring and control shall be electronic SMART type with HART communication protocol. 09

c) *Process Platform:* Instruments connected to the Control Room Panels and Unit/ Remote Control Panels shall be electronic SMART type with HART communication protocol.

d) In modification jobs, wherever Foundation Fieldbus (FF) type Instruments are existing, new FF instruments shall be used only for monitoring purpose to match the existing philosophy, if specifically mentioned in the SOW. However, in control loops, only HART based instruments shall be used.

e) In modification jobs, wherever Wireless type Instruments exist, new wireless instruments shall be used for monitoring purpose only, if specifically mentioned in the SOW.

f) In general final control elements at field (located in the hazardous area) shall be pneumatic type. Hydraulic/ Electric actuators shall be provided wherever specified in the bid scope of work.

g) Instrument ranges shall be selected such that the normal operating point is between 30% to 70% of span for linear scale and 60% to 80% of span for square root scale.

h) If static pressure measurement for pressure compensation is required for a compressible fluid, it shall be taken upstream of the measuring element.

i) Field switches and status alarms shall be electrical or pneumatic as per the scope of work. 09

j) In an effort to provide standardization, it is the Contractor's responsibility to ensure that the various equipment/ package sub-vendors supply the same type and make of instruments and accessories. Vendor shall be responsible for complete design, sizing, installation and workability of the systems. Vendor specifications and relevant calculations/ sizing sheets shall be submitted along with Purchase Specification (PS) for Company's approval.

k) Hand-held Intrinsically Safe universal type calibrator/ HART communicator/ Industrial Laptops/ Smart Tabs loaded with licensed software shall be supplied for online diagnostics, configuration or calibration of electronic instruments as per the quantities specified in SOW.

3.6.4.1.4.1 Pneumatic Field Instrumentation

a) The instrument air supply shall conform to ISA 7.0.01 "Quality Standard for Instrument Air".

09 b) Instrument Air/ Gas piping and branch connections shall be as per ANNEXURE-II attached in this document.

c) Each pneumatic instrument supply shall be provided with independent filter regulator. For pneumatic instruments, dry instrument gas/ air supply shall be as follows :

Minimum	Normal	Maximum
5.5 kg/cm^{2}g	7.5 kg/cm^{2}g	10.5 kg/cm^{2}g

All related equipment shall be suitable for operating in the above specified range.

09 d) For pneumatic control applications, the actuating signal range shall be 0.2 to 1 kg/cm^{2}g or 0.4 to 2.0 kg/cm^{2}g.

e) For pneumatic On-Off applications, the actuating signal shall be 0 or 5.5 kg/cm^{2}g, however the logic signal shall be 3.5 kg/cm^{2}g.

09 f) For platforms with sour gas service specified with "Electro - Hydraulic System", the hydraulic power shall be utilized for control, logic and actuation to the extent envisaged in the project SOW and its P&IDs.

3.6.4.1.4.2 Electronic Field Instruments

a) All Electronic transmitters shall be 24 VDC loop-powered type with 4-20 mA SMART analog output signal. Electronic Transmitters shall have inbuilt digital output LCD display in engineering units.

Where this is not possible, a suitable separate intrinsically safe local loop indicator/ display unit shall be provided.

All field transmitter enclosures shall be dual chamber type.

b) All electronic transmitters shall be minimum SIL-2 certified.

09 c) For monitoring and alarm, electronic transmitters with 4-20 mA output shall be used.

d) In case of process platforms, electronic transmitters with 4-20 mA output shall be used instead of switches for trips. The ESD PLC shall convert the 4-20 mA signals into digital input for the intended application.

e) All local electrical control and field alarm switches shall be hermetically sealed dual SPDT/ DPDT Microswitch activated switches having gold plated contacts.

f) Shutdown, Blowdown and Deluge Valves shall be provided with Proximity type switches for valve position indication in RTU/ PCS/ DCS.

g) For Process Platforms and Smart Wellhead Platforms with External Power, all Solenoid Valves (SOV) and relays shall be continuous rated type. SOVs shall preferably be of universal type.

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For solar-powered wellhead platforms, solenoid valves shall be pulse-operated type.

h) All the electronic instrumentation systems shall be provided with lightening and spike/ surge protection circuits. The OEM's recommendations shall be adhered to for design and implementation of the "Earth" system.

i) Unless otherwise specified in the scope of work for the project, signals for RTU shall be isolated 4-20mA DC for analog loops or dry contacts or isolated pulse signals as relevant to the service. These signals shall be terminated in RTU marshalling panel and made available in the RTU.

3.6.4.1.5 Control Room Instrumentation

a) Centralized control and monitoring and data/ alarm/ event logging/ trending etc. for all field instrumentation and packaged equipment shall be done through Distributed Control System (DCS) in Process Complex, through RTU in solar powered wellheads and through Process Control System (PCS) in Smart Wellhead Platforms with External Power.

All the package instrumentation signals along with necessary serial/ digital communication interfaces, isolated 4-20 mA DC (load resistance minimum 500 ohms), isolated dry contacts, signals from Unit Control Panels (UCP) etc. as per SOW shall be provided by the Packager-Vendor.

b) All signals to and from the Central Control Room shall be electrical/ electronic.

09

- The standard analog signal shall be 'SMART' 4-20 mA using 2-wire system, standard thermocouple, RTD output, and/ or suitable pulse signal etc.
- The standard digital signal (inputs/ outputs) shall be 24 VDC.
- Serial communication with packages and devices shall be as per Basic Bid Work.

c) DCS shall be OPC enabled and shall be equipped with Antivirus and Firewall to protect the system against external intrusion or access. OPC requirements are specified in the Functional Specifications for DCS.

d) If PLC based control system is envisaged for monitoring and control on Wellhead Platforms, then OPC enabled system shall be provided in HMIs. Communication of PLC with other system(s) shall be preferably Ethernet based and not on "proprietary" protocols. If PLC communicates on proprietary protocols, then *"in built" OPC enabling is mandatory* for such system. PLC shall be OPC enabled for reading/ writing into it from SCADA with secure communication.

Associated HMI/ Master Terminal Unit (MTU)/ Workstations at Process Platform shall be OPC enabled for data transfer and connectivity to DCS/ SCADA.

e) All platforms shall have Fire & Gas Detection System conforming to FS-3500 (Scheme A or B) and shall initiate ESD/ FSD signal for shutdown in case of confirmed gas/ fire.

f) For Smart Wellhead Platforms with External Power, the PLC-HMIs shall continue to operate standalone, even in case of communication break down with the associated process platform.

g) Process PLCs, Safety PLCs (ESD and F&G), UCP PLCs of packaged equipment etc. shall have provision of interfaces like MODBUS RS-485, TCP/ IP and Ethernet/ IP protocols for communication with other systems.

h) Relays utilized in 24 VDC control logic circuits shall have gold plated contacts rated 0.5A at 24 VDC and shall be hermetically sealed. Those interfacing with field equipment shall be rated 24 VDC (2A)/ 110 VAC (5A).

09 i) For Process and Wellhead platforms, all the electronic cards (CPU, Controllers, I/O, Power Supply, Communication etc.) and HMI of DCS, OPC servers, ESD PLC, F&G PLC, F&G Controller, High Integrity Pressure Protection System (HIPPS), Unit Control Panels (UCP) and Local Control Panels (LCP) of Packaged Equipment etc. shall be G3/ GX compliant with conformal protective coating (Harsh environment) requirements as per ANSI/ ISA 71.04, IEC 61086-1 and IEC 60654-4/ equivalent national or international standards.

j) All instrument panels and cabinets in the safe area (Control room, Switchgear room, and pressurized rooms) shall be 'general purpose' enclosures with IP-42 protection or higher.

k) Control Room having DCS and other control panels shall have air quality as per ISA 71.01 and ISA 71.04 standards.

l) The DCS and PLCs shall be rated to perform, without being de-rated, in conditions arising out of absence/ failure of the HVAC, in the control room/ controlled atmosphere, for at least 8 Hours. Temperature may rise to 60^0C in rooms housing the panels and cabinets.

m) PLC shall be suitable for operating in temperature range of 0^o C to 60^o C and in Relative humidity of up to 99%.

n) In order to remove dissipated heat effectively from the cabinets/ panels housing electronic cards, controllers and PLCs, suitable ventilation shall be provided by means of atleast two louvers with filters and ball bearing fans in the panels.

o) All the system and marshalling cabinets for DCS, ESD PLC, Fire & Gas System, UCP, RCP etc. shall be free standing cubicles preferably with standard dimensions of 800/ 1200 mm x 800 mm x 2100 mm (W x D x H).

Clear working spaces of 1000 mm in rear and 1500mm in front of the panels shall be provided for door opening and maintenance purpose.

Cabinets shall be placed such that aesthetics of the control room is maintained.

p) Unit Control Panels (UCP) of packaged equipment shall be free standing and as per the equipment package Vendor's standards. For each unit, individual panel shall be preferred.

However, all the electronic signals from these UCPs e.g. DI, DO, AI, AO etc. shall communicate with the DCS/ RTU/ PCS or any HMI/ Control System located in the control room through serial or TCP/ IP interface for 'monitoring'. 'Trip' from platform ESD System to package UCPs shall be hardwired. Critical alarms to DCS from UCPs and start/ stop signals from DCS to UCPs shall also be hardwired.

q) All front mounted user interfaces like displays, keypads, lamps, push buttons, selector switches etc. on Control Panels shall be arranged in a logical and aesthetic manner for easy visibility, operation and maintenance. User-interfaces shall be mounted above 750mm from the panel-base.

Each user interface on the front and rear side shall have a name plate indicating their description.

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All lamps on Control Panels shall be provided with lamp test facility.

r) All fused and non-fused DIN Rail Mounted Terminal Blocks (TBs) used in DCS, ESD and F&G system panels, marshalling cabinets and field junction boxes shall be screw less Push-in type to facilitate easy tool-less insertion of conductors. TB connection shall be designed to withstand deck vibrations for maintaining reliable contact.

Stacked type/ multi-tier TBs shall not be used.

The TBs shall be preferably of Phoenix/ Wago/ Weidmüller/ Allen Bradley make.

s) Screw less Push-in type Busbar TBs (Wago or equivalent) shall be used for terminating 24 VDC or 110 VDC wire terminals or shield (screen) connections instead of Busbar plate in System and Marshalling Cabinets of DCS, ESD and F&G System.

t) Nickel plated brass lugs shall be used for wiring terminations. The connecting wires shall have double cross ferruling using tube ferrules with source and destination marking on both ends.

u) Bottom seal shall be used for cable entry in System and Marshalling Cabinets of DCS, ESD and F&G System in place of traditional cable glands/ gland plates to achieve easy, quick, light weight, flexible and high density cable installation. This removable Panel Cable Sealing System shall be designed to ensure bonding, grounding and EMI protection as per requirements of OEMs of DCS/ ESD/ F&G System. Sealing system Vendor

shall be from Multi Cable Transit (MCT) Vendor list in Suggested Vendor List (SVL).

09 v) MCT blocks shall be used for cable entry in the Control Room, TIC/ E&I Room, Non-pressurized equipment rooms, Enclosure area etc. as passive fire and pressure protection seal and shall be re-usable. It shall be designed to protect against fire, water, smoke, dust, gas, rodents and other hazards.

w) Rechargeable Portable gas detectors shall be provided as per the quantity specified in the SOW.

3.6.4.1.6 Safety Instrumented System

In general, the following safety systems shall be provided:

a) **Emergency Shutdown (ESD) System**: The ESD system shall initiate automatic process shutdown in case of abnormal condition of the specified process parameter.

09 b) **Fire & Gas (F&G) System:** The F&G System shall initiate appropriate actions viz. alarms and/or shutdown upon detection of pre-set levels of Hydrocarbon (HC) and/ or H2S accumulation and Fire. The F&G system shall conform to the Functional Specification FS-3500.

On Wellhead Platforms: Fire and Gas detectors, fusible plugs and ESD & FSD pull stations (Pneumatic/ Electric) shall be provided with Control Cards or PLC as per the scope of work.

On Process Platforms: SIL-3 PLC based Fire and Gas Detection System shall be provided. For details, refer FS-3500.

c) **Manual ESD & FSD Stations**: The ESD & FSD stations shall be provided at all strategic locations on the platform for manual initiation of ESD or FSD. All the ESD & FSD stations shall be located in places which are accessible on main walkways and stairs and provided with Acrylic covers to prevent accidental actuations.

d) **HIPPS** system shall be provided for pipeline protection and integrity, wherever specified in the Basic Bid Work.

e) All shutdown and alarm switches shall be "Fail Safe" and the targeted abnormal conditions shall cause a loss of actuating signal to the final control element.

09 f) Automatic trip circuits shall meet the following requirements:

i) Parameters used for shutdown shall be sensed by independent/ individual sensors at independent tapping points. Such sensors and tapping points shall not be shared by any other loop.

ii) All trips shall have a pre-warning alarm and an alarm to indicate the trip conditions.

iii) Adequate trip bypass facilities are to be provided with warning and visual indication in the operator console/ annunciation panel to indicate that the trip has been bypassed.

3.6.4.1.7 Optical Communication Cable

09 If Fiber Optic Cable is envisaged in Scope of Work for communication between Platforms/ Wellhead Platform to Main Process Platform, it shall comply following specifications as a minimum:

a) The optical fiber shall be encapsulated in steel tube or armored during normal handling which shall be filled with jelly to provide high degree of protection from pressure and impact. Adequate free length of optical fiber shall be provided to offset longitudinal stress.

b) Suitable material like high density polyethylene shall be used to protect optical fiber from abrasion. Optical Fiber shall be protected from water ingress and shall have adaptable 'ohmic' resistance. Splicing and preparation of optical fibers for termination shall be done with standard and approved/ certified equipment. All optical fiber cores coming shall be first terminated at fiber termination cabinet. Terminals and adapters used shall conform to the relevant specifications. Patch cable/ extension cable from the Splitter JB (first terminal box) to the terminal equipment (PLC cabinet, DCS cabinet etc.) shall be redundant, routed separately and adequately protected against physical damage/ fiber breaks. Each fiber within the cable shall be uniquely colored and tagged for identification.

Splice boxes (DIN Rail mounted), data connectors, patch cables, couplings, pigtails etc. shall be Pheonix or equivalent make.

09 c) The cable shall be chemical resistant, termite & rodent proof and moisture proof. It shall be flame retardant/ fire resistant depending on application as defined in FS-3508.

d) Optical equipment e.g. Fiber Optic interface modules/ communication modules, standard connectors, switches, network components etc. shall be provided in a redundant configuration for end-to-end communication as per scope of work to ensure effective connectivity.

e) Specific to topside or sub-sea requirement, following shall also be complied:

1. For topside use:

Optical Fiber cable laid between platforms in the same complex shall be multi-mode type. For further details, refer Instrumentation FS-3508 (Instrumentation Cables).

2. For sub-sea use:

i) Optical fiber cable in composite cable shall have high strength, low loss and should be designed to meet both mechanical, optical

requirement and shall be suitable for laying up to 100 meters of water depth.

ii) The optical fiber cable shall be corrugated steel tape armored comprising of minimum of 12 Nos. single mode fibers. Number of fibers of each kind and their construction shall be selected so as not to load the individual fibers beyond 50% of their bandwidth capacity and 50% of the number of each type of fibers shall be available spare in the cable for future use.

For further details, refer Electrical FS-4020 (Subsea Composite Cable).

3.6.4.2 INSTRUMENT POWER SYSTEM PHILOSOPHY

3.6.4.2.1 Pneumatic/ Hydraulic Supply

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a) For Pneumatic Instruments, dry instrument gas/ air supply shall be as per Clause 3.6.4.1.4.1 above.

09 b) Hydraulic power supply for hydraulic systems shall be as per the relevant Functional Specifications FS-3511, FS-3506 etc. as applicable.

3.6.4.2.2 Electric Power Supply

a) Components of power supply system shall be of the highest available quality for reliability and long service life.

b) Contractor shall provide Uninterruptible Power Supply (UPS) for all field instruments, DCS, ESD system, F&G system, control room instrumentation etc.

c) Power distribution to all instruments shall be through independent *indicating cartridge type* fuses for isolating the instruments during maintenance.

d) Dual redundant power supply units shall be fed from different power distribution boards. Where dual redundant power supply in cards/ modules/ sub-systems is specified, the supply shall be fed from different power supply units.

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e) In general, the following Power Supplies shall be used:

1. For Process Platforms and Smart Wellhead Platforms with External Power through cable:

i. 230/ 110 VAC ± 5%, 50Hz ± 1% (UPS with 'Maintenance free' Battery Back-up) for all Instrumentation & Control System(s) e.g. DCS, ESD System, F&G System, etc. However, the Instrumentation & Control System(s) using the UPS supply shall be suitable for 230/ 110 VAC ± 10%, 50 Hz ± 3%.

ii. 24 VDC ± 5% as applicable for field instruments, interrogation voltage (Logic/ Interlock systems), Solenoid valves, Detectors in Fire & Gas system, etc. This voltage may be derived from the '230/ 110

VAC' UPS voltage by the respective System Vendors in their system panels for use.

iii. F&G system shall have floating ground power supply (with its '–ve' not earthed).

iv. Non-UPS power 230/ 110V AC ± 5%, 50Hz ± 1% for backlit level gauge illumination and panel interior lighting.

v. 230 VAC ± 5%, 50Hz ± 1% (Non-UPS Utility Power) for maintenance activities, miscellaneous requirements, Utility sockets, etc.

vi. Independent battery back-up shall be as per FS-3500.

09 **2. For Conventional Wellhead Platforms:**

i. 24 VDC ± 5% Battery Negative earthed for field instruments, solenoid valves, status lamps and Process PLC (if specified in SOW).

ii. 24 VDC ± 5% for F&G System. F&G system shall have floating ground power supply (with its '–ve' not earthed).

iii. Independent battery back-up shall be as per FS-3500.

f) Any other voltage levels required by Instrumentation Package Vendor shall be obtained/ converted from power supply mentioned at Clause e) above in respective panels itself and distributed within system battery limit.

g) AC/ DC distribution board with Supply type (AC/ DC with rating) tag shall be provided with suitable isolator/ switches and fuse units for each feeder.

h) For details of electric supply system, refer Electrical SOW and Electrical Design Criteria.

3.6.4.3 INSTRUMENT EARTHING SYSTEM PHILOSOPHY

09 Following Instrumentation items or equipment, but not limited to the list, shall require different types of Earthing:

- Field Instruments,
- Control System Cabinets,
- Analyzer Housing/ Cabinet,
- Junction Boxes,
- Enclosures,
- GI Cable Trays
- Consoles.

Mainly three separate Earthing Systems shall be provided for Instrumentation.

a) **Electrical Safety Earth –** Bonded to the site structure and utilized for electrical safety of metal enclosures and chassis on all instruments and electrical components.

Cable armors shall be earthed at both ends for lightning protection/ shock prevention i.e. at the Junction Box and at the Control System end. Since the Junction Box body and Control System cabinet body is connected to the site structure, cable armors are ultimately connected to the Electrical Safety Earth.

09 b) **Instrument Clean Earth** – The primary aim of the Instrument Clean Earth is to provide a low impedance path to the noise currents induced by RFI/ EMI. Shields of Single pair/ Multi pair Analog Instrument Signal Cables shall be connected to this Earth.

Instrument Cable Screens/ Shields shall be cut at field sensor/ instrument end and properly taped/ heat shrinked to insulate from the site structure and other metal works and bonded to the main electrical earthing system at a single point to avoid creation of grounding loops. It shall be properly terminated at Marshalling Cabinet side.

09 c) **Intrinsically Safe Earth** – This Earth shall be provided for IS instruments only and shall be utilized for termination of IS active barrier Earth connections. It shall be insulated from the site structure and other metal work and bonded to the main Electrical earthing System at a single point.

3.6.4.4 EQUIPMENT PROTECTION PHILOSOPHY

3.6.4.4.1 Environmental Protection

09 a) All instruments/ equipment and installation material shall be suitable for the overall climatic & ambient conditions and the local environment specified in the Basic Bid work. The conditions will include exposure to moist salt laden atmosphere, sea water splash, sunlight, monsoon rainfall, high ambient temperature and humidity, wind, fungal growth, vibration and shock, EMI and RFI. All equipment shall be capable to withstand these conditions during shipment, storage and installation prior to commissioning. Instrumentation shall withstand not only the quoted environmental conditions, but also the periodic testing of the Deluge or Fire Hose System.

b) As all of the Company's sites are subject to seismic activity as indicated in the General Basic Bid Work and General Design Criteria, all instrument/ electrical frames, panel and racks shall be securely fixed in position.

09 c) In view of the highly corrosive ambient conditions, all internal and external parts which are not inherently corrosion resistant by choice of material shall be prepared and finished by plating or paint finish in accordance with FS-2005 (Structural Functional Specification for Protective Coating). Seals and purges shall be used as necessary, to ensure high resistance to corrosion and provide reliable instrument performance.

09 d) All field instruments shall be provided with necessary weather and anti-corrosion protection. All field instruments shall be provided with plastic bags (min. 1.5 mm thick) to protect them during handling, installation and commissioning. The bags shall be kept in place at all times except during

work on the devices. Drying agent (desiccant) shall be put inside the bag and shall be replaced after its color change. Additional protection by other means such as canvas shall be provided to prevent damage caused by welding during construction work at nearby location. Labels and tags that may be exposed to paint spray shall be temporarily masked with a transparent material during construction activities, which shall be removed at the time of hand over of the work. Plastic plugs/ caps shall be fitted to all instrument tubing and air, process and cable entry ports until final connections are made.

e) Corrosion allowance shall be considered in addition to the design thickness of all plates/ sheets and materials used for installation, hand-wheel of valves, weather hoods, mini shelters, stanchions, supports etc. to arrive at the minimum thickness requirement, as the same are exposed to harsh marine environment on the platform. Painting shall be done in line with the painting specifications enclosed in the bid document.

3.6.4.4.2 Ingress Protection

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a) All electronic field instruments shall have ingress protection to IP-65 or better. All pneumatic field instruments used in control applications shall have ingress protection to IP-55 or better. All instruments installed inside pressurized equipment/ control rooms shall have ingress protection to IP-42 as a minimum.

b) Control Panels shall be suitable for Hazardous Area Classification. However, in general:

i. Control Panels in open areas shall be IP-65 rated or higher.

ii. Control Panels in non-pressurized equipment rooms and enclosed areas shall be IP-54 rated or higher.

iii. Control Panels in pressurized equipment rooms and control rooms shall be IP-42 rated or higher.

3.6.4.4.3 Tropicalization

All electrical components shall be tropicalized to protect against humidity, moisture and fungal growth by means of hermetically sealed units, protective coating on circuit boards, gold plated edge connectors, etc.

3.6.4.4.4 Hazardous Area Instrumentation

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a) The Hazardous Area shall be classified in accordance with API RP 505 or API RP 500. All instruments which are mounted outside of normally pressurized control/ equipment rooms shall be certified by statutory bodies such as IEC/ FM/ UL/ BASEEFA/ CENELEC/ CSA/ PTB/ ATEX/ PESO for

use in Class I, Div. 1, Group C & D, T3 Hazardous Area, *irrespective of instrument's location.*

09 b) "Intrinsically safe" protection shall be provided for all process transmitters in hazardous areas using external barriers.

c) All other instrument loops e.g. solenoid valves, gas detectors, UV, IR fire sensors, intercom/ paging system etc. shall be explosion proof/ flame proof conforming to Ex-d or NEMA-7.

d) Where instruments are not manufactured with enclosures suitable and certified for Class I, Div. 1, Group C & D or Class I, Zone 1, Group IIA and IIB, enclosures suitable and certified for Class I, Div. 2 Group C & D or Class I, Zone 2, Group IIA and IIB may be accepted for installation in areas classified as 'Class I, Div.2 Group C & D or Class I, Zone 2, Group IIA and IIB'.

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However, instruments certified for use in Class I, Div. 2, Group C & D or Class I, Zone 2, Group IIA and IIB areas shall not be installed in areas classified as 'Class I, Div.1, Group C & D or Class I, Zone 1, Group IIA and IIB'.

e) Local Indicator, Transmitter, Input/ Output to peripheral devices like laptop computers & networks etc. shall be intrinsically safe.

f) All 'enclosure type' shall be certified by statutory bodies like IEC/ FM/ UL/ BASEEFA/ CENELEC/ CSA/ PTB/ ATEX/ PESO etc. for use in the area applicable.

g) Intrinsic safety approval shall be based on entity concept and necessary compatibility checks shall be carried out by Contractor before selecting any loop component. Isolating barriers shall be normally active type unless otherwise specified. It shall be plug-in type, mounted on modular back plane termination units. Each individual input and output in a loop shall have its separate barrier. No barrier shall be shared between any two loops.

09 3.6.4.4.5 Radio Frequency (RF) and Electromagnetic (EM) Interferences

All equipment shall remain unaffected and immune to RF-EM interference as per IEC 61000-4-3. Levels of permissible RFI and EMI shall be as per IEC 61000-4-3. Band-pass and/ or band stop filters shall be fitted, as necessary.

09 3.6.4.4.6 Sealing

a) Seal systems shall be used to isolate instrument from the process fluid encountered in the following services:

- Process fluids that vaporize, condense or solidify under operating pressure and ambient temperature.
- Process fluids that will subject the sensor to high temperature.

- Corrosive process conditions.
- Viscous liquids.

b) Sealing may be accomplished with diaphragm seals.

c) Any other protection, if required, shall also be provided.

3.6.4.5 MATERIAL SELECTION PHILOSOPHY

09 a) All materials and equipment furnished shall be new and unused, free of defects, highest grade & quality and in current manufacturing range for the required service.

b) Material of Construction (MOC) of instruments shall be chosen as per temperature, pressure, corrosion conditions and other process requirements.

c) Suitability of material for the corrosive effects of the harsh offshore atmosphere.

09 d) Suitability of wetted parts for the specified process conditions, with SS-316 as a minimum, for outside pressurized rooms and Monel for salt water service. SS316 shall be with carbon content of 0.08% or better to improve resistance to corrosion.

e) Galvanic compatibility between dissimilar materials, with isolating bushes, plates, used where necessary to prevent corrosion due to galvanic action.

09 f) Wherever process condition demands NACE quality Instrument tubing and fittings, the same shall be provided as per latest NACE standard material. For non-NACE application, material for tubing and fitting shall be appropriate for relevant process fluid with SS-316 as minimum.

g) MOC of Instrument Enclosures and Junction Boxes (JB) shall be SS-316/ epoxy coated copper free (i.e. less than 0.4% copper by mass) die-cast Aluminum.

h) Moulded polyester parts shall be Anti-Static for Hazardous Area locations, and in general, be constructed from UV-stabilized glass reinforced polyester. Impact Resistance shall be in accordance with IEC-62262 & IEC-60068-2-75.

i) All proposed Vendor standard plastic components, if any, shall be non-toxic and fire resistant, UV stabilized and compatible with the environmental conditions.

j) All spindles, bushings, bolting, screws, brackets, etc. shall be manufactured from a suitable grade of stainless steel as minimum. All bolts and screws shall have a flat SS-316 washer under the nut, and the thread length shall be such that there is complete full engagement of the nut, with a minimum of two threads protruding.

k) Supports for instruments and any other item directly welded to the structure shall be painted conforming to latest revision of Functional Specification FS-2005 (Structural FS for Protective Coating).

l) All wetted parts of instruments which are in contact with process fluid containing CO_2 in excess of 2 PSI partial pressure shall be as follows:

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Fluid Temperature	Material to be used
<71°C	ASTM A182-F316 (316SS)
>71°C	ASTM A182-F51 (2205 duplex steel)

m) The MOC for the wetted parts and the body of all individual instruments shall be as per the Material Selection Chart wherever given in the Annexure of the relevant Functional Specification. The MOC of all parts shall be as indicated in Material Selection Chart or better suitable for the process conditions.

Where such chart is not available/ piping class is not covered in the respective FS, Vendor shall follow material specified in PMS and/ or P&ID and suitable for the fluid composition and the process conditions. Proposed MOC shall be subject to Company's review and approval.

n) The details of various piping classes mentioned in these Material Selection Charts shall be as per Piping Material Specifications (PMS) enclosed with the 'Piping' section of the bid document.

3.6.4.6 INSTRUMENT INSTALLATION PHILOSOPHY

a) All work shall be of the highest quality craftsmanship and shall conform to the best applicable engineering practices and relevant codes referred in the bid document. Installation of field mounted instruments for process measurement in general shall be in accordance with API RP 551.

b) All instruments and accessories shall be installed in a neat workman like manner for ease of operation and maintenance.

c) The Contractor shall prepare hook-up and installation detail drawings for each type of instruments, junction boxes, control panels etc. Installation shall be carried out in accordance with approved drawings.

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d) The Contractor shall install instruments and equipment with due consideration to the following:

i. All process instruments shall be located as close as possible to the point of measurement and accessible from the deck, ladder or a platform.

ii. Instruments shall be mounted/ connected so as not to stress vessel nozzles or pipe tapping.

iii. Instruments e.g. PT, DPT, FT etc. shall not be installed by taking support of its impulse piping/ tubing or electrical connections. Pressure gauges and Temperature indicators shall be installed directly on process piping.

iv. All cables/ tubing shall be properly laid on cable/ tube trays which shall be supported at regular intervals. Instruments, tubing, cables and cable ladder shall not be fixed to gratings or handrails unless approved by the Company.

v. Fittings such as instrument isolating valves and instrument air or gas regulators shall be supported either on the stanchion or close-coupled to the instrument in a manner that no undue stress is imposed on the tubing or instrument.

vi. Visibility and accessibility shall be ensured for maintenance and operations purpose.

vii. Positioning of equipment/ instruments shall not constitute a safety or operational hazard.

viii. Wherever possible, instruments and junction boxes shall be installed such that they are protected from the effects of rain and sun, ensuring easy access and visibility. Where this is not possible, a fixed cover or hood shall be provided to protect instruments, without impairing access or visibility.

ix. All the instruments and detectors in the well bay area shall be protected by fixed cover or hood.

09 x. Field mounted Local Control Panels shall also be weather protected by minimum 3mm thick stainless steel canopy.

xi. All instruments shall be installed and fastened such that deck vibration does not affect their performance.

xii. Easy access and provision for lifting heavy items e.g. valves, pressure relief valves, orifice etc. shall be provided.

xiii. Instruments requiring frequent routine access (including hand-valves, manual resets, manual switches, etc.) shall be mounted approximately 1.4m above the Top of Plating (TOP)/ Top of Grating (TOG) for ease of accessibility

xiv. Instruments shall be properly supported on brackets or mounted on sub-plates, or placed on a suitable pedestal, pipe stand or structural support. Pipe or structural stands may be welded directly onto platform plate, with a suitable penetration in the grating, where applicable. Where platform's structural member is used for the mounting of instruments/ accessories, the same shall be done at yard before the painting/ finishing of the structural member so that welding at offshore site should not make it susceptible to corrosion in marine environment.

e) Instrument stanchions, gauge boards, junction boxes and panels shall be mounted as per approved drawings and with consideration for:

i. The most direct routes for tubing and piping to and from the above mounting arrangements, using common tubing runs and avoiding crossovers.

ii. Ease of inter-connections between instruments.

iii. Ease of access for on-site calibration and/ or removal of instruments

iv. Minimum interference between tubing, piping and cabling to instruments.

f) SS-316 filter regulator with gauge shall be provided with each instrument requiring regulated gas or air supply.

09 g) The Contractor shall install drain and vent connections with due consideration to the following:

i. All liquid drains of field instruments, especially for level instruments, shall be taken to the deck level facing downwards and routed to the nearest process/ deck drain.

ii. All vents shall be suitably routed to safe location away from the operator/ maintenance personnel operating on the instrument. All atmospheric vent ports of venting instruments and pilot valves shall be fitted with bug screens.

iii. All instrument drains/ vents not routed to deck drains/ safe locations shall be suitably plugged.

iv. All vent/ drains on high pressure applications shall have seal welded connections and shall be tested suitably.

h) The instrument mounting details (on stanchions/ deck plate, gratings, structural member etc.) shall be as per the standard engineering practices and installation standards, and the same shall be submitted for information with all necessary details of location and mounting.

i) Working platforms with access ladders shall be provided for elevated inaccessible instruments.

j) Wherever Piping or Vessel is insulated, electrical heat tracing shall be provided for Level Instruments too to avoid oil congealing or condensate formation.

k) Additional installation requirements of individual instruments, if any, shall be as per the relevant Functional Specifications.

09 l) **Stand Pipe/ Bridle Philosophy for Level Instruments**

Stand pipes/ bridles shall normally be avoided. They can be considered only in small sized vessels/ boots having space constraint for nozzle tapping.

i. In general, level instruments for alarm and trips should be mounted on separate stand pipes.

ii. The material and rating of the nozzle isolation valves and pipe fittings shall match with the pipe specification of the vessel. The pipe diameter of the bridle should be minimum 2" NB.

iii. Thermal insulation and heat tracing for standpipes/ bridles, sensor cages/ chambers shall be provided as per the associated piping

iv. Standpipe shall not be used for Emulsions and induced emulsions formed

by foams agitation in highly agitated liquids.

v. No level instrument shall be installed inside the standpipe itself.

vi. Standpipe connection from the bottom of the vessel shall be avoided.

vii. On horizontal vessels having boot, separate standpipe shall be used for boot interface level measurement in addition to standpipe for the horizontal vessel.

09 viii. Each instrument shall have its own isolation valves, vents and drain to facilitate maintenance. Lock Open (LO) isolation valves shall also be provided between the vessel and bridle. The distance between the standpipe and the vessel nozzle should not exceed 1.5 m.

ix. A vent valve shall be provided on top of standpipe to vent any air or gas in upper dead part of the standpipe. Minimum ¾" drain valve shall also be provided at the bottom.

x. Level instruments shall be mounted on bridle in such a way that no gas trap or liquid trap should occur when level in the vessel crosses the tapping level.

xi. Suitable strain relief measures e.g. bridle supports shall be provided to avoid unacceptable loads on the vessel nozzles due to long bridles, thermal expansion or heavy instruments mounting.

xii. Standpipes shall be fabricated as per level sketches.

3.6.4.7 INSTRUMENT INSPECTION & TESTING PHILOSOPHY

3.6.4.7.1 General

a) The Vendor's quality plan shall include a comprehensive fully documented Inspection and Testing Plan (ITP) specific to the project.

b) The Contractor/ Vendor shall consider all the testing requirements as mentioned in the individual IRTs for each type of instrument.

c) The procedures shall include inspection specifically for compliance with hazardous areas requirements, including valid certificates, without which no circuit or loop shall be energized.

09 d) Company Representative/ ONGC Certification Agency (CA) TPI shall check instruments, control panels and instrumentation-packages prior to shipment for compliance of Functional Specifications and Inspection Requirement Tables (IRTs).

e) All testing, calibration and pre-commissioning shall be done by the Contractor. The Contractor shall also provide assistance as required during commissioning activities.

For major package equipment e.g. DCS, ESD System, F&G System, HIPPS etc. the Contractor shall carry out site testing, pre-commissioning and commissioning activities in the supervision of OEM/ Packager.

f) The Contractor shall provide suitable workshop facilities and shall provide all necessary test and calibration instruments and equipment.

g) The Company reserves the right to reject any or all test and calibration work if found not complying with the specifications. The Contractor shall complete and submit documentation for all calibration, testing and pre-commissioning.

h) The Contractor shall in the presence of the Company Representative/ ONGC CA TPI, verify by inspection, calibration and loop testing that all instrumentation in field and control room including local and remote control panels are complete and operable as per specification requirement. All testing and calibration shall be subject to approval by the Company Representative/ ONGC CA/ TPI.

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i) In addition to calibration/ testing, loop checking, setting for safety devices like process switches, safety valves etc. and simulation testing of all interlock and shutdown systems, done at fabrication yards, the same shall also be carried out at offshore site as well.

j) Major Equipment e.g. DCS, ESD System, F&G System and HIPPS shall be inspected during Factory Acceptance Test (FAT) based on the approved FAT document. Site Acceptance Test (SAT) shall be done at site during commissioning as per approved SAT document.

k) Logic as per approved Cause & Effect Matrix, Safe Chart, Schematics etc. shall be demonstrated and checked by simulation during FAT. Changes in graphics, if any, shall be made based on user-feedback during FAT/ SAT.

l) In general, the pipes and tubes shall be cleaned before testing. They shall then be subject to hydro-test (or other applicable tests) and then blow dried.

3.6.4.7.2 Flushing of Lines

The Contractor shall remove in line instruments like flow meter, control valves/ safety valves, if necessary, and provide spool pieces/ flanges prior to flushing of lines.

3.6.4.7.3 Instrument Air/ Gas Supply Lines

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a) Instrument air/ gas piping and tubing shall be disconnected upstream of all air filter regulators before testing and then piping and tubing shall be hydro tested. Then it shall be blown down to remove hydro test water, slag and mill scale.

b) Instrument air/ gas supply lines shall be blown with instrument air prior to connecting to instruments. Instrument air/ gas mains shall be isolated from the instrument and pressurized to 1.5 times maximum working pressure with instrument air, they shall be isolated from the pressure source and the pressure reading on test gauge shall not fall by more than one psig in ten minutes.

c) Nitrogen shall be used at wellhead platforms where Instrument Air is not available.

3.6.4.7.4 Instrument Signal Lines

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a) Instrument signal lines shall be blown with instrument air/ gas prior to connecting to instruments.

b) Nitrogen shall be used at wellhead platforms where Instrument Air is not available.

c) All air/ gas tubing shall be tested and inspected by one of the methods given in ISA RP 7.1 “Pneumatic Control Circuit Pressure test”.

3.6.4.7.5 Impulse Lines

a) All process impulse lines shall be disconnected and flushed with potable water. Air lines shall be blown down with filtered air. Hydraulic lines shall be flushed with hydraulic oil.

b) After flushing, process impulse lines shall be isolated from the instrument and pressurized hydraulically to 1.5 times maximum working pressure corrected for ambient temperature. They shall then be isolated from the pressure source and the pressure reading on a test gauge shall not fall at a rate exceeding one psig/ hour.

3.6.4.7.6 Direct Mounted Level Instruments

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For the direct mounted level instruments, the associated equipment/ vessels/ tanks shall be pressurized to the maximum operating pressure gradually and steadily with the instrument on line. The equipment/ vessels/ tanks shall then be isolated from main pressure source. The pressure shall not fall at a rate exceeding one psig/ hr.

3.6.4.7.7 Wiring

a) Wiring shall be checked to ensure that it is correctly connected and properly grounded. Insulation test shall be carried out on all wiring taking necessary precautions. Correct connections of all switches shall be checked.

b) Tests for Cables and wiring shall be as per the specifications elaborated in the Electrical section of the bid document. Wiring shall be ‘Megger’ tested as a minimum for the rated voltage.

3.6.4.7.8 Calibration & Testing

a) During pre-commissioning/ commissioning, the Calibration/ Testing of instruments shall be done with the instrument or system in place, wherever possible. Otherwise instrument calibration shall be done prior to installation.

b) All analog instruments shall be calibrated in both upscale and downscale directions. Five (5) Point calibration shall be done at 0, 25, 50, 75 and 100 per cent of the instrument range upscale (rising) and 100, 75, 50, 25 and 0 percent of the instrument range downscale (falling). Upon completion of these tests, the instruments shall be drained, the components removed and the shipping stops put in place.

5 point recalibration shall also be checked during pre-commissioning/ commissioning.

c) Factory calibrated flow meters shall be installed and fastened as per Vendor guidelines after the associated piping has been flushed and blow dried.

d) During pre-commissioning/ commissioning, all Control Valves prior to stroke checking shall be cleaned thoroughly. The full stroke of valve shall be checked for opening and closing in the range of 0%, 25%, 50%, 75% and 100% (rising & falling).

09 e) Pressure safety valves shall be set/ tested by using dry air/ nitrogen for gas service and water for liquid service.

f) All trip switches shall be calibrated over the entire range and checked for set points & reset points as per the alarm & trip schedule.

g) Solenoid valves shall be checked functionally for its operation, auto/manual reset and seat leakages.

h) If Process SOW or P&IDs mention Partial Stroke Testing (PST) of Shutdown Valves (SDV), PST check shall be demonstrated during FAT/ SAT.

i) Tight Shut Off (TSO) Control Valves and Shut Down Valves shall be checked for seat leak test and gland leak test.

j) All special instruments like analyzer shall be checked and calibrated as per manufacturer's instruction. Prior to testing, all analyzer sample lines shall be thoroughly cleaned by Carbon Tetrachloride or any other cleaning liquid. After cleaning, these lines shall be thoroughly purged with dry nitrogen.

k) **Testing and Calibration Report**

i. The Contractor shall provide records of all instrument calibration/ testing in prescribed format (Annexure-I).

ii. Calibration Certificates of all Master calibration equipment used for site calibrations like pressure gauges, HART calibrator, multifunction calibrator, multimeter etc. shall be traceable to PTB (Physikalisch-Technische Bundesanstalt)/ NMI (Netherlands Measurement Institute, National Metrology Institute, National Measurement Institute)/ OIML

(Organisation of Legal Metrology)/ NEL (National Engineering Laboratory)/ NABL (National Accreditation Board for Testing & Calibration Laboratories) etc. accredited laboratories certified in accordance with ISO/ IEC 17025 (General requirements for the competence of testing and calibration laboratories) standard.
The Contractor shall submit these documents along with Annexure-I.

3.6.4.8 INSTRUMENT SPARES PHILOSOPHY

09 **3.6.4.8.1** For all major equipment, the Contractor shall include Commissioning Spares (Annexure - VI) and Mandatory Spares in the Purchase Specification (PS) as a part of the supplied package/ system/ instrument.

The Contractor shall also furnish list of recommended spares for one year's trouble free operation in prescribed format (Annexure - VII) along with the prices for Company's review.

09 **3.6.4.8.2** Original Equipment Manufacturer (OEM)/ Packager shall provide guarantee for supply of spares for at least 10 years for electronic instrumentation system(s) for the entire life cycle of the item.

In case, Vendor intends to phase out the models of the supplied major systems such as DCS/ PLC/ F&G System etc. any time after 07 years, they shall intimate the Company well in advance and ensure availability of spares for supplied systems for the next 03 years.

If the Vendor intends to phase out the model of the supplied system before 07 years have elapsed, they shall intimate the Company well in advance and ensure availability of spares for the balance period of 07 years and an additional 03 years.

3.6.4.8.3 All spare parts shall comply with the same standards and specifications as the original equipment/ system and shall be fully interchangeable with original parts.

3.6.4.8.4 All spare parts shall be marked with the manufacturer's part number and description.

3.6.4.8.5 All spare parts shall be packed separately from the main equipment/ system/ instrument and shall be properly protected to prevent deterioration and damage during shipment and storage.

09 **3.6.4.8.6** The spares requirement of individual systems shall be as per the relevant Functional Specifications enclosed elsewhere in the bid document. However, unless otherwise mentioned, the following shall be the basis for finalizing the minimum requirements of spares for instrumentation:

a) Mandatory spare instruments shall be 10% minimum or 1 (one), whichever is higher, of each type and range for PG, DPG, TG, T/C, SOV, RTD, AFR & I/P Convertor/ Smart Positioner. It does not include PSV, CV, field transmitters, field switches, flow meters and LG.

The above is applicable for both field instruments and instrumentation/ mechanical/ electrical packaged equipment.

The mandatory spare instruments shall be supplied for each Platform under New as well as Modification scope.

b) It is Vendor's responsibility to have commissioning spares as required for commissioning of plant and shall be included in their firm scope of supplies/ work.

c) For control panels, installed spares shall be 20% minimum or 1 (one), whichever is higher, for Indicating Lamps, Annunciation Windows, Selector Switches, Push Buttons, Relays, Power Feeders (AC & DC) (in ACDB & DCDB), Switch Fuse units (for loop power supply). These spares shall be wired up to the terminal blocks (TB) in the system and marshalling cabinets.

d) Loose spares for panels shall be 5% minimum or 1 (one), whichever is higher. Items to be considered as loose spares are indicating lamps, selector switches, push buttons, relays, MCBs, fuse and non-fuse type field TBs, Push-Pull Valves, Logic valves of each type, timers etc.

Same is applicable for packaged equipment.

09 e) For Wellhead Shutdown Panel, minimum 1 No. Well Control Module & Minimum 1 No. Test Separator Shutdown Module shall be provided or as specified in scope of work.

f) Installed I/O Card modules shall be 20% minimum or 1 (one), whichever is higher, for card based F&G systems.

g) Control panels shall have additional 20% free space available for future expansion.

h) For well head panels, 10% spare instrument air header branch lines and 15% spare bulk heads (for each size) shall be provided in each panel.

i) In junction boxes (JB), 20% wired spare terminals shall be kept as installed spares.

Also, while choosing multi-core/ multi pair cables, minimum, 20% cores/ pairs shall be kept as spare for future use. The entire cable shall be terminated at both ends (JB and Panel) including spare cores/ pairs. Spare cable entries shall be plugged/ blinded.

j) For cable tray sizing, cable tray width shall be sized so that 50% space is free and available for future use.

k) Consumable spares for a minimum of 06 months duration after commissioning which shall include printer paper, toner cartridges etc. shall be provided.

09 **3.6.4.9 PHILOSOPHY FOR FUTURE FACILITIES**

Provisions to operate, monitor, control and shutdown shall be made in all control systems such as control panels (system and marshalling) and shutdown panels

etc. for future facilities, wherever shown in P&ID and/ or indicated in the Basic Bid Work. All panel/ cabinet mounted instruments and accessories required for this purpose shall be supplied and installed by Contractor.

3.6.5 INSTRUMENT PROCESS CONNECTIONS

a) Process connections for instruments on vessels/ stand-pipes/ piping/ tanks/ equipment etc. shall be as per Annexure - XVI.

b) First isolation/ block valve in piping scope and impulse piping (if used) of all instruments shall be as per the piping specifications and shall meet the line or vessel specification.

The first block valve shall be installed as close as possible to the process tapping. Unions between the process tapping and the first block valve are not permitted.

c) Instrument isolation valves and manifolds shall be provided as per hook-up drawings. Vent and drain valves shall be provided in the vent tubing/ piping and drain.

09 d) Tap-offs from lines or vessels for all direct connected pressure instruments shall meet the line or vessel specification. After individual piping block valve, all Pressure instruments shall have 2-valve manifold and Differential Pressure instruments & Multivariable Transmitters (MVT) shall have 5-valve manifold.

For DP level instruments, the manifold arrangement shall be such that the filled leg and dry leg are not equalized during isolation and bleeding of the legs.

e) A maximum of 02 (two) pressure instruments may be connected to the same pressure tapping, if their function is not safety-related. Instruments for ESD systems shall each have their own individual tappings.

f) For Thermowell installation, minimum piping size shall be 4". For line size below 4", expander-reducer can be used. Flange connection and electrical connections shall be as per respective functional specifications.

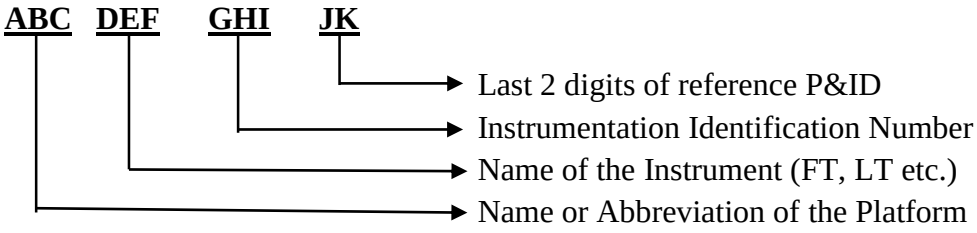
g) Tubing and fittings for air/ gas supply, transmission and control signals, and process impulse lines shall be as per FS-3507 (Instrumentation Bulk Materials). Threaded end connections shall be as per ANSI B 2.1.

h) Flanged connections shall be as per ANSI B 16.5. The Contractor shall provide suitably rated flange adaptors to allow tubing connections where required.

3.6.6 GENERAL REQUIREMENTS

3.6.6.1 TAGGING AND NAME PLATES

09 a) The Contractor shall assign individual tag numbers in accordance with the Company's established system to all instruments. The tag numbers shall in general be as follows:



Fire & Gas devices shall be tag numbered as specified in FS-3500.

b) The tag number of every instrument shall appear on all relevant drawings and documents that pertain to the specific instrument.

c) Nameplates and identification tags shall be provided to properly identify each piece of instrumentation/ equipment including instruments, Junction boxes, bulkhead connections, cables, wiring etc.

d) All instruments shall have SS tag or nameplate of a permanent type with identification number and service description.

e) All panel name plates shall be made of Stainless Steel or Traffolyte with black engraved lettering on white background.

The name plates shall be fastened with SS 316 screws/ riveted.

f) Name plates of front mounted panel instruments e.g. Flow computer shall be made of Traffolyte with black engraved lettering on white background. Also, name plates shall be provided inside the panel on the rear of instruments for clear identification.

09 g) All wiring, cables, tubing, junction boxes etc. shall be properly tagged. Plastic adhesive tapes shall not be used.

All wiring shall be tagged with ferrules at both ends.

All tubings and cables tag plates shall be of SS316 and properly binded with SS316 wire.

h) Labels shall be positioned directly on the equipment or immediately adjacent for best visibility.

i) Typical instrument label details for sour service shall be as per Annexure – III.

09 3.6.6.2 DOCUMENTATION

The Contractor shall provide final documentation in hard copies and electronic form as per the Volume-I of the Bid.

All the AS-BUILT documents shall be verified by ONGC CA/ TPI and duly signed by LSTK Contractor and their DEC.

09 3.6.6.2.1 Project Documentation

a) Design Engineering Consultant (DEC) shall prepare the Project DCI/ MCI along with its category (Approval/ Information) in line with the project scope of work and Standard DCI/ MCI (Appendix C8 of Bid), including any additional documents, and obtain approval from Company.

b) The Contractor shall be responsible for the authenticity of the Vendor/ Sub-Vendor drawings and data.

c) Detailed test schedules for FAT & SAT shall be submitted for Company's approval one month before the testing for major packaged equipment like DCS, ESD System, F&G System, HIPPS etc. For details refer respective FS.

d) Any additional data/ drawings required during detailed engineering shall be furnished by Vendor/ Contractor.

e) All drawings and literatures shall be in English language and in the Metric Measurement System.

09 3.6.6.2.2 Instrument Index

The Contractor shall provide Instrument Index. The index shall include the following information, as applicable:

1. Instrument Tag Number
2. Instrument type/ description
3. Service Description
4. Process Fluid
5. P&ID No./ F&G Detector Layout No.
6. Location of instrument (Field, Control Panel etc.)
7. Vessel/ Line/ Equipment details
8. Operating Pressure/ Temperature/ Flow/ Level parameters
9. Signal (Analog/ Digital/ Serial)
10. Type of Signal (IS/ NIS)
11. Manufacturer/ Model
12. Instrument Range
13. Calibrated Range
14. Purchase Specification No.
15. Data Sheet Number
16. Hookup Drawing No.
17. Installation Drawing No.

18. Remarks

09 **3.6.6.2.3 Purchase Specification (PS)/ MCI**

Purchase Specifications (PS) shall be submitted for Company's review/ approval.

09 **3.6.6.2.3.1 Deliverables**

PS shall have the following documents (as applicable). The PS shall be indexed with page numbers and duly endorsed by LSTK Contractor, DEC and Vendor:

1. Cover sheet mentioning the document number and Vendor details
2. Form-A - Technical Questionnaire (Annexure-IV)
3. Form-B - Compliance to Bid Specifications (Annexure-V)
4. Form-C - List of Commissioning Spares (Annexure-VI)
5. Form-D - List of One year Operational Spares (Annexure-VII)
6. Form-E - List of Special Tools & Tackles (Annexure-VIII)
7. Form-F - List of Construction Spares & Consumables (Annexure-IX)
8. Form-G - Vendor Weight Control Datasheet (Annexure-X)
9. Form-H - Equipment Weight Certificate (Annexure-XI)
10. Form-I - Vendor DCI/ MCI or Vendor Data Schedule (Annexure-XII)
11. Form-J - Vendor Document Review (Annexure-XIII)
12. Form-K - Utility Requirement (Annexure-XIV)
13. Responsibility/ Work Scope Matrix showing responsibilities of LSTK Contractor, DEC and Vendor (Annexure-XV)
14. Instrumentation Design Criteria, related Functional Specifications and IRTs etc. duly signed and stamped by the Vendor.
15. Vendor's Quality Assurance Plan/ Inspection Test Plan approved by ONGC CA/ TPI in line with IRT.
16. Process Datasheets
17. Instrument Datasheets, Specifications and Model De-codification
18. Sizing calculations
19. Job Specification
20. Hardware & Software Functional Design Specification
21. System and Network Architecture
22. Bill of Material
23. General Arrangement Drawing
24. Internal Arrangement Drawing for Panels
25. Power wiring diagram
26. Cable termination drawing for panels
27. Wiring diagram
28. IO Allocation
29. Controller Loading Calculations
30. Power Consumption Calculations
31. Heat Load Calculations
32. Instrument air/ gas consumption details

33. Grounding details
34. Panel wiring table list
35. Reliability/ PFD calculations
36. Loop wiring diagrams
37. Loop schematics
38. Functional schematic & Logic Diagrams (as applicable)
39. Catalogues
40. Installation, Operation and Maintenance Manual (IOM)
41. Weather protection, hazardous area classification & SIL certificates
42. Test certificates, as applicable

09 **3.6.6.2.3.2 Instrument Data Sheet**

Contractor shall submit Instrument Datasheets endorsed by Vendor and approved by DEC in Purchase Specifications.

a) The datasheet shall be as per the formats specified in respective Functional Specifications (FS).

b) Where datasheets are not provided in the respective FS, the datasheets shall be in accordance with relevant ISA standards.

c) When a number of instruments are identical in their requirements, they can have a common datasheet with all the associated tag numbers annexed with the datasheet.

09 **3.6.6.2.3.3 Instrument Sizing Calculations**

The following Sizing Calculations endorsed by Vendor and approved by DEC shall be submitted in Purchase Specifications:

1. Control valve sizing
2. Flow Meter sizing
3. SAPCV sizing
4. Pressure Safety Valve (PSV) and Temperature Safety Valve (TSV) sizing
5. Shutdown/ Blowdown valve torque calculation and stroking time
6. Thermowell Wake Frequency
7. Restriction orifice (RO) sizing
8. Hydraulic Design Calculations including reservoir & accumulator sizing

09 **3.6.6.2.3.4 Manufacturing Record Book (MRB)**

In addition to PS deliverables mentioned in Clause 3.6.6.2.3.1 above, following documents as a minimum shall also be included in Manufacturing Record Book (MRB) as a part of AS-BUILT documentation:

1. Instrument calibration procedures
2. Proposed maintenance schedule of all major components, for the design life of the system, and their replacement requirements
3. Calibration reports and Loop checking reports

4. Test Certificates
5. Material Take Off

3.6.6.3 REVIEW AND APPROVAL

09

3.6.6.3.1 The Contractor and Design Engineering Consultant (DEC) shall review, scrutinize and approve all Vendor & Sub-Vendor documentation submitted as part of detailed engineering for all instruments, instrumentation packages, control systems etc.

Only DEC approved drawings and documents shall be submitted for Company's review.

3.6.6.3.2 For other details of review and approval processes, refer General Conditions of Contract (GCC) of Bid Volume-I.

3.6.6.4 VENDOR SELECTION AND PRE-QUALIFICATION

09

3.6.6.4.1 Selection of Instruments/ Control Systems

All instruments/ control systems offered shall be of proven design and in current manufacturing range.

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3.6.6.4.2 Selection of Vendors

a) Contractor shall select Vendors from the Suggested Vendors List (SVL) of OES.
b) Existence of the Vendor's name in the suggested vendor list does not absolve the Contactor from supplying equipment of proven design as per the Company specification.

09

3.6.6.4.3 New proposed Vendors/ Pre-Qualification of new Vendors

a) Where the Vendor is not listed in SVL, Contractor shall follow provisions of Bid Volume-I. Contractor shall request acceptance of such new Vendor with proper justification.
b) Contractor shall submit Pre-Qualification Documents (PQD) comprising of Vendor Checklist and documentation for such acceptance.
c) The Contractor shall not place PO to new Vendor prior to acceptance by Company.

3.6.6.5 PACKAGING & FORWARDING/ SHIPMENT/ TRANSPORTATION

09

a) Instruments or parts, which can be damaged during shipment, shall be packed separately in the original manufacturer's boxes. Each crate, bag, box, pallet

or package shall be identified with Purchase Order Number and Packing List in a weather proof envelope.

b) Open ports or connections shall be properly sealed. Threaded connections shall be protected with forged steel or moulded threaded plugs/ caps.

09

c) All flanged openings shall be protected with wood or steel closures attached by proper bolting and sealed with a plastic compound to protect it from entry of foreign material and to protect the flange faces.

d) All mechanical or machined surfaces subject to atmospheric corrosion prior to installation on site shall be treated with an easily removable rust preventive coating.

09

e) A desiccant shall be provided inside all box enclosures to prevent moisture damage due to high humidity.

f) A suitable desiccant shall also be placed inside those control panels or junction boxes, which contain electronic or electrical components during shipment.

g) Any equipment or its component exceeding 15 kg shall be supplied with lifting lugs or eye bolts. The lugs or eyebolts should be positioned such that the component can be readily slung from a point over its center of gravity.

09

h) All items shall be properly packed, made sea worthy and protected from damage during shipment. Any instrument which could possibly be damaged during shipment shall be disassembled and removed. The disassembled instruments shall be packed and shipped with the associated system/ package. All openings shall be suitably sealed.

It is Contractor's responsibility to reassemble the disassembled instruments during installation of system/ package.

i) The electronic and electrical components removed from package-mounted panel shall be shipped independently. The shipping notice shall be written on the box or crate with red letters such as "Handle with Care"/ "Keep Dry"/ "Fragile" etc.

j) The Contractor shall be solely responsible for the adequacy of the Preparation for shipment. Contractor shall also state recommendations for long term storage (up to 12 months) for both indoors and open air storage.

k) Nucleonic/ Radioactive material handling:

09

i. Packing, transport, replacement and disposal of nucleonic materials shall be the responsibility of Vendor and Contractor.

ii. Vendor and Contractor shall comply with the safety norms as per international standards & safety regulations and Bhabha Atomic Research Center (BARC, India) guidelines.

iii. Certification from Regulating Authority for safe exposure limit for human safety and operation of the 'nucleonic source' shall be furnished.

iv. Vendor shall furnish all procedures for safe packing, handling and storage along with proper tools and SOPs/ instructions.

09 v. Procedure for safe disposal of radioactive material and its container shall be approved by statutory body of country of origin as well as by statutory body in India.

09 3.6.6.6 RECEIPT AND STORAGE

a) The Contactor shall develop a comprehensive plan for receipt, storage and release of all equipment, instruments etc. including items supplied by Company or other Contractors/ Sub-Contractors. This plan shall provide the status and location of any item under its custody.

b) The Contractor shall be fully responsible for equipment, instruments etc. under its custody. The Contractor shall provide safe and secure facilities to provide a storage environment which shall protect the equipment, instruments etc. as per Vendor's requirements. Any equipment, instruments etc. damaged under the custody of Contractor shall be repaired, reinstated or replaced by them.

09 3.6.7 ANNEXURES

Following annexures are *typical formats* which shall be submitted during design and execution stage.

1. ANNEXURE - I : Test And Calibration Form
2. ANNEXURE - II : Pneumatic Instruments – Air Supply Branching Connections
3. ANNEXURE - III : Label Details For Instruments/ Systems Handling H_2S
4. ANNEXURE - IV: FORM -A - Technical Questionnaire
5. ANNEXURE - V : FORM-B - Compliance To Bid Specifications
6. ANNEXURE - VI : FORM-C – List Of Commissioning Spares
7. ANNEXURE - VII : FORM-D – List Of One Year Operational Spares
8. ANNEXURE - VIII: FORM-E - List of Special Tools & Tackles
9. ANNEXURE - IX: FORM-F - List of Construction Spares & Consumables
10. ANNEXURE - X: FORM-G - Vendor Weight Control Datasheet
11. ANNEXURE - XI: FORM-H - Equipment Weight Certificate
12. ANNEXURE - XII: FORM-I - Vendor DCI/ MCI or Vendor Data Schedule
13. ANNEXURE - XIII: FORM-J - Vendor Document Review
14. ANNEXURE - XIV: FORM-K - Utility Requirement
15. ANNEXURE - XV: Responsibility/ Work Scope Matrix
16. ANNEXURE - XVI : Instrument Process Connection

09 ANNEXURE - I : TEST AND CALIBRATION FORM

TEST AND CALIBRATION REPORT

TRANSMITTER (PRESSURE/ DIFFERENTIAL PRESSURE/ TEMPERATURE/ FLOW/ LEVEL)

						CAL DATE	
						REV No.	
TAG No.		INSTRUMENT TYPE		P&ID No.		SERVICE	
INST RANGE		CAL RANGE		MAKE		MODEL	

CALIBRATION TEST

INPUT %		APPLIED PRESSURE/ RESISTANCE ETC.	STANDARD OUTPUT (IN mA)	ACTUAL OUTPUT (IN mA)	TRANSMITTER READING	DCS READING	ERROR	REMARKS
RISING	0							
	25							
	50							
	75							
	100							
FALLING	100							
	75							
	50							
	25							
	0							

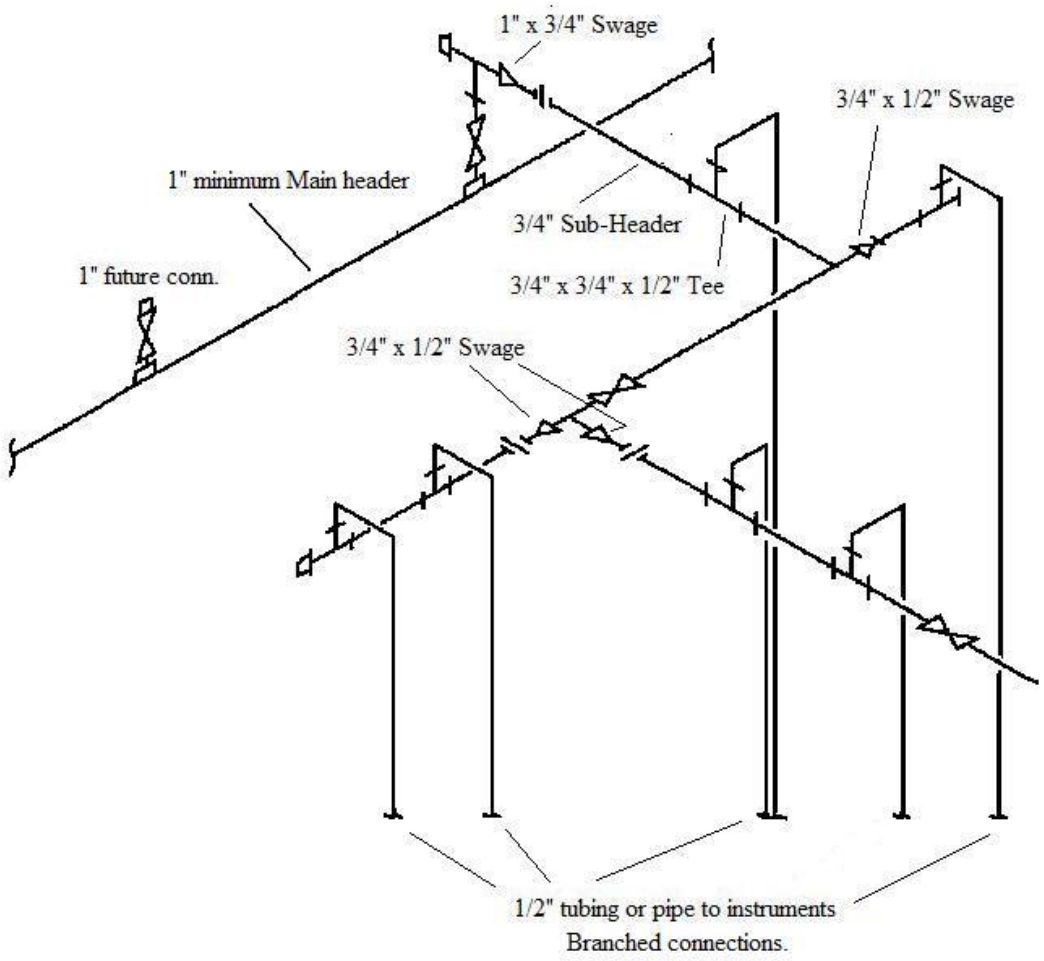
MASTER EQUIPMENT DETAIL

SNo.	NAME	MAKE	MODEL No.	CERTIFICATE No.	CERTIFYING AGENCY	CAL DATE	VALID UPTO	
1	CALIBRATOR							
2								
3								

	CALIBRATED BY	WITNESSED BY	REMARKS
NAME			
SIGN			

Note : Calibration Certificates of all Master calibration equipment shall be attached.

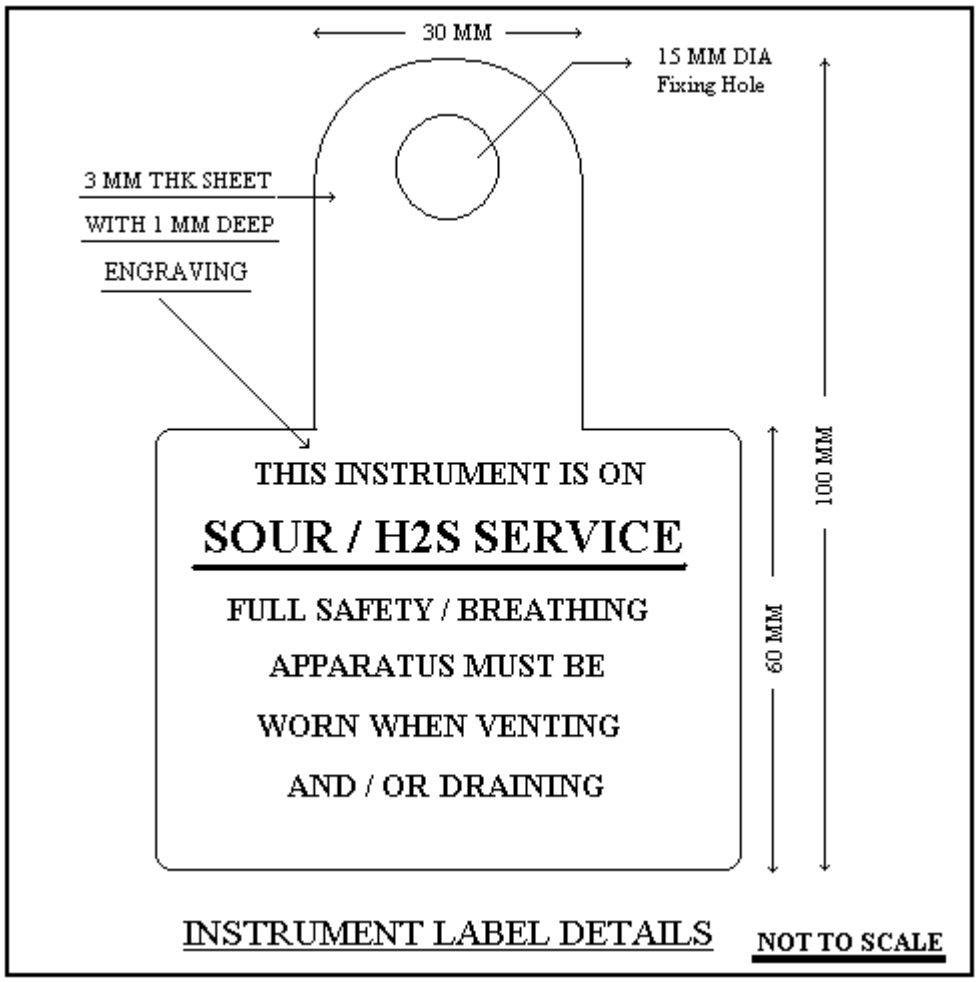
ANNEXURE - II : PNEUMATIC INSTRUMENTS – AIR SUPPLY BRANCHING CONNECTIONS



Notes:

1. All connections shall be taken from the top of the header. Branch connections from headers/ sub-headers shall be with SS 316 instrument isolation valves for air tubing to individual instruments. Other details shall be as per FS-3507.
2. ANSI Class# ratings for air piping and pipe fittings on the air headers shall be as per the Piping Specification, including that of the first block valve and swage/ pipe nipples.

ANNEXURE - III : LABEL DETAILS FOR INSTRUMENTS/ SYSTEMS HANDLING H_2S



09 ANNEXURE - IV : FORM-A: TECHNICAL QUESTIONNAIRE

TECHNICAL QUESTIONNAIRE (FORM-A)		
Project:	**Doc. No.:**	
This questionnaire shall be completely filled in and submitted to the Contractor by the Vendor. Vendor shall sign/ date and affix his seal to this form.		
A) SCOPE OF SUPPLY	Indicate answers by ✓ in the appropriate ☐ boxes.	
(I) Is the vendor in total agreement with his scope of supply as per Contractor's Material Requisition (MR)?	☐ **YES**	☐ **NO**
(II) If NOT in total agreement, indicate in which of the following deviation is required:		
a) Data Sheets	☐ **YES**	☐ **NO**
b) Specifications	☐ **YES**	☐ **NO**
c) Contractor's Vendor Data Requirement Schedule (VDRS)	☐ **YES**	☐ **NO**
d) QA/ QC requirements (Inspection & Testing Requirement)	☐ **YES**	☐ **NO**
If answer is 'YES' for any of the above in A (II), exact deviation required shall be described in detail indicating the appropriate clause(s) in attached Form-B. If no deviation is indicated, then vendor to specify NIL in Form-B and it will be assumed that all the requirements of the scope shall be complied with and no deviations whatsoever shall be accepted after the placement of order. Vendor shall submit filled in Form-B, duly signed & stamped.		
1. Whether total system supply is in Vendor's scope, are there any items vendor wishes to add to achieve total system completion in all respects? (If 'Yes' details to be given)	☐ **YES**	☐ **NO**
2. Does the bid include		
a) Commissioning spares	☐ **YES**	☐ **NO** ☐ **NA**
b) Maintenance/ operational spares (1 year)	☐ **YES**	☐ **NO** ☐ **NA**
d) Special tools & Tackles	☐ **YES**	☐ **NO** ☐ **NA**
e) Construction spares & consumables	☐ **YES**	☐ **NO** ☐ **NA**
3. Others:		
a) Are Mounting accessories, if any, included in bid?	☐ **YES**	☐ **NO** ☐ **NA**
b) Are Special installation procedures, if any, included in bid?	☐ **YES**	☐ **NO** ☐ **NA**
c) Is Supervision of installation in Vendor scope?	☐ **YES**	☐ **NO** ☐ **NA**
d) Is Installation in Vendor scope?	☐ **YES**	☐ **NO** ☐ **NA**
e) Is Supervision of commissioning in Vendor scope?	☐ **YES**	☐ **NO** ☐ **NA**
f) Is Commissioning in Vendor scope?	☐ **YES**	☐ **NO** ☐ **NA**
4. Does the bid include:		
a) Power requirement	☐ **YES**	☐ **NO** ☐ **NA**

b) Special Cable requirement	☐ YES ☐ NO ☐ NA
c) Special Calibration equipment	☐ YES ☐ NO ☐ NA
(Extra charges, if any, for the above to be indicated in the bid)	

B) INSPECTION/ CERTIFICATE

1. Has Vendor indicated in bid detailed inspection procedures being carried out by him?	☐ YES ☐ NO ☐ NA
2. Any special inspection to be carried out other than those required as per bid?	☐ YES ☐ NO ☐ NA
3. Can Vendor furnish certificate from competent authorities as indicated below after order placement:	☐ YES ☐ NO ☐ NA

SNo.	CERTIFICATION FOR	CODE No.	NAME OF CERT. AUTHORITY	
a)	Explosion proof housing			☐ YES ☐ NO ☐ NA
b)	Intrinsically Safe			☐ YES ☐ NO ☐ NA
c)	Weatherproof housing			☐ YES ☐ NO ☐ NA
d)	Indian Boiler Regulation approval			☐ YES ☐ NO ☐ NA
e)	Fire Safe Valves			☐ YES ☐ NO ☐ NA
f)	Flame Retardant Cables			☐ YES ☐ NO ☐ NA
g)	Fire Resistant Cables			☐ YES ☐ NO ☐ NA
h)	Others (please specify)			☐ YES ☐ NO ☐ NA

C) DOCUMENTATION (VENDOR OFFER)

Referring to Contractor's Vendor Document Requirement Schedule (VDRS), Vendor offer shall include the following:

1. Cover sheet indicating doc. number, revision number, equipment/ package description and Vendor's name with date & details.	☐ YES
2. Scope of supply sheet with sub-vendor list	☐ YES
3. Filled in Form-A "Technical Questionnaire", duly signed & stamped by Vendor	☐ YES
4. Filled in Form-B "Deviation List", duly signed & stamped by Vendor	☐ YES
5. Filled in Form-C "List of Commissioning Spares", duly signed & stamped by Vendor	☐ YES
6. Filled in Form-D "List of One Year Operational Spares", duly signed & stamped by Vendor.	☐ YES
7. Filled in Form-E "List of Special Tools & Tackles", duly signed & stamped by Vendor.	☐ YES
8. Filled in Form-F "List of Construction spares & Consumables", duly signed & stamped by Vendor.	☐ YES
9. Filled in Form-G "Vendor weight control data sheet", duly signed & stamped by Vendor	☐ YES

10. Catalogue /Leaflets/ Technical literature for the offered Instrument/ System	☐ **YES**
11. Filled in data sheets, duly signed & stamped by Vendor.	☐ **YES**
12. Typical GA drawing showing overall dimension, C.O.G & weight details for the Instrument/ System.	☐ **YES**
13. Other documents/ drawings shall as per Contractor's VDRS	☐ **YES**
D) VENDOR TO CONFIRM THE COMPLIANCE TO THE FOLLOWING SPECIFICATION/ DESIGN DOCUMENTS	
1. CONTRACTOR'S DATASHEET FOR APPLICABLE INSTRUMENT/ SYSTEM	☐ **YES**
2. APPLICABLE FUNCTIONAL SPECIFICATIONS	☐ **YES**
3. INSTRUMENT PROCESS DATASHEETS	☐ **YES**
4. INSTRUMENTATION DESIGN CRITERIA	☐ **YES**
5. PIPING MATERIAL SPECIFICATION	☐ **YES**
6. FUNCTIONAL SPECIFICATION FOR PROTECTIVE COATING	☐ **YES**

VENDOR'S SEAL/ STAMP **VENDOR'S SIGNATURE & DATE**

09 ANNEXURE - V : FORM-B - COMPLIANCE TO BID SPECIFICATIONS

COMPLIANCE TO BID SPECIFICATIONS (FORM-B)

Project:	**Doc. No.:**

SNo.	Document Clause No.	Requirement	Vendor's Deviation With Reasons

Notes:

1. Vendor should fill in this form, the deviations, if any, to the Bid requirements. In case of no deviation, Vendor should fill "NIL" in the 'Deviation' column.
2. Deviations, not specified in this Form-B, but indicated elsewhere in the Vendor documents shall not be considered for review/ approval and it will be presumed that Vendor will comply all requirements stipulated in Bid.
3. Where minor deviations persist, List of Clarifications shall be submitted for approval.

VENDOR'S SIGNATURE, SEAL & DATE **DEC'S SIGNATURE, SEAL & DATE**

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - VI : FORM-C – LIST OF COMMISSIONING SPARES

LIST OF COMMISSIONING SPARES (FORM-C)

Project:	**Doc. No.:**

SNo.	Part Number	Description	Quantity	Unit price *	Total price *

***Currency :**

Note:

Vendor should state in this form a list of commissioning spares, which are required for the purpose of commissioning of the offered Instrument/ System. If commissioning spares are not required, then Vendor should fill "NIL" in the respective column.

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

ANNEXURE - VII : FORM-D – LIST OF ONE YEAR OPERATIONAL SPARES

09

LIST OF ONE YEAR OPERATIONAL SPARES (FORM-D)

Project:	**Doc. No.:**

SNo.	Part Number	Description	Quantity	Unit price *	Total price *

***Currency :**

Note:

Vendor should state in this form a list of ONE YEAR operational spares, which are essential for one year operation/ maintenance of the offered Instrument/ System. If operational spares are not required, then Vendor should fill "NIL" in the respective column.

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - VIII : FORM-E – LIST OF SPECIAL TOOLS & TACKLES

LIST OF SPECIAL TOOLS & TACKLES (FORM-E)

Project:	**Doc. No.:**

SNo.	Item Description	Quantity	Unit Rate *	Total price *	Remarks

***Currency:**

Note:

Vendor should state in this form, list of special Tools and Tackles which are essential for the regular operation and maintenance of the offered Instrument/ System. If NO such special tools & tackles are required, then vendor should fill in "NIL" in the respective column.

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - IX : FORM-F – LIST OF CONSTRUCTION SPARES & CONSUMABLES

LIST OF CONSTRUCTION SPARES & CONSUMABLES (FORM-F)

Project:	**Doc. No.:**

SNo.	Item Description	Quantity	Unit Rate *	Total price *	Remarks

***Currency:**

Notes:

1. Any loose items and/ or equipment, such as gaskets, gland packing, seals etc. used during final testing but needs to be changed before commissioning shall be properly identified, packed, labelled and supplied along with main equipment.
2. Vendor shall supply lubricants, chemicals and such consumables along with main equipment.
3. Calibration aids, gases etc. that are in Vendor's scope of supply shall also be supplied along with main equipment.
4. Spares/ consumables as specified in Note 1, 2 & 3, which are required for the offered equipment/ package, shall be listed in this form by vendor. If NO such spares/ consumables, etc., are required for the offered equipment/ package design, then vendor should fill in "NIL" in the respective column.

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - X : FORM-G – VENDOR WEIGHT CONTROL DATA SHEET

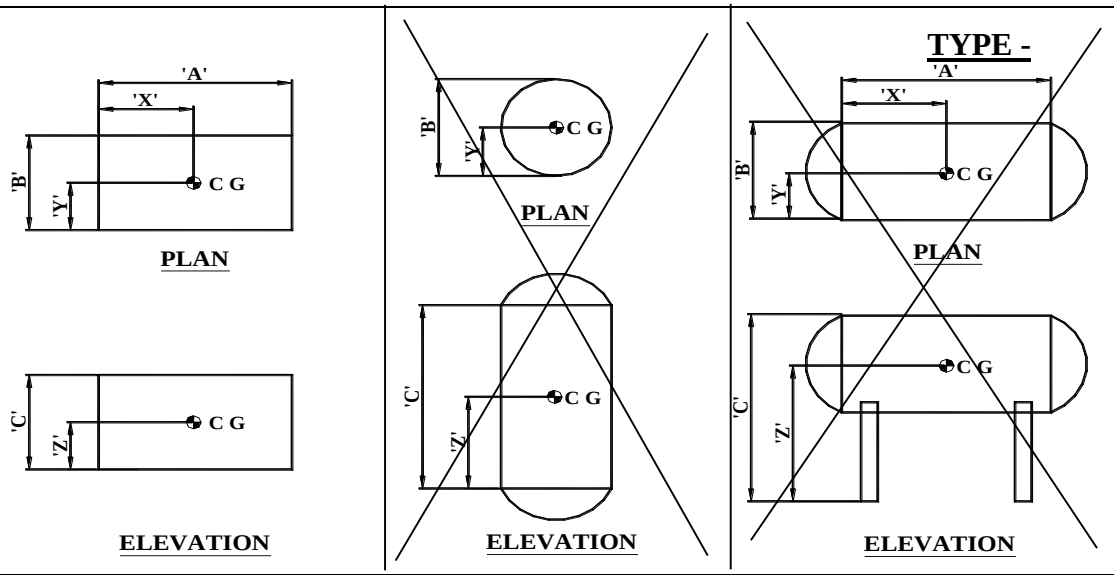
VENDOR WEIGHT CONTROL DATA SHEET (FORM-G)

Project:	**Doc. No.:**

EQUIPMENT DESCRIPTION:

FILL APPROPRIATE CIRCLE
TOLERANCE CODE : ● ● PRELIMINARY ○ DESIGN (CALCULATED) ○ M.T.O WEIGHTED ○ TOLERANCE ±

EQUIPMENT ARRANGEMENT TYPES



ITEM / TAG No.	TYPE (I / II / III)	* WEIGHTS (kgs)			* DIMENSION (in mm)			* C.O.G (Dry / Opt.) (In mm)		
		DRY	OPT.	TEST	A	B	C	X	Y	Z

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - XI : FORM-H – EQUIPMENT WEIGHT CERTIFICATE

EQUIPMENT WEIGHT CERTIFICATE (FORM-H)		
Project:	**Doc. No.:**	
VENDOR	**CONTACT :** **TEL. No.**	**REF. DWG. No.**
METHOD OF WEIGHING: (e.g. Scales, Load cells etc.) **Vendor to prescribe method used.**		
	DATE OF LAST CALIBRATION	**SPECIFIED ACCURACY REQUIREMENT** **± %**
	NOTE:-	
RESULT OF WEIGHING **TOTAL EQUIPMENT DRY WEIGHT** **(Excluding packing, temporary protection etc.)**		**METRIC TONNES**
ALLOCATED WEIGHT **(Weight estimate agreed by Contractor and Vendor).**		**METRIC TONNES**
REASONS FOR VARIATION BETWEEN ALLOCATED WEIGHT AND CERTIFIED WEIGHT		

WEIGHING ADDRESS	WITNESSED BY	
	FOR VENDOR	**FOR CONTRACTOR**
	Representative	**Representative**
Date:	**Signature**	**Signature**

09 ANNEXURE - XII : FORM-I – VENDOR DCI/ MCI

VENDOR DCI/ MCI (FORM-I)

Project:			**Doc. No.:**			
SNo.	**DRAWING/ DOCUMENT No.**	**DESCRIPTION**	**SCH. / ACT.**	**REV x**	**REV x**	**REMARKS**
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			
			SCH.			
			ACT.			

VENDOR'S SIGNATURE, SEAL & DATE

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - XIII : FORM-J – VENDOR DOCUMENT REVIEW

VENDOR DOCUMENT REVIEW (FORM-J)	
Project:	**Doc. No.:**
CONTRACTOR'S NAME	
DEC NAME	

APPROVAL STATUS

[ ] A APPROVED

[ ] B APPROVED WITH COMMENTS, REVISE AND RESUBMIT FOR APPROVAL. WORK MAY PROCEED SUBJECT TO INCORPORATION OF CHANGES INDICATED.

[ ] C NOT APPROVED, REVISE AND RESUBMIT FOR APPROVAL.

[ ] D RECEIVED FOR INFORMATION ONLY

Permission to proceed does not constitute acceptance of design and calculations, test methods or materials developed or selected by Vendor and does not relieve Vendor from full compliance with contractual or other obligations nor detract from any of Contractor's/ Company's rights.

CONTRACTOR'S SIGNATURE, SEAL & DATE

09 ANNEXURE - XIV : FORM-K – UTILITIES REQUIREMENT

UTILITIES REQUIREMENT (FORM-K)

Project:	**Doc. No.:**

The Vendor should state in this form, list of utilities required for the normal operation of the system covered in the requisition.

Utilities	Available	Required by Vendor
Electric Power (Non-UPS) AC:		
DC:		
For Instruments		
Electric Power (UPS) AC:		
DC:		
For Instruments		
Instrument Air Pressure – Min / Nor / Max / Design Temperature – Min / Nor / Max / Design Consumption	Unit Kg/cm^2g °C	
Utility Air Pressure – Min / Nor / Max / Design Temperature – Min / Nor / Max / Design Consumption	Unit Kg/cm^2g °C	
Water Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	
Steam Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	
Fuel Gas Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	
Fuel Oil Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	
Inert Gas Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	
Cooling Water Pressure – Operating / Design Temperature – Operating / Design Consumption	Unit Kg/cm^2g °C	

VENDOR'S SIGNATURE, SEAL & DATE **CONTRACTOR'S SIGNATURE, SEAL & DATE**

09 ANNEXURE - XV : RESPONSIBILITY/ WORK SCOPE MATRIX

WORK SCOPE MATRIX	
Project:	**Doc. No.:**

The Responsibility/ Work Scope Matrix shall be filled by LSTK Contractor to indicate responsibilities of DEC, Vendor/ Sub-vendor and LSTK Contractor.

S/ N	Description/ Activity	Responsibility			Remarks
		Contractor	DEC	Vendor	
1.	Design & Engineering				
2.	Manufacturing				
3.	Inspection				
4.	Testing				
5.	FAT & SAT				
6.	Installation				
7.	Installation Assistance				
8.	Commissioning				
9.	Commissioning Assistance				
10.	Training				
11.	Documentation				
CONTRACTOR'S SIGNATURE, SEAL & DATE		**DEC'S SIGNATURE, SEAL & DATE**		**VENDOR'S SIGNATURE, SEAL & DATE**	

ANNEXURE - XVI : INSTRUMENT PROCESS CONNECTION : 1 of 2

SNo.	INSTRUMENT TYPE	VESSEL/ STAND PIPE CONNECTION	FIRST BLOCK VALVE	INSTRUMENT CONNECTION
1	EXTERNAL DISPLACER LEVEL INSTRUMENT ON VESSEL	2" FLGD	2" FLGD	2" FLGD
2	EXTERNAL DISPLACER LEVEL INSTRUMENT ON STANDPIPE	2" FLGD	2" FLGD	2" FLGD
3	INTERNAL DISPLACER LEVEL INSTRUMENT	4" FLGD	-	4" FLGD
4	EXTERNALBALL FLOAT LEVEL INSTRUMENT ON VESSEL	2" FLGD	2" FLGD	1" SW
5	EXTERNAL BALL FLOAT LEVEL INSTRUMENT ON STANDPIPE	1" SW/ FLGD	1" SW/ FLGD	1" SW
6	INTERNAL BALL FLOAT LEVEL INSTRUMENT	4" FLGD	-	4" FLGD
7	LEVEL GAUGE ON VESSEL	2" FLGD	2" FLGD	¾" SCRD
8	LEVEL GAUGE ON STANDPIPE	¾" SW/ FLGD*	¾" SW/ FLGD	¾" SCRD
9	D.P INSTRUMENT ON VESSEL	1½" FLGD	1½" FLGD	½" SCRD
10	D.P INSTRUMENT ON STAND PIPE	¾" SW/ FLGD*	¾" SW/ FLGD	½" SCRD
11	DIAPHRAGM SEAL D.P INSTRUMENT ON VESSEL	3" FLGD	3" FLGD	3" FLGD
12	EXTENDED D.P INSTRUMENT ON VESSEL	4" FLGD	-	4" FLGD
13	D.P TUBE LEVEL INSTRUMENT	1½" FLGD	½" SW (BY INST)	½" SCRD
14	TANK LEVEL INSTRUMENT (MECH.), [PRESSURIZED]	1½" FLGD	1½" FLNGD (BY INST)	1½" SCRD
15	TANK LEVEL INSTRUMENT (SERVO), [PRESSURIZED]	6" FLGD	6" FLGD (BY INST)	6" FLGD
16	PRESSURE INSTRUMENTS ON VESSEL	1½" FLGD	1½" FLGD	½" SCRD
17	DIAPHRAGM SEAL PRESSURE INSTRUMENT ON VESSEL (SCRD)	1½" FLGD	1½" FLGD	½" SCRD
18	DIAPHRAGM SEAL PRESSURE INSTRUMENT ON VESSEL (FLGD)	1½" FLGD	1½" FLGD	1½" FLGD
19	THERMOWELL	1½" FLGD	-	1½" FLGD
20	STANDPIPE	2" FLGD MIN.	-	2" FLGD
21	TANK LEVEL INSTRUMENT (MECH.), [ATMOSPHERIC]	1½" FLGD	-	1½" SCRD
22	TANK LEVEL INSTRUMENT (SERVO), [ATMOSPHERIC]	6" FLGD	-	6" FLGD

NOTES:

1. **SCRD: SCREWED; FLGD: FLANGED; SW: SOCKET WELD**
2. FOR ANY OTHER INSTRUMENT NOT REFERRED ABOVE, THE CONNECTION DETAILS SHALL BE AS PER INDIVIDUAL REQUIREMENT.

3. ALL FLANGE/ SW RATING SHALL BE AS PER PIPING SPECIFICATION.
4. IN CASE OF DIRECT MOUNTED FLANGED INSTRUMENTS AND WHERE FLANGED FIRST ISOLATION VALVES ARE PROVIDED, BOLTING AND GASKET SHALL BE IN PIPING SCOPE.
5. * MEANS AS PER PIPING SPECIFICATIONS.

ANNEXURE - XVI : INSTRUMENT PROCESS CONNECTION : 2 of 2

SNo	TYPE OF INSTRUMENTS	WHERE PIPING CLASS PRESCRIBES SCREWED CONNECTIONS			WHERE PIPING CLASS PRESCRIBES SOCKET WELD CONNECTIONS			WHERE PIPING CLASS PRESCRIBES FLANGED CONNECTIONS		
		Process Connection	1st Block Valve	Instrument Connection	Process Connection	1st Block Valve	Instrument Connection	Process Connection	1st Block Valve	Instrument Connection
1	ORIFICE FLOW METER	½" SCRD	½" SCRD	½" SCRD	½" SCRD *	½" SW	½" SCRD	½" SCRD *	½" FLGD	½" SCRD
2	ANNUBAR	1½" FLGD	1½" FLGD (BY INSTT)	1½" FLGD 4	1½" FLGD	1½" FLGD (BY INSTT)	1½" FLGD 4	1½" FLGD	1½" FLGD (BY INSTT)	1½" FLGD 4
3	PRESSURE INSTRUMENT	¾" SCRD	¾" SCRD	½" SCRD	¾" SW	¾" SW	½" SCRD	¾" FLGD	¾" FLGD	½" SCRD
4	DIAPHRAGM SEAL PRESSURE INSTRUMENT (SCRD)	¾" SCRD	¾" SCRD	½" SCRD	¾" SW	¾" SW	½" SCRD	¾" FLGD	¾" FLGD	½" SCRD
5	DIAPHRAGM SEAL PRESSURE INSTRUMENT (FLGD)	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD	1½" FLGD
6	THERMOWELL	1½" FLGD	-	1½" FLGD	1½" FLGD	-	1½" FLGD	1½" FLGD	-	1½" FLGD

NOTES:

1. **SCRD: SCREWED; FLGD: FLANGED; SW: SOCKET WELD**
2. FOR ANY OTHER INSTRUMENT NOT REFERRED ABOVE, THE CONNECTION DETAILS SHALL BE AS PER INDIVIDUAL REQUIREMENT.
3. ALL FLANGE/ SW RATING SHALL BE AS PER PIPING SPECIFICATION.
4. IN CASE OF DIRECT MOUNTED FLANGED INSTRUMENTS AND WHERE FLANGED FIRST ISOLATION VALVES ARE PROVIDED, BOLTING AND GASKET SHALL BE IN PIPING SCOPE.
5. INSTALLATION OF ALL INLINE INSTRUMENTS SHALL BE IN PIPING SCOPE.
6. * MEANS SEAL WELDING REQUIRED
7. $ MEANS CONNECTIONS FOR DP INSTRUMENTS IS ½ "SCRD.

ADDENDUM TO FUNCTIONAL SPECIFICATION FOR WELDING & NDT

CSR	HN	TMK	SC	2	28.07.2025	Addendum to 2009F Rev 8
Prepared By	Peer-Reviewed By	Reviewed By	Approved By	No. of Pages	Date	Rev.

AMENDMENTS/ADDITIONS

Following Amendment(s)/additional requirement(s) shall be applicable:

PL4.6 Production Welding

For pre-heating of line pipe in offshore installation barge, it is recommended that pre-heating manually by torch is to be avoided. Instead, it is recommended that suitable methods such as electrical induction or electrical resistance heating etc. are to be used.

हौज एवं हौज कपलिंग के लिए तकनीकी टिप्पणियाँ

TECHNICAL NOTES FOR HOSES & HOSE COUPLINGS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
3	28.03.23	REVISED & ISSUED AS STANDARD SPECIFICATION	PK	SH	SB	SM
2	30.03.18	REVISED & ISSUED AS STANDARD SPECIFICATION	PK	SH	MI	RN
1	28.03.13	REVISED & ISSUED AS STANDARD SPECIFICATION	UK	SH	RN ATD	DM
0	29.06.09	ISSUED AS STANDARD SPECIFICATION	PK	RN	SC	ND

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Abbreviations:

AARH	:	Arithmetic Average Roughness Height
ASME	:	American Society of Mechanical Engineers
CS	:	Carbon Steel
eDMS	:	Electronic Document Management System
MR	:	Material Requisition
PMI	:	Positive Material Identification

Piping Standards Committee

Convenor: Mr. G. Balaji

Members:
Mr. Udayan Chakravarty
Mr. R. Giridhar
Mr. K Anjaneyulu (SMED)
Mr. Rajesh Chitara (SMMS)
Mr. SC Maity (Structural)
Mr. Harsh Vardhan (Projects)
Mr. Pankaj Kumar Rai (Construction)

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2.0 TECHNICAL REQUIREMENTS 4
3.0 DOCUMENTATION 4
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7.0 ATTACHMANTS FOR MR 5
8.0 REFERENCES 5

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1.0 PURPOSE & SCOPE

These Technical Notes shall form part of Material Requisition as well as the Purchase Requisition for Hoses & Hose couplings. These shall also serve as reference for Materials Engineer and Inspection. These notes cover the technical requirements, documentation, inspection, testing, marking and dispatch requirements for hoses and hose couplings. Supplier shall strictly comply with MR/PR stipulations and no deviations shall be permitted.

All codes and standards for manufacture, testing, inspection etc. shall be of latest editions as on issue date of RFQ.

2.0 TECHNICAL REQUIREMENTS

2.1 Hose material and Ends thereof shall be as per Material Requisition (MR).

2.1.1 The material composition, physical properties, heat treatment, mandatory tests, dimensions and tolerances and marking shall conform to the applicable codes/standards/specifications.

2.2 Hoses shall be suitable for outdoor installation in an atmosphere containing hydrocarbon vapour, steam etc. The outer cover shall be resistant to ageing, abrasion & weather effects etc.

2.3 Flanges for flanged hoses shall be in accordance with ASME B16.5 unless specified otherwise in the requisition. Flange finish shall be as specified in the requisition. The interpretation for range of face finish shall be as follows:

Serrated/Smooth/125AARH: 125 to 250 micro inch AARH.

2.4 Hose end connections shall be supplied as per MR.

2.5 CS Fittings (hose couplings, clamps, claws, nozzles etc.) shall be cadmium chromium Plated.

2.6 Manufacturer shall guarantee suitability of hoses for the service and working conditions (design pressure, design temperature) as specified in Data sheet of hoses and hose couplings.

3.0 DOCUMENTATION

3.1 No documents shall be submitted along with the offer. However, vendor shall submit the following documents for records through EIL eDMS:

- Duly signed and stamped data sheet for hoses & hose couplings.
- Vendor's catalogue/ drawings showing detailed dimensions along with material for different parts.
- Material test certificates as applicable; test reports of mandatory tests, hydrotest and special tests as per applicable specifications and standards.

3.2 In addition to submissions through eDMS, the following documents shall be submitted in hard copies (6 sets) and soft copies (2 CDs/DVDs) along with the delivery of respective items:

- Vendor's catalogue/ drawings showing detailed dimensions along with material for different parts.

4.0 TESTING & INSPECTION

4.1 All hoses shall be tested at 2 times the design Pressure along with end connections.

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4.2 Steam resistance test, vacuum test and electrical continuity test shall be carried out as applicable in Clauses 4.2.1 & 4.2.2 and shall fulfill the requirements of the applicable design codes mentioned in MR. In cases where these are not specified, tests shall be as per specifications followed by the manufacturer.

4.2.1 Electrical continuity test shall be conducted on all individual non-metallic hoses.

4.2.2 Vacuum test shall be conducted on all individual metallic and rubber hoses intended for steam service. For rubber hoses intended for steam service, steam resistance test shall also be conducted.

4.3 Flanges shall be so aligned so that the deviations measured in any direction do not exceed 0.5°.

4.4 The length of the finished hose assembly shall not differ from the specified length by more than ±1%. Hose assembly shall be measured after being subjected to the various tests. Tolerances shall be as per specified design code.

4.5 Positive Material Identification (PMI) shall be performed as per the scope and procedures as defined in the 'Specification for Positive Material Identification (PMI) at Supplier's Works (No. 6-81-0001).

4.6 Refer Specification no 6-81-0135 for 'Inspection & Test Plan for Hose Pipes and Couplings'.

5.0 MARKING & DESPATCH

5.1 All hoses along with their couplings shall be dry, clean and free from moisture, dirt and loose foreign material of any kind. All hoses shall be suitably packed and protected to avoid damage during transportation, shipment and storage.

5.2 Each hose shall have a weatherproof tag attached with a corrosion resistant metal wire. The weatherproof tag shall be embossed with item code, size of hose and length of hose. In addition, all hoses shall be clearly marked with service and maximum working pressure at both ends.

6.0 ATTACHMENTS

6-44-0064-A1 : Sketches for hoses and hose couplings (End details)

7.0 ATTACHMENTS FOR MR

Format No.3-1643-0095: Data sheet for hoses and hose couplings

8.0 REFERENCES

6-81-0001	Specification for Positive Material Identification (PMI) at Supplier's Works
6-81-0135	Inspection & Test Plan for Hose Pipes and Couplings
6-78-0001	Specification for Quality Management System Requirements from Bidders
6-78-0003:	Specification for documentation requirements from suppliers

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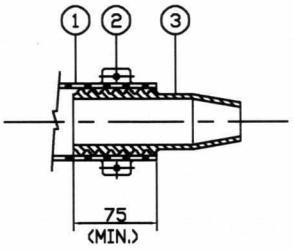
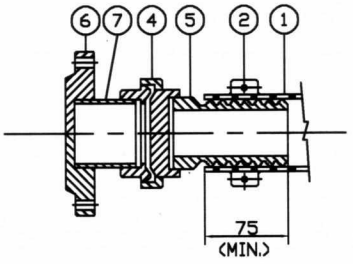
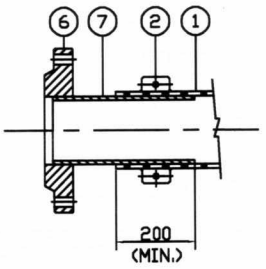


TYPE-1

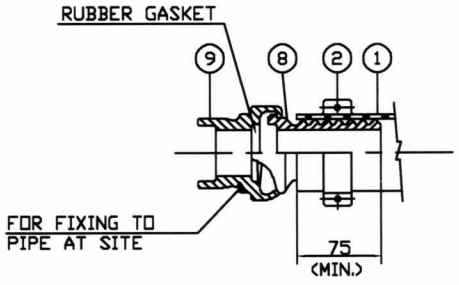


TYPE-2

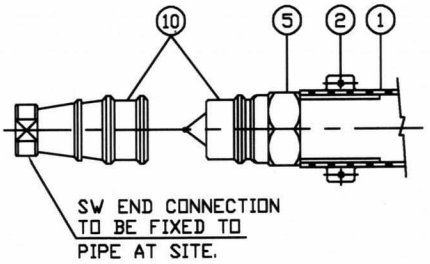


TYPE-3A & 3B

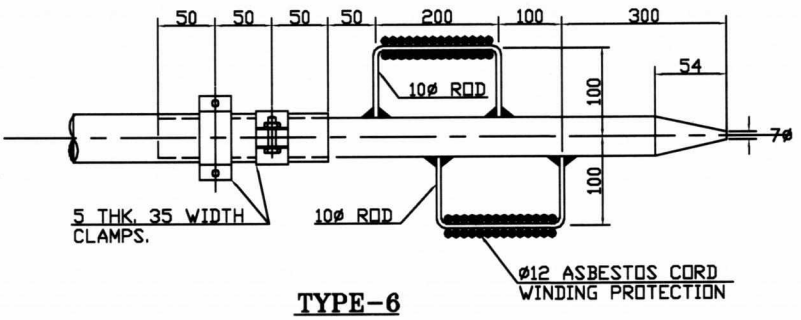
NOTE: FOR TYPE 3B HOSE TO BE FULLY VULCANIZED OVER NIPPLE & NO CLAMP



TYPE-4



TYPE-5



TYPE-6

(FOR STEAM HOSE CONNECTION)
PIPE OD 33.4 & 4.5 THK.

ITEM No.	DESCRIPTION
1	HOSE PIPE AS SPECIFIED
2	HOSE PIPE CLAMP WITH BOLTS MINIMUM 6mm THK. STEEL PLATE (IS2062) IN TWO PARTS BOLT/ NUTS A307 GR'B'
3	PIPE NIPPLE MATERIAL AS SPECIFIED (SEAMLESS)
4	3000 LB SW UNION, ENDS TO SUIT ITEM ⑤ AND ⑦. MATERIAL AS SPECIFIED
5	SEAMLESS PIPE NIPPLE WITH SERRATIONS TO SUIT ITEMS ① AND ④. MATERIAL AS SPECIFIED
6	END FLANGE MATERIAL AND DETAIL AS SPECIFIED
7	PIPE MATERIAL AS SPECIFIED WITH SERRATIONS
8	QUICK CLAW COUPLING (HOSE END) MATERIAL AS SPECIFIED BOTH COUPLINGS TO BE SUPPLIED IN TYPE 4
9	QUICK CLAW COUPLING WITH SOCKET WELD PIPE END. MATERIAL AS SPECIFIED
10	QUICK ACTING, QUICK RELEASE COUPLER/ ADAPTOR ASSEMBLY WITH BUILT IN CHECK VALVE BOTH COUPLER AND ADAPTOR SHALL BE SUPPLIED IN TYPE 5.

NOTE:-

ALL DIMENSIONS ARE IN MM.

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